PART 5

Written paper: 800 single best answers

Twenty-four blocks weighted to the blueprint. Work a block under timed conditions at 80 seconds a question, then mark it from Part 8.

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PART 5 — WRITTEN PAPER: 800 SINGLE BEST ANSWER QUESTIONS

How to use this bank. Work through a block at a time under timed conditions — allow **80 seconds per question**, which is the real pace of the written papers. **The answers and explanations for all 800 questions are at the back of this notebook, in Part 8**, together with a complete answer-key grid for rapid marking. Resist the temptation to look up each answer as you go: mark a whole block, then read the explanations for every question in it, including the ones you got right.

What this bank is pitched at. These questions are written to a standard comparable with the American Board of Oral Implantology / Implant Dentistry Part I written paper, and calibrated so that an experienced general dentist with a special interest in implants, who has revised properly, should expect to score somewhere around **55 per cent** on a first pass. If you are scoring in the fifties, you are where a passing candidate sits. If you are scoring in the thirties, the explanations are the revision.

Three features of the construction are worth knowing before you start, because they change how you should answer.

The five options are deliberately of equal weight. There is no long, hedged, obviously-correct option to fall back on, and no absurd option to eliminate for free. Most distractors are **true statements that do not answer the question asked**, or the right answer to a slightly different question, or the answer a competent clinician would give if they missed one datum in the stem. Read the stem twice and find the datum that decides it.

Some questions have more than one right item. Where it is honest, options take the form *A and B only* or *A, B and C* — and where a stem lists propositions (i), (ii) and (iii), the options combine them. About one question in four is built this way. The combination is the key roughly half the time it appears, so it can be neither assumed nor discounted; when a combination is wrong it usually fails on exactly one item, and the explanation names which.

The stems carry numbers, and the numbers decide. Ages, doses, HbA1c, INR, pack-years, millimetres of bone, insertion torque, ISQ, months since extraction, millimetres of bone loss. A large minority of questions turn on **anatomy, physiology, pharmacology or materials science** examined through a clinical decision rather than as free-standing recall — and a good number turn on a single value crossing a published threshold. Knowing the list of risk factors is not enough; you have to know the number.

Blueprint weighting mirrors the real paper: procedures, planning and complications carry the most questions.

Block	Questions	Blueprint area
A	1–28	History, risk assessment and medical considerations
B	29–48	Examination, occlusion and space
C	49–70	Investigations, imaging, guides and the wax-up
D	71–86	Surgical anatomy
E	87–106	Osseointegration, bone biology and healing
F	107–132	Materials, implant design and surfaces
G	133–160	Treatment planning, SAC, options, consent and referral
H	161–180	Prescribing, local anaesthesia, drug interactions, sedation
I	181–218	Surgical placement, flaps, timing, loading, suturing, asepsis
J	219–246	Minor hard and soft tissue augmentation
K	247–280	Restoring implants, occlusion and the laboratory
L	281–300	Complications, peri-implant disease, maintenance and governance
M	301–340	History, risk and medical considerations — extended
N	341–375	Examination, occlusion and space — extended

Block	Questions	Blueprint area
O	376–410	Investigations, imaging, guides and the wax-up — extended
P	411–440	Surgical anatomy — extended
Q	441–475	Osseointegration, bone biology and healing — extended
R	476–515	Materials, implant design and surfaces — extended
S	516–560	Planning, SAC, consent and referral — extended
T	561–590	Prescribing, local anaesthesia and sedation — extended
U	591–650	Surgical placement — extended
V	651–700	Minor augmentation — extended
W	701–760	Restoring implants, occlusion and the laboratory — extended
X	761–800	Complications, peri-implant disease, maintenance and governance — extended

Block A — History, risk assessment and medical considerations (Q1–28)

1 A 47-year-old man completed treatment for generalised stage III grade B periodontitis four years ago and attends six-monthly. His full-mouth plaque score is 18 per cent and bleeding on probing 22 per cent, but two upper molars retain 6 mm pockets that bleed. Radiographs show stable horizontal bone loss of about 40 per cent. He requests an implant at the LL6. Which finding most increases his risk of peri-implantitis?

A. The residual 6 mm pockets that bleed on probing
B. The full-mouth bleeding score of 22 per cent
C. The full-mouth plaque score of 18 per cent
D. The 40 per cent radiographic bone loss
E. The grade B rate of progression

2 A 66-year-old woman needs an implant at the LR5. She has taken alendronate 70 mg weekly for three years for osteoporosis and prednisolone 10 mg daily for two years for polymyalgia rheumatica. She has never smoked, has no diabetes and has had two uneventful extractions since starting the alendronate. Which feature of her history places her in the higher-risk group for MRONJ?

A. The three-year duration of alendronate therapy
B. The concurrent long-term prednisolone
C. The two previous extractions during therapy
D. The weekly rather than daily bisphosphonate regimen
E. Her age at the time of surgery

3 A 63-year-old man with multiple myeloma has received intravenous zoledronic acid monthly for eighteen months. He has lost the LL6 and LL7 and asks for implants rather than a partial denture. His oral hygiene is good, he does not smoke, and his haematologist confirms his disease is in plateau phase. What is the most appropriate response?

A. Decline implant treatment and plan a removable prosthesis
B. Refer to an oral and maxillofacial unit for implant placement
C. Place implants with antibiotic cover and a flapless technique
D. Await a three-month interruption of zoledronic acid, then place
E. Place a single implant at the LL6 to reduce surgical burden

4 A 72-year-old man on warfarin for atrial fibrillation is due for a single implant at the LR4 next Tuesday. His anticoagulation has been stable for two years with no dose changes. His INR was 3.8 when measured at the anticoagulant clinic four days ago. He takes no antiplatelet drug. What is the most appropriate course?

A. Proceed as planned, since the INR is below 4.0
B. Postpone until the INR has been reduced below 3.0
C. Omit warfarin for two days before surgery
D. Repeat the INR within 72 hours of surgery and decide then
E. Arrange bridging with low molecular weight heparin

5 A 68-year-old woman takes apixaban 5 mg twice daily following a pulmonary embolism. She is booked for placement of two implants in the lower left quadrant with flap elevation, first appointment of the morning. She takes no other drug affecting haemostasis and her renal function is normal. Which is the most appropriate management of her medication?

A. Continue both doses unchanged on the day of surgery
B. Miss the morning dose and take the evening dose as usual
C. Omit apixaban for 48 hours before surgery
D. Check the INR on the morning of surgery and proceed if below 4.0
E. Substitute low molecular weight heparin for three days

6 A 59-year-old man takes clopidogrel 75 mg daily, and no other antiplatelet drug, following a coronary stent inserted two years ago. He is booked for a single implant at the LR5 with a small flap, and a colleague suggests he omit the clopidogrel for 24 hours beforehand to reduce bleeding. His platelet count and clotting screen are normal. Which statement best explains why that suggestion would achieve nothing useful?

A. Clopidogrel inhibits the P2Y12 receptor irreversibly for the platelet's lifespan
B. Clopidogrel is a prodrug with an active metabolite of long half-life
C. Clopidogrel acts on cyclo-oxygenase, so recovery requires new platelets
D. Clopidogrel has an elimination half-life of several days
E. Clopidogrel prolongs the INR, which takes days to normalise

7 A 54-year-old man with a body mass index of 32 attends for implant assessment. He has no diagnosed medical condition and takes no medication. As part of your work-up his general practitioner sends a recent HbA1c of 50 mmol/mol, taken when he reported tiredness. He is asymptomatic otherwise, does not smoke, and his periodontal tissues are healthy. What does this result most appropriately prompt?

A. Referral to his general practitioner, as this is in the diabetic range
B. Reassurance, as this is within the non-diabetic reference range
C. Advice on diet, as this indicates pre-diabetes only
D. Postponement of all treatment until the value falls below 42 mmol/mol
E. Repeat testing after surgery to monitor the effect of healing

8 A 51-year-old woman who smokes 20 cigarettes a day wants an implant at the UR4. The site has a 4 mm buccal dehiscence that will require simultaneous guided bone regeneration with a particulate xenograft and a resorbable membrane. She declines cessation but accepts a reduction to ten a day. Which statement most accurately conveys the effect of her smoking on this plan?

A. Membrane exposure and graft failure are the outcomes most affected
B. The main effect is on implant survival in the native bone
C. Halving consumption removes most of the excess risk
D. Smoking mainly affects healing of the soft tissue incision line
E. The effect is confined to maxillary rather than mandibular sites

9 A 64-year-old man had 60 Gy of radiotherapy to a field including the left mandible for a tonsillar carcinoma three years ago. He is disease-free and has stopped smoking. A panoramic radiograph shows a coarse, patchy trabecular pattern in the left body of the mandible. He asks for an implant to replace the LL6. What is the most appropriate course?

A. Place the implant after a course of hyperbaric oxygen therapy
B. Place the implant with a two-week course of antibiotics
C. Decline permanently, as irradiated bone cannot osseointegrate
D. Refer for specialist assessment within the oncology and restorative team
E. Proceed as normal, since three years have elapsed since treatment

10 A 55 kg woman is having two implants placed under local anaesthesia with 4 per cent articaine with adrenaline 1:100,000, supplied in 2.2 ml cartridges. You intend to work within the recommended maximum of 7 mg/kg of articaine. Disregarding the adrenaline, what is the largest number of whole cartridges you may use during this appointment?

A. Two cartridges
B. Three cartridges
C. Four cartridges
D. Six cartridges
E. Eight cartridges

11 A 58-year-old man takes propranolol 80 mg twice daily for essential tremor. Ten minutes after three cartridges of 2 per cent lidocaine with adrenaline 1:80,000 have been given for placement of an implant at the UL6, he feels unwell; his blood pressure has risen from 132/78 to 186/106 mmHg and his pulse has fallen from 76 to 52 beats per minute. What is the most likely explanation?

A. An adrenaline-induced tachyarrhythmia masked by the beta blocker
B. A vasovagal reaction to the series of injections
C. Unopposed alpha-adrenergic vasoconstriction with reflex bradycardia
D. A direct pressor action of lidocaine on vascular smooth muscle
E. An endogenous catecholamine surge driven by anxiety

12 A 17-year-old girl has a congenitally absent UR2, currently restored with a resin-bonded bridge that she dislikes. Orthodontic treatment finished nine months ago and the space is ideal at 6.5 mm. Her mother wants the implant placed during the school holidays. Which of the following would appropriately inform the decision that facial growth has ceased?

A. Superimposed lateral cephalograms taken 6 to 12 months apart
B. Serial standing height measurements over the same period
C. Attainment of the age of 18 years
D. A and B only
E. A, B and C

13 A 31-year-old woman who is 22 weeks pregnant asks to begin implant treatment for a missing LL5 lost to a fracture. The pregnancy is uncomplicated, she is well, and she has been told the second trimester is the safest time for dental care. She has no pain and is wearing an acrylic partial denture without difficulty. What is the strongest justification for deferring the surgery?

A. The fetal dose from the small-volume CBCT that planning requires
B. Placental transfer of amide local anaesthetic to the fetus
C. The risk of supine hypotensive syndrome during a long appointment
D. The teratogenic potential of the analgesics likely to be prescribed
E. The treatment is elective and can be provided later without detriment

14 A 38-year-old man asks you to replace an implant crown at the UR1 placed three years ago elsewhere. The restoration is well made, the tissues are healthy and the gingival margin sits 0.5 mm coronal to the contralateral tooth, a discrepancy invisible at conversational distance. He has consulted four clinicians, brings his own photographs with measurements, and has made formal complaints about two of them. What is the most appropriate course?

A. Remake the crown with a modified emergence profile
B. Undertake a connective tissue graft to level the margin
C. Decline further intervention and discuss referral for psychological assessment
D. Seek a second opinion from an implant colleague before deciding
E. Proceed after a written agreement that no further work will follow

15 You are consenting a 44-year-old woman, a professional singer, for an implant at the LL5, where the CBCT shows 4 mm of bone above the mandibular canal. You have described the surgery, the healing period, the alternatives and the cost, and she has asked no questions about complications. Which principle should determine the risks you go on to discuss?

A. The risks a responsible body of implant dentists would disclose
B. The risks a reasonable person in her position would find significant
C. The risks occurring more often than one case in a hundred
D. The risks she specifically raises during the consultation
E. The risks listed in the implant manufacturer's documentation

16 A 79-year-old woman with early Alzheimer's disease attends to discuss two implants to retain a lower overdenture. She converses appropriately, knows why she has come, but forgets parts of the discussion when they are revisited later in the appointment. Her daughter says, "She won't remember anyway — just do what's best and I'll sign for her." What should you do first?

A. Accept the daughter's consent as her nearest relative
B. Proceed on a best-interests basis and record the reasoning
C. Ask her general practitioner to provide written consent
D. Assess her capacity for this particular decision at this time
E. Defer treatment until a lasting power of attorney is registered

17 A 56-year-old woman has taken sertraline 100 mg daily for six years for depression that is now stable. She is otherwise healthy, does not smoke, and is planning a single implant at the UR5. Consider these statements: (i) selective serotonin reuptake inhibitor use has been associated with increased implant failure in retrospective studies; (ii) her medication should be stopped before surgery; (iii) SSRIs can prolong bleeding by depleting platelet serotonin. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

18 A 61-year-old man had a myocardial infarction four months ago and received a drug-eluting stent. He takes aspirin 75 mg and ticagrelor 90 mg twice daily, and his cardiologist has told him dual therapy will continue for a year. He is well, walks two miles daily, and wants a single implant at the LR6 as soon as possible. What is the most appropriate course?

A. Proceed now, stopping the ticagrelor for five days beforehand
B. Defer the elective surgery and liaise with his cardiologist
C. Proceed now, relying on local haemostatic measures alone
D. Proceed now with antibiotic prophylaxis against stent infection
E. Defer for a further eight months until antiplatelet therapy ends

19 A 60-year-old woman with a mechanical aortic valve replacement, performed six years ago, is booked for implant placement at the UL4. She is on warfarin with a stable INR and has never had infective endocarditis. Which of the following apply to her antibiotic management under current UK guidance?

A. Routine antibiotic prophylaxis against infective endocarditis is not recommended
B. Chlorhexidine mouthwash should be given as prophylaxis before the procedure
C. Amoxicillin 3 g should be given orally one hour before the procedure
D. A and B only
E. A, B and C

20 A 65-year-old man takes prednisolone 10 mg daily, and has done for two years, for chronic obstructive pulmonary disease. He is listed for placement of a single implant at the LL4 under local anaesthesia, expected to take forty minutes. His blood pressure is 128/76 mmHg and he is systemically well. What is the most appropriate management of his corticosteroid on the day?

A. Double his usual dose on the morning of surgery
B. Give 100 mg of hydrocortisone intravenously before starting
C. Give no supplement, ensuring his usual morning dose is taken
D. Withhold the morning dose to reduce the risk of infection
E. Convert him to intramuscular hydrocortisone for 24 hours

21 A 48-year-old healthy non-smoker is having a single implant placed at the LR5 in native bone, with no augmentation planned and a healed site. She has no penicillin allergy and no cardiac history. You are deciding whether to prescribe antibiotics. Which statement best represents the current evidence for systemic antibiotics in this situation?

A. A single preoperative dose reduces early implant failure
B. A five-day postoperative course reduces early implant failure
C. Antibiotics reduce marginal bone loss at one year
D. Any benefit is confined to sites that have been grafted
E. Antibiotics confer no benefit at any stage of the protocol

22 A 62-year-old woman with rheumatoid arthritis takes methotrexate 15 mg once weekly with folic acid. Following uneventful placement of an implant at the UL5 she asks what she may take for discomfort, and mentions that ibuprofen has always suited her. What is the most important reason for advising her against it?

A. Non-steroidal drugs inhibit prostaglandin-driven bone formation around the implant
B. Non-steroidal drugs reduce renal clearance of methotrexate and raise its toxicity
C. Non-steroidal drugs displace methotrexate from binding sites in synovium
D. Non-steroidal drugs increase gastric ulceration in rheumatoid disease
E. Non-steroidal drugs antagonise the anti-inflammatory action of methotrexate

23 A 44-year-old man is being planned for an implant at the LR6. He has bilateral masseteric hypertrophy, a well-defined linea alba, and wear facets on the lower canines and premolars that his diet does not explain. His partner reports night-time grinding. Bone volume, periodontal condition and hygiene are all good and he does not smoke. Which measure is most likely to protect the restoration?

A. Reducing the buccolingual width of the occlusal table
B. Providing group function on the implant crown
C. Selecting a zirconia rather than titanium abutment
D. Providing a hard full-coverage splint for night-time wear
E. Increasing crown height to distribute the occlusal load

24 A 71-year-old woman has a DEXA T-score of −2.8 at the femoral neck. She has declined all antiresorptive medication and takes only calcium and vitamin D. She has no other medical problems, does not smoke, and wants an implant at the LR6. Consider these statements: (i) skeletal bone mineral density correlates poorly with bone quality at the implant site; (ii) osteoporosis alone is not a contraindication to osseointegration; (iii) the drug history matters more to the plan than the diagnosis. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

25 A 68-year-old man with stage 4 chronic kidney disease attends for implant assessment. He takes eight regular medicines and has been under two hospital teams for a decade. You write to his general practitioner and receive a one-line reply: "This patient is fit for dental treatment." You remain uncertain how his renal disease and his medication affect your plan. What should you do next?

A. Proceed, since written medical clearance has been obtained
B. Ask the patient to describe each of his conditions in turn
C. Write again with specific questions about the issues that affect the plan
D. Refer him to a hospital department for all implant treatment
E. Arrange the relevant blood tests yourself before planning further

26 A 43-year-old woman needs a small buccal augmentation at the UR3 at the time of implant placement. Her religious beliefs prohibit the use of any porcine or bovine derived material, and she asks what her options are. She is otherwise healthy and understands that a second surgical site may be needed. Which of the following may be offered consistently with her wishes?

A. Autogenous bone harvested locally
B. A beta-tricalcium phosphate alloplast
C. Freeze-dried human allograft
D. A and B only
E. All of the above

27 A 58-year-old man returns after five years with two implants at the LL5 and LL6. He has not attended for maintenance in three years, smokes five cigarettes a day, and has never had periodontitis. His plaque score is 62 per cent, both implants bleed on probing with 6 mm pockets, and radiographs show 2.5 mm of bone loss at each. Which feature of his history best explains the deterioration?

A. Smoking five cigarettes a day
B. Three years without supportive maintenance
C. The absence of buccal keratinised mucosa
D. The crowns being cement-retained rather than screw-retained
E. Gingivitis recorded before the implants were placed

28 An 83-year-old man with Parkinson's disease has a resting tremor of both hands and lives alone. He manages a complete upper denture well but cannot control the lower, and asks whether implants would help. He is medically stable on levodopa, attends regularly and is strongly motivated. Consider these statements: (i) his age alone is not a reason to decline; (ii) his ability to clean around the abutments is central to the decision; (iii) two unsplinted attachments may be easier for him to manage than a bar. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

Block B — Examination, occlusion and space (Q29–48)

29 A 34-year-old woman lost the UR4 to a vertical root fracture two years ago. The adjacent crowns are upright, the interdental space measures 7 mm at the contact points and 6.5 mm at the bone crest on a parallel periapical radiograph, and the ridge is 7 mm wide buccolingually. Bone height is ample and the occlusion is stable. Which is the most appropriate plan?

A. A 4.1 mm diameter implant placed centrally in the space
B. A 3.3 mm diameter implant placed centrally in the space
C. Orthodontic space opening followed by a 4.1 mm implant
D. A resin-bonded bridge cantilevered from the UR3
E. A 5.0 mm diameter implant to match the emergence profile

30 A 29-year-old man has lost the UR1 and UR2 in a cycling accident. Both sites have adequate bone, he has a high lip line, and he is anxious about the appearance of the gum between the two replacement teeth. You are deciding how to configure the units. Which of the following would improve the prospects for the papilla between them?

A. Separating the two implants by at least 3 mm at the crest
B. Placing a single implant at the UR1 with a cantilevered UR2 pontic
C. Placing both implants 1 mm deeper to allow greater tissue thickness
D. A and B only
E. All of the above

31 Before planning an implant at the UL1 you measure a parallel periapical radiograph. The distance from the planned contact point with the UL2 to the crest of the interproximal bone is 7 mm. The adjacent teeth are periodontally sound, the phenotype is thin and the patient displays the full height of the papillae when she smiles. What should you tell her to expect?

A. An open embrasure, which the crown form will have to disguise
B. Complete papillary fill within twelve months of restoration
C. Complete fill once a connective tissue graft has been placed
D. Papillary fill determined by the depth of implant placement
E. Papillary fill determined by the abutment emergence profile

32 A 74-year-old man is to have two implants placed in the anterior mandible to retain his lower complete denture. On a trial denture set-up at the correct vertical dimension, the distance from the mucosal crest to the occlusal plane in the canine region measures 9 mm. His existing dentures are otherwise satisfactory and his oral hygiene is adequate. Which prosthetic design is most appropriate?

A. A cast bar with two distal cantilevers and clip retention
B. A milled bar with a rigid attachment and reciprocal arms
C. Two unsplinted stud attachments on the implants
D. A fixed screw-retained prosthesis on the two implants
E. Magnets seated on the implant abutments for retention

33 A 62-year-old woman with a heavily worn dentition is being planned for two implants in the lower posterior sextants. With marks placed on the nose and chin, the distance at rest measures 68 mm and in maximum intercuspation 62 mm. She has no temporomandibular symptoms and her speech is normal. Restorative space at the implant sites is limited. What does this measurement most support?

A. Reducing the occlusal vertical dimension to gain restorative space
B. Accepting the vertical dimension and using shorter crowns
C. Surgical crown lengthening of the opposing teeth
D. Alveoloplasty at the implant sites before placement
E. A modest increase in vertical dimension, trialled reversibly

34 A 47-year-old man is to have an implant at the LR6, lost eight years ago. The UR6 has over-erupted by about 3 mm, and a periapical radiograph shows that its cemento-enamel junction, gingival margin and alveolar crest have all migrated with it. There is 3.5 mm of interocclusal space at the site. The tooth is unrestored, vital and symptom-free. Which is the most appropriate way to create restorative space?

A. Orthodontic intrusion of the UR6 with temporary anchorage
B. Enameloplasty of the UR6 occlusal surface under water spray
C. Elective root canal treatment and a shortened crown on the UR6
D. Placing the implant and restoring with a 3.5 mm high crown
E. Extraction of the UR6 and replacement with a second implant

35 A single implant is to be restored at the LL6 in a 51-year-old woman with a complete natural dentition, canine guidance in both excursions and no parafunction. The implant is 4.8 mm in diameter, well positioned axially, and the adjacent teeth are unrestored and periodontally healthy. Consider these design features: (i) contact only under firm closure; (ii) no contact on the implant crown in lateral excursion; (iii) a narrowed occlusal table. Which are appropriate?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

36 You are checking the occlusion on a newly fitted implant crown at the UR5. In light closure, articulating foil marks the adjacent natural premolar and canine but not the implant crown; in firm closure the implant crown marks lightly. Which physiological difference between an implant and a natural tooth best justifies aiming for exactly this result?

A. The implant lacks proprioceptive innervation, so occlusal awareness is absent
B. The implant moves axially by only a few micrometres, natural teeth by far more
C. The implant is ankylosed, so its crown will over-erupt if left clear of contact
D. The implant transmits load to cortical rather than cancellous bone
E. The implant has a higher elastic modulus than dentine, so it deflects less

37 A 69-year-old woman has worn a complete upper denture for eleven years and asks about a fixed implant-supported replacement. Her existing denture has a full labial flange, her upper lip is thin, and at rest she shows 2 mm of incisal edge. She has enough bone for six implants without augmentation. Which assessment best establishes whether a fixed prosthesis can satisfy her?

A. Mounting the existing denture on a semi-adjustable articulator
B. A CBCT with a radiographic guide duplicating her denture
C. A trial of a duplicate denture with the labial flange removed
D. Measuring the interocclusal distance in the anterior region
E. Photographs at rest and at full smile with the denture in place

38 A 40-year-old woman requests an implant to replace the UL1, extracted three weeks ago after trauma. She has a high smile line displaying 3 mm of gingiva, a thin phenotype, and rectangular crown forms. A periapical radiograph shows the interproximal bone on the mesial of the UL2 to be 4 mm apical to the level on the adjacent UR1, following a previous root fracture. Which finding most increases the aesthetic risk?

A. The 3 mm of gingival display on full smile
B. The thin gingival phenotype
C. The rectangular crown form
D. The bone level on the adjacent UL2
E. The three-week interval since extraction

39 A 43-year-old man attends for assessment before two posterior implants. He denies any awareness of clenching or grinding and sleeps alone. On examination you are looking for objective evidence of parafunction before finalising the prosthetic design. Which of the following findings would support that diagnosis?

A. Bilateral masseteric hypertrophy on clenching
B. Generalised buccal gingival recession
C. Non-carious cervical lesions on the premolars
D. A and B only
E. A, B and C

40 A 55-year-old woman needs an implant at the LL6, lost twelve years ago. The ridge has resorbed lingually so that its centre lies about 3 mm palatal to the buccal cusps of the opposing UL6. The ridge is 7 mm wide with 13 mm of bone above the canal, the opposing tooth is unrestored, and her remaining occlusion is stable in group function. What is the most appropriate approach?

A. Angle the implant buccally to achieve a conventional relationship
B. Place the implant centrally and restore with a buccal cantilever
C. Graft the ridge buccally with a block before placement
D. Reduce the palatal cusp of the UL6 to obtain clearance
E. Place the implant centrally and restore in a crossbite relationship

41 A 52-year-old woman is to have an implant placed at the LR5, where the CBCT shows 3 mm of bone above the roof of the mandibular canal. Surgery is uneventful. Four days later she telephones to report numbness of the lower lip on that side. Which record would most help you to establish what has happened?

A. A signed consent form listing altered sensation as a risk
B. The radiology report describing the canal position on the CBCT
C. A note of the drill lengths and depth stops used at surgery
D. A documented pre-operative test of lip and chin sensation
E. A clinical photograph of the surgical site taken at closure

42 A 46-year-old man asks about an implant to replace the UR6, extracted a year ago. Screening gives BPE scores of 4 in the upper right and lower left sextants and 2 elsewhere, with generalised marginal bleeding and visible calculus. He smokes five cigarettes a day and has not seen a hygienist. He is keen for the implant to be placed before a holiday in two months. What is the most appropriate next step?

A. Place the implant now and treat the periodontal disease afterwards
B. Take full-mouth periapical radiographs before any pocket charting
C. Chart six-point pockets, provide non-surgical therapy and re-evaluate
D. Extract all teeth with probing depths greater than 6 mm
E. Refer him to a periodontist before undertaking any assessment

43 A 58-year-old man has an interincisal opening of 32 mm, stable for years after facial trauma in his twenties. He wants implants at the LL6 and LL7. Both sites have ample bone. The shortest drill and handpiece assembly in your kit requires 38 mm of vertical clearance to seat at the second molar, and 27 mm at the first molar. He has no temporomandibular symptoms. What is the most appropriate plan?

A. Proceed at the LL7 using a surgical guide to control angulation
B. Restrict the plan to the LL6 and accept a shortened dental arch
C. Proceed at the LL7 through a flapless approach to save space
D. Refer him for placement of the LL7 implant under general anaesthesia
E. Begin jaw-opening exercises and reassess the LL7 in six months

44 A CBCT cross-section at the LL6 site shows a pronounced submandibular fossa: the lingual plate is undercut below the crest, so that the ridge overhangs the concavity by about 3 mm. The crest is 6 mm wide, there is 14 mm of bone above the canal, and the mucosa is healthy. You intend to place a 4.1 mm implant. Which of the following would reduce the risk of perforating the lingual plate?

A. Directing the osteotomy buccally, away from the concavity
B. Confirming the ridge width on a panoramic radiograph
C. Using a flapless approach to preserve the periosteal blood supply
D. A and B only
E. A, B and C

45 A 36-year-old woman has lost the UR1. CBCT shows a nasopalatine canal 5 mm in diameter whose incisive foramen lies immediately palatal to the planned implant position, leaving only 3 mm of bone between the canal and the labial plate at the level of the intended platform. The labial plate is intact and 1 mm thick, and the ridge is otherwise adequate in height. What is the most appropriate plan?

A. Place the implant into the canal after enucleating its contents
B. Position the implant labial to the canal and augment the labial plate
C. Place a narrow implant palatal to the canal to gain stability
D. Abandon implant treatment and provide a resin-bonded bridge
E. Expand the canal with osteotomes and place a wide implant

46 A mandibular fixed prosthesis is planned on five implants placed between the mental foramina. On the planning cast, the anteroposterior spread — the distance from the centre of the most anterior implant to a line joining the centres of the two most distal implants — measures 10 mm. Occlusal forces are judged average and the opposing arch is a complete denture. Applying the conventional guideline, what is the greatest distal cantilever that should be planned?

A. 5 mm on each side
B. 10 mm on each side
C. 12 mm on each side
D. 15 mm on each side
E. 20 mm on each side

47 You are deciding how much occlusal record-taking a case justifies. Mounting on a semi-adjustable articulator with a facebow transfer costs an extra appointment and adds laboratory expense, and is not required for every restoration. In which of the following situations is it most clearly indicated?

A. A single implant crown at the LR5 with a stable intercuspal position
B. A single implant crown at the UR6 opposing an intact natural arch
C. Two implant crowns replacing lower premolars in group function
D. An implant crown at the UL1 with unaltered anterior guidance
E. A reorganised occlusion restored at an increased vertical dimension

48 A 60-year-old man has an implant at the LL6 restored two years ago. The buccal keratinised mucosa measures 1 mm, the peri-implant tissues bleed on probing at 4 mm, and he reports that brushing the area is uncomfortable so he avoids it. Consider these statements: (i) sites with less than 2 mm of keratinised mucosa show more plaque and bleeding; (ii) an implant should not be restored until a graft has provided 2 mm; (iii) patients report more discomfort brushing such sites. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

Block C — Investigations, imaging, guides and the diagnostic wax-up (Q49–70)

49 You refer a patient to a colleague at a neighbouring practice, who owns the only cone beam unit in the area, for cross-sectional imaging of an LR6 site. She reads your referral and clinical details, decides the examination is warranted, selects the field of view and exposure factors, and operates the machine herself. A maxillofacial radiologist later provides the written report. Under IR(ME)R, who carries the duty of justification?

A. You, because the clinical information came from you
B. Your colleague, acting as practitioner for this exposure
C. The radiologist who provides the written report
D. The employer, through the written procedures for the unit
E. Both of you jointly, as referrer and practitioner

50 You are planning a single implant at the LR6 where a panoramic radiograph suggests the mandibular canal lies close to the intended apex. Your unit offers a 5 × 5 cm volume reconstructed at 0.3 mm, 0.2 mm or 0.125 mm voxels, and the manufacturer's 0.125 mm protocol runs a longer exposure at the same tube current. What would selecting the finest voxel achieve here?

A. Lower image noise for the same patient dose
B. Reduced streak artefact from the adjacent metal-ceramic crown
C. Improved spatial resolution at the cost of a higher patient dose
D. Grey values corresponding more closely to Hounsfield units
E. A more accurate surface model for manufacture of a guide

51 A colleague measures the grey value in the trabecular bone of a planned LR6 osteotomy on a cone beam volume, obtains a figure of about 320, tells the patient the bone is "type 4" and plans an extended healing period on that basis. What is the strongest objection to using the number in this way?

A. Grey values fall as the field of view is reduced, so small volumes read low
B. Grey values describe mineral content rather than trabecular architecture
C. Density classifications were derived histologically rather than radiographically
D. Grey values are uncalibrated and vary with unit, field of view and position
E. The measurement should be taken in the cortical plate, not the trabecular core

52 A cone beam volume of the lower left quadrant is taken to plan an implant at the LL6. The patient has metal-ceramic crowns on the LL5 and LL7. The reconstruction shows bright streaks radiating from both crowns and a broad dark band running between them, crossing the crest of the planned site and obscuring the crestal cortex. Which physical process best explains the dark band?

A. Preferential attenuation of lower-energy photons within the dense restorations
B. Scattered radiation reaching the detector from outside the primary beam
C. Movement of the patient during rotation of the source and detector
D. Too few basis projections being acquired for the reconstruction
E. Variation in the response of individual detector elements

53 An 82-year-old man with a resting tremor has a small-volume cone beam examination for an implant at the LR5. On review the cortical outlines of the mandibular canal are doubled, the trabecular pattern is unsharp throughout the volume, and the crestal margin cannot be located confidently. What is the most appropriate course?

A. Measure from the more sharply defined of the two canal outlines
B. Reconstruct the existing data at a larger voxel to remove the doubling
C. Plan from a panoramic radiograph, allowing a 4 mm margin above the canal
D. Accept the volume and place an implant 2 mm shorter than planned
E. Justify and repeat the examination with head support and a shorter protocol

54 A cone beam volume taken to plan an implant at the UL6 includes the left antrum. There is a dome-shaped, homogeneous soft-tissue opacity about 10 mm high arising from the sinus floor, with no corticated margin and no expansion of the bony walls. The membrane elsewhere is of normal thickness, the ostium appears patent, and the patient has no nasal symptoms. Which is the most likely explanation?

A. Chronic rhinosinusitis
B. An antral pseudocyst
C. A mucocele
D. Odontogenic sinusitis
E. An antrolith

55 A cone beam examination taken for an implant at the LL6 includes the lower border of the mandible. Below the mandibular canal, near the angle, there is a well-defined, corticated, ovoid radiolucency about 12 mm across, with thinning and concavity of the lingual cortex and an intact buccal cortex. The patient is asymptomatic and the overlying mucosa is normal. What is the most appropriate course?

A. Arrange an incisional biopsy before implant surgery
B. Refer on a suspected cancer pathway
C. Repeat the examination in twelve months to confirm stability
D. Record it as a developmental lingual depression and proceed
E. Abandon implant treatment in that quadrant

56 You are deciding which of next week's patients need cross-sectional imaging. The first has an LR6 site where a panoramic radiograph, corrected for magnification, suggests fewer than 2 mm would remain between the planned apex and the canal. The second has a UR2 site with a clinically knife-edged ridge whose width cannot be judged. The third has a UL5 site with 14 mm of height on a periapical and a broad ridge. For which is cone beam imaging justified?

A. The LR6 site with less than 2 mm above the canal
B. The UR2 site with a knife-edged ridge
C. The UL5 site with 14 mm of height
D. A and B only
E. All of the above

57 A panoramic radiograph taken to assess an LR6 site shows 13 mm from the alveolar crest to the superior border of the mandibular canal. A 5 mm steel ball bearing carried on a stent in the same region images 6.3 mm in the vertical dimension. You intend to leave 2 mm of clearance above the canal. What is the longest implant this measurement supports?

A. 6 mm
B. 8 mm
C. 10 mm
D. 11 mm
E. 13 mm

58 You are taking the baseline periapical radiograph at the fit of a screw-retained crown on an implant at the LR5, intending to compare marginal bone levels with it at future reviews over many years. Which single feature of the image most determines whether that comparison will be valid?

A. Use of a beam-aiming holder rather than a bisecting-angle technique
B. Rectangular collimation matched to the receptor
C. Clear resolution of the implant threads on both proximal surfaces
D. Inclusion of the whole implant apex within the image
E. A digital receptor rather than conventional film

59 An implant at the UR5 was restored twelve months ago with a screw-retained crown. Probing depths are 3 mm with no bleeding, the patient is symptom-free, and you decide to take a periapical radiograph at this review. Which of the following are sound reasons for taking it?

A. Recording the bone level after initial remodelling as a reference
B. Detecting residual cement beneath the mucosa
C. Confirming that osseointegration has been achieved
D. A and B only
E. A, B and C

60 A patient declines cone beam imaging before a planned implant at the UR4. A periapical radiograph shows adequate height and no pathology, but the buccopalatal width cannot be judged clinically because the mucosa is thick. You consider ridge mapping with a calibrated caliper through anaesthetised mucosa. What is its principal limitation compared with cross-sectional imaging?

A. It over-estimates width by the thickness of the mucosa
B. It cannot be related to the planned tooth position
C. It yields no information about the density of the bone
D. It requires a flap to be raised in order to be accurate
E. It samples only the points probed, missing deeper concavities

61 You plan a fully guided placement at the LR6 with a tooth-supported guide, leaving 2 mm between the planned apex and the roof of the mandibular canal. The patient has read that computer-guided surgery is accurate to a fraction of a millimetre and asks whether this removes the risk of nerve injury. Which statement most accurately conveys the residual risk?

A. Tooth-supported guides remove lateral error but not depth error
B. Reported deviations are averages, and individual cases deviate further
C. The residual error lies almost entirely in the surface scan
D. Guided placement is accurate to within 0.5 mm in the great majority of cases
E. A 2 mm margin makes nerve injury essentially impossible

62 Consider these contributors to the difference between the planned and the achieved implant position in static guided surgery: (i) clearance between the drill shank and the sleeve; (ii) movement of the guide on its supporting tissues while drilling; (iii) error in matching the surface scan to the cone beam volume where restorations produce artefact. Which contribute?

A. (i) only
B. (i) and (ii) only
C. (ii) and (iii) only
D. (i) and (iii) only
E. (i), (ii) and (iii)

63 You are specifying a surgical guide for an implant at the LL6 in a patient with limited mouth opening. The planning software offers sleeves of 3 mm or 5 mm in length, and offsets of 2 mm or 8 mm between the base of the sleeve and the crest. Which combination would most reduce angular deviation of the osteotomy, accepting that access must still be checked?

A. The 5 mm sleeve at a 2 mm offset from the crest
B. The 5 mm sleeve at an 8 mm offset from the crest
C. The 3 mm sleeve at a 2 mm offset from the crest
D. The 3 mm sleeve at an 8 mm offset from the crest
E. Either sleeve, since the offset alone determines deviation

64 A fully guided, flapless placement is planned at the LL3 in a dense anterior mandible with a thick cortical plate. The cross-sections show a broad ridge with no lingual concavity and generous width both sides of the planned axis. Four successive drills will pass through a 5 mm sleeve seated 2 mm above the crest. Which is the most important hazard of this arrangement?

A. Fracture of the buccal plate, because the guide fixes the entry point
B. Perforation of the lingual plate, because no flap is raised
C. Overheating of the bone, because irrigant is obstructed by the sleeve
D. Loss of primary stability, because guided drills cut oversize
E. Contamination of the osteotomy from the surface of the guide

65 You are considering a flapless approach at the UR4 with a tooth-supported guide, both adjacent teeth being sound and unrestored. The cone beam volume shows 7 mm of ridge width, a buccal plate about 1 mm thick over the planned implant surface, and a 3 mm band of keratinised mucosa across the site. Which of the following are important limitations of working without a flap here?

A. A buccal fenestration would go unrecognised
B. Primary stability would be harder to obtain
C. The guide would require fixation pins to stay stable
D. A and B only
E. A, B and C

66 A 34-year-old woman lost the UR1 to trauma two years ago. She has a high smile line displaying 2 mm of gingiva, and the ridge is deficient by about 3 mm vertically and 4 mm horizontally. A mounted diagnostic wax-up placed at the correct incisal edge position leaves the cervical margin of the planned UR1 some 3 mm apical to that of the UL1. What does the wax-up most usefully establish?

A. That the implant should sit 3 mm apical to the adjacent cemento-enamel junction
B. That a longer clinical crown will be acceptable at her lip line
C. That the deficiency is best treated by connective tissue grafting at placement
D. That the implant should be angled palatally to permit screw retention
E. That vertical augmentation or gingival-coloured prosthetic material will be needed

67 An edentulous 70-year-old woman wants a mandibular implant overdenture. Duplicate dentures mounted at the planned vertical dimension show 6 mm from the crest of the ridge to the occlusal surfaces of the lower denture teeth in the canine regions. The ridge is broad, access is good and she has no medical contraindication. Which is the most appropriate way of dealing with this finding?

A. Select the lowest-profile stud attachment available
B. Plan a crestal bone reduction at placement, prescribed from the wax-up
C. Increase the occlusal vertical dimension by 4 mm
D. Provide a bar-retained rather than a stud-retained overdenture
E. Thin the denture base over the attachment housings

68 A patient with an overerupted UL6 is being planned for an implant at the LL6. You mount study casts on a semi-adjustable articulator and prescribe a diagnostic wax-up. Consider what the mounted wax-up can establish: (i) the interocclusal space available for the restoration; (ii) whether the planned tooth position lies over the ridge; (iii) the density of the bone at the implant site. Which can it establish?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

69 A cone beam examination taken before placing an implant at the LR6 shows two distinct corticated canals in the posterior body of the mandible: the main mandibular canal, and a smaller accessory canal lying about 3 mm above and buccal to it, both running forward and merging just distal to the mental foramen. Which is the most important consequence for the plan?

A. Local anaesthesia may be incomplete, requiring a supplementary technique
B. The finding should be reported but does not alter the osteotomy depth
C. The osteotomy should be moved mesially, away from the division
D. The safety margin must be measured from the more coronal of the two canals
E. The examination should be repeated at a finer voxel to confirm the variant

70 You have completed an implant case at the UR1 and wish to show the clinical photographs, the cone beam images and the radiographs at a local study evening for dentists. Two years ago the patient signed the practice's standard consent form for treatment and for photographs to be taken "for the clinical record". She has since moved and is not contactable at the address you hold. Which applies?

A. Specific consent for this use, additional to that for the clinical record
B. Consent for the record extends to teaching within the profession
C. Cropping the images so that the face is not shown removes the need
D. A and B only
E. All of the above

Block D — Surgical anatomy (Q71–86)

71 A cone beam examination is taken before placing two implants in the interforaminal region of a 64-year-old man. On the panoramic reconstruction of the volume the mandibular canal appears to continue about 4 mm anterior to the right mental foramen before fading. You need to know how far forward the nerve actually runs. Which is the most reliable way to establish this?

A. Measure the apparent anterior extension on the panoramic reconstruction
B. Add a standard 2 mm to the anterior border of the radiographic foramen
C. Identify the foramen on a paralleling periapical and measure forward from it
D. Trace the canal on consecutive cross-sections to the point where it turns back
E. Palpate the foramen and mark the overlying mucosa before incising

72 In a 68-year-old woman being planned for two implants to retain a mandibular overdenture, the anterior loop measured on cross-sections extends 3 mm anterior to the right mental foramen, and a mandibular incisive canal about 2 mm in diameter continues forward from it towards the midline. You intend to keep 2 mm of clearance from neural structures. How far anterior to the mental foramen must the right implant be placed?

A. 2 mm
B. 3 mm
C. 5 mm
D. 8 mm
E. It may be placed at the level of the foramen

73 You are raising a full-thickness flap with a vertical relieving incision to augment the buccal plate at the LL5, a site immediately anterior to the mental foramen. Knowing the direction in which the mental nerve leaves the foramen determines where you place the incision and how you retract. In which direction does the nerve most commonly emerge?

A. Anteriorly and superiorly
B. Posteriorly and superiorly
C. Directly buccally, perpendicular to the cortex
D. Anteriorly and inferiorly
E. Posteriorly and inferiorly

74 Two implants are placed in the interforaminal region of an atrophic mandible for an overdenture. During preparation of the right osteotomy, sited 4 mm anterior to the mental foramen, there is brisk bleeding from the depth of the site which ceases when the implant is seated. Both implants are stable at 40 Ncm. What is the most likely consequence for the patient?

A. Anaesthesia of the lower lip and chin on that side
B. A rapidly expanding swelling in the floor of the mouth
C. Loss of vitality of the lower incisors on that side
D. Persistent neuropathic pain at the mental foramen
E. No deficit, or altered sensation limited to the anterior labial gingiva

75 A cross-section at the LR6 shows the alveolar crest inclined lingually, a submandibular fossa whose deepest point lies about 7 mm below the crest, and only 4 mm of bone between the intended apex and the lingual cortex. Which of the following would increase the risk of lingual plate perforation at this site?

A. Directing the osteotomy along the axis suggested by the crest
B. Working flapless, so that the lingual plate cannot be palpated
C. Choosing a longer implant to improve primary stability
D. A and B only
E. All of the above

76 Ten minutes after completing an osteotomy at the LL4, a 46-year-old man's floor of mouth is visibly swollen, his tongue is being displaced upwards and backwards, and his voice has changed. He is talking in short sentences, his oxygen saturation is 97 per cent on air and there is no external swelling. What is the most appropriate immediate action?

A. Apply bimanual pressure and arrange emergency transfer to hospital
B. Apply bimanual pressure and observe him in the surgery for an hour
C. Seat the implant to tamponade the site and suture the flap
D. Infiltrate local anaesthetic with adrenaline into the floor of the mouth
E. Prescribe tranexamic acid mouthwash and antibiotics, and review tomorrow

77 An implant is being placed at the LL7 under an inferior alveolar nerve block. The patient's lower lip is profoundly numb and the buccal soft tissues are anaesthetised, but she reports sharp, well-localised pain as the drill engages the lingual aspect of the crest. Aspiration was negative and the block was given fifteen minutes ago. What is the most likely explanation?

A. Incomplete deposition around the inferior alveolar nerve
B. Sensory supply from the long buccal nerve
C. Accessory supply from the mylohyoid nerve
D. A bifid canal carrying a second neurovascular bundle
E. Referred sensation from the lingual nerve

78 You need to release a flap to obtain tension-free closure over a graft at the LL7, and a colleague suggests a vertical relieving incision on the lingual aspect distal to the tooth. Which consideration most strongly argues against that suggestion?

A. The lingual artery lies immediately deep to the periosteum here
B. The lingual nerve lies at or above the crest in a minority of patients
C. The submandibular duct crosses the region at crestal level
D. The mylohyoid muscle prevents advancement of a lingual flap
E. The lingual plate has no periosteal blood supply at this level

79 A UR1 is to be replaced in a 52-year-old man. The cone beam volume shows a nasopalatine canal 5 mm in diameter whose labial wall is 1.5 mm thick, and the ridge is deficient labially, so that the prosthetically correct implant position would place about a third of the implant surface within the canal. What is the most likely outcome if it is placed as planned?

A. Fibrous encapsulation of the engaged surface and failure to integrate
B. Permanent anaesthesia of the anterior palatal mucosa
C. Severe haemorrhage from the nasopalatine artery
D. Perforation into the floor of the nose
E. Loss of vitality of the adjacent central incisor

80 Consider these statements about the nasopalatine canal: (i) it enlarges as the anterior maxilla resorbs after tooth loss; (ii) it transmits terminal branches of the sphenopalatine artery; (iii) its diameter is assessed most reliably on a paralleling periapical radiograph. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (ii) and (iii) only
D. (i) and (iii) only
E. (i), (ii) and (iii)

81 You are harvesting a subepithelial connective tissue graft from the palate to thicken the peri-implant mucosa at a UR2 implant. The palatal vault is noticeably shallow and the mucosa over the donor area is 3.5 mm thick. Which factor most limits the apico-coronal height of graft that can be taken safely?

A. The thickness of the palatal mucosa over the donor area
B. The position of the greater palatine foramen relative to the molars
C. The width of the recipient site to be augmented
D. The depth of the vault, which sets the distance to the bundle
E. The extent of the rugae, which limit the anterior incision

82 A cone beam volume taken before referring a patient for a lateral window sinus floor elevation at the UR6 shows a well-corticated channel about 2 mm across running horizontally within the lateral wall of the antrum, roughly 17 mm above the alveolar crest. Which vessel does this most likely represent?

A. The intraosseous anastomosis of the posterior superior alveolar and infraorbital arteries
B. The greater palatine artery within its bony canal
C. A terminal branch of the sphenopalatine artery
D. The infraorbital nerve within the infraorbital canal
E. The buccal artery entering the wall of the antrum

83 Three weeks after a sinus floor elevation a patient develops persistent left-sided nasal obstruction, purulent nasal discharge and a feeling of pressure below the eye. The graft appears undisturbed and the wound has healed. Which anatomical feature best explains why swelling of the sinus lining leads to this picture?

A. The floor of the antrum lies below the level of the nasal floor
B. The lining is ciliated epithelium beating towards the antral floor
C. The antrum drains through the inferior meatus of the nose
D. The blood supply of the membrane is effectively end-arterial
E. The ostium lies high on the medial wall, so drainage is not gravity-assisted

84 Consider these statements: (i) a maxillary sinus septum most often arises from the floor and runs in a buccopalatal direction; (ii) the thickness of the sinus membrane is assessed most reliably on the panoramic reconstruction of a cone beam volume; (iii) the sinus is lined by a non-ciliated stratified squamous epithelium. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (i) and (iii) only
E. (i), (ii) and (iii)

85 A 62-year-old non-smoker has an implant placed at the UL6 in a site healed for a year, with 11 mm of height beneath the sinus floor and 8 mm of width. The drills meet very little resistance, the osteotomy feels hollow, and the final insertion torque is 12 Ncm. Which anatomical feature best explains this finding?

A. Pneumatisation of the antrum into the alveolar process
B. Residual bundle bone from the extraction socket
C. A thin cortical plate with sparse, widely spaced trabeculae
D. A thick cortex overlying dense trabecular bone
E. Perforation of the plate by the greater palatine canal

86 An 80-year-old woman cannot control her lower complete denture. A panoramic radiograph shows a mandible about 8 mm in total vertical height in the premolar regions, with both mental foramina lying at the level of the alveolar crest and a thin, pencil-like lower border. She is medically well, motivated and attends regularly. Which features place this case beyond a general dentist with a special interest?

A. The residual mandibular height of about 8 mm
B. The crestal position of the mental foramina
C. Her age of 80 years
D. A and B only
E. A, B and C

Block E — Osseointegration, bone biology and healing (Q87–106)

87 A 36-year-old woman has an unrestorable oblique root fracture of the UR1. CBCT shows an intact buccal wall 0.7 mm thick, 4 mm of bone apical to the socket and no fenestration; the phenotype is thick and the gingival margins of the two central incisors are level. You plan immediate placement with a connective tissue graft and a screw-retained provisional. Which statement best justifies warning her that the buccal profile will still change?

A. The buccal wall is largely bundle bone, which depends on the periodontal ligament
B. Immediate implants achieve lower primary stability than those in healed sites
C. Connective tissue grafts lose most of their volume during the first year
D. A flapless approach limits access for grafting the buccal gap
E. Thick phenotypes remodel more after extraction than thin ones

88 Consider these statements about the bone that forms around an implant during the first eight weeks of healing: (i) woven bone is deposited far more rapidly than lamellar bone but is less mineralised and less stiff; (ii) the rise in secondary stability during the second month reflects mineralisation and lamellar replacement of that woven bone; (iii) lamellar bone is laid down directly on the implant surface within the first week. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

89 Two implants are retrieved from an experimental study four weeks after placement, one with a machined surface and one moderately rough, both placed into standardised osteotomies with a 0.5 mm circumferential gap. Undecalcified sections are prepared to compare the pattern of bone formation at the two interfaces. Which observation would best demonstrate that contact osteogenesis has occurred at the roughened implant?

A. New bone advancing from the cut osteotomy wall towards the implant
B. A fibrin scaffold spanning the healing chamber without mineralised tissue
C. Bone matrix deposited on the implant surface with no intervening old bone
D. Osteoclastic resorption lacunae along the wall of the osteotomy
E. Marrow spaces adjacent to the implant repopulated with adipocytes

90 A 57-year-old man has two implants placed at one visit. The LR6, in dense bone, seats at 60 Ncm with a resonance frequency reading of ISQ 80; the LR4, in noticeably softer bone, seats at 25 Ncm with ISQ 68. At three weeks both are asymptomatic and neither moves, but the LR6 now reads 66 and the LR4 reads 70. Which best explains the difference between the two trajectories?

A. Faster contact osteogenesis at the lower-torque site
B. Greater woven bone deposition at the lower-torque site
C. Transducer orientation differing between the two readings
D. Remodelling of the bone compressed at the higher-torque site
E. Micromotion of the higher-torque implant beneath its healing cap

91 A 52-year-old non-smoker has an implant placed immediately into the socket of a fractured UL2. Insertion torque is 32 Ncm and the ISQ is 72. She asks for a fixed provisional at the same visit rather than a removable appliance. The adjacent teeth are unrestored, her overbite is 4 mm with a normal overjet, and she reports no parafunction. Which single measure most reduces the risk that fibrous tissue rather than bone forms at the interface?

A. Prescribing a soft diet for the first six weeks
B. Choosing a screw-retained rather than a cemented provisional
C. Splinting the provisional to both adjacent teeth with composite
D. Reducing the subgingival emergence profile of the provisional
E. Relieving the provisional from contact in centric and all excursions

92 A 61-year-old man has an implant placed in the dense anterior mandible. The osteotomy is prepared with a single drill that has been autoclaved and reused over two years, run at 1200 revolutions per minute with continuous external irrigation and firm apical pressure, and the implant seats at 55 Ncm. He is a healthy non-smoker. At eight weeks the implant is mobile and a thin uniform radiolucent line surrounds it. Which factor is most likely to have caused this?

A. Repeated use of a worn drill in dense cortical bone
B. External rather than internal irrigation of the osteotomy
C. The drill speed of 1200 revolutions per minute
D. The insertion torque of 55 Ncm at seating
E. The absence of a countersink at the crestal cortex

93 A 62-year-old woman has implants placed at the UL6, where CBCT shows a thin crestal cortex and widely spaced trabeculae, and at the LL6, where the cortex measures 2 mm. Both are placed by the same technique with the same implant system, and both are stable at surgery. You allow eight weeks of submerged healing in the mandible but four months in the maxilla. Which statement best explains the longer maxillary healing period?

A. Trabecular bone is less vascular than cortical bone, so it heals more slowly
B. Cortical bone must undergo osteonal remodelling before it can carry load
C. Less bone contacts the implant at placement, so more must be earned biologically
D. The maxillary periosteum supplies fewer osteoprogenitor cells than the mandibular
E. Woven bone mineralises more slowly in the posterior maxilla than in the mandible

94 A 58-year-old man has an implant placed at the UR6 in a site whose CBCT showed a thin cortex and widely spaced trabeculae. Despite under-preparation of the osteotomy and a tapered implant, it seats at 12 Ncm with an ISQ of 52, though it is stable to finger pressure and correctly positioned. He is healthy, does not smoke, has taken two weeks of leave for the treatment and would like the crown as soon as possible. What is the most appropriate course?

A. Load with a non-occluding provisional crown at this visit
B. Place a healing abutment and restore the implant at six weeks
C. Remove the implant and graft the site for placement in six months
D. Submerge the implant and extend the healing period to six months
E. Place a second implant at the UR5 and splint the two restorations

95 A bone-level implant at the LR5 was deliberately placed 3 mm below the crest to obtain an emergence profile, and the abutment was removed and replaced four times during the restorative phase. Six months after loading, standardised radiographs show 1.8 mm of bone loss to the level of the first thread on both mesial and distal surfaces. Probing depths are 4 mm, there is no bleeding on probing and no suppuration. What is the most likely explanation?

A. Peri-implantitis arising from residual cement at the crown margin
B. Re-establishment of the supracrestal soft tissue attachment
C. Compression necrosis from an excessive insertion torque
D. Overload transmitted through an excessive crown height
E. Loss of bundle bone following the original extraction

96 You are explaining to a patient why the tissues around her new implant will need cleaning at least as carefully as those around her own teeth, and she asks how they differ. Which of the following genuinely distinguish the peri-implant supracrestal tissues from those of a natural tooth?

A. Collagen fibres run parallel to the surface rather than inserting into it
B. The epithelial attachment is formed without hemidesmosomes
C. The connective tissue compartment is more richly vascularised
D. A and B only
E. A, B and C

97 Six days after implant placement with simultaneous guided bone regeneration at the UR4, using a particulate xenograft and a native collagen membrane, a 3 mm dehiscence has opened over the crest. The membrane is visible at the base of it, the mucosal margins are mildly erythematous but not friable, there is no suppuration and the patient has no pain and no swelling. What is the most appropriate management?

A. Re-suture the flap under local anaesthesia at this visit
B. Remove the membrane and irrigate the graft with saline
C. Apply chlorhexidine gel and review weekly until the site epithelialises
D. Prescribe a seven-day course of amoxicillin and review in one week
E. Raise a coronally advanced flap to obtain primary closure

98 A 59-year-old man with type 2 diabetes and an HbA1c of 64 mmol/mol asks why you want his control improved before an implant is placed, pointing out that he has healed well after two extractions. He takes metformin, does not smoke, and his periodontal tissues are stable. Which mechanism best explains the delayed osseointegration reported in poorly controlled diabetes?

A. Insulin deficiency prevents mineralisation of the woven bone matrix
B. Hyperglycaemia depletes the circulating pool of osteoprogenitor cells
C. Neutrophil dysfunction allows bacterial colonisation of the osteotomy
D. Microangiopathy prevents a fibrin clot from adhering to the implant
E. Advanced glycation end-products impair collagen turnover and bone formation

99 A 74-year-old woman took alendronate 70 mg weekly for six years and stopped two years ago. An implant placed at the LL6 four years ago integrated uneventfully and has functioned since. She now presents, after a period of local infection treated by her general dental practitioner, with 8 mm of exposed non-healing bone alongside the implant, which itself remains immobile. Which statement best explains why integration succeeded but the site has now failed?

A. The drug inhibits osteoblasts, so the interface was never fully mineralised
B. Formation is largely preserved while the remodelling that repairs bone is suppressed
C. Bisphosphonate concentration in the jaws rises in the years after surgery
D. Peri-implant mucosa is the principal target of nitrogen-containing bisphosphonates
E. Suppressed osteoclasts allow the implant to intrude under occlusal load

100 A specimen retrieved two weeks after placement of a moderately rough implant, in a human biopsy study, shows a narrow band of bone with empty osteocyte lacunae along the wall of the osteotomy, being removed by multinucleate cells, with woven bone forming in the chambers between the threads. The osteotomy had been prepared at low speed with sharp drills and copious irrigation, and the implant seated at 30 Ncm. What does this finding most likely represent?

A. Overheating of the bone beyond the thermal threshold during preparation
B. A foreign body reaction to titanium particles shed during insertion
C. Cement line matrix laid down immediately before contact osteogenesis
D. Unavoidable surgical injury whose removal underlies the fall in stability
E. Compression necrosis produced by an excessive insertion torque

101 Ten weeks after placement of an implant at the LL5, a periapical radiograph shows a continuous radiolucent line about 0.5 mm wide following the entire outline of the implant body. The overlying mucosa is healthy and the patient reports no pain, but when you engage the cover screw the implant turns within the site. There is no suppuration and no swelling. What is the most appropriate course?

A. Repeat the radiograph with a paralleling technique before deciding
B. Remove the implant and allow the site to heal before reassessment
C. Leave the site undisturbed and re-examine in a further eight weeks
D. Debride the site and place a wider implant at the same appointment
E. Restore the implant so that functional loading remodels the interface

102 Consider these statements about occlusal load and the bone surrounding an osseointegrated implant: (i) bone adapts to mechanical strain within a physiological window, and is lost both above and below that window; (ii) occlusal overload in the absence of inflammation is an established cause of peri-implantitis; (iii) functional strain concentrates in the crestal bone, where the implant is most rigidly supported. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

103 A colleague has fixed a cortico-cancellous ramus block graft to a deficient UR2 site with two positional screws, and asks you why he should wait about four months before placing the implant when particulate autogenous bone in a similar defect is often considered ready sooner. Healing has been uneventful and the patient is a healthy non-smoker. Which statement best explains the difference?

A. Particulate grafts are osteoinductive whereas block grafts are only osteoconductive
B. A block relies on the overlying periosteum alone for its entire blood supply
C. Particulate bone unites directly with the recipient cortex without resorption
D. Fixation screws delay revascularisation by compressing the graft interface
E. Its cortical component must be resorbed before it can be replaced by new bone

104 An implant at the UL5 was loaded fourteen months ago and the patient attends for review. Standardised radiographs show 1.3 mm of marginal bone loss during the first year of function and a further 0.1 mm since. The implant is immobile, there is no peri-implant radiolucency and no pain or altered sensation, probing depths are 3 mm and there is no bleeding on probing. How should this implant be classified?

A. Failing, because loss exceeded 1 mm during the first year of loading
B. Successful, because the loss falls within the accepted criteria
C. Surviving but not successful, because of the first-year bone loss
D. Peri-implant mucositis, requiring reinforcement of supportive therapy
E. Peri-implantitis, requiring surgical intervention at this visit

105 A 66-year-old woman takes prednisolone 15 mg daily, and has done for eighteen months, for giant cell arteritis. She is having an implant placed at the UR5 with simultaneous guided bone regeneration for a 3 mm buccal dehiscence. She is a non-smoker, takes no antiresorptive drug and her HbA1c is 41 mmol/mol. Which of the following are established consequences of her corticosteroid therapy for this wound?

A. Impaired fibroblast proliferation with reduced collagen synthesis
B. Inhibition of osteoclast activity comparable with an antiresorptive
C. Failure of the fibrin clot to adhere to the implant surface
D. A and B only
E. All of the above

106 An implant is placed immediately into the socket of an UR4 that was lost to a vertical root fracture. It is correctly positioned against the palatal wall, the buccal plate is intact and 1.2 mm thick, and a horizontal gap of 2 mm remains between the implant and the inner surface of that wall. The patient is a healthy non-smoker with a thick phenotype. Which of the following are supported by the evidence on defects of this kind?

A. A gap of this size will largely fill with bone without being grafted
B. Grafting the gap reduces the horizontal dimensional change of the ridge
C. The buccal wall resorbs to some degree whether or not the gap is grafted
D. A and B only
E. All of the above

Block F — Materials, implant design and surfaces (Q107–132)

107 A 24-year-old woman has a congenitally absent UR2. The mesiodistal space is 6 mm, the ridge measures 5 mm wide at crest level on CBCT with 12 mm of height, and you intend to place a 3.3 mm diameter implant without augmentation. The system offers this diameter in grade 4 commercially pure titanium and in a titanium–zirconium alloy. Which property best justifies choosing the alloy?

A. Its higher tensile and fatigue strength at the same diameter
B. Its lower elastic modulus, closer to that of cortical bone
C. Its greater surface energy following acid etching
D. Its resistance to low-temperature degradation in a moist environment
E. Its reduced tendency to galvanic corrosion against gold alloys

108 A colleague looking at an implant catalogue notices that the implants are made of grade 4 commercially pure titanium while the abutment screws are grade 5, and asks what the grading actually describes. Both components are used in the same system and both carry the same surface passivation. Which statement best explains why grade 4 commercially pure titanium is substantially stronger than grade 1?

A. It contains a small percentage of aluminium as a strengthening element
B. It has a higher content of interstitial oxygen, iron, carbon and nitrogen
C. It is work-hardened during machining of the thread profile
D. It carries a thicker and more stable titanium dioxide surface layer
E. It has a finer grain structure produced by subsequent heat treatment

109 The abutment screw retaining a molar crown is manufactured from Ti-6Al-4V rather than commercially pure titanium, while the implant it engages is made of the pure metal. The crown is a single unit with an even occlusion and no cantilever, and the manufacturer specifies a tightening torque of 35 Ncm. Which is the principal reason for the alloy being chosen for this component?

A. The alloy has a lower coefficient of friction against the implant threads
B. The alloy forms a more stable oxide layer within the peri-implant sulcus
C. The alloy has a much higher yield strength, allowing greater preload
D. The alloy has a lower elastic modulus, so the screw elongates further
E. The alloy resists galvanic corrosion against a zirconia abutment

110 During a maintenance appointment a colleague debrides exposed threads at a peri-implantitis site using a stainless steel curette, having found the plastic instruments ineffective against the mineralised deposit. The implant has a moderately rough surface and 3 mm of the body is exposed above the bone. Which statement best describes the consequence for the implant surface?

A. The passive layer is permanently removed, allowing corrosion to proceed
B. Galvanic corrosion between steel and titanium dissolves the exposed threads
C. The calcium phosphate coating applied at manufacture is stripped away
D. Steel is transferred onto the surface and the topography is scored
E. The titanium is work-hardened, making the implant neck brittle

111 Histological studies of tissue retrieved from around failing implants frequently demonstrate titanium particles within the peri-implant connective tissue and adjacent macrophages, and their significance is currently debated. Which of the following are recognised sources of such particles?

A. Friction between implant and bone during insertion of the fixture
B. Instrumentation of an exposed implant surface with a metal curette
C. Micromovement at the implant–abutment interface during function
D. A and B only
E. All of the above

112 A manufacturer offers the same implant body with two surface treatments: a moderately rough sandblasted and acid-etched surface with an average roughness (Sa) of about 1.5 micrometres, and a rougher plasma-sprayed variant with an Sa of about 3.5 micrometres. The patient is a treated periodontitis case with a history of one implant lost to peri-implantitis. Which best describes the consequence of choosing the rougher surface?

A. Greater bone-to-implant contact with no change in biofilm retention
B. Faster osseointegration and less marginal bone loss during the first year
C. Little further gain in integration, with greater biofilm retention if exposed
D. Improved wettability and greater adhesion of the fibrin clot at surgery
E. Reduced ion release, because the oxide layer is correspondingly thicker

113 You are placing an implant in a posterior maxillary site of low density and wish to load at six to eight weeks rather than at three months. The representative recommends a chemically modified hydrophilic version of the manufacturer's usual sandblasted, acid-etched surface, supplied in a sealed saline ampoule. Which of the following are established properties of that modification?

A. Higher surface energy, so blood and plasma proteins spread across it
B. A greater surface roughness than the conventional sandblasted surface
C. Antibacterial activity that reduces the rate of early peri-implant infection
D. A and B only
E. A, B and C

114 A 76-year-old man attends with an implant placed eighteen years ago that carries a plasma-sprayed hydroxyapatite coating. Probing depths are 7 mm with bleeding, the radiograph shows bone loss to half the implant length, and the exposed surface has an irregular, flaking appearance where it can be seen. Which property of the coating best explains its historical association with this pattern of failure?

A. Hydroxyapatite is radiolucent, so marginal bone loss appears exaggerated
B. The coating raises the surface roughness beyond 10 micrometres
C. Hydroxyapatite can be removed only by osteoclastic resorption
D. The coating may delaminate from the substrate and dissolve unevenly
E. Plasma spraying thins the underlying titanium dioxide layer

115 A peri-implantitis lesion at the LL6 shows 4 mm of horizontal bone loss with no intrabony component, 3 mm of keratinised mucosa, and a moderately rough surface that will remain supracrestal after treatment. The patient is a non-smoker in regular maintenance, and a resective, apically repositioned flap has been agreed after full discussion of the aesthetic consequences. Which adjunct is most likely to reduce recurrent biofilm accumulation on the exposed surface?

A. Air polishing the exposed threads with glycine powder at each review
B. Implantoplasty of the exposed threads with diamond and polishing burs
C. Application of citric acid to the exposed surface for two minutes
D. Local delivery of minocycline microspheres following debridement
E. A seven-day course of amoxicillin with metronidazole after surgery

116 A monolithic zirconia crown on an implant at the LR6 requires occlusal adjustment at delivery. It is reduced with a coarse diamond under water and the patient is dismissed without the adjusted surface being polished. Which statement best describes what grinding has done to the material?

A. Grinding has removed the glaze, exposing porous feldspathic ceramic beneath
B. Grinding has annealed the surface, relieving residual compressive stress
C. Grinding has removed the yttria stabiliser, lowering the flexural strength
D. Grinding has coarsened the crystal size, making the ceramic more translucent
E. Grinding has transformed tetragonal grains to monoclinic, with local expansion

117 A one-piece zirconia abutment supporting a cemented crown at the UR1 fractures at the level of the implant platform nine years after placement. There is no history of trauma or parafunction, the occlusion shows light contact in centric with canine guidance, the screw is intact and the bone level is unchanged. Which mechanism best explains the failure?

A. Slow tetragonal-to-monoclinic transformation at the surface in a moist environment
B. Fatigue of the titanium screw transferring bending load into the ceramic
C. Thermal cycling causing differential expansion at the ceramic-to-metal interface
D. Hydrolysis of the yttria stabiliser into a soluble hydroxide over time
E. Creep of the ceramic under the sustained clamping force of the screw

118 A one-piece zirconia abutment is proposed for a 3.3 mm diameter implant at the UR2 in a patient with a thin phenotype and a high smile line. The laboratory offers as an alternative a zirconia superstructure bonded to a titanium base, and asks which you prefer. Both would be screw-retained and the occlusal scheme is favourable. Which is the strongest argument for the titanium base?

A. Zirconia cannot be bonded reliably to the overlying crown material
B. Zirconia is radiolucent, so residual cement could not be identified
C. Zirconia in direct contact with the connection wears the titanium implant
D. Zirconia has a lower flexural strength than lithium disilicate
E. Zirconia cannot be prepared to a margin below the mucosal level

119 A 41-year-old woman requests a metal-free restoration for a missing UR1 and has read about one-piece zirconia implants. The site is healed with adequate bone volume, but the ridge angulation would require about 20 degrees of prosthetic correction, and she has a thin phenotype with a high smile line. Which of the following are legitimate reservations about a one-piece zirconia implant here?

A. Angulation cannot be compensated at abutment level as it could with titanium
B. The transmucosal portion is exposed to loading throughout the healing period
C. Zirconia fails to osseointegrate in experimental and animal models
D. A and B only
E. A, B and C

120 A single crown is being delivered on an implant at the LL5 and the manufacturer specifies a tightening torque of 35 Ncm for the abutment screw, applied with a calibrated torque driver. The components are new and the fit is passive. Which best describes what that applied torque is intended to achieve?

A. Compression of the abutment into the implant until the mating surfaces yield
B. Elastic elongation of the screw, clamping the joint under tension
C. Complete conversion into frictional resistance at the threads and screw head
D. Seating of a conical abutment by overcoming the wedging effect of the taper
E. A clamping force matched to the maximum occlusal force expected on the crown

121 A screw-retained crown on an external hexagon implant at the LR6 has loosened three times in a year. Each time the same screw has been re-tightened with a calibrated driver to the specified 30 Ncm. The occlusion is even with no contact in excursion, the crown is not cantilevered, and a periapical radiograph shows a completely seated, passive fit with stable bone levels. What is the most appropriate next step?

A. Fit a new screw, torque to specification, and re-torque after ten minutes
B. Apply a thread-locking adhesive to the existing screw before tightening
C. Cement the crown to the abutment so the joint cannot be disturbed
D. Reduce the occlusal table and re-torque the existing screw as before
E. Replace the restoration with a cemented crown on a new abutment

122 Consider these statements about the difference in elastic modulus between titanium, at roughly 110 GPa, and cortical bone, at roughly 15 GPa: (i) the mismatch concentrates functional strain in the crestal bone; (ii) it is part of the rationale for seeking designs or materials that behave more like a periodontal ligament; (iii) zirconia is preferable in this respect because its modulus lies closer to that of bone. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

123 Two implant systems are being compared for a single molar restoration in a patient with a heavy occlusion. One uses an external hexagon connection, the other an internal conical connection with a matching abutment. Published data show a lower incidence of screw loosening with the conical design. Which feature best explains that difference?

A. The screw is longer, so more preload is generated at the same applied torque
B. The taper is sealed with cement, excluding bacteria from the internal joint
C. A higher tightening torque may safely be applied to the abutment screw
D. The design is platform switched, moving the joint horizontally from bone
E. The taper carries lateral load by friction, so less is transferred to the screw

124 A patient's two implants in the lower right quadrant are restored with abutments narrower than the implant platforms, and she asks why the radiographs of these look different from those of an implant placed for her ten years earlier, where the abutment matched the platform. Both sets of implants are healthy and in regular maintenance. Which statement best explains the crestal bone response to the newer design?

A. The inflammatory infiltrate at the microgap is displaced horizontally inwards
B. The narrower abutment reduces the bending moment on the implant neck
C. The reduced abutment diameter increases the width of keratinised mucosa
D. The mismatch increases the surface area available for bone-to-implant contact
E. The supracrestal attachment is then obliged to form in a vertical direction only

125 The residual bone above the mandibular canal at the LR6 measures 10 mm on CBCT and the ridge is 9 mm wide at crest level, with a dense cortical plate and the opposing dentition natural. You may place a 4.1 mm diameter implant of 8 mm length, leaving 2 mm above the canal, or a 5.0 mm implant of 6 mm length. Which statement best supports choosing the wider, shorter implant?

A. A wider implant achieves a higher insertion torque in any bone type
B. Shorter implants distribute occlusal load more evenly along the body
C. Resistance to bending increases far more steeply with diameter than with length
D. Crown-to-implant ratio is what determines subsequent marginal bone loss
E. Implant length contributes little to the surface area available for bone contact

126 Two designs are offered for a posterior implant to be placed in a site of moderate density: one with a conventional V-shaped thread profile and one with a square, or power, thread. The intended restoration is a single crown with an axially directed occlusion. Which statement best describes the mechanical rationale for the square thread?

A. It reduces the insertion torque required in dense cortical bone
B. It increases the available surface area approximately threefold
C. It resists tensile forces more effectively beneath a cantilever
D. It converts more of the occlusal load into compressive stress in bone
E. It allows a shallower osteotomy to be prepared for the same stability

127 A UR1 lost to a vertical root fracture is to be extracted and the socket preserved, with implant placement planned at four months. The buccal plate is intact but under 1 mm thick, the phenotype is thin, and the principal aim is to maintain ridge contour beneath the future restoration. The patient accepts a graft of animal origin. Which material's resorption profile best matches that aim?

A. Autogenous particulate harvested from the adjacent ramus
B. Deproteinised bovine bone mineral with a collagen matrix
C. Beta-tricalcium phosphate granules
D. Calcium sulphate paste as a socket filler
E. Demineralised freeze-dried bone allograft

128 A trephine core is taken through the lateral window site nine months after a sinus floor augmentation performed with deproteinised bovine bone mineral, at the time of implant placement. Histology shows about 25 per cent new vital bone, residual graft particles surrounded by new bone and osteoid with osteoblasts on their surfaces, and no inflammatory infiltrate. What does this finding indicate?

A. Failure of the graft to remodel, requiring a second augmentation procedure
B. Fibrous encapsulation of the particles, which contraindicates implant placement
C. Inadequate vital bone, so placement should be deferred for a further six months
D. Contamination of the graft by sinus mucosa displaced during the procedure
E. Normal healing of a slowly substituted xenograft, suitable for implant placement

129 A native, non-cross-linked porcine collagen membrane has been used to cover a particulate graft at a dehiscence defect. The flap has been advanced by periosteal release and closed without tension, and the patient asks how long the membrane will remain in place. Which of the following are true of a membrane of this type?

A. It is degraded by host enzymes over a matter of a few weeks
B. It maintains space over a large defect without additional support
C. It is removed at a second procedure approximately four weeks later
D. A and B only
E. All of the above

130 A ridge augmentation is planned at the LR5 in a patient in whom the flap will be difficult to advance and where primary closure may not be maintained, and a high-density PTFE membrane is selected in preference to a resorbable collagen barrier. The graft is particulate and the site is not infected. Which property of the membrane makes it appropriate to this problem?

A. It is cross-linked, so it resists collagenase at an exposed site
B. It integrates with the flap margins, sealing the wound periphery
C. It resorbs slowly enough to maintain space for about six months
D. It releases antimicrobial ions throughout the healing period
E. Its pore size is too small to permit bacterial penetration

131 A crown is to be cemented on an implant abutment at the UR4 whose margin lies 2 mm subgingivally on the palatal aspect and 1 mm buccally, in a patient with a history of periodontitis. You are deciding how to reduce the risk of residual cement in the sulcus. Which of the following actually reduce the amount of cement left behind?

A. Choosing a luting cement with a film thickness below 25 micrometres
B. Choosing a cement of high radiopacity, so that excess can be identified
C. Extruding the excess extra-orally on a replica of the abutment before seating
D. A and B only
E. A, B and C

132 A patient presents with bleeding on probing, 6 mm pockets and 2 mm of bone loss at a cemented implant crown placed elsewhere three years ago, and reports the symptoms began within months of the crown being fitted. The abutment margin lies 2 mm subgingivally and the periapical radiograph shows no radiopaque material at the margin. Which statement best guides your interpretation of that radiograph?

A. Absence of a radiopacity on the film excludes retained cement as the cause
B. The radiopacity of a cement depends on its film thickness rather than its composition
C. Cone beam imaging would reliably demonstrate any retained fragment
D. Some resin cements are barely more radiopaque than dentine and may be invisible
E. Cement is always visible on a periapical film if the angulation is correct

Block G — Treatment planning, SAC, options, consent and referral (Q133–160)

133 A 34-year-old woman has an unrestorable oblique root fracture of the UR1. Her smile exposes 3 mm of gingiva, her phenotype is thin and scalloped, and the adjacent teeth are unrestored. CBCT shows an intact facial wall measuring 1.1 mm, no periapical infection and 5 mm of bone apical to the socket. She wishes the tooth removed and an implant placed and provisionalised at the same visit. Using the SAC classification, how is this case best described?

A. Advanced surgically and straightforward restoratively
B. Advanced surgically and advanced restoratively
C. Complex surgically and straightforward restoratively
D. Complex surgically and advanced restoratively
E. Complex surgically and complex restoratively

134 A 46-year-old man wants a single implant at the LL6, lost a year ago. The site has 12 mm of bone above the mandibular canal, a crestal width of 8 mm, sound unrestored adjacent teeth and a low lip line. He smokes 20 cigarettes a day. He completed treatment for generalised stage III periodontitis eighteen months ago but two sites still probe 5 mm and bleed. Which features raise this case above straightforward?

A. His smoking habit
B. The residual bleeding pockets of 5 mm
C. The crestal width of 8 mm
D. A and B only
E. A, B and C

135 A 27-year-old woman has a congenitally absent UR2. Orthodontic treatment finished two years ago and she wears a bonded retainer. The mesiodistal space measures 5.5 mm at the alveolar crest and 6 mm between the contact points, the ridge is 5 mm wide, the adjacent teeth are unrestored and her lip line is high. Which single feature most strongly favours a cantilever resin-bonded bridge over an implant?

A. The space of 5.5 mm at the crest
B. The ridge width of 5 mm
C. The unrestored state of the adjacent teeth
D. The high lip line
E. The recent completion of orthodontics

136 A 71-year-old man has been missing the LL6, LL7 and LR7 for eight years. He has occluding pairs of teeth to the second premolars on both sides, an intact upper arch with no over-eruption or drifting, no symptoms and a plaque score of 11 per cent. He eats a normal diet and asks whether he should have implants "before the bone disappears". Consider these statements: (i) an arch shortened to the premolars gives adequate function for most patients; (ii) eight symptom-free years without drifting support leaving the arch as it is; (iii) molar loss predictably leads to temporomandibular disorder. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

137 A 55-year-old non-smoker has a vertical root fracture of the UR4 with a buccal sinus tract. CBCT shows the buccal plate is already absent over the coronal two-thirds of the root, leaving a defect about 6 mm deep once the tooth is removed. Her lip line is high and her phenotype thin. She wants a fixed replacement. Which plan is most appropriate?

A. Immediate placement with simultaneous guided bone regeneration
B. Immediate placement with a simultaneous connective tissue graft
C. Extraction with grafting of the defect and placement at four months
D. Extraction with early placement at six to eight weeks
E. Extraction with unassisted healing and placement at six months

138 You are consenting a 38-year-old professional clarinettist for a single implant at the LL5. CBCT shows 3.5 mm of bone between the planned apex of an 8 mm implant and the roof of the mandibular canal. She has asked about the fee and the healing time but nothing about complications. Which risk most clearly meets the test of materiality for this particular patient?

A. Loosening of the abutment screw during function
B. Permanent altered sensation of the lower lip and chin
C. Failure of osseointegration in the first three months
D. Chipping of ceramic on the definitive crown
E. Recession exposing the abutment margin

139 During placement of an implant at the UL4 in a fully alert patient under local anaesthesia you find a 4 mm buccal fenestration exposing implant threads. Simultaneous guided bone regeneration with a bovine xenograft and a porcine collagen membrane would deal with it, but grafting was never discussed and the consent documentation covers implant placement alone. Which is the most appropriate course?

A. Graft as planned and explain what was done at the review visit
B. Place the implant unaugmented and reassess the defect at six weeks
C. Remove the implant, close the flap and re-consent at a later visit
D. Graft using only bone collected from the osteotomy, avoiding the issue
E. Stop, describe the finding and the options, and proceed only with her agreement

140 A 74-year-old man with moderate vascular dementia lives in residential care. He cannot retain or weigh information about implant treatment, and a properly conducted assessment concludes he lacks capacity for this decision. His lower complete denture, made eleven years ago, is unstable. His daughter, who holds a registered lasting power of attorney for health and welfare, asks for two implants to retain it. What is the most appropriate course?

A. Place two implants, since the attorney has consented on his behalf
B. Apply to the Court of Protection before providing any treatment
C. Provide a new conventional lower denture and reassess the need
D. Provide no treatment, as he cannot consent to any intervention
E. Place two implants under intravenous sedation to limit distress

141 A 52-year-old woman has congenitally absent UR2 and UL2 restored with resin-bonded bridges that have repeatedly debonded. She has a high lip line, previous orthodontic space opening, 4 mm of horizontal ridge deficiency at each site and adequate bone height. She wants fixed replacements and you have placed sixty straightforward posterior implants. To whom is referral most appropriately directed?

A. An oral surgeon, for ridge augmentation before you restore
B. A periodontist, for soft tissue augmentation at both sites
C. An oral and maxillofacial surgeon, for a mandibular block graft
D. A consultant in restorative dentistry, for joint planning
E. A general dental practitioner holding an implant diploma

142 A 60-year-old woman wants a single implant at the LR5, lost six months ago after a root fracture. CBCT shows 11 mm of bone above the mandibular canal and 7.5 mm of crestal width. Her type 2 diabetes is controlled on metformin with an HbA1c of 48 mmol/mol, she has never smoked, her periodontal tissues are stable and she is mildly anxious but accepted a surgical extraction under local anaesthesia last year. What is the most appropriate course?

A. Place the implant yourself under local anaesthesia
B. Refer for placement because of the diabetes
C. Refer for placement under intravenous sedation
D. Defer until the HbA1c falls below 42 mmol/mol
E. Refer for a restorative opinion on the prosthetic design

143 Six weeks after placing an implant at the UR3 you find the buccal mucosa thin and greyish over the abutment and recommend a connective tissue graft, adding £600 to the £2,400 written estimate the patient accepted before treatment began. She has already paid two instalments and is keen to proceed at the next visit. What should you do before providing the graft?

A. Provide the graft and add the fee to the final invoice
B. Provide the graft without charge, as the need arose during your treatment
C. Explain the additional cost verbally and record that you have done so
D. Ask her to countersign the original estimate to acknowledge the change
E. Give her a revised written estimate and obtain her agreement to it

144 A 68-year-old woman is to have two implants placed in the interforaminal region to retain a new lower complete denture with individual stud attachments. She is medically well, cleans effectively and attends regularly. As part of consenting her, which event should you tell her is the most likely to require attention during the first five years?

A. Loss of one of the two implants
B. Replacement of the retentive inserts in the denture
C. Fracture of the acrylic denture base
D. Surgical treatment of peri-implantitis
E. Replacement of both implant abutments

145 A 47-year-old healthy non-smoker with a stable periodontium asks what he is committing himself to over the next twenty years if he accepts a single implant crown at the UR5. Which of the following should form part of that discussion?

A. Professional maintenance will be needed for as long as the implant is in place
B. Screws and ceramic may require replacement during the life of the restoration
C. The implant itself will need replacing after about ten years
D. A and B only
E. All of the above

146 A 64-year-old edentulous man wants a fixed maxillary prosthesis. His complete upper denture has a labial flange, and with the flange removed his upper lip visibly falls in. CBCT shows adequate bone volume for implants, but the crest lies about 8 mm palatal to the planned incisal position. He has read about fixed teeth in a day and is impatient to begin. What is the most appropriate next step?

A. Make a trial denture set-up without a flange and assess the appearance
B. Plan an implant-retained overdenture with a labial flange
C. Plan six implants for a fixed prosthesis with pink ceramic
D. Order a further CBCT with a radiographic stent in place
E. Refer for assessment for a fixed full-arch reconstruction

147 A 41-year-old woman will lose the UR1 to external cervical resorption. CBCT shows an intact facial wall 0.7 mm thick, no periapical pathology and 6 mm of sound bone apical and palatal to the socket. Her lip line is medium and her phenotype thick, and she would prefer not to wear a removable provisional. Which placement protocol is best supported?

A. Immediate placement with an immediate provisional crown
B. Immediate placement with a healing abutment
C. Early placement at six weeks after soft tissue healing
D. Early placement at twelve weeks after partial bone healing
E. Late placement at six months into a healed ridge

148 A 57-year-old man has an LR6 root-treated twelve years ago and crowned. It is symptom-free, but there is recurrent caries at the distal margin extending 2 mm below the alveolar crest, a grade II buccal furcation involvement, 45 per cent bone loss at the tooth and a defective crown margin. The rest of his periodontium is stable and well maintained. Which finding most strongly favours extraction and implant replacement?

A. The root filling placed twelve years ago
B. The grade II buccal furcation involvement
C. The 45 per cent bone loss at the tooth
D. The defective margin of the existing crown
E. The caries extending 2 mm below the crest

149 A 54-year-old man is missing the LR5 and LR6 and has a sound, recently placed full crown on the LR4. To avoid the cost of a second implant he asks whether one implant at the LR6 could be joined to the LR4 to carry a three-unit fixed prosthesis. Which statement best explains why this design is generally avoided?

A. The implant and the tooth have different coefficients of thermal expansion
B. The implant is displaced a few micrometres and the tooth up to a hundred
C. The implant cannot be splinted to a crown that has already been cemented
D. The tooth abutment will be devitalised by the additional occlusal load
E. The prosthesis cannot be retrieved once it has been cemented in place

150 A 58-year-old edentulous man asks for a fixed lower prosthesis on four implants loaded on the day of surgery, having read about it online. He has 13 mm of bone height anterior to the mental foramina, adequate width and no medical problems. You have placed about eighty single implants and have never provided a full-arch reconstruction. Which of the following are appropriate parts of your response?

A. Explaining that a two-implant overdenture is an alternative worth considering
B. Referring him for assessment for the fixed prosthesis he has asked about
C. Providing the four implants with immediate loading under mentorship
D. A and B only
E. A, B and C

151 A 70-year-old woman has worn complete dentures for six years. Her upper denture is comfortable; her lower is retentive only while she keeps her tongue still. She is fit, cleans well and attends regularly. Consider these statements: (i) two implants retaining a mandibular overdenture should be considered the first choice of treatment for the edentulous mandible; (ii) four implants give significantly greater patient satisfaction than two in the mandible; (iii) two implants are equally established as the standard for the edentulous maxilla. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (i) and (iii) only
E. (i), (ii) and (iii)

152 A 63-year-old non-smoker is missing the UL6. CBCT shows 7 mm of residual bone height beneath the sinus floor, a crestal width of 8 mm, a membrane of normal thickness, no septum and no sinus pathology. She is medically well, has a stable periodontium and wants a fixed replacement. Which option delivers the outcome she wants with the least exposure to complications?

A. A lateral window elevation with placement at six months
B. A transcrestal elevation with a 10 mm implant placed simultaneously
C. A 6 mm implant placed without sinus floor elevation
D. A cantilever pontic from a new crown on the UL5
E. A cobalt-chromium removable partial denture

153 You are planning a single implant to replace the UR1 in a 44-year-old woman. Her lip line is high, her phenotype thin, her crown forms triangular and her stated expectations high. The interproximal bone on the adjacent UR2 lies 7 mm below the planned contact point. Which finding most reliably predicts that the papilla will be deficient?

A. The high lip line
B. The thin phenotype
C. The triangular crown form
D. Her high aesthetic expectations
E. The 7 mm from contact point to interproximal bone

154 A 36-year-old woman has lost the UR1 and UR2 in a fall. The ridge is intact with adequate height and width, the adjacent UR3 and UL1 are unrestored with normal bone levels, her phenotype is thick and her lip line high. She wants a fixed replacement. Which design is most likely to preserve the soft tissue between the two replaced teeth?

A. Two adjacent implants restored with individual crowns
B. One implant at the UR1 with a cantilever pontic replacing the UR2
C. Two adjacent implants splinted as a single restoration
D. One implant at the UR2 with a cantilever pontic replacing the UR1
E. Two adjacent implants with platform-switched abutments

155 A 29-year-old man is to have the UR1 removed and replaced with an implant. The tooth has a horizontal root fracture and the buccal wall is intact but thin. He is getting married in ten weeks, has seen photographs of completed cases online and asks whether the definitive crown can be fitted before the wedding. What is the most appropriate response?

A. Extract and place immediately, fitting the definitive crown at eight weeks
B. Extract, place at six weeks and load conventionally after the wedding
C. Explain that a provisional will be in place for the wedding and plan around it
D. Defer all treatment until after the wedding to avoid disappointment
E. Provide a resin-bonded bridge and reconsider implants in a year

156 Two years after you placed an implant at the UL5 a patient complains that she was never told a bone graft might be needed and objects to the additional fee she paid on the day of surgery. Her record contains a signed practice consent form listing "implant placement, with grafting if required", a copy of the practice information leaflet, the receipt she signed, and a clinical entry for the surgical visit describing the procedure performed. Which of these would most strongly support your position?

A. A contemporaneous note of the planning discussion and the options offered
B. The signed consent form listing grafting as a possibility
C. The practice information leaflet issued at the first visit
D. A statement from the nurse who assisted at the surgery
E. The signed receipt showing she paid without objection at the time

157 A 24-year-old woman lost the LR6 two years ago and has had no replacement. The LR7 has tipped mesially by about 20 degrees, the UR6 has over-erupted 2 mm, the mesiodistal space measures 8 mm at the alveolar crest and 6 mm between the occlusal contact points. She is caries-free with excellent hygiene and now wants an implant. Which finding most affects the feasibility of the treatment she is requesting?

A. The 8 mm of space at the alveolar crest
B. The mesial tipping of the LR7
C. The 2 mm of over-eruption of the UR6
D. The two-year interval since the extraction
E. The 6 mm of space between the contact points

158 Consider these statements about the SAC classification as it is used in implant treatment planning: (i) a case is classified separately for its surgical and its restorative aspects, and the two classifications need not agree; (ii) the classification depends on the procedure and is independent of the individual patient; (iii) once assigned at the planning stage the classification is not revised. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (i) and (iii) only
E. (i), (ii) and (iii)

159 A small-field CBCT taken to plan an implant at the LL6 also includes the lower part of the left maxillary sinus, where you notice a smooth dome-shaped opacity about 12 mm high arising from the sinus floor. The patient has no nasal or sinus symptoms and no facial pain. Which of the following is required of you?

A. Reporting on the whole of the imaged volume, not only the implant site
B. Referring the finding for an urgent suspected cancer opinion
C. Repeating the examination with a larger field to include both sinuses
D. A and B only
E. A, B and C

160 A 46-year-old man wants two implants and can readily afford the surgery, but tells you he will not pay for hygiene appointments or annual reviews because his previous dentist "never needed to see him". He smokes ten cigarettes a day, his plaque score is 55 per cent and he has generalised gingivitis with no attachment loss. He is otherwise healthy and the sites are anatomically straightforward. What is the most appropriate course?

A. Proceed, having recorded that he declined maintenance
B. Proceed with a screw-retained design to simplify later access
C. Address the hygiene and the maintenance commitment before deciding
D. Decline and provide a removable partial denture instead
E. Refer him to a colleague who includes maintenance in the fee

Block H — Prescribing, local anaesthesia, drug interactions and sedation (Q161–180)

161 A 66-year-old man with stable angina and a drug-eluting stent placed three years ago attends for an implant at the LR5. His cardiologist advises keeping the total adrenaline dose within 0.04 mg for the appointment. You have 2 per cent lidocaine with adrenaline 1:80,000 in 2.2 ml cartridges. Disregarding the lidocaine itself, what is the largest number of whole cartridges you may give?

A. One cartridge
B. Two cartridges
C. Three cartridges
D. Four cartridges
E. Six cartridges

162 You are choosing between 2 per cent lidocaine with adrenaline 1:80,000 and plain 2 per cent lidocaine for a long implant appointment in a 55-year-old woman with well-controlled hypertension. The maximum recommended dose of lidocaine is lower for the plain solution than for the solution containing adrenaline. Which explanation best accounts for that difference?

A. Adrenaline reduces the fraction of lidocaine that circulates unbound in plasma
B. Adrenaline accelerates the hepatic metabolism of lidocaine
C. Adrenaline lowers tissue pH, so less lidocaine is present in the ionised form
D. Adrenaline slows systemic absorption, so the peak plasma concentration is lower
E. Adrenaline shortens the elimination half-life of lidocaine

163 A 74-year-old man with well-compensated alcoholic liver disease requires three hours of implant surgery under local anaesthesia. You are considering 4 per cent articaine with adrenaline in preference to 2 per cent lidocaine with adrenaline. Which of the following is a genuine pharmacological advantage of articaine in this patient?

A. Most of a dose is inactivated by plasma esterases rather than in the liver
B. It is excreted unchanged by the kidney, bypassing hepatic metabolism entirely
C. It carries a lower risk of methaemoglobinaemia than lidocaine
D. A and B only
E. A and C only

164 A 24-year-old woman has received seven cartridges of 3 per cent prilocaine with felypressin during a long surgical appointment, preceded by a topical benzocaine gel. Ninety minutes later she is agitated and dusky, her pulse oximeter reads 85 per cent and does not improve with high-flow oxygen, her chest is clear and a venous sample looks unusually dark. What is the most likely diagnosis?

A. Local anaesthetic systemic toxicity
B. Methaemoglobinaemia
C. Hyperventilation with respiratory alkalosis
D. Anaphylaxis to the anaesthetic solution
E. Silent aspiration of blood or debris

165 During buccal infiltration a patient becomes light-headed, describes tingling around her lips and a metallic taste, and says the room sounds as though it is ringing. She is speaking normally, her airway is clear, her pulse is 88 and regular and her blood pressure is 128/80 mmHg. What do these features most likely represent?

A. A vasovagal episode provoked by the injection
B. An immediate hypersensitivity reaction to a preservative
C. Early central nervous system signs of local anaesthetic toxicity
D. A panic attack with hyperventilation
E. An adrenaline reaction from an intravascular injection

166 A 29-year-old woman who is 26 weeks pregnant needs an unplanned procedure to remove a painful failed implant. The pregnancy is uncomplicated and she is otherwise well. A colleague suggests 3 per cent prilocaine with felypressin so as to avoid giving her adrenaline. What is the most appropriate response?

A. Agree, since adrenaline crosses the placenta and reduces uterine perfusion
B. Agree, but limit the total dose to two cartridges
C. Use plain lidocaine, so that no vasoconstrictor is given at all
D. Use articaine with adrenaline, which crosses the placenta least
E. Use lidocaine with adrenaline, since felypressin is the agent to avoid

167 A 61-year-old man is listed for implant placement and his record carries the entry "penicillin allergy". On questioning he says that about thirty years ago a course of penicillin gave him nausea, loose stools and abdominal cramps lasting two days, with no rash, swelling, wheeze or need for treatment, and he has avoided penicillins since. Which is the most appropriate conclusion?

A. He has a delayed hypersensitivity and should avoid all beta-lactam drugs
B. The reaction described is intolerance, and amoxicillin may be prescribed
C. He should receive a macrolide whenever an antibiotic is indicated
D. He should be referred for allergy testing before any antibiotic is prescribed
E. He may be given cefalexin, since cephalosporin cross-reactivity is low

168 A 58-year-old man taking simvastatin 40 mg at night and apixaban 5 mg twice daily develops a spreading infection around a recently placed implant, with fever and trismus. His record documents an anaphylactic reaction to amoxicillin. Which antibiotic should be avoided because of his regular medication?

A. Metronidazole
B. Doxycycline
C. Clarithromycin
D. Clindamycin
E. Cefalexin

169 A 70-year-old woman on warfarin for atrial fibrillation, with an INR that has been stable at about 2.4 for three years, is prescribed metronidazole 400 mg three times daily for seven days after implant surgery complicated by infection. Four days later she has bleeding gums, extensive bruising and an INR of 6.1. Which mechanism best explains what has happened?

A. Metronidazole displaces warfarin from binding sites on plasma albumin
B. Metronidazole suppresses vitamin K synthesis by the gut flora
C. Metronidazole reduces the renal clearance of warfarin
D. Metronidazole inhibits the hepatic metabolism of S-warfarin
E. Metronidazole exerts a direct antiplatelet effect of its own

170 You are prescribing analgesia after implant surgery. Consider these three patients: (i) a 68-year-old man with stage 4 chronic kidney disease taking ramipril and furosemide; (ii) a 44-year-old woman with asthma who developed wheeze and facial flushing after taking aspirin; (iii) a 55-year-old man with hypertension controlled on amlodipine alone. For which of them is ibuprofen contraindicated?

A. (i) only
B. (ii) only
C. (i) and (iii) only
D. (i) and (ii) only
E. (i), (ii) and (iii)

171 A healthy 39-year-old man weighing 80 kg has had two implants placed with a small flap and no augmentation, under local anaesthesia that is still profound. He takes no medication, has no gastric or respiratory history and asks what he should take when he gets home. Which regimen is best supported?

A. Paracetamol 1 g four times daily, with codeine 30 mg added as required
B. Ibuprofen 400 mg with paracetamol 1 g, taken regularly for 48 hours
C. Ibuprofen 400 mg as required, with codeine 30 mg if that is inadequate
D. Paracetamol 1 g four times daily alone for the first 48 hours
E. Dihydrocodeine 30 mg with paracetamol 1 g as required

172 A 62-year-old woman with type 2 diabetes controlled on metformin, whose HbA1c is 52 mmol/mol, is having two implants placed with a small connective tissue graft. A colleague suggests a single 8 mg oral dose of dexamethasone before surgery to limit post-operative swelling. Which consideration is most important before agreeing?

A. A single dose will suppress her hypothalamic-pituitary-adrenal axis
B. Corticosteroids impair osseointegration at this dose
C. Dexamethasone interacts with metformin and reduces its absorption
D. Antibiotic cover becomes necessary once a corticosteroid has been given
E. Her blood glucose will rise for a day or two and she must be told

173 A 46-year-old man is extremely anxious about implant surgery and asks for intravenous sedation at your practice, where you hold a sedation qualification and provide midazolam sedation regularly. His body mass index is 43, he has obstructive sleep apnoea treated with nocturnal continuous positive airway pressure, he takes two antihypertensives and his exercise tolerance is one flight of stairs. What is the most appropriate course?

A. Provide midazolam sedation using a reduced initial increment
B. Provide inhalation sedation with nitrous oxide and oxygen instead
C. Refer for sedation assessment in a specialist or hospital setting
D. Provide oral midazolam the night before and treat under local anaesthesia
E. Provide midazolam sedation with continuous capnography in place

174 A 78-year-old woman becomes over-sedated during implant surgery under intravenous midazolam. Flumazenil is given and she responds promptly, becoming alert and cooperative within a minute. She is otherwise well and her family are waiting to take her home. Which is the most important consequence for her subsequent care?

A. She may become re-sedated as the flumazenil is eliminated
B. She will be resistant to benzodiazepines at future appointments
C. She is at high risk of a withdrawal seizure from the flumazenil
D. She must be admitted for overnight observation as a matter of course
E. She will need a reduced dose of local anaesthetic for the rest of the visit

175 A 55-year-old man is booked for intravenous midazolam sedation for implant surgery. He takes modified-release morphine 30 mg twice daily for chronic back pain and has taken his morning dose. He is otherwise fit, his airway assessment is unremarkable and his oxygen saturation on air is 98 per cent. Which is the most important implication for the sedation?

A. He will need a larger dose of midazolam because of opioid tolerance
B. Respiratory depression may occur at a lower midazolam dose than expected
C. Flumazenil will reverse both drugs if he becomes over-sedated
D. The morphine will prolong recovery of psychomotor function only
E. Midazolam will potentiate his analgesia and reduce the anaesthetic needed

176 You are considering inhalation sedation with nitrous oxide and oxygen for an anxious patient before implant surgery. Which of the following are recognised reasons to avoid or postpone the technique?
A. Middle ear surgery performed in the past week
B. Untreated vitamin B12 deficiency
C. A blocked nose from an acute upper respiratory infection
D. A and B only
E. All of the above

177 A 72-year-old man had a total hip replacement fourteen months ago. His orthopaedic clinic letter, written at discharge, states that he should have "antibiotic cover for dental work". He is otherwise healthy, takes no regular medication and is attending for placement of a single implant at the LR6 in a healed, ungrafted site. What is the most appropriate course?
A. Give amoxicillin 3 g one hour before, as the surgeon has requested
B. Give a five-day course of amoxicillin after the procedure
C. Postpone until the orthopaedic team confirms the advice in writing
D. Explain that the prosthetic joint itself requires no antibiotic cover
E. Give clindamycin 600 mg as an alternative prophylactic regimen

178 A 52-year-old woman rinses with 0.2 per cent chlorhexidine immediately before implant surgery. Within five minutes she has widespread urticaria, facial swelling, audible wheeze and a systolic blood pressure of 74 mmHg. She has used chlorhexidine-containing mouthwash and skin preparations previously without any reaction. Which statement is correct?
A. Chlorhexidine is not a recognised cause of anaphylaxis
B. Sensitisation requires previous parenteral exposure to chlorhexidine
C. Previous uneventful exposure is expected, since sensitisation must precede reaction
D. An antihistamine should be given before adrenaline in a conscious patient
E. The picture is most likely a chemical irritant response to the rinse

179 A 34-year-old man attends for implant placement. He is restless and talkative, his pulse is 108 and regular and his blood pressure 168/96 mmHg. On direct questioning he says he used cocaine "a couple of hours ago" and does so most weekends. He is otherwise healthy, has eaten normally and insists that he wants to go ahead today. What is the most appropriate course?
A. Proceed using plain lidocaine, with no vasoconstrictor
B. Proceed using prilocaine with felypressin
C. Proceed once his pulse and blood pressure have settled in the chair
D. Proceed with lidocaine and adrenaline, limited to two cartridges
E. Postpone the elective surgery and arrange to see him another day

180 Four days after uneventful implant placement at the LL6, a patient telephones reporting discomfort. Consider these three situations: (i) a diffuse swelling extending to the submandibular region with trismus and a temperature of 38.2°C; (ii) a localised fluctuant swelling over the implant, which you incise and drain at the same visit, with no systemic upset; (iii) persistent aching at the site with no swelling, no bleeding and a normal temperature. In which is a therapeutic antibiotic indicated?
A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

Block I — Surgical placement, flaps, timing, loading, suturing and asepsis (Q181–218)

181 A 34-year-old woman lost the UR1 and UR2 in a cycling accident two years ago. The ridge is healed, the phenotype is thick, and the interproximal bone crest lies 4 mm below the planned contact points. The mesiodistal space available means the two implant platforms would sit 2 mm apart. What is the principal disadvantage of that spacing?

A. The interimplant papilla will be governed by the attachment level of the adjacent teeth
B. Primary stability will be reduced because the interseptal bone is thin
C. The crestal remodelling around each implant will overlap and the crest will fall
D. The embrasure will be too narrow to admit an interdental brush
E. The screw access channels will emerge through the incisal edges

182 A 41-year-old man has an implant placed at the UL1. At the end of surgery the platform lies 1 mm apical to the planned mid-facial mucosal margin rather than the 3 mm intended. Bone is intact circumferentially, the platform sits 1.5 mm palatal to the emergence of the adjacent incisors, and insertion torque was 40 Ncm. Which consequence is most likely?

A. An abrupt emergence profile with the restorative margin visible through the mucosa
B. Excess cement lodged in a sulcus too deep to inspect reliably
C. Remodelling to the first thread producing a mid-facial dehiscence
D. Loss of the papilla shared with the adjacent central incisor
E. Fracture of the abutment screw under eccentric anterior loading

183 A CBCT for a planned UR1 implant shows a ridge 7 mm wide at the crest. The implant is planned in the correct prosthetic position, 1.5 mm palatal to the emergence of the adjacent incisors, and a 4.1 mm platform in that position leaves 0.6 mm of bone buccally and 2.4 mm palatally. The mucosa is thin and the smile line is high. What is the strongest reason to modify the plan?

A. A 4.1 mm platform cannot be accommodated within a 7 mm ridge
B. A buccal wall below 1 mm will resorb and the mid-facial mucosa will recede
C. The implant will emerge buccal to the cingula of the adjacent incisors
D. Insertion torque will be excessive against the dense palatal cortex
E. The palatal plate is likely to fenestrate during final drilling

184 A 52-year-old woman is to have a single implant at the UL2. Periapical radiographs show the interproximal bone crest on the UL1 lying 7 mm apical to the planned contact point, and the crest on the UL3 lying 5 mm apical to its contact point. Both teeth are vital with normal attachment and no restorations. What should she be told about the papillae?

A. Both papillae should fill, since a single implant lies between two natural teeth
B. The mesial papilla should fill and the distal is likely to remain deficient
C. Neither papilla will fill, because papilla height follows the implant crest
D. The distal papilla should fill and the mesial is likely to remain deficient
E. Papilla height cannot be predicted until the provisional has been contoured

185 You are planning a buccal mucoperiosteal flap from the UR4 to the UR6 to expose a 4 mm horizontal ridge defect for guided bone regeneration, with coronal advancement over a membrane and tension-free primary closure. The overlying mucosa is thin. Which of the following statements about that flap are correct?

A. Perfusion enters from the base through supraperiosteal vessels running coronally
B. Each vertical relieving incision interrupts collateral supply to the flap it bounds
C. Relieving incisions should be sited at line angles over sound bone
D. A and C only
E. All of the above

186 Six weeks after implant placement with contour grafting at the UR1, the mid-facial margin of the adjacent UR2 has receded 1.5 mm. The flap was raised full thickness through sulcular incisions around the implant site and both neighbours, with two vertical reliefs beyond the mucogingival junction. The UR2 is vital, unrestored, thin-phenotype, and was not grafted. What is the most likely explanation?

A. Elevation of a thin buccal plate and its periosteum over the UR2 root
B. Suture tension transmitted through the interdental papillae
C. Traumatic contact on the UR2 from the implant provisional crown
D. Toothbrush abrasion during the period of altered oral hygiene
E. Resorption of xenograft particles placed over the UR2 root surface

187 Guided bone regeneration at the LR4 to LR6 requires about 8 mm of coronal flap advancement if closure is to be passive over a titanium-reinforced membrane. The mental foramen is identified on CBCT at the level of the LR5 apex, 9 mm below the crest, and the flap has been raised to expose it. Which description of the periosteal releasing incision is correct?

A. A full-thickness incision through mucosa and periosteum at the flap base
B. A series of vertical scores through the periosteum across the crest
C. A shallow incision dividing periosteum only, followed by blunt dissection
D. Sharp dissection within the mucosa, leaving the periosteum on the flap
E. Division of the buccinator attachment at the mucogingival junction

188 A colleague places most of his implants flaplessly with a tissue punch, arguing that patients report less swelling, that healing is quicker and that the periosteum is left undisturbed. He uses a fully guided template in every case. Which of the following are genuinely given up by that approach?

A. Direct inspection of the buccal plate for fenestration or dehiscence
B. The opportunity to graft a defect discovered during placement
C. Periosteal blood supply to the crestal bone
D. A and B only
E. All of the above

189 A 47-year-old non-smoker is to have an implant at the LR5, planned on a merged CBCT and intraoral scan with a fully guided template. The crest is 7 mm wide with a 6 mm band of keratinised mucosa, there is 13 mm of bone above the mandibular canal, and the cross-sectional image shows a buccal concavity beginning 4 mm below the crest. Which finding most argues against a flapless approach?

A. The crestal width of 7 mm
B. The buccal concavity beginning 4 mm below the crest
C. The 6 mm band of keratinised mucosa
D. The 13 mm of bone above the mandibular canal
E. The use of a template rather than freehand placement

190 A UR4 is extracted because of a vertical root fracture, and the socket walls are found to be intact. The patient returns six weeks later: the mucosa has closed over the socket but the radiograph shows only early trabecular fill. An implant is placed on that day and left submerged, and a screw-retained crown is fitted in full occlusal contact twelve weeks after placement. How is this protocol best described?

A. Type 1 placement with early loading
B. Type 2 placement with early loading
C. Type 3 placement with conventional loading
D. Type 3 placement with early loading
E. Type 2 placement with conventional loading

191 A UR1 with a horizontal root fracture is removed and an implant placed in the socket the same morning. That afternoon a laboratory-made provisional crown is fitted, contoured to support the mucosa and relieved so that it holds no contact in intercuspation or in any excursion. The patient wears it continuously. Which term describes this?

A. Type 1 placement with immediate loading
B. Type 1 placement with early loading
C. Type 1 placement with immediate restoration
D. Type 2 placement with immediate restoration
E. Immediate loading, because the provisional is in function

192 A 60-year-old woman has an implant placed at the LR5 seven months after the tooth was extracted, into a fully healed ridge with no residual socket outline. Insertion torque was 40 Ncm and a healing abutment was fitted at surgery. The definitive screw-retained crown was delivered in full occlusal contact six weeks after placement. Which describes the protocol?

A. Type 3 placement with early loading
B. Type 4 placement with conventional loading
C. Type 4 placement with immediate restoration
D. Type 4 placement with early loading
E. Type 3 placement with conventional loading

193 A 55-year-old man has a single implant placed at the UR5 in a healed ridge. Insertion torque was 45 Ncm and the ISQ 76. The planned crown height is 10 mm, the peri-implant mucosa is 3 mm thick, he has never smoked and his oral hygiene is excellent. His partner reports that he grinds his teeth every night. He would like the crown fitted within the week. Which factor most argues against immediate loading?

A. The maxillary rather than mandibular location of the site
B. The unsplinted single-unit design of the restoration
C. The planned crown height of 10 mm
D. The 3 mm thickness of the peri-implant mucosa
E. The nightly grinding reported by his partner

194 An implant 4.1 mm in diameter is placed in a healed LR5 site and the surgical motor records an insertion torque of 45 Ncm. Preparation felt firm throughout and the crest was well corticated. The operator wishes to know what this single figure entitles him to conclude. Which of the following may legitimately be inferred from that reading?

A. The implant has sufficient primary stability for an immediate restoration
B. The bone at the site is of correspondingly high mineral density
C. Osseointegration is correspondingly more likely to be achieved
D. A and B only
E. A, B and C

195 An implant is placed in the posterior maxilla and monitored by resonance frequency analysis, using a magnetic transducer screwed to the implant. Consider these statements: (i) the value obtained depends on the direction in which the transducer is read; (ii) a rise from ISQ 55 at placement to ISQ 68 at eight weeks indicates developing secondary stability; (iii) an ISQ of 55 at placement precludes any loading protocol. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

196 An implant is placed in the posterior maxilla through a deliberately undersized osteotomy in bone that offered little resistance to the pilot drill. The insertion torque reaches 40 Ncm, but the resonance frequency reading taken immediately afterwards is only ISQ 54. The operator is surprised that the two indices disagree so plainly. Which statement best explains the discrepancy?

A. Undersized preparation raises the ISQ but not the insertion torque
B. Insertion torque is measured dynamically and ISQ statically
C. Torque reflects frictional resistance, ISQ the stiffness of the implant–bone unit
D. The magnetic peg was not fully seated when the reading was taken
E. Hydrophilic implant surfaces give artificially low readings at placement

197 An implant placed with 40 Ncm in a healed LR6 site recorded ISQ 74 at surgery. At three weeks the patient is comfortable, the mucosa around the healing abutment is quiet, she wears no denture and there is no radiographic change. The ISQ has fallen to 65. What is the most likely explanation?

A. Early peri-implant infection reducing bone-to-implant contact
B. Crestal remodelling to the level of the first thread
C. Micromotion transmitted through the healing abutment
D. Resorption of compressed peri-implant bone before woven bone bridges the gap
E. Failure of the implant surface to adsorb plasma proteins at placement

198 A single implant at the LL5 was placed with an insertion torque of 25 Ncm and provisionalised in full occlusal contact the same day. At six weeks the crown is mobile, the patient reports discomfort on biting, and a periapical radiograph shows a thin continuous radiolucent line around the entire implant with no crestal saucerisation. Which mechanism best explains this?

A. Interfacial micromotion beyond about 100 micrometres produced fibrous repair
B. Compression necrosis of the crestal cortex from excessive insertion torque
C. Bacterial contamination of the osteotomy prevented clot stabilisation
D. Heat generated during adjustment of the provisional damaged the interface
E. Occlusal overload fractured the woven bone bridging the healing chamber

199 A 13 mm osteotomy is prepared in the dense anterior mandible. The drill is taken to full depth in a single continuous pass at 1500 revolutions per minute, with external irrigation that the assistant interrupts twice to aspirate. Above which threshold of thermal insult is bone expected to undergo necrosis and fibrous repair rather than osseointegration?

A. 40 degrees Celsius maintained for five minutes
B. 47 degrees Celsius maintained for one minute
C. 56 degrees Celsius maintained for one minute
D. 60 degrees Celsius maintained for thirty seconds
E. 70 degrees Celsius reached momentarily

200 A 49-year-old man is having two implants placed in the dense anterior mandible, each requiring a 13 mm osteotomy through a heavily corticated crest. Your nurse asks how the drilling protocol is designed to protect the bone. Which of the following measures genuinely reduce the temperature generated at the osteotomy wall?

A. Copious irrigation delivered to the cutting tip throughout preparation
B. Reaching the final diameter in a single step rather than in graded stages
C. Increasing the drill speed from 800 to 2000 revolutions per minute
D. A and B only
E. A, B and C

201 A 63-year-old woman is having an implant placed at the UR6. CBCT showed a thin cortex over sparse trabeculae, and at surgery the pilot drill advances under its own weight with almost no resistance. There is 11 mm of bone below the sinus floor and the ridge is 8 mm wide. Which modification is most likely to improve primary stability?

A. Increasing drill speed to reduce lateral compression of the walls
B. Tapping the osteotomy fully before inserting the implant
C. Stopping short of the final drill diameter and omitting the countersink
D. Selecting a shorter, wider implant to engage the crestal cortex
E. Abandoning placement and grafting the site before re-entry

202 An implant is being seated in the dense anterior mandible of a 58-year-old man. With about 2 mm still to go the ratchet reads 65 Ncm and further rotation is difficult. The osteotomy was prepared to the manufacturer's recommended final drill, the crest is heavily corticated, and no countersink was used. What is the most appropriate action?

A. Complete seating, since high torque predicts reliable osseointegration
B. Complete seating and load the implant immediately, given the stability
C. Leave the implant 2 mm supracrestal and restore to that level
D. Reverse the implant, relieve the crest with a countersink, then reseat it
E. Remove the implant and replace it with one of a larger diameter

203 An implant 8.5 mm long is inserted at the LL6 in soft bone. As the platform reaches the crest it begins to spin freely, and the motor records 10 Ncm. CBCT had shown 11 mm of bone above the mandibular canal, the ridge measures 9 mm wide, and wider implants of the same length are open on the tray. Which is the best way to salvage the site?

A. Replace it with a wider implant of the same length in the same osteotomy
B. Remove the implant, graft the osteotomy and re-enter after four months
C. Advance the implant a further 2 mm to engage bone nearer the canal
D. Fit a healing abutment and allow eight weeks of undisturbed healing
E. Splint the implant to the adjacent teeth with a composite bridge

204 After the pilot drill at the LR5 a direction indicator is placed and a periapical radiograph taken. The axis is seen to converge distally, and the tip of the indicator lies 0.9 mm from the mesial root of the LR6. Nothing beyond the pilot osteotomy has been prepared and the LR6 is vital and unrestored. What should you do before widening the osteotomy?

A. Continue the sequence and place a shorter implant to increase clearance
B. Redirect the pilot osteotomy with a side-cutting bur against a guide
C. Place the planned implant and accept the proximity to the root
D. Abandon the site, graft the osteotomy and reassess in four months
E. Refer for orthodontic uprighting of the LR6 before proceeding

205 During preparation at the LL4 the drill loses resistance abruptly at 11 mm and brisk bleeding wells from the osteotomy. Within a few minutes the floor of the mouth is swelling and the tongue is being displaced upwards; the patient is anxious and her voice has altered. She takes apixaban, correctly managed with a single missed dose. What is the most appropriate immediate action?

A. Insert the planned implant to tamponade the bleeding, then observe
B. Pack the osteotomy with collagen and discharge with written instructions
C. Prescribe tranexamic acid mouthwash and review the following morning
D. Complete the osteotomy, place a shorter implant and review at 24 hours
E. Apply pressure, secure the airway and arrange immediate hospital transfer

206 CBCT for an implant at the LR6 shows 12.5 mm from the crest to the superior border of the mandibular canal, measured on the cross-sectional image at the planned site. The ridge is 8 mm wide with a flat crest and there is no lingual concavity. You intend to keep a 2 mm safety margin above the canal. Which is the longest implant you should plan?

A. 8 mm
B. 9 mm
C. 10 mm
D. 11.5 mm
E. 13 mm

207 A 4 mm horizontal defect at the LR4 has been grafted with a particulate xenograft and covered with a dense polytetrafluoroethylene membrane. A periosteal release has produced passive coronal advancement, and the site will be left submerged for six months before re-entry. Which suture material is most appropriate for the closure?

A. 3/0 braided silk
B. 4/0 braided polyglactin 910
C. 5/0 plain gut
D. 4/0 polytetrafluoroethylene monofilament
E. 3/0 chromic gut

208 Five days after guided bone regeneration at the UR4 the incision line has opened over the membrane and graft particles are visible. The flap had been advanced without a periosteal release and closed with eight interrupted sutures tied firmly enough to blanch the margins. The patient does not smoke and wears no denture. Which change would most reduce the risk at the next attempt?

A. A periosteal release with horizontal mattress sutures set back from the margins
B. A greater number of interrupted sutures placed closer together
C. A thicker suture on a larger needle to resist tearing through the flap
D. Knots tied more firmly so that the margins are held in contact
E. Retention of the sutures for twenty-one days rather than ten

209 Non-resorbable sutures over a grafted, coronally advanced flap are conventionally retained for 10 to 14 days, whereas those closing a simple crestal incision at a non-augmented site are removed at about 7 days. Which is the strongest justification for the longer period over the graft?

A. Suture tracks epithelialise after five days and become plaque-retentive
B. Wound tensile strength develops well behind epithelial closure
C. Resorbable membranes need mechanical support until they hydrate
D. Graft particles migrate until the periosteum reattaches to the flap
E. The released flap remains ischaemic throughout the first week

210 Your practice sterilises instruments in a non-vacuum, downward-displacement (type N) benchtop steriliser. You are about to buy an implant surgical kit containing internally irrigated drills, a ratchet with a hollow shaft and a contra-angle surgical handpiece. Which statement about processing this equipment is correct?

A. A type N cycle is adequate provided the holding time is extended
B. Internally irrigated drills should be chemically disinfected instead
C. Hollow and lumened items require a vacuum (type B) steriliser
D. The handpiece may be processed in a washer-disinfector alone
E. Using single-use drills removes the need for a vacuum steriliser

211 A colleague proposes preparing implant osteotomies with a conventional air-driven high-speed handpiece, pointing out that it is autoclavable, familiar and already in the surgery. He intends to use copious external saline irrigation, a graded drill sequence and a light touch. Which is the principal objection to that plan?

A. Torque falls under load, so the drill stalls in dense cortical bone
B. Rotational speed cannot be reduced to the range used for osteotomy
C. Internal irrigation cannot be delivered through the handpiece head
D. Exhaust air is driven into the tissues, risking emphysema and infection
E. Autoclaving degrades the bearings faster than in a surgical motor

212 An implant is placed at the UR5 with a 3 mm buccal dehiscence, grafted with a particulate xenograft and covered by a non-resorbable membrane. Insertion torque was 40 Ncm and the mucosa is thick with a wide band of keratinised tissue. You are deciding between fitting a healing abutment and burying the implant. Which of the following are valid arguments for submerging it?

A. A transmucosal component gives bacteria a route to the membrane
B. Submergence preserves the width of the keratinised mucosa
C. Submerged implants show higher survival than one-stage placement
D. A and B only
E. All of the above

213 A 66-year-old woman has an implant placed at the UL5 with simultaneous contour grafting under a resorbable membrane. She wears an acrylic partial denture whose saddle rests directly over the site, and she expects to wear it whenever she leaves the house. The closure is passive and she does not smoke. Which instruction most reduces the risk of membrane exposure?

A. Leave the denture out for 48 hours, then relieve it fully and soft line it
B. Wear the denture continuously so that it splints and protects the wound
C. Add clasps so that load is carried by the abutment teeth instead
D. Reline the denture with self-curing hard acrylic at the review visit
E. Reduce the occlusal surfaces to eliminate contact in lateral excursions

214 A UR1 with a horizontal root fracture is removed atraumatically and the buccal plate is found intact, measuring 0.8 mm on the CBCT. An implant is placed engaging palatal and apical bone in the correct prosthetic position, and a 2.4 mm horizontal gap remains between the implant and the inner surface of the buccal plate at the level of the shoulder. How should that gap be managed?

A. Leave it to fill with clot, since defects below 3 mm heal spontaneously
B. Graft it with a slowly resorbing xenograft particulate
C. Fill it with autogenous bone collected from the osteotomy
D. Cover it with a resorbable membrane without any graft material
E. Replace the implant with a wider one that eliminates the gap

215 During immediate placement at the UR1, the osteotomy is initiated against the palatal wall of the socket rather than in its centre, using a precision drill to score the palatal bone before the pilot drill is engaged, and the axis is directed to engage bone apical and palatal to the socket. Which is the strongest reason for doing so?

A. It enlarges the horizontal gap that will subsequently be grafted
B. It avoids the incisive canal, which lies buccal to the socket apex
C. It keeps the implant away from the buccal plate and into stable bone
D. It allows a screw-retained rather than a cement-retained restoration
E. It reduces the risk of perforating the floor of the nose

216 A 44-year-old man needs the UR2 extracted after a failed root canal retreatment. CBCT shows an intact buccal plate 1.1 mm thick, a well-circumscribed periapical radiolucency 6 mm in diameter with no suppuration or sinus tract, and 5 mm of bone apical to the socket. The mucosal phenotype is thin and he has a high smile line. Which feature most argues against immediate placement?

A. The periapical radiolucency of 6 mm
B. The buccal plate thickness of 1.1 mm
C. The 5 mm of bone apical to the socket
D. The position of the tooth within the aesthetic zone
E. The thin phenotype combined with a high smile line

217 Consider these statements about immediate implant placement into an intact anterior maxillary socket: (i) the buccal wall resorbs whether or not an implant is placed; (ii) grafting the horizontal gap reduces but does not abolish the buccal contour change; (iii) sockets with a buccal wall thicker than 1 mm resorb less than those with a thinner wall. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (ii) and (iii) only
D. (i) and (iii) only
E. (i), (ii) and (iii)

218 You have placed an implant at the LR6, with a particulate xenograft and a resorbable membrane over a small buccal dehiscence, and closed with monofilament sutures. Your nurse is completing the operative record while you write the post-operative instructions. Which of the following must be recorded in order to satisfy medical device traceability requirements?

A. The manufacturer, reference and lot number of the implant
B. The lot numbers of the graft material and the membrane
C. The insertion torque achieved at placement
D. A and B only
E. A, B and C

Block J — Minor hard and soft tissue augmentation (Q219–246)

219 A 54-year-old non-smoker is having an implant placed at the UR4. On flap elevation there is a 3 mm buccal dehiscence, with the implant otherwise correctly positioned and stable at 35 Ncm. You plan to graft with bone chips collected from the osteotomy, mixed in equal parts with deproteinised bovine bone mineral and covered by a native collagen membrane. What is the strongest justification for including the autogenous chips?

A. They provide a mineralised scaffold along which vessels and osteoblasts advance
B. They resist substitution and so hold the augmented contour for years
C. They are the only component of the mixture carrying viable osteogenic cells
D. They release bone morphogenetic protein once the mineral phase is removed
E. They supply the rigidity that the particulate substitute lacks

220 A 39-year-old woman is having an implant at the UR1 with a 2 mm horizontal contour deficiency on the labial aspect. The soft tissue phenotype is thin and she has a high lip line, and you want the augmented profile to remain stable when she is reviewed at five years. Which property of the grafting material most determines that outcome?

A. A high proportion of vital bone within the regenerate at six months
B. A high osteoinductive potential within the grafted particles
C. A resorption rate matched to the rate of new bone formation
D. A large particle size that increases the interparticle porosity
E. A negligible substitution rate, so particles persist within the regenerate

221 You are selecting a material to graft a 3 mm buccal dehiscence at the UL4 in a healthy 45-year-old, and a representative tells you that the product he supplies is osteoinductive and will therefore integrate faster than the xenograft you currently use. Which of the following materials can properly be described in that way?

A. Demineralised freeze-dried bone allograft
B. Deproteinised bovine bone mineral
C. Beta-tricalcium phosphate
D. A and B only
E. A, B and C

222 A staged horizontal augmentation grafted with demineralised freeze-dried bone allograft under a native collagen membrane is reopened at six months. The membrane has long since resorbed and considerably less width has been gained than was planned, with the surface of the regenerate collapsed towards the original ridge. Which property of the material best explains the loss of contour?

A. Its human origin provokes a low-grade immune response to the particles
B. Demineralisation leaves particles that resist compression poorly
C. Its osteoinductive potential is exhausted within the first two weeks
D. Its particle size is too small to preserve interparticle porosity
E. It is replaced by fibrous connective tissue rather than by bone

223 A 47-year-old man is having a staged horizontal augmentation at the LR5 and LR6 with a particulate graft under a collagen membrane. Once the graft is in place the buccal flap will only reach the lingual flap under obvious tension, and you make a periosteal releasing incision before suturing. Which consequence of closing under tension does that manoeuvre most directly prevent?

A. Compression of the graft particles beneath the membrane
B. Migration of epithelium beneath the margin of the membrane
C. Displacement of the membrane from the underlying cortical bone
D. Ischaemic breakdown of the incision line with membrane exposure
E. Contraction of the graft volume during early remodelling

224 During guided bone regeneration at the UL5 the collagen membrane is fixed with two titanium pins buccally and tucked beneath the palatal flap, so that neither the membrane nor the graft beneath it can move as the patient functions. Which biological event does immobilising the graft most directly prevent?

A. Formation of fibrous tissue within the graft in place of bone
B. Resorption of the membrane before regeneration is complete
C. Loss of the space maintained beneath the membrane
D. Ingrowth of epithelium along the outer surface of the membrane
E. Premature vascular invasion of the graft from the periosteum

225 Twelve days after simultaneous guided bone regeneration at the UL4 with a xenograft and a native collagen membrane, a 3 mm dehiscence has opened over the crest and the membrane is visible. The patient has no pain, there is no suppuration, and plaque is accumulating at the exposed margin. What is the most appropriate management?

A. Remove the exposed portion of membrane and re-suture the flap
B. Leave the membrane, apply chlorhexidine gel and review weekly
C. Prescribe amoxicillin for seven days and review at one week
D. Debride the exposed membrane surface and place an additional suture
E. Re-enter the site, remove the graft and allow it to heal

226 A grafted extraction socket at the UR5 is to be covered by a barrier that will be left deliberately exposed to the mouth rather than closed primarily, so that the keratinised tissue is not advanced across the ridge. A dense polytetrafluoroethylene membrane is selected. What is the principal reason for that choice?

A. It resorbs slowly enough to protect the graft for six months
B. It integrates with the overlying mucosa and seals the socket
C. Its submicron porosity resists bacterial penetration while it is exposed
D. It is cell-occlusive whereas a collagen membrane is not
E. Its stiffness maintains space over the socket without a graft

227 A native collagen membrane placed over a 4 mm dehiscence at the UR4 becomes exposed at three weeks. The patient is comfortable, there is no suppuration, and she has maintained the site with chlorhexidine as instructed. Which of the following apply to this situation?

A. The membrane should be removed at once to protect the graft beneath it
B. The bone gain finally achieved at the site is likely to be reduced
C. A seven-day course of antibiotics will preserve the grafted volume
D. B and C only
E. All of the above

228 A 58-year-old man has been edentulous at the LR6 and LR7 for eight years. CBCT shows 8 mm of bone above the superior border of the mandibular canal, a ridge 7 mm wide, and a crest that lies 4 mm apical to the bone level at the adjacent LR5. He is a non-smoker with good plaque control and wants a fixed replacement. What is the most appropriate plan?

A. Two 6 mm implants placed in the existing bone and splinted
B. Vertical augmentation with a titanium-reinforced membrane, then implants
C. Referral for an autogenous block graft before implant placement
D. Two 10 mm implants placed with the apices adjacent to the canal
E. A cantilevered bridge from the LR5 with no implant treatment

229 Consider these statements about alveolar ridge preservation with a xenograft and a collagen matrix following extraction of an upper premolar: (i) it reduces horizontal ridge resorption but does not prevent it; (ii) it cannot restore bone where the buccal plate is already absent; (iii) it slows the formation of vital bone within the socket. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

230 A 50-year-old man is to have an upper first premolar extracted and asks what the ridge will be like in six months if nothing is placed in the socket. He is a non-smoker with a healthy periodontium and an intact buccal plate. Which figures best represent the evidence?

A. Horizontal and vertical loss of about 2 mm each
B. Vertical loss of about 3 to 4 mm and horizontal loss of about 1 mm
C. Horizontal loss of about 3 to 4 mm and vertical loss of about 1 mm
D. Horizontal loss of about 1 mm, with vertical change negligible
E. Loss of about half the ridge width, most of it occurring after six months

231 At placement of a 10 mm implant at the UL5 the buccal plate is found to be absent over the coronal 5 mm, leaving threads exposed within a broad, shallow buccal concavity. The implant is in the correct prosthetic position and stable at 40 Ncm, and you intend to graft and place a membrane. Which feature of the defect most determines whether the augmentation will succeed?

A. The proportion of the implant surface that remains covered by bone
B. The thickness of the residual palatal cortical plate
C. The buccolingual width of the ridge measured at the crest
D. The number of bony walls available to contain the graft
E. The height of the graft above the adjacent bone crest

232 An immediate implant is placed at the UR2 following a vertical root fracture. The buccal plate is intact and measures 1 mm thick, the horizontal gap between the implant and the inner surface of that plate is 2.5 mm, and insertion torque is 30 Ncm. You fill the gap with a low-substitution xenograft. What is the strongest justification for doing so?

A. It increases the primary stability recorded at placement
B. It prevents epithelium growing down into the residual gap
C. It accelerates the formation of vital bone within the gap
D. It reduces bacterial colonisation of the implant surface
E. It compensates for the modelling that the buccal wall will undergo

233 A flap raised at the UL3 during implant placement reveals a window of exposed implant surface 4 mm by 2 mm at the apex of the fixture, with 5 mm of intact labial bone coronal to it. The implant is in the planned restorative position and is stable. What is the most likely explanation for this finding?

A. The implant chosen was wider than the ridge at the crest
B. The osteotomy followed the crestal axis into the labial concavity
C. The implant was positioned too far palatally within the ridge
D. Insufficient bone was removed from the crest before drilling
E. The pilot drill was deflected by the dense labial cortical plate

234 A 49-year-old woman has 4 mm of ridge width at the crest of the UL5, widening to 7 mm at 5 mm apical to the crest, with 12 mm of available height. You have planned to place the implant and graft simultaneously. Which finding at surgery would most strongly indicate staging the augmentation instead?

A. An implant that reaches only 15 Ncm of insertion torque
B. A buccal dehiscence measuring 4 mm vertically
C. Five implant threads exposed on the buccal aspect
D. A residual buccal plate measuring less than 1 mm thick
E. A graft that cannot be covered without releasing the flap

235 A 61-year-old woman lost the UR1, UR2 and UL1 in a road traffic accident four years ago and wears a partial denture. CBCT shows a ridge 3 mm wide at the crest, with the crest lying 6 mm apical to the bone level at the adjacent canines. She displays 3 mm of gingiva when she smiles. She is healthy, does not smoke, and asks for fixed implant-supported teeth. What is the most appropriate course?

A. Stage a horizontal augmentation with particulate graft and a collagen membrane
B. Place three narrow-diameter implants with simultaneous grafting
C. Split the ridge and place the implants with simultaneous augmentation
D. Continue with the denture and review the site annually
E. Refer for assessment of combined vertical and horizontal reconstruction

236 A block graft is to be harvested from the external oblique ridge of the mandible to augment an atrophic maxilla. The patient has full dentition including third molars, and the planned block measures 15 mm by 8 mm. Which structure is at greatest risk during this harvest?

A. The lingual nerve at the level of the third molar
B. The long buccal nerve crossing the anterior border of the ramus
C. The inferior alveolar nerve within the mandibular canal
D. The facial artery at the antegonial notch
E. The masseteric branch of the maxillary artery

237 During a staged augmentation at the LL4 and LL5 you plan a periosteal releasing incision in order to close the buccal flap without tension over the graft. Which anatomical consideration most limits how that release may be made at this site?

A. The mentalis muscle inserting close to the crest anteriorly
B. The mental nerve emerging and looping forward beneath the periosteum
C. The buccinator attaching at the level of the external oblique ridge
D. The mylohyoid ridge tethering the lingual flap
E. The facial artery crossing the lower border at the antegonial notch

238 A 57-year-old man has an implant crown at the LR6 with 1 mm of keratinised mucosa buccally, mobile mucosa reaching the margin, repeated bleeding on probing and discomfort when he brushes the site. He has no history of periodontitis and does not smoke. Which of the following statements are supported by the evidence on keratinised mucosal width around implants?

A. A width below 2 mm is associated with more plaque accumulation and bleeding
B. A width below 2 mm is associated with discomfort during brushing
C. Grafting to increase the width reliably arrests marginal bone loss
D. A and B only
E. A, B and C

239 A 62-year-old woman has an implant-retained crown at the LL6 with 1 mm of keratinised mucosa, a shallow vestibule, and frenal and mucosal pull at the mucosal margin. She reports that brushing the site is uncomfortable and that it bleeds when she does. The crown is screw-retained and correctly contoured, and the bone level is stable. What is the most appropriate surgical management?

A. A subepithelial connective tissue graft placed through a buccal tunnel
B. A xenogeneic collagen matrix beneath a coronally advanced flap
C. A pedicle flap rotated from the lingual aspect
D. An apically positioned flap with a free gingival graft
E. A coronally advanced flap with an enamel matrix derivative

240 A subepithelial connective tissue graft is to be harvested from the lateral palate between the distal aspect of the canine and the mesial aspect of the first molar. The patient has a shallow, flat palatal vault. Why does the form of the vault matter to the way you plan this harvest?

A. The greater palatine neurovascular bundle lies closer to the gingival margin
B. The masticatory mucosa in the donor zone is more glandular and fatty
C. The mucosa is thinner and will yield a smaller volume of graft
D. The incisive canal lies closer to the anterior limit of the harvest
E. The greater palatine foramen is displaced forward to the second premolar

241 A connective tissue graft measuring 12 mm by 8 mm is being harvested from the lateral palate through a single horizontal incision placed 3 mm apical to the gingival margin, and the overlying partial-thickness flap is deliberately kept to a thickness of about 1 to 1.5 mm as the dissection proceeds. What is the principal reason for that dimension?

A. To limit bleeding from the cut margins of the wound
B. To reduce the risk of injuring the greater palatine artery
C. To retain a blood supply sufficient to prevent flap necrosis
D. To leave enough keratinised tissue for a later second harvest
E. To allow the flap to be sutured back without tension

242 You are considering the maxillary tuberosity rather than the lateral palate as the donor site for a connective tissue graft to thicken the tissue at a single implant site. The patient has erupted upper third molars. Which of the following statements about the tuberosity as a donor site are correct?

A. The tissue is denser in collagen and shrinks less after grafting
B. The volume available is larger and more predictable than the palate
C. The graft is harvested already de-epithelialised and needs no trimming
D. A and B only
E. A, B and C

243 At the placement of an implant at the LR5 the vertical thickness of the mucosa over the crest, measured with a probe through the anaesthetised tissue before flap elevation, is 1.5 mm. The bone volume is generous, the implant is placed at the correct depth and reaches 45 Ncm, and a healing abutment is fitted. Which consequence is this measurement most likely to have?

A. Reduced primary stability at the time of placement
B. Delayed epithelial healing around the healing abutment
C. A higher risk of residual cement at the crown margin
D. Recession of the buccal mucosa within the first year
E. Greater early crestal bone loss as the attachment establishes

244 A 35-year-old woman is having an implant placed at the UR1. The labial plate is intact but measures 0.8 mm on the CBCT, the phenotype is thin and the papillae are long. A connective tissue graft is planned to increase buccal volume, and a submerged healing protocol is intended. Which timing offers the most predictable aesthetic result?

A. At the time of extraction, before the site has been assessed
B. At the time of implant placement, beneath the elevated flap
C. At second-stage surgery when the implant is uncovered
D. After the definitive crown has been fitted and reviewed
E. When recession of 1 mm or more has become apparent

245 A free gingival graft has been secured buccal to an implant at the LL6 on a periosteal bed, with pressure applied for five minutes and the patient instructed not to disturb the site. Which statement best describes how the graft survives the first days after placement?

A. Capillaries from the recipient bed perfuse it within twelve hours
B. Vessels within the graft anastomose with the bed within 24 hours
C. It is nourished by diffusion from saliva across its epithelial surface
D. Plasmatic circulation sustains it until revascularisation begins
E. The retained periosteum of the bed perfuses it directly from day one

246 A representative recommends that you add leucocyte- and platelet-rich fibrin to your dehiscence grafting, and shows you case photographs. You have no experience of the material and want to establish what may fairly be claimed for it. Which of the following statements about it are accurate?

A. It is an autologous fibrin matrix containing platelets and leucocytes
B. Its best-documented clinical benefit is in soft tissue healing
C. It is osteoinductive and can replace the particulate graft in a dehiscence
D. A and B only
E. All of the above

Block K — Restoring implants, occlusion and the laboratory (Q247–280)

247 A colleague has placed three implants at the LR4, LR6 and LR7, and the cross-sectional views show that they diverge from one another by up to 15 degrees. The restoring dentist intends a closed-tray transfer impression in a stock tray with a putty-and-wash silicone. Which feature of that technique most limits the accuracy of the resulting cast?

A. The transfer copings are relocated into the set impression by hand
B. The impression material must recover elastically as the tray is withdrawn
C. The transfer copings are shorter than the corresponding open-tray copings
D. The analogues are connected to the copings outside the mouth
E. The tray is not perforated over the implant positions

248 Four implants have been placed in an edentulous mandible for a bar-retained overdenture. The technician asks for an open-tray impression with the copings splinted in autopolymerising acrylic resin, and specifies that the splint should be cut through between the copings once it has set and rejoined with small increments of fresh resin. What is the purpose of that instruction?

A. It allows each coping to be checked radiographically for complete seating
B. It compensates for the setting expansion of the die stone
C. It confines polymerisation shrinkage to a very small volume of resin
D. It reduces the force needed to remove the impression from the mouth
E. It allows a coping to be repositioned if it rotates during removal

249 A master cast for a full-arch mandibular framework has been poured from a splinted open-tray impression of five implants. Before the framework is milled, the technician returns a verification jig — resin cylinders joined by a sectioned and rejoined resin index — and asks that it be tried in the mouth. Which of the following does the jig establish?

A. That the analogue positions on the cast reproduce the implant positions in the mouth
B. That the definitive framework will seat passively on the implants
C. That the interocclusal record has been taken at the correct vertical dimension
D. A and B only
E. A, B and C

250 A ten-unit screw-retained framework is tried in. When only the most distal screw is tightened, the framework can be rocked and a gap of about 0.3 mm opens at the opposite terminal abutment; the same is seen in reverse when only the other terminal screw is tightened. The patient reports no discomfort and the framework seats fully when all screws are tightened. What is the most appropriate next step?

A. Tighten the remaining screws in sequence and confirm seating radiographically
B. Relieve the intaglio surface of the framework over the terminal abutment
C. Accept the fit, as this magnitude lies within the tolerance quoted for passivity
D. Section the framework at the discrepancy, lute it in the mouth and rejoin it
E. Record a new interocclusal registration and rearticulate the master cast

251 A full-arch maxillary framework made from an intraoral scan of six implants will not seat, and the deviation is being reviewed. Which of the following increase the deviation of a full-arch implant scan?

A. An edentulous span longer than about 20 mm between adjacent scan bodies
B. Marked divergence between adjacent scan bodies
C. A matt polymer scan body rather than a machined titanium one
D. A and B only
E. A, B and C

252 Three implants in the lower left quadrant are to be recorded with an open-tray impression, the copings splinted with resin, in a rigid custom tray. You are choosing between an irreversible hydrocolloid, a polyether and a putty-and-wash addition silicone. Which property of the material matters most in this particular application?

A. Hydrophilicity, so the material adapts to a moist sulcus
B. Rigidity when set, so the material resists deformation as the tray is withdrawn
C. Tear strength, so thin sections survive removal from undercuts
D. Extended working time, so the copings can be splinted before setting
E. Dimensional stability, so pouring may be delayed for several days

253 A screw-retained crown has been fitted at the UR5 and the patient has no symptoms. A periapical radiograph is taken with a paralleling holder, with the beam perpendicular to the long axis of the implant, and shows a wedge-shaped radiolucency about 0.3 mm wide between abutment and implant on the mesial, with the interface closed on the distal. Which is the most likely explanation?

A. Early crestal remodelling has exposed the first thread mesially
B. The connection incorporates a platform switch at the abutment margin
C. Excess cement is trapped at the mesial margin of the abutment
D. The image is geometrically distorted because of beam angulation
E. The abutment is not fully seated on the implant platform

254 A patient having a single crown at the UL6 asks whether it is better for the crown to be screwed or cemented, and you explain what the comparative studies show. Which statement best represents the findings of systematic reviews comparing the two retention modes?

A. Survival is similar, with more technical complications reported for screw retention
B. Survival is similar, with more technical complications reported for cement retention
C. Survival is higher with cement retention, with fewer biological complications
D. Survival is higher with screw retention, with fewer technical complications
E. Survival and complication rates are equivalent between the two modes

255 A cement-retained crown is planned at the UL5 on a stock abutment whose mesial margin will lie about 2 mm below the mucosal margin. Which of the following reduce the quantity of cement extruded below the margin?

A. Loading the crown extra-orally on a replica abutment before seating
B. Siting the margin no more than 1 mm below the mucosal margin
C. Selecting a radiopaque rather than a radiolucent cement
D. A and B only
E. All of the above

256 An implant at the UR1 has been placed with its long axis emerging about 20 degrees labial to the ideal, through the labial surface of the planned crown. The alternatives are a cemented crown on a custom abutment with a margin 1.5 mm subgingival, or a restoration on an angulated screw channel abutment. Which is the principal advantage of the angulated screw channel here?

A. It allows a higher torque to be applied through the angled driver
B. It preserves retrievability without creating a subgingival cement margin
C. It reduces the diameter of the access opening on the labial surface
D. It compensates for a small discrepancy in the fit of the restoration
E. It permits a deeper abutment margin without the risk of cement excess

257 A screw-retained provisional crown has been in place at the UR1 for eight weeks. The tissues are healthy and the phenotype is thick, but the mucosal zenith sits about 1 mm coronal to that of the UL1, making the crown look short. Which modification of the provisional is most likely to correct this?

A. Remove material from the critical contour so that the margin migrates apically
B. Add material to the subcritical contour to support the interdental papillae
C. Add material to the critical contour to displace the margin apically
D. Remove material from the subcritical contour to allow the tissue to thicken
E. Reduce the overall height of contour to relieve pressure on the free margin

258 An LR6 has been restored on a 4.1 mm bone-level implant with a screw-retained monolithic crown whose occlusal table is 11 mm wide. The subgingival profile widens rapidly and convexly from the platform. The patient brushes and uses interdental brushes well, but the mesial surface bleeds on probing with a 5 mm pocket and 1 mm of bone loss at two years. Which feature of the restoration most likely accounts for this?

A. The 4.1 mm platform beneath an 11 mm occlusal table
B. A crown-to-implant ratio approaching 2 to 1
C. The screw access channel opening in the central fossa
D. An occlusal table wider buccolingually than the platform
E. An emergence angle of about 45 degrees with a convex profile

259 A single screw-retained crown is being fitted at the LR6 in a healthy 45-year-old with a full natural dentition, no parafunction and stable posterior support. Shimstock foil 8 to 12 micrometres thick is being used to check the occlusion. How should the intercuspal contacts on the crown be adjusted?

A. Shimstock is held equally by the crown and the adjacent teeth in light closure
B. Shimstock is held by the adjacent teeth in light closure and by the crown on firm clenching
C. Shimstock pulls freely at the crown both in light closure and on firm clenching
D. Shimstock is held by the crown in light closure and by the adjacent teeth on clenching
E. Shimstock is held at the crown in lateral excursion but not in intercuspal position

260 A trainee asks why the crown on a posterior implant is deliberately left in slightly lighter contact than the adjacent natural teeth, when the aim of the restoration is to restore function. Which statement best explains the practice?

A. The adjacent teeth intrude under load, so equal contacts transfer force to the implant
B. The implant intrudes further than the teeth, so the contacts equalise under function
C. The periodontal ligament transmits occlusal load more slowly than surrounding bone
D. The implant has a higher tactile threshold, so overload is not perceived by the patient
E. The implant crown wears the opposing enamel faster than a natural tooth would

261 A 39-year-old has lost the UR3 to a vertical root fracture and an implant has been restored there. The rest of the dentition is intact and unworn, the occlusion is class I with canine guidance on the left, and there is no evidence of parafunction. Which excursive scheme is most appropriate for the implant crown?

A. Canine guidance carried entirely on the implant crown
B. Group function on the posterior teeth with the implant free of contact
C. Guidance shared with the adjacent natural teeth on as shallow a path as aesthetics allow
D. Balanced articulation so that the non-working side shares the excursion
E. The crown left free of contact in intercuspal position and in all excursions

262 Two implants at the LR4 and LR5 are splinted and carry a distal cantilever pontic at the LR6, with the occlusal table narrowed and the cantilever centre of load about 8 mm distal to the LR5 platform. A load is applied to the pontic. Which statement describes the resulting forces at the two implants?

A. Both implants are loaded in compression in proportion to their distance from the load
B. The load is shared equally between them because they are rigidly splinted
C. The mesial implant is loaded in compression and the distal implant in tension
D. The distal implant is loaded in compression and the mesial implant in tension
E. The vertical load is converted into a purely horizontal force at both platforms

263 A full-arch fixed mandibular prosthesis is planned on five implants. The anteroposterior spread, measured from the centre of the most anterior implant to the line joining the distal aspects of the terminal implants, is 12 mm, and the patient wishes to have first molars. Which of the following are established guides to the acceptable length of a distal cantilever?

A. Not more than about 1.5 times the anteroposterior spread
B. Not more than the length of the shortest supporting implant
C. Not more than half the buccolingual width of the occlusal table
D. A and B only
E. A, B and C

264 A 6 mm implant has been placed at the LR6 to avoid a sinus lift, and the restoration will have a crown height of 12 mm from the platform, giving a crown-to-implant ratio of 2 to 1. The patient has read that this ratio predicts failure and asks whether the plan should change. Which statement best represents the evidence?

A. Marginal bone loss increases in proportion to the crown height
B. Technical complications increase, while marginal bone loss does not
C. Survival falls below 85 per cent once the ratio exceeds 2 to 1
D. The ratio predicts failure only where the restoration is cement-retained
E. The ratio ceases to matter once the implant is splinted to a neighbour

265 An implant is placed immediately into the socket of a UR1 with an intact buccal plate, a 2 mm gap to the buccal wall grafted, and an insertion torque of 40 Ncm. An immediate screw-retained provisional crown is planned. Which of the following are requirements of that provisional?

A. Freedom from contact in intercuspal position and in all excursions
B. Screw retention rather than cementation of the provisional
C. Primary stability at placement of at least about 35 Ncm
D. A and B only
E. All of the above

266 After ten weeks of conditioning with a provisional crown, the peri-implant mucosa at the UR1 has a well-formed emergence profile that must be reproduced in the definitive restoration. The case is being recorded digitally with an intraoral scanner. Which of the following will transfer the conditioned profile to the laboratory?

A. Scanning the mucosal profile immediately on removing the provisional, then placing the scan body
B. Recording the site with a stock impression coping in a closed tray
C. Having the profile copied from the master cast poured before conditioning
D. A and B only
E. A, B and C

267 A three-unit metal-ceramic implant-supported prosthesis is planned for a 52-year-old with heavy masseters and a history of ceramic fractures on natural teeth. The laboratory asks how the prosthesis should be specified to reduce the risk of the commonest technical complication. Which feature is most likely to achieve that?

A. A connector cross-section increased to about 12 square millimetres
B. A framework in zirconia rather than in cobalt-chromium
C. An occlusal surface formed in the framework material without veneering ceramic
D. A veneering ceramic fired at a lower temperature than usual
E. A thicker layer of veneering ceramic over the supporting framework

268 A zirconia crown is planned at the UR4 on a bone-level titanium implant, and the choice lies between a one-piece zirconia abutment engaging the implant connection directly and a zirconia superstructure bonded to a prefabricated titanium base. Which is the principal advantage of using the titanium base?

A. It increases the flexural strength of the zirconia above the mucosal margin
B. It allows a higher torque to be applied to the abutment screw
C. It improves the shade of the mucosal margin in a thin phenotype
D. It maintains a titanium-to-titanium interface at the implant connection
E. It corrects an unfavourable implant angulation of up to 25 degrees

269 Two nominally identical titanium abutment screws are tightened to 35 Ncm with a calibrated wrench. One is new; the other has been placed and removed repeatedly and its threads are worn and dry. Measurement shows appreciably different clamping force in the two joints. Which best explains the discrepancy?

A. The screw elongates plastically, and yield strength varies between screws
B. Most of the applied torque is consumed by friction, which differs between joints
C. The torque wrench records peak rather than mean torque during tightening
D. Preload depends on thread pitch, which alters as the threads wear
E. The elastic modulus of the alloy varies with the temperature of the joint

270 A provisional abutment at the LL5 has been unscrewed and replaced eight times over four months while the emergence profile was developed. The definitive crown is now ready to fit, and the provisional screw appears undamaged. What should be done with the screw at the definitive appointment?

A. Fit the new screw supplied with the definitive components and torque it as specified
B. Torque the existing screw about 10 per cent above the manufacturer's value
C. Apply a proprietary lubricant to the existing screw before torquing it
D. Torque the existing screw and repeat the torque after ten minutes
E. Replace the existing screw only if it fails to reach the set torque

271 An abutment screw whose manufacturer specifies 35 Ncm is inadvertently torqued to 55 Ncm because the wrench was set incorrectly, and the error is noticed immediately afterwards. The crown appears fully seated and the patient is comfortable. Which is the most likely consequence of what has happened?

A. The abutment seats more deeply into the conical connection, improving the seal
B. Preload falls immediately as the mating surfaces burnish under the excess load
C. The internal threads strip and the implant becomes mobile within the bone
D. The crestal bone is compressed and remodels apically over the first year
E. The screw is stretched beyond its elastic limit and may later fail by fatigue

272 A patient is referred having had three episodes of abutment screw loosening on a single molar crown placed elsewhere, on an implant with an external hexagon connection. The occlusion has been checked and is correct, and the crown is being remade. A second implant is now planned at the contralateral molar site. Which feature would most reduce the risk of the same problem occurring there?

A. A wider platform diameter beneath the same occlusal table
B. A gold alloy rather than titanium alloy abutment screw
C. An internal conical connection engaging over several millimetres
D. A narrower buccolingual occlusal table on the definitive crown
E. Cement rather than screw retention of the definitive crown

273 A custom zirconia abutment and a screw-retained crown are being prescribed for the UL5, from a conventional impression sent to an external laboratory. Which of the following must appear in the prescription or accompany the work?

A. The implant system, platform diameter and connection type
B. The intended position of the abutment margin relative to the mucosal margin
C. A statement that the impression has been disinfected
D. A and B only
E. All of the above

274 A patient collects her new implant crown and asks for "the paperwork for the device", explaining that she keeps records of all her medical implants. The crown was made by an external laboratory on a custom abutment. Which document must the laboratory have provided with the finished restoration?

A. A copy of the technician's professional registration
B. A statement identifying the device, its materials and the prescribing clinician
C. The implant manufacturer's certificate of conformity for the abutment
D. A and B only
E. All of the above

275 An implant at the UL6 is to be restored. The interocclusal space from the implant platform to the opposing palatal cusp is 4.5 mm, the restoration margin would lie about 2.5 mm below the mucosal margin mesially, and the implant axis emerges through the buccal cusp of the planned crown. Which of these findings favour screw retention?

A. The interocclusal space of 4.5 mm
B. The margin position 2.5 mm below the mucosal margin
C. The implant axis emerging through the buccal cusp
D. A and B only
E. A, B and C

276 A mandibular full-arch fixed prosthesis on five implants is being designed for a 68-year-old who will continue to wear a complete maxillary denture. The ridges are favourable, there is no parafunction, and the denture is well made and retentive. Which occlusal scheme is most appropriate for the reconstruction?

A. Bilateral balanced articulation with lingualised occlusal contacts
B. Canine guidance carried on the mandibular prosthesis
C. Group function on the working side with non-working disclusion
D. Mutually protected occlusion with anterior guidance
E. Contacts confined to the premolar regions in intercuspal position

277 A bone-level implant at the LR5 is being uncovered at second stage. The definitive abutment is already available from the planning records, and the alternative is to place a healing abutment now and connect the definitive abutment at the impression appointment six weeks later. Which is the strongest argument for placing the definitive abutment today?

A. The mucosal margin stabilises sooner because the emergence profile is final
B. The definitive abutment achieves a higher preload at the first torque application
C. Using fewer components reduces the risk of selecting the wrong platform
D. A provisional restoration can then be cemented rather than screwed
E. Repeated disconnection disturbs the mucosal attachment and adds crestal bone loss

278 Four years after fitting, a metal-ceramic implant-supported prosthesis from the LR5 to the LR7 has a chip of veneering ceramic from the buccal cusp of the LR6 pontic, about 2 mm across, with no metal exposed. The patient has no symptoms and no aesthetic concern. Examination shows a working-side excursive contact on that cusp, and the framework and screws are sound. What is the most appropriate management?

A. Repair the defect with bonded composite and leave the occlusion unaltered
B. Smooth and polish the fracture and eliminate the excursive contact
C. Remake the prosthesis in monolithic zirconia and provide a night-time splint
D. Remove the prosthesis and replace the abutment screws before repairing it
E. Reduce the opposing cusp to eliminate the contact in intercuspal position

279 An abutment screw has fractured at the LL5, leaving the apical fragment within the internal threads of the implant, its fractured surface lying just below the platform. The implant is osseointegrated, the mucosa is healthy and there is no mobility. Which is the most appropriate first attempt at retrieval?

A. Cut a slot in the fragment with a small round bur and use a flat-bladed driver
B. Recut the internal threads with the manufacturer's tap before attempting retrieval
C. Attempt counter-rotation of the fragment with an ultrasonic tip under magnification
D. Section the implant collar to gain direct access to the fragment
E. Restore the implant with a cemented crown over the retained fragment

280 A colleague states that a full-arch framework must fit within 10 micrometres to be considered passive, quoting the figure originally proposed in the early osseointegration literature, and asks whether a framework that fits well clinically but has a measured discrepancy of 40 micrometres should be remade. Which statement best represents the current position on passive fit?

A. A misfit below 10 micrometres is required and is routinely achieved clinically
B. A misfit below 30 micrometres is required for screw-retained frameworks
C. A misfit below 150 micrometres is acceptable where the framework is cemented
D. Absolute passivity is achievable only with a cast gold alloy framework
E. No threshold of misfit has been shown to cause biological failure

Block L — Complications, peri-implant disease, maintenance and governance (Q281–300)

281 A 63-year-old man attends five years after an implant was placed at the LR6. Probing depths are 5 mm circumferentially with no bleeding on gentle probing, no suppuration and no discomfort. The radiograph shows the bone level 2.5 mm apical to the implant platform, unchanged from the film taken at one year. His plaque score is 9 per cent. Which is the correct diagnosis?

A. Peri-implant mucositis
B. Peri-implantitis
C. Peri-implant health at an implant with reduced bone support
D. Peri-implantitis that has been successfully treated
E. Peri-implant health cannot be diagnosed at a 5 mm probing depth

282 A 48-year-old woman has an implant crown at the UL4 fitted two years ago. Probing depths have increased from 3 mm to 5 mm, all four surfaces bleed on gentle probing, the mucosa is swollen and there is no suppuration. The radiograph shows the bone level at the first thread and unchanged from the film taken at fitting. Which finding excludes a diagnosis of peri-implantitis?

A. The absence of suppuration from the sulcus
B. The probing depth of 5 mm rather than 6 mm
C. The short interval of two years since restoration
D. The bone level unchanged from the baseline film
E. The absence of any mobility of the implant

283 A patient transfers into your practice with an implant placed abroad six years ago and no previous radiographs available. There is bleeding and suppuration on gentle probing at the UR5, a probing depth of 7 mm mesially, and the bone level lies 3.5 mm apical to the most coronal intraosseous part of the implant. Which combination of thresholds allows peri-implantitis to be diagnosed when no baseline data exist?

A. Bleeding with a probing depth of 4 mm or more and bone loss of 1 mm or more
B. Bleeding with a probing depth of 5 mm or more and bone loss of 2 mm or more
C. Bleeding with a probing depth of 5 mm or more and bone loss of 3 mm or more
D. Bleeding with a probing depth of 6 mm or more and bone loss of 2 mm or more
E. Bleeding with a probing depth of 6 mm or more and bone loss of 3 mm or more

284 You are auditing the maintenance records of the practice and find that no probing depths have been recorded at any implant. A colleague tells you that he does not probe around implants, on the grounds that probing tears the mucosal seal and drives bacteria into the peri-implant tissues, and that he relies on radiographs instead. Which statement best answers that argument?

A. Probing at about 0.25 N is safe, and the seal re-establishes within days
B. Probing is safe provided that a plastic probe is used at the abutment
C. Probing is unnecessary because radiographs detect the disease earlier
D. Probing is safe only where a band of keratinised mucosa is present
E. Probing should be limited to sites showing visible signs of inflammation

285 A patient who has agreed to two implants asks how common problems with the gums around implants are, and whether you can put a number on it. Which of the following figures may fairly be quoted from the systematic review evidence?

A. Peri-implant mucositis affects roughly 40 to 45 per cent of patients
B. Peri-implantitis affects roughly 20 per cent of patients
C. Reported peri-implantitis prevalence ranges from about 1 to 47 per cent
D. A and B only
E. All of the above

286 A patient asks why you are proposing three-monthly rather than annual reviews once her implants are restored, and what it is about her that concerns you. In ranking the factors that predict peri-implantitis, which has the strongest and most consistent evidence supporting it?

A. Absence of keratinised peri-implant mucosa
B. A history of treated periodontitis
C. Occlusal overload of the implant restoration
D. A moderately rough rather than a machined implant surface
E. Long-term systemic bisphosphonate therapy

287 Eight months after a crown was cemented at the UR5 with a resin cement, the site bleeds and suppurates with a 6 mm pocket. The crown margin lies 2.5 mm submucosally on the palatal aspect. A periapical radiograph shows 2 mm of bone loss and no radiopaque material at the margin. Which statement best explains why that radiograph does not exclude residual cement?

A. Cement is displaced apically beyond the field of a periapical view
B. The bone loss obscures the crown margin on a two-dimensional image
C. Suppuration dissolves the cement before the radiograph is taken
D. Many dental cements are insufficiently radiopaque to be visible
E. Excess cement on the buccal surface is superimposed on the implant

288 A 60-year-old woman has peri-implantitis at the LL5, with a 7 mm pocket, bleeding and suppuration, and 3 mm of radiographic bone loss. You begin with non-surgical therapy, reinforcement of hygiene and correction of an over-contoured crown. Which of the following does the evidence support?

A. Mechanical debridement alone resolves the majority of such lesions
B. Bleeding and probing depths improve, but deep pockets frequently persist
C. Adjunctive systemic antibiotics remove the need for later surgery
D. B and C only
E. A, B and C

289 Consider these statements about peri-implant mucositis: (i) experimentally induced mucositis resolves when effective plaque control is reinstituted; (ii) resolution is complete within three weeks in most sites; (iii) mucositis is regarded as the precursor lesion of peri-implantitis. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

290 After non-surgical therapy an implant at the LR6 still has a 7 mm pocket with suppuration and bleeding. A flap is raised and the defect is found to be a circumferential intrabony crater 4 mm deep with all four bony walls intact, and the site is not visible when the patient smiles. The implant is stable and the prosthesis is screw-retained and cleansable. Which approach is best supported by the evidence?

A. Access flap with surface decontamination and no graft
B. Implantoplasty of the exposed threads and apical repositioning
C. Explantation followed by grafting of the site
D. Systemic antibiotics with repeated non-surgical debridement
E. Reconstructive therapy with a bone substitute after decontamination

291 A 66-year-old man attends complaining of discomfort on biting at the UL6. The implant crown moves when the crown is held between finger and thumb, probing depths are 4 mm with bleeding, and a periapical radiograph shows a continuous radiolucent line around the entire outline of the implant. He is otherwise well. What is the most appropriate course?

A. Remove the implant and allow the site to heal before reassessing
B. Remove the crown and re-torque the abutment screw to specification
C. Treat as peri-implantitis with surgical debridement and grafting
D. Splint the implant to the adjacent teeth and review in three months
E. Prescribe antibiotics and review the mobility in six weeks

292 A 58-year-old woman restored with three implants five years ago attends "when she remembers", which has averaged once every fourteen months. Her plaque control is moderate, she has never had periodontitis and does not smoke, and there is bleeding at two of the three implants with probing depths of 4 mm and no bone loss. What recall interval should you recommend, and on what basis?

A. Twelve months, since she has no history of periodontitis
B. No longer than five to six months, beyond which peri-implantitis is more frequent
C. Three months for life, irrespective of her risk profile
D. Twenty-four months, with a radiograph taken at each visit
E. Attendance only when she notices bleeding or discomfort

293 You are removing biofilm from a titanium abutment and from two exposed implant threads at the LL6 during a supportive care visit for a 59-year-old patient whose peri-implantitis was treated surgically a year ago. The site now bleeds at one point, with a probing depth of 5 mm and a stable bone level. Which of the following are appropriate for this purpose?

A. Air polishing with a glycine or erythritol powder
B. Titanium or PEEK curettes shaped to the abutment
C. A stainless steel ultrasonic tip used at reduced power
D. A and B only
E. All of the above

294 A patient with peri-implant mucositis at two implants has been debrided, shown how to clean the sites with interdental brushes, and reviewed. She asks whether a chlorhexidine mouthwash would help her, and offers to use it indefinitely. Which statement best represents the evidence?

A. It substitutes for interdental cleaning where access is difficult
B. It resolves peri-implant mucositis reliably when used for two weeks
C. It reduces marginal bone loss when used long term around implants
D. It should be prescribed routinely at every supportive care visit
E. It adds little to genuinely effective mechanical biofilm removal

295 A 52-year-old man telephones on the morning after placement of an implant at the LR5 to report numbness of the right lower lip and chin that has not worn off. The post-operative periapical radiograph, reviewed now, shows the implant apex within the outline of the mandibular canal. He is systemically well and in no pain. What should you do first?

A. Prescribe a corticosteroid and a non-steroidal analgesic and review in a week
B. Reassure him that most such disturbances recover within six months
C. Refer him urgently to an oral and maxillofacial unit for assessment
D. Remove or back out the implant, and explain to him what has happened
E. Map the affected area, photograph it and review him at two weeks

296 A manufacturer issues a field safety notice for a specific lot of abutment screws distributed over an eighteen-month period, and asks that patients treated with them be identified and reviewed. Which entry in your records is what allows you to do that?

A. The lot numbers of every component recorded at the time of fitting
B. The dates on which each implant restoration was fitted
C. The invoice records held by your supplier for that period
D. The implant record cards issued to each of the patients
E. The radiographs taken at the fit of each definitive prosthesis

297 A 45-year-old man has a single implant crown at the UR4 fitted eighteen months ago. He is asymptomatic, has no bleeding on probing, probing depths of 3 mm and good plaque control. His previous dentist took a periapical radiograph at fitting and another at one year, both of which you hold. Which is the most appropriate radiographic policy from here?

A. A periapical every twelve months to detect early bone loss
B. Further radiographs only when the clinical findings indicate them
C. A panoramic radiograph every two years for comparison
D. A small-field CBCT to establish a three-dimensional baseline
E. A periapical every six months until the fifth year

298 A patient telephones the practice three weeks after her implant crown at the UR2 was fitted to say she is unhappy with its appearance and is considering making a formal complaint. You examined her at the fit appointment and consider the shade and contour a good match, and the tissues are healthy. What is the most appropriate first action?

A. Arrange to see her, listen to her concern and examine the restoration
B. Write setting out why the crown meets the accepted standard of care
C. Notify your indemnity organisation before you respond to her at all
D. Offer to remake the crown at no charge in order to resolve the matter
E. Refer her to a colleague for an independent opinion on the aesthetics

299 Five years after placement, a patient with peri-implantitis at the LL6 states that she was never told this could happen and that she would not have had the implant if she had known. Your notes from the consenting appointment are brief. Which of the following would best support your position?

A. A signed consent form listing the procedure and the fee agreed
B. A practice leaflet on implant maintenance given before treatment
C. A contemporaneous note of the risks discussed and of her questions
D. A and B only
E. A, B and C

300 A 64-year-old man has advanced peri-implantitis at two adjacent implants supporting a splinted fixed prosthesis at the UR1 and UR2, with 8 mm pockets, suppuration, 6 mm of radiographic bone loss at each, and 1 mm of keratinised mucosa. He displays the gingival margins fully when he smiles. Non-surgical therapy has reduced the bleeding but not the pocket depths. What is the most appropriate course?

A. Undertake access surgery with implantoplasty and apical repositioning
B. Repeat non-surgical debridement with an adjunctive local antibiotic
C. Refer for specialist assessment, with explantation among the options
D. Explant both implants and place two replacements at the same visit
E. Maintain at three-monthly intervals and accept slow progression

Block M — History, risk assessment and medical considerations, extended (Q301–340)

301 A 58-year-old man has had type 2 diabetes for eleven years. His HbA1c is 64 mmol/mol, he has background retinopathy, and he wants two implants in the lower left quadrant. You are explaining to a trainee why poor glycaemic control is associated with more early implant failures. Which of the following contribute to that association?

A. Accumulation of advanced glycation end-products within bone collagen
B. Impaired neutrophil chemotaxis and phagocytosis at high glucose concentrations
C. Microvascular disease reducing perfusion of the osteotomy walls
D. A and B only
E. All of the above

302 A 61-year-old woman with type 2 diabetes attends for implant assessment. Her HbA1c four weeks ago was 61 mmol/mol, her general practitioner has just added a second oral agent, and she has no retinopathy, nephropathy or neuropathy. She is a non-smoker with healthy periodontal tissues and wants two implants in the upper premolar region. What is the most appropriate course?

A. Proceed, since the value lies below the 64 mmol/mol threshold for elective surgery
B. Proceed with a single preoperative antibiotic dose and delayed loading
C. Defer, and reassess after a further three months of glycaemic management
D. Defer until the value has fallen below the 48 mmol/mol diagnostic threshold
E. Proceed, since duration of diabetes predicts failure better than the current value

303 A 55-year-old man with type 2 diabetes had an HbA1c of 68 mmol/mol three months ago. His medication was changed four weeks ago and he asks whether a repeat test now would allow surgery to be booked. He is otherwise well and his renal function and haemoglobin are normal. Which property of the test most affects the answer you give?

A. It is unreliable unless the sample is taken after an overnight fast
B. It reflects only the two weeks immediately preceding the sample
C. It rises transiently for six weeks after any change of therapy
D. It reflects average glycaemia over the preceding two to three months
E. It is invalid in any patient taking metformin

304 A 67-year-old woman with type 2 diabetes is booked for a two-hour implant appointment starting at 11 am. She takes metformin 1 g twice daily, gliclazide 80 mg twice daily and sitagliptin 100 mg daily, and she has been told to eat and medicate as normal. Which of her medicines could cause hypoglycaemia during the appointment?

A. The metformin
B. The gliclazide
C. The sitagliptin
D. A and B only
E. A, B and C

305 A 49-year-old man with type 2 diabetes of six years' duration has an HbA1c of 50 mmol/mol on metformin alone, no complications and excellent oral hygiene. He asks whether his diabetes will affect how long his implants last. Which statement best represents the evidence you should give him?

A. Survival is comparable, but peri-implant disease is more likely and maintenance must be closer
B. Survival is reduced by about half, irrespective of the quality of his control
C. Survival is comparable and the risk of peri-implant disease is not increased
D. Survival is reduced only where the implants are placed in the maxilla
E. Survival is comparable provided a submerged healing protocol is used

306 A 71-year-old woman who has received denosumab 60 mg every six months for two years for osteoporosis had an upper premolar extracted ten weeks ago. Bone remains exposed at the socket, it is tender, and there is no history of radiotherapy to the head and neck and no evidence of metastatic disease. Which diagnosis do these findings support?

A. Osteoradionecrosis of the maxilla
B. Chronic suppurative osteomyelitis
C. Medication-related osteonecrosis of the jaw
D. Delayed socket healing not yet meeting the diagnostic criteria
E. A sequestrum of the buccal plate following extraction

307 A 69-year-old man taking an oral antiresorptive is being assessed before implant placement, and his physician offers to arrange investigations to quantify the risk of osteonecrosis. Which of the following should inform your risk assessment?

A. The indication for which the antiresorptive was prescribed
B. The serum C-terminal telopeptide concentration
C. The femoral neck T-score on densitometry
D. A and B only
E. All of the above

308 In a teaching session you are contrasting alendronate with denosumab for a group of trainees, and you are asked how alendronate produces its antiresorptive effect at cellular level. Which statement describes its action?

A. It binds RANKL and prevents osteoclast differentiation and survival
B. It inhibits farnesyl pyrophosphate synthase within the osteoclast
C. It inhibits carbonic anhydrase at the osteoclast ruffled border
D. It is incorporated into ATP analogues that inhibit sclerostin
E. It suppresses osteoblast apoptosis and thereby increases bone formation

309 A 68-year-old woman taking alendronate 70 mg weekly for four years for osteoporosis asks how her risk of osteonecrosis compares with that of her brother, who receives intravenous zoledronic acid every four weeks for metastatic prostate cancer. Neither has other risk factors. Which pair of estimates is closest to the published figures?

A. About 1 per cent for her and about 10 per cent for him
B. About 5 per cent for both, since the two drugs act by the same mechanism
C. Under 1 per cent for both, since implant sites heal by primary intention
D. Well under 1 per cent for her and of the order of 1 per cent or more for him
E. The risk cannot be quantified usefully for either of them

310 A 71-year-old man with atrial fibrillation takes rivaroxaban 20 mg once daily with his evening meal. He is booked for two implants with flap elevation at 9 am. His renal function is normal, he takes no antiplatelet drug, and he has had no bleeding problems. What is the most appropriate management of his anticoagulation?

A. Proceed without altering his medication, treating early and using local measures
B. Advise him to delay his evening dose until four hours after haemostasis
C. Advise him to miss the evening dose on the day before surgery
D. Advise him to omit the drug for 48 hours and restart afterwards
E. Check his INR on the morning of surgery and proceed if it is below 4.0

311 A 78-year-old man takes dabigatran 110 mg twice daily for atrial fibrillation. His estimated glomerular filtration rate has fallen over two years from 62 to 34 ml/min/1.73m². He is due for a single implant with a small flap. Which pharmacological property of dabigatran most explains the concern raised by that change?

A. It is a prodrug requiring activation by esterases within the renal tubule
B. It is highly protein-bound and is displaced from albumin in uraemia
C. It is excreted largely unchanged by the kidney, so it accumulates
D. It is metabolised by CYP3A4, whose activity falls in renal impairment
E. Its effect is monitored by the INR, which is unreliable in renal impairment

312 A 74-year-old woman on warfarin for a metallic mitral valve has an INR of 3.6 taken yesterday, stable for two years on the same dose. A colleague proposes that she omit tomorrow's dose so that the INR falls before implant placement the following day. Which statement best explains why that would not achieve what he intends?

A. Warfarin acts directly on factor Xa, so the INR responds within a few hours
B. Warfarin has a half-life of about 36 to 42 hours, so one omitted dose changes little
C. The INR reflects the intrinsic pathway, which warfarin does not influence
D. Warfarin inhibits vitamin K epoxide reductase, so its effect reverses immediately
E. Prothrombin has a half-life of four hours, so the INR falls rapidly once dosing stops

313 A 64-year-old man takes aspirin 75 mg daily and ticagrelor 90 mg twice daily following an acute coronary syndrome treated by stenting eight months ago. A colleague suggests that the ticagrelor be stopped for seven to ten days, as he would for clopidogrel. Which pharmacological difference between the two drugs is most relevant to that suggestion?

A. Ticagrelor is a prodrug requiring hepatic activation by CYP2C19
B. Ticagrelor inhibits cyclo-oxygenase-1 in addition to the P2Y12 receptor
C. Ticagrelor is cleared renally, so its duration is shortened in renal impairment
D. Ticagrelor acts directly on the glycoprotein IIb/IIIa receptor
E. Ticagrelor binds the P2Y12 receptor reversibly, so function returns as it is cleared

314 A 70-year-old man takes aspirin 75 mg daily alone for secondary prevention after a myocardial infarction six years ago. He is otherwise well, takes no other drug affecting haemostasis, and is booked for a single implant at the LR5 with a small flap. Which of the following are appropriate in managing his treatment?

A. Continuing the aspirin without interruption
B. Treating early in the day and early in the week
C. Suturing the flap and packing with an oxidised cellulose dressing
D. A and B only
E. All of the above

315 A 62-year-old man was treated for an oropharyngeal carcinoma five years ago with intensity-modulated radiotherapy prescribed to 66 Gy. He is disease-free and has stopped smoking. The radiotherapy planning record shows that the anterior mandible, where two implants are proposed to retain an overdenture, received 28 Gy. Which figure should most influence your assessment of the risk?

A. The 66 Gy prescribed to the primary tumour volume
B. The 28 Gy delivered to the proposed implant site
C. The five years that have elapsed since treatment
D. The number of fractions in which the dose was delivered
E. The total volume of mandible lying within the treated field

316 A 58-year-old woman had 64 Gy to the left maxilla and neck for a salivary gland malignancy three years ago and has two implants placed in a hospital unit at that time. She now has sticky saliva, several new cervical carious lesions, and mild mucositis around both implants. Which is the most important element of her ongoing maintenance?

A. Prescribing 5000 ppm fluoride toothpaste and reviewing her at three-monthly intervals
B. Prescribing a chlorhexidine mouthwash for indefinite daily use
C. Arranging hyperbaric oxygen before any further dental intervention
D. Replacing her remaining teeth with implants to eliminate the caries risk
E. Providing a soft splint to protect the atrophic irradiated mucosa

317 A 54-year-old man received a renal transplant three years ago and takes tacrolimus, mycophenolate mofetil and prednisolone 5 mg daily. His graft function is stable with an eGFR of 58 ml/min/1.73m^2. Placement of a single implant at the LL6 has gone uneventfully and he asks what he may take for discomfort. Which is the most important prescribing consideration?

A. Prescribing a seven-day course of amoxicillin to cover his immunosuppression
B. Doubling his prednisolone on the day of surgery and the day after
C. Avoiding paracetamol because of its hepatic metabolism
D. Avoiding non-steroidal anti-inflammatory drugs because of nephrotoxicity
E. Avoiding articaine because of its renal excretion

318 A 49-year-old woman is receiving adjuvant chemotherapy for breast cancer in cycles of 21 days, with three cycles remaining. She is well between cycles, has good oral hygiene, and wants to start implant treatment for a missing LR6 so that it is finished sooner. What is the most appropriate course?

A. Treat in the week immediately after each cycle, when her counts are highest
B. Treat 10 to 14 days after a cycle, timing surgery to the neutrophil nadir
C. Defer until chemotherapy is complete and her counts have recovered
D. Treat at any point in the cycle under antibiotic cover
E. Defer until she has completed five years of endocrine therapy

319 You are reviewing three patients before implant surgery: a woman with a mechanical mitral valve replacement, a man who had infective endocarditis eight years ago, and a man with an isolated atrial septal defect closed surgically in childhood with no residual shunt. Which of them fall within the group identified in national guidance as at increased risk of infective endocarditis?

A. The woman with the mechanical mitral valve
B. The man with previous infective endocarditis
C. The man with the repaired atrial septal defect
D. A and B only
E. A, B and C

320 A 73-year-old man with an implantable cardioverter defibrillator, fitted after a ventricular arrhythmia, attends for second-stage surgery at which hyperplastic mucosa over two implants must be removed. He is otherwise well and his device was checked last month. Which precaution is most relevant?

A. Deactivating the device beforehand through the cardiology team
B. Using a scalpel or a laser rather than monopolar electrosurgery
C. Avoiding local anaesthetic containing adrenaline in any concentration
D. Avoiding all ultrasonic instruments, including the scaler
E. Avoiding magnetic attachments in the definitive prosthesis

321 A 66-year-old man attends for placement of a single implant. He is visibly anxious, has taken his amlodipine as usual, and has no chest pain, headache or visual disturbance. His blood pressure, measured as he sits down, is 178/104 mmHg; it was 146/88 mmHg at his assessment appointment two months ago. What should you do first?

A. Rest him for ten minutes and repeat the measurement before deciding
B. Defer the procedure and refer for medical assessment on this reading
C. Proceed using a local anaesthetic solution without a vasoconstrictor
D. Proceed but limit the appointment to no more than thirty minutes
E. Proceed and re-check the blood pressure at the end of the appointment

322 A 51-year-old woman requesting implants has smoked 25 cigarettes a day since the age of 27 and stopped two years ago at the age of 49. She has never smoked anything else. You wish to record her cumulative tobacco exposure in the notes as a pack-year figure. Which value should you record?

A. 12 pack-years
B. 24 pack-years
C. 30 pack-years
D. 48 pack-years
E. 55 pack-years

323 A 44-year-old man who smoked 20 cigarettes a day for eighteen years switched entirely to a nicotine-containing e-cigarette nine months ago. He is being planned for an implant at the UR5 with a small buccal augmentation, and asks whether his risk is now the same as a non-smoker's. Which statement is the most accurate response?

A. His risk is now equivalent to that of someone who has never smoked
B. His risk is unchanged, because the combustion products are the same
C. Vaping is contraindicated during healing under current dental guidance
D. Risk relates to carbon monoxide alone, which vapour does not contain
E. Nicotine-mediated vasoconstriction persists and the peri-implant evidence is immature

324 A 47-year-old woman who smokes 15 cigarettes a day has agreed to stop before implant placement with simultaneous guided bone regeneration, and asks exactly what she is being asked to do. Which of the following form the cessation protocol conventionally advised around implant surgery?

A. Stopping tobacco for at least one week before surgery
B. Abstaining for at least eight weeks after surgery
C. Avoiding nicotine replacement therapy throughout the healing period
D. A and B only
E. A, B and C

325 A 43-year-old man reports that he grinds his teeth. His partner has heard the noise at night. Examination shows wear facets on the canines and premolars that his diet does not explain, bilateral masseteric hypertrophy and a well-defined linea alba. No sleep study has been performed. How should his bruxism be classified for the purpose of planning?

A. Possible sleep bruxism, on the basis of self-report alone
B. Probable sleep bruxism, on self-report together with clinical signs
C. Definite sleep bruxism, since a partner has witnessed the grinding
D. Awake bruxism, since the wear involves the canines and premolars
E. Unclassifiable until an ambulatory electromyographic recording is obtained

326 A 45-year-old man with confirmed parafunction, masseteric hypertrophy and previous fracture of two porcelain crowns is to have the LR5 and LR6 replaced with implants. Bone volume is generous, his periodontal condition is stable and he does not smoke. Which feature of the proposed design would be least appropriate for him?

A. A distal cantilever pontic supported by the two splinted implants
B. A monolithic zirconia occlusal surface on both units
C. Screw retention rather than cementation of the prosthesis
D. An occlusal table narrowed in the buccolingual dimension
E. A hard full-coverage splint provided for night-time wear

327 An implant is being placed at the UR6 site. The CBCT showed a thin cortex and sparse trabeculae, the drills met almost no resistance, the final insertion torque was 12 Ncm and the resonance frequency reading is 52 ISQ. The implant is stable to finger pressure and the site is otherwise healthy. Which modification of the protocol is most appropriate?

A. Torque the implant to 35 Ncm to improve its primary stability
B. Load it immediately with a provisional kept free of occlusal contact
C. Place a wider and shorter implant in the same osteotomy and load it early
D. Bury the implant and re-enter it at six weeks for impressions
E. Accept the reduced stability and extend the healing period before loading

328 A 74-year-old woman took alendronate 70 mg weekly for six years and was changed fourteen months ago to teriparatide 20 micrograms subcutaneously each day. She also takes calcium with colecalciferol. She has never taken corticosteroids and has no malignancy. Which of her medicines contribute to her risk of osteonecrosis of the jaw?

A. The calcium with colecalciferol
B. The six years of alendronate
C. The teriparatide
D. B and C only
E. All of the above

329 A 31-year-old woman who is 34 weeks pregnant develops a painful infection around a failing implant that must be managed now rather than deferred. The pregnancy has been uncomplicated, she takes no regular medication and has no allergies. She needs analgesia for the next few days. Which of the following may appropriately be prescribed?

A. Ibuprofen 400 mg three times daily
B. Codeine 30 mg with paracetamol
C. Paracetamol 1 g four times daily
D. B and C only
E. A, B and C

330 A 19-year-old man lost the UR1 in a cycling accident at 16 and wears a resin-bonded bridge. His standing height has been unchanged for 18 months, and cephalograms taken a year apart superimpose without vertical change. He is keen to proceed with an implant. Which point must still be explained before he consents?

A. The implant will erupt with the adjacent teeth as the alveolus continues to develop
B. The risk applies only if treatment is undertaken before the age of 21
C. Infraocclusion, should it occur, is corrected by orthodontic extrusion of the implant
D. The problem is confined to the mandible, where growth continues for longest
E. Continued dentoalveolar development may leave the crown in infraocclusion years later

331 A 14-year-old girl has congenitally absent UR2 and UL2. Orthodontic treatment is nearly complete and 6 mm of space has been created at each site, with good bone width and a high smile line. Her parents ask what should be provided until implants can be considered. Which is the most appropriate interim restoration?

A. Resin-bonded bridges cantilevered from the adjacent central incisors
B. A removable acrylic partial denture worn until growth has ceased
C. Implants placed now, with the crowns replaced in adulthood
D. Orthodontic space closure with canine substitution and later reassessment
E. Composite build-ups bonded to the mesial surfaces of the canines

332 A 46-year-old man with generalised tooth surface loss requests a full implant-supported reconstruction. He brings a photograph of a television presenter, says that his appearance has cost him promotion twice and that the treatment will "sort out" his career and his marriage, and has not asked about cost or duration. He has no psychiatric history and cleans his teeth well. Which feature of the consultation most predicts dissatisfaction with the outcome?

A. The photograph he has brought as a reference for the appearance
B. His limited understanding of the cost and duration of the treatment
C. The severity of the tooth surface loss he wishes to have restored
D. His belief that the result will change his career and his marriage
E. His failure to ask any question about the risks of the treatment

333 A 24-year-old woman attends for replacement of two lower molars lost to caries. She has marked palatal erosion of the upper anterior teeth, a body mass index of 16.5, and after careful questioning discloses self-induced vomiting several times a week for four years. She is under no medical care. What is the most appropriate course?

A. Provide a fluoride tray and dietary advice, then proceed with the implants
B. Defer implant treatment, provide prevention and support her access to medical care
C. Proceed with the lower implants and defer the anterior restorations only
D. Decline treatment permanently and discharge her back to her practitioner
E. Require a psychiatric report before providing any dental treatment at all

334 A 58-year-old man with a heavily restored but restorable UR1 insists that it be extracted and replaced with an implant, having understood your explanation that the tooth has a reasonable prognosis and that an implant carries risks the tooth does not. He answers questions coherently, weighs the information and gives his reasons. Which principle governs your response?

A. Capacity is absent where a decision conflicts directly with clinical advice
B. A second opinion must be obtained before any unwise decision is acted upon
C. A decision others consider unwise does not of itself indicate a lack of capacity
D. A clinician must comply with the decision of any patient who has capacity
E. The decision should be referred to a best-interests meeting for resolution

335 A 16-year-old man attends alone to discuss replacing a UR1 lost in a rugby injury. He understands the options, asks sensible questions and wants to proceed with planning. His mother has not come and has not been consulted, and he asks whether his own agreement is sufficient. Which statement best describes his legal position in England?

A. Parental consent is required for any treatment until he reaches 18
B. His competence must first be established through a Gillick assessment
C. He may consent to examination and radiographs but not to surgery
D. His consent must be countersigned by a parent for a private contract
E. He is presumed competent to consent to his own treatment

336 A 63-year-old man develops a spreading infection five days after implant placement. He has a documented history of anaphylaxis to penicillin, and his regular medicines are simvastatin 40 mg at night, ramipril 5 mg and metformin. You are considering clarithromycin 500 mg twice daily. Which is the most important reason to reconsider that choice?

A. Clarithromycin reduces the antihypertensive effect of ramipril
B. Clarithromycin reduces the absorption of statins from the gut
C. Clarithromycin prolongs the QT interval, an effect statins potentiate
D. Clarithromycin markedly raises simvastatin levels and risks rhabdomyolysis
E. Clarithromycin is contraindicated in all patients with penicillin allergy

337 A 76-year-old woman on warfarin, with a stable INR of 2.6, has angular cheilitis and denture-related stomatitis beneath an implant-retained overdenture. A colleague suggests miconazole oromucosal gel, reasoning that a topical preparation cannot interact with an oral anticoagulant. Which statement best explains why that reasoning fails?

A. Miconazole is inactivated by saliva and would be ineffective in the mouth
B. Miconazole is absorbed sufficiently to inhibit warfarin metabolism and raise the INR
C. Miconazole displaces warfarin from albumin, but only transiently
D. Miconazole reduces the vitamin K produced by the intestinal flora
E. Miconazole has no interaction with warfarin when applied topically

338 A 58-year-old woman reports being "allergic to penicillin". Questioning establishes that on one occasion an itchy raised rash appeared over her trunk within half an hour of the first capsule, with swelling of the lips, and that on a different course some years earlier she had loose stools from the third day. Which features indicate an IgE-mediated allergy?

A. The urticarial rash appearing within thirty minutes
B. The swelling of the lips at the same time
C. The loose stools beginning on the third day
D. A and B only
E. A, B and C

339 A 41-year-old man has taken antiretroviral therapy for HIV for nine years. His most recent CD4 count is 620 cells per microlitre, his viral load is undetectable, he does not smoke and his periodontal tissues are healthy. He wants a single implant at the LL5 and is anxious that he will be refused. What is the most appropriate course?

A. Proceed only with a five-day postoperative course of antibiotics
B. Decline, because osseointegration is unpredictable in HIV infection
C. Proceed as for any comparable patient, with routine liaison and maintenance
D. Refer him to a hospital unit because of the transmission risk to staff
E. Proceed only once the CD4 count has risen above 800 cells per microlitre

340 A 70-year-old man takes ramipril 10 mg daily and indapamide 2.5 mg daily for hypertension, with no other regular medication. He telephones two days after uneventful implant placement asking whether he may take ibuprofen 400 mg three times daily for a dull ache, as it has always suited him. Which is the most important reason to advise against it?

A. Ibuprofen reduces the antiplatelet effect of the indapamide
B. Ibuprofen is contraindicated alongside any antihypertensive drug
C. Ibuprofen and ramipril together raise the serum potassium
D. Ibuprofen increases the risk of angio-oedema with ACE inhibitors
E. The combination of an ACE inhibitor, a diuretic and a non-steroidal drug risks acute kidney injury

Block N — Examination, occlusion and space, extended (Q341–375)

341 A 34-year-old woman has been missing the LR5 for eight years. The distance between the contact points of the LR4 and LR6 is 8.5 mm, but a paralleling periapical shows their roots converging, leaving 5.0 mm between them at the level of the planned implant platform. The ridge is 7 mm wide and 13 mm of bone lies above the canal. What is the most appropriate course?

A. Place a 4.1 mm implant centred between the converging roots
B. Place a 3.3 mm implant centred between the converging roots
C. Refer for orthodontic uprighting of the adjacent roots first
D. Place a 4.1 mm implant angled away from the LR6 root
E. Restore the space with a resin-retained cantilever bridge

342 A 39-year-old man is to have an implant at the UR2. On the diagnostic wax-up the contact point with the UR1 will lie 7 mm coronal to the interproximal bone crest of the UR1, measured on a paralleling periapical. His phenotype is of medium thickness, the crown forms are ovoid and he has a medium smile line. Which is the most likely soft-tissue outcome?

A. Complete papilla fill in almost every case
B. Complete papilla fill in about half of cases
C. Failure of complete papilla fill in the majority of cases
D. Failure of papilla fill whatever the contact point position
E. An outcome unrelated to the contact-point-to-crest distance

343 A 44-year-old woman is having implants placed at the UR1 and UR2. The plan is altered at surgery so that the two implant shoulders lie 2.5 mm apart rather than the 3.5 mm originally drawn. Both implants achieve 35 Ncm of insertion torque and the buccal plate is intact. Which is the most likely consequence?

A. Approximately twice as much lateral crestal bone loss between them
B. Reduced primary stability of the more distal implant
C. Repeated loosening of both abutment screws in function
D. Loss of the buccal plate over both implants within a year
E. Failure of osseointegration of one of the two implants

344 You are planning a single implant at the LL5 and deciding how close the implant shoulder may be placed to the root of the LL6. Which of the following justify keeping at least 1.5 mm between them?

A. Crestal remodelling around the implant extends laterally about 1.4 mm
B. The interproximal papilla requires 3 mm of horizontal space
C. The papilla is supplied from the periosteum overlying the implant
D. A and B only
E. All of the above

345 A CBCT cross-section at the UR5 shows a crest 6.5 mm wide, narrowing to 4 mm at a point 6 mm apical to it because of a pronounced buccal concavity. The sinus floor lies 14 mm above the crest and the bone is well corticated. A 4.1 mm by 10 mm implant is planned along the crestal axis. Which complication is this anatomy most likely to produce?

A. A buccal fenestration over the apical part of the implant
B. A buccal dehiscence extending from the crest apically
C. Perforation of the floor of the maxillary sinus
D. Loss of the interproximal crest at the UR4
E. Compression necrosis of the palatal cortical plate

346 A 57-year-old man needs an implant at the LR5. The crest is knife-edged and 5.5 mm wide, widening to 8.5 mm at a level 4 mm apical to it. Sixteen millimetres of bone lie above the canal and the interocclusal space is generous. You intend to place a 4.1 mm implant with at least 1 mm of bone on each side. Which of the following would achieve that?

A. Reducing the crest by 3 mm to reach the wider bone
B. Expanding the ridge with osteotomes at the time of placement
C. Staged guided bone regeneration before placement
D. A and B only
E. All of the above

347 A 62-year-old man is to have an implant crown at the LR6. Mounted casts show 4.5 mm from the planned implant platform to the opposing occlusal surface, the opposing molar being unopposed elsewhere and slightly extruded. The laboratory proposes a cement-retained crown on a prefabricated abutment. Which complication is most likely?

A. Fracture of the abutment screw under function
B. Chipping of the occlusal ceramic within the first year
C. Repeated loss of the cemented crown
D. Food impaction beneath the proximal contact
E. Loss of interproximal papilla height over time

348 A 71-year-old edentulous woman is to have two interforaminal implants to retain a lower overdenture with individual stud attachments. Mounted casts show 7 mm from the planned implant platforms to the opposing occlusal plane. She has a well-formed ridge, adequate keratinised mucosa and a stable upper complete denture. Which complication is most likely?

A. Wear of the nylon retention inserts within a year
B. Fracture of the denture base over the attachment housings
C. Hyperplasia of the mucosa around the abutments
D. Loosening of the abutment screws in function
E. Accelerated resorption of the posterior mandibular ridge

349 You are planning to increase the occlusal vertical dimension by 4 mm, measured between the upper and lower incisal edges, in a patient who will also receive implant crowns at the LR6 and LL6 where restorative space is currently 4 mm. Approximately how much additional interocclusal space will this manoeuvre create at the first molars?

A. About 1 mm
B. About 2 mm
C. About 4 mm
D. About 6 mm
E. About 8 mm

350 A 55-year-old man has severe palatal wear of the upper incisors. Photographs taken ten years ago show an unchanged lower facial height, and the freeway space measured with a Willis gauge is 2 mm. He is to have implants at the LR6 and LL6 and asks whether his bite has collapsed. Which conclusion best fits these findings?

A. Dentoalveolar compensation has maintained the vertical dimension
B. The vertical dimension has been lost with the worn tooth substance
C. The mandible has autorotated forwards to close the space
D. The wear is erosive rather than attritional in origin
E. The vertical dimension has been increased by the wear pattern

351 An unopposed UR6 has over-erupted by 4 mm, carrying its gingival margin and its alveolar crest with it, and now occupies the space needed to restore an implant at the LR6. The tooth is sound, unrestored, vital and periodontally healthy. Which of the following could create the space required?

A. Enameloplasty of the occlusal surface
B. Elective root treatment and crown shortening
C. Orthodontic intrusion of the UR6
D. B and C only
E. A, B and C

352 A 48-year-old man lost the LR6 six years ago. The LR7 has tilted mesially by about 25 degrees, has a 7 mm probing depth on its mesial surface that bleeds, and a periapical shows an angular bony defect there. The LR5 is upright and sound, the plaque score is 30 per cent and he does not smoke. He wants an implant at the LR6. What is the most appropriate sequence?

A. Place the implant, then upright the LR7 orthodontically
B. Upright the LR7 orthodontically, then place the implant
C. Extract the LR7 and place implants at the LR6 and LR7
D. Place the implant and provide a crown on the LR7
E. Treat the periodontal defect, then upright, then place

353 You are adjusting a new implant crown at the LR6, which stands between two sound natural teeth in a patient with no parafunction. Shimstock is held by the adjacent teeth under light closure but by the implant crown only when the patient clenches firmly. What is the physiological basis for this adjustment?

A. Natural teeth intrude within the periodontal ligament under load, whereas the implant does not
B. The implant has no periodontal mechanoreceptors and cannot register the contact
C. Bone around an osseointegrated implant deforms more slowly than the ligament
D. Occlusal forces generated on implant crowns exceed those on natural teeth
E. The periodontal ligament allows natural teeth to migrate mesially with time

354 A 46-year-old woman is to have an implant at the UR3. The canine has been missing for many years and group function has developed on that side; the remaining teeth are unrestored, the overbite is average and there is no parafunction. Which occlusal scheme is most appropriate for the implant crown?

A. Reproduce canine guidance on the implant crown alone
B. Share the excursive contacts between the implant and the adjacent teeth
C. Provide a non-working contact on the implant to balance the arch
D. Leave the implant crown free of any excursive contact at all
E. Steepen the guidance angle to reduce the number of contacts

355 An implant crown at the LL6 has had its abutment screw retightened twice in a year. The crown is intact, the implant is osseointegrated, the proximal contacts are sound and radiographs show no bone loss. On excursion to the right, the mesiolingual cusp of the crown contacts heavily. Which explanation best accounts for the repeated loosening?

A. The contact represents an interference on the working side
B. The contact applies a lateral force at a distance from the implant axis
C. The screw has deformed plastically and lost its elastic recovery
D. The crown has been cemented with an excessive film thickness
E. The screw was never retorqued after the initial settling period

356 A 60-year-old man has a 2 mm slide from centric relation to intercuspation, a stable intercuspal position and no joint or muscle symptoms. He has lost both lower first molars and all four second molars, his lower facial height has reduced and the freeway space measures 6 mm. Implant crowns are planned at the LR6 and LL6. Which of the following would justify reorganising his occlusion rather than conforming to it?

A. The 2 mm slide from centric relation to intercuspation
B. The need to increase the vertical dimension when restoring
C. The absence of joint and muscle symptoms
D. A and B only
E. A, B and C

357 A 42-year-old woman attends three years after an implant crown was fitted at the LR6. She now packs food mesially and shimstock passes freely between the crown and the LR5, although the contact was tight when the crown was fitted. Bone levels are unchanged from the baseline film, there is no bleeding and the crown is screw-retained and sound. What is the most likely explanation?

A. Occlusal wear of the proximal surface of the crown
B. Distal drift of the natural dentition with increasing age
C. Continued eruption of the natural teeth relative to the implant
D. Loss of the seal at the screw access channel
E. Rotation of the crown on the implant platform under load

358 A 34-year-old man with a Class II division 2 incisal relationship has a 7 mm overbite and 1 mm of overjet. An implant is planned at the UR1, and the CBCT shows that the available bone dictates an axis along which the screw access channel would emerge in the middle of the palatal surface, within the area contacted by the lower incisor. Which is the most appropriate solution?

A. Place the implant more labially to relocate the access channel
B. Provide a cement-retained crown on a prefabricated abutment
C. Reduce the overbite orthodontically before placing the implant
D. Restore with an angulated screw channel abutment and crown
E. Accept the position and seal the channel with composite resin

359 A 45-year-old man attends for implant assessment and reports jaw stiffness on waking. He has bilateral masseteric hypertrophy, a well-defined linea alba and scalloped indentations along the lateral borders of the tongue, but no wear facets that match in any excursive position and no cervical abfraction. His diet is unremarkable. Which conclusion best fits these findings?

A. The mucosal findings represent normal anatomical variation
B. Tooth wear has been masked by compensatory continued eruption
C. The parafunction is predominantly clenching rather than grinding
D. The mucosal changes indicate low-grade chronic candidal infection
E. The habit occurs during waking hours rather than during sleep

360 A 29-year-old woman needs an implant at the UR1 following trauma. She has a high smile line displaying 3 mm of gingiva, a thin scalloped phenotype with the root outlines visible through the mucosa, triangular crown forms, a 7 mm mesiodistal gap and unrestored adjacent teeth. Which of these findings most alters the surgical technique you would plan?

A. The 3 mm of gingival display on full smile
B. The thin scalloped gingival phenotype
C. The triangular crown form of the neighbouring teeth
D. The 7 mm mesiodistal width of the edentulous gap
E. The unrestored state of the adjacent teeth

361 A 66-year-old man has an implant planned at the UR1, where 2 mm of vertical ridge height has been lost, so the mucosal margin of the crown will sit about 2 mm apical to that of the UL1. He declines any grafting. On full animated smile his upper lip covers the gingival margins and the cervical 2 mm of the crowns. Which finding most supports accepting the discrepancy?

A. The 2 mm vertical deficiency of the residual ridge
B. His refusal of any further grafting procedure
C. The lip position observed on full animated smile
D. The single-tooth nature of the proposed restoration
E. The absence of restorations on the adjacent UL1

362 A 58-year-old woman with an edentulous maxilla wants a fixed implant reconstruction. The anterior ridge has lost about 6 mm of vertical height, her existing complete denture is well made and comfortable, and on full smile she displays the whole of the denture teeth and 2 mm of the labial flange. Which finding most argues against a fixed prosthesis without further planning?

A. The extent of the edentulous span to be restored
B. The 6 mm loss of anterior ridge height
C. Her stated preference for a fixed restoration
D. The absence of natural teeth to provide guidance
E. The display of the prosthesis-to-tissue junction on smiling

363 A 61-year-old woman has an implant at the LL6 with 1 mm of keratinised mucosa buccally, a shallow vestibule and a mucosal margin that blanches when the cheek is retracted. She finds brushing the site uncomfortable and avoids it. Which of the following consequences of the narrow keratinised band are supported by the evidence?

A. Greater plaque accumulation with mucosal inflammation
B. A reduced ten-year survival rate for the implant
C. An established increase in the incidence of peri-implantitis
D. A and B only
E. All of the above

364 A 45-year-old man has generalised interproximal bone loss reaching 60 per cent of the root length at the worst sites, probing depths up to 7 mm, and has lost three teeth as a direct result of periodontitis. He smokes 15 cigarettes a day and his HbA1c is 38 mmol/mol. He requests implants to replace two of the missing teeth. Which is the correct diagnosis under the 2017 classification?

A. Stage II grade C periodontitis
B. Stage III grade C periodontitis
C. Stage III grade B periodontitis
D. Stage IV grade B periodontitis
E. Stage IV grade C periodontitis

365 A 39-year-old man has 3 mm of interproximal recession at several sites with probing depths of 3 mm, giving interdental attachment loss of 6 mm at the UR6, the UL5 and the LR6. Bleeding on probing is 15 per cent, he has never smoked, and no restoration margin or caries underlies any of the affected sites. Which of the following would establish a diagnosis of periodontitis under the 2017 classification?

A. Interdental attachment loss at two or more non-adjacent teeth
B. Buccal attachment loss of 3 mm or more with pocketing over 3 mm at two teeth
C. Radiographic bone loss in the absence of measured attachment loss
D. A and B only
E. A, B and C

366 At a review two years after fitting, an implant crown at the UL5 probes 5 mm circumferentially with no bleeding and no suppuration, and the radiographic bone level is unchanged from the film taken at fitting. The mucosa is thick, the crown emerges from a deeply placed platform and the patient reports no symptoms. Which is the most appropriate interpretation of the probing depth?

A. It indicates peri-implant mucositis and warrants debridement
B. It satisfies the case definition of peri-implantitis at this site
C. It reflects the depth of the platform beneath thick mucosa
D. It demonstrates loss of the supracrestal tissue attachment
E. It is invalid because probing disrupts the mucosal seal

367 Before placing an implant at the LR5, with its planned apex 3 mm above the mandibular canal, you carry out and record a light-touch and sharp–blunt examination of the lower lip and chin on both sides, mapping any area of altered response. What is the most important reason for doing this?

A. To complete the documentation required by the consent process
B. To identify patients in whom surgery should not be attempted
C. To provide evidence in the event of a later complaint or claim
D. To determine the safety margin required from the canal roof
E. To establish whether a deficit found later pre-dated the surgery

368 A 74-year-old woman has a two-implant mandibular overdenture that she dislodges repeatedly when speaking and eating, although it is firmly retained when fully seated and the nylon inserts were renewed last month. Her tongue is broad with scalloped margins and rests over the lingual flange, and the lower posterior teeth are set lingual to the crest of the ridge. What is the most likely explanation?

A. Wear of the nylon inserts within the attachment housings
B. Insufficient interocclusal space for the attachment housings
C. Divergence of the implants beyond the tolerance of the attachments
D. Inadequate distal extension of the denture base over the pads
E. Placement of the teeth and flange outside the neutral zone

369 Two weeks after an implant crown was fitted at the LR6, a 52-year-old man repeatedly bites his cheek on that side. The crown holds shimstock only on firm closure, carries no excursive contact, and the buccal mucosa shows a linear ulcer opposite the buccal cusps. The contralateral side is unaffected and he had no such problem before treatment. What is the most likely cause?

A. Excessive height of the buccal cusps on the new crown
B. A reduced vertical dimension allowing the cheek to fold inwards
C. A working-side interference on the implant restoration
D. Loss of horizontal overlap with the opposing buccal cusp
E. Buccal positioning of the implant relative to the crest

370 A 68-year-old man wears a well-made complete upper denture opposing a natural lower arch with intact anterior teeth, and is to have an implant crown at the LL6. He has worn the same denture for five years and it is stable, and the anterior maxillary ridge is already flattened. Which of the following are recognised consequences of this arrangement?

A. Accelerated resorption of the anterior maxillary ridge
B. Tipping and instability of the upper denture in function
C. Higher occlusal forces on the implant than against natural teeth
D. A and B only
E. A, B and C

371 A 33-year-old woman has lost the UR1 and UR2 to trauma. The combined mesiodistal space is 13 mm, the ridge is 6.5 mm wide with about 2 mm of vertical deficiency, her phenotype is thin and she has a high smile line. She has been offered two implants elsewhere. Which plan best protects the papilla between the two units?

A. Two implants placed 3 mm apart, each with its own crown
B. Two implants placed 3 mm apart with splinted crowns
C. Two implants with a connective tissue graft at second stage
D. One implant at the UR2 with a cantilever pontic at the UR1
E. One implant at the UR1 with a cantilever pontic at the UR2

372 An implant at the UR1 has been placed with its shoulder 4 mm apical to the intended mucosal margin of the crown, on the reasoning that a deeper position gives a more gradual emergence profile. The buccal plate is 2 mm thick, integration is uneventful and a cement-retained crown is planned. Which consequence is most likely?

A. Loss of primary stability during the healing period
B. Fracture of the abutment screw under functional load
C. Mid-facial mucosal recession within the first year
D. Retention of excess cement at the crown margin
E. Inability to seat the impression coping completely

373 At placement of an implant at the UR1 the buccal plate measures 0.8 mm at the crest, the mucosa is thin, and the shoulder sits 1 mm palatal to the emergence line of the adjacent teeth. Insertion torque is 40 Ncm, healing is uneventful and the crown is fitted at four months. Which outcome is most likely to be seen by one year?

A. Peri-implantitis with bleeding and suppuration on probing
B. Fracture of the thin buccal plate under occlusal loading
C. Loss of the interproximal papillae on both sides of the crown
D. Loss of osseointegration around the coronal threads
E. Mid-facial recession exposing the restorative margin

374 A 24-year-old woman has a 9 mm mesiodistal space at the UR2 following orthodontic treatment completed elsewhere; the contralateral lateral incisor is 6.5 mm wide. The ridge is 7 mm wide, her phenotype is thin and she has a high smile line. Which of the following would give an acceptable aesthetic result?

A. Orthodontic redistribution of the space before placement
B. A wide-platform implant restored to the full 9 mm width
C. A standard implant with a 6.5 mm crown and wide embrasures
D. A and B only
E. All of the above

375 A 63-year-old woman has lost the LR7. The LR6 is sound and in occlusion with the UR6, the rest of the arch is intact and unrestored, she chews comfortably and has no symptoms. Her maximum interincisal opening measures 30 mm and the ridge at the LR7 site is 6 mm wide with 12 mm of bone above the canal. Which of the following support a decision not to replace the tooth?

A. A dentition shortened to the first molars gives adequate function
B. The vertical access needed for the drill exceeds her opening
C. Implants placed at second molar sites have markedly poorer survival
D. A and B only
E. A, B and C

Block O — Investigations, imaging, guides and the wax-up, extended (Q376–410)

376 A general dental practitioner refers a patient to you for a CBCT before placing an implant at the LL6, writing only "CBCT lower left please". You hold the practitioner role under IR(ME)R for this exposure. The patient has travelled a long distance and is in the waiting room, and your previous scan of the day over-ran. What should you do first?

A. Acquire the scan, since the referral came from a registered dentist
B. Acquire the scan using the smallest field of view the machine offers
C. Obtain the clinical information needed to justify the exposure
D. Return the referral and decline to assess the patient today
E. Take a panoramic radiograph in place of the requested scan

377 Three requests for CBCT reach you on the same list, each from an experienced implant colleague. Which of the following would you justify?

A. A mandibular first molar site where the canal cannot be traced on the panoramic
B. An anterior maxillary site where the ridge is palpably concave buccally
C. A maxillary second premolar site with 12 mm of bone below the sinus floor
D. A and B only
E. A, B and C

378 You are setting the acquisition for a CBCT to plan a single implant at the LR5. The machine allows the field of view, the tube current, the tube voltage, the number of basis projections and the voxel size all to be varied independently. Which single change would most reduce the effective dose to this patient?

A. Reducing the field of view to the smallest that includes the site
B. Reducing the tube current for the acquisition
C. Increasing the tube voltage while holding the current constant
D. Reducing the number of basis projections acquired
E. Increasing the voxel size used for the reconstruction

379 A colleague asks why your implant planning scans are acquired at a voxel size of about 0.2 mm when the machine offers 0.08 mm, and suggests the finer setting must give a better image. Which is the most important reason for the setting you use?

A. Smaller voxels cannot resolve the cortical outline of the canal
B. Smaller voxels increase image noise unless the exposure is raised
C. Smaller voxels worsen beam hardening around metal restorations
D. Smaller voxels prolong reconstruction beyond clinical usefulness
E. Smaller voxels restrict the field of view that can be acquired

380 A 16-year-old girl and a 65-year-old man each require a small-volume CBCT of the anterior maxilla, at identical exposure settings, before planning an implant for a missing lateral incisor. Both are otherwise well. Why is the risk attached to the exposure greater for the younger patient?

A. Effective dose is higher in a smaller head at the same settings
B. The salivary glands lie closer to the beam at that age
C. Deterministic effects occur at lower doses in growing tissue
D. Dividing tissues are more radiosensitive and life expectancy is longer
E. The thyroid gland is larger and absorbs more of the primary beam

381 A panoramic radiograph taken on a machine with a stated magnification factor of 1.25 shows 14 mm of bone between the alveolar crest and the superior border of the mandibular canal at the LL6. No calibration object was included in the exposure and the patient was correctly positioned. What is the most appropriate estimate of the true bone height?

A. About 9 mm
B. About 10 mm
C. About 11 mm
D. About 14 mm
E. About 18 mm

382 You wish to use the same panoramic radiograph to measure the mesiodistal distance between the LL5 and the LL7 before planning an implant at the LL6. The image is well positioned and of good density. Which statement best describes the reliability of that measurement?

A. Horizontal magnification varies with the object's position in the focal trough
B. Horizontal magnification is constant and equal to the vertical factor
C. Horizontal measurement is reliable if a calibration sphere is included
D. Horizontal distortion is abolished by correct positioning of the patient
E. Horizontal magnification is smaller than the vertical factor in all machines

383 A panoramic radiograph taken at assessment shows a poorly defined radiopacity projected over the left ascending ramus, sitting higher than, and blurred in comparison with, a well-defined opacity of similar shape on the right side. The patient wore a metallic earring in the right ear throughout the exposure. She has no history of trauma, salivary symptoms or neck swelling. What does the left-sided opacity most likely represent?

A. A sialolith within the left submandibular duct
B. A calcified lymph node in the left side of the neck
C. Blurring caused by patient movement during the exposure
D. Superimposition of the cervical spine on the ramus
E. A ghost image produced by the right-sided earring

384 A CBCT is taken before placing an implant at the LR4. On the cross-sectional images a round, well-corticated canal about 2 mm in diameter is seen running horizontally within the body of the mandible, 4 mm anterior to the mental foramen and 9 mm below the crest, with no connection to the apices of the adjacent teeth. The patient has normal lip sensation. What does this most likely represent?

A. The anterior loop of the mental nerve
B. A nutrient canal within the lingual cortical plate
C. An accessory mental foramen with its short canal
D. A residual periapical lesion at the LR3 apex
E. The mandibular incisive canal

385 You are reading the cross-sectional images at a planned LL6 site before placing a 4.1 mm by 10 mm implant in a healed ridge. Which of the following measurements must be taken from the cross-section itself?

A. The ridge width at the level of the planned implant platform
B. The depth of the lingual concavity below the crest
C. The distance from the crest to the superior border of the canal
D. A and B only
E. All of the above

386 On the CBCT for a planned implant at the LL6 the mandibular canal has no discernible cortical border over a 10 mm length beneath the site, although a slightly darker band consistent with the neurovascular bundle can be followed on several adjacent slices. The remaining bone appears normal and the patient has no sensory symptoms. What is the most appropriate response?

A. Report the canal as absent at this level and plan accordingly
B. Repeat the acquisition at the smallest voxel size available
C. Measure to the mental foramen and extrapolate the canal's course
D. Measure from the most coronal point at which the bundle is identifiable
E. Restrict the implant to 8 mm in length whatever the measurement shows

387 On the CBCT of a planned UR6 site the crestal cortex is barely discernible and the trabecular pattern is fine and sparse. Residual height below the sinus floor is 11 mm and the ridge is 6.5 mm wide. At surgery the pilot drill meets almost no resistance through its whole length. Which modification to the protocol is most appropriate?

A. Place a wider implant of the same length in the prepared site
B. Tap the osteotomy fully before inserting the implant
C. Under-prepare the osteotomy and extend the healing period
D. Load the implant immediately to stimulate bone formation
E. Increase irrigation and reduce the speed of the remaining drills

388 You are planning an implant at the LR5. The patient has metal-ceramic crowns on the LR6 and the LR7, and the CBCT taken elsewhere shows dark bands and bright streaks crossing the intended site at the level of the crest, obscuring the bone entirely. The scan is otherwise well positioned and free of movement. Which change is most likely to produce a diagnostic image of the site?

A. Increasing the tube current for the repeat acquisition
B. Reducing the voxel size to the smallest available
C. Enlarging the field of view to include both arches
D. Reducing the number of basis projections acquired
E. Repositioning so the crowns lie outside the plane of interest

389 An 82-year-old man with an essential tremor of the head has a CBCT for implant planning at the LL5. The reconstruction shows double contours at every cortical border and generalised unsharpness throughout the volume. He has no metal restorations, the collimation was correct and the exposure factors were those used routinely. Which change is most likely to yield a diagnostic scan on repeat?

A. Selecting an acquisition protocol with a shorter scan time
B. Selecting the smallest voxel size the machine offers
C. Increasing the tube current for the repeat acquisition
D. Enlarging the field of view to include the whole mandible
E. Applying the metal artefact reduction algorithm on reconstruction

390 A colleague tells you that he routinely selects the largest field of view his CBCT machine offers, on the grounds that it saves having to repeat a scan if something else turns up. Which of the following are consequences of that policy?

A. A greater effective dose to the patient at the same exposure factors
B. A duty to report on the whole of the imaged volume
C. A larger voxel, and so less detail, for a given detector size
D. A and B only
E. All of the above

391 You are planning a fully guided implant at the LR6, where 13 mm of bone lies above a clearly corticated mandibular canal. Published deviations for static guided surgery are of the order of 1 mm at the entry point and 1.5 mm at the apex, with angular deviations of some 3 to 5 degrees and occasional larger outliers. What minimum safety margin should you plan between the implant apex and the superior border of the canal?

A. 0.5 mm
B. 1 mm
C. 2 mm
D. 4 mm
E. 6 mm

392 A fully edentulous mandible is to be treated with four implants placed through a mucosa-supported guide produced from a dual-scan dataset. The mucosa over the anterior ridge is thick and displaceable, the ridge is well formed and fixation pins are planned. Which single feature most limits the accuracy achievable?

A. Displacement of the mucosa as the guide is seated
B. The tolerance between the drill and the metal sleeve
C. The voxel size at which the CBCT was acquired
D. Registration of the two scans on the fiducial markers
E. The dimensional tolerance of the printing process

393 You are considering a bone-supported guide for a partially edentulous posterior mandible in which two implants are planned. Which of the following are true of that choice?

A. It requires elevation of a broad flap before the guide will seat
B. It cannot be used where teeth remain in the same quadrant
C. It is less accurate than a mucosa-supported guide in the same case
D. A and B only
E. All of the above

394 A patient needs an implant at the LR7 and you are weighing dynamic navigation against a conventional static guide. Her maximum opening is 36 mm, the canal is 4 mm below the planned apex and the bone is well corticated. Which is the principal advantage of navigation in this particular site?

A. It removes the need for a preoperative cross-sectional scan
B. It abolishes the need for a safety margin from the canal
C. It compensates automatically for movement of the patient
D. It needs no vertical clearance for a guide and drill sleeve
E. It is more accurate than static guidance in all published series

395 During a dynamic navigation case the tracking arrays are mounted on the handpiece and on a rigid splint clipped to the patient's remaining teeth, and registration has been verified before drilling. Which event would most degrade the accuracy of the resulting implant position?

A. A change in the patient's head position during the osteotomy
B. Loosening of the splint that carries the patient's array
C. Movement of the assistant briefly across the camera's view
D. A reduction in irrigation flow during the osteotomy
E. A change in drill speed from the value planned

396 A 58-year-old woman needs an implant at the LR7. Her maximum interincisal opening is 38 mm, and the fully guided kit requires 42 mm of clearance between the guide sleeve and the opposing arch to accommodate the longest drill and the handpiece. The bone is 12 mm above a well-corticated canal and the ridge is 8 mm wide. Which is the most appropriate response?

A. Place the implant freehand after seating and removing the guide
B. Shorten the planned implant so that shorter drills may be used
C. Move the planned site forward to the LR6 to gain access
D. Use a guide for the pilot drill only and complete freehand
E. Grind the guide sleeve down to recover the missing clearance

397 A diagnostic wax-up has been completed on articulated casts for a patient missing the UR4 and UR5. Which of the following does the wax-up itself establish?

A. The clearance between the planned implants and the sinus floor
B. The restorative space required at each of the two planned sites
C. The density of the bone at the crest of each planned site
D. A and B only
E. A, B and C

398 An edentulous 66-year-old man wants a fixed maxillary implant prosthesis. A trial denture judged satisfactory for tooth position, lip support and phonetics shows the distance from the residual crest to the planned incisal edge of the UR1 to be 18 mm. He displays no gingiva on smiling, and the ridge is of adequate width throughout. What does this measurement most strongly indicate?

A. That the anterior implants must be angulated palatally
B. That a labial flange will be needed to support the lip
C. That vertical ridge augmentation should precede placement
D. That an implant-retained overdenture should be provided
E. That the prosthesis must include replacement of soft tissue

399 A radiographic guide is being made for an edentulous maxilla by duplicating the patient's existing complete denture, which is stable and correctly extended. The technician asks whether to mark the planned tooth positions with small metal spheres or to duplicate the teeth in acrylic containing barium sulphate. Which is the most important reason for choosing the barium-containing acrylic?

A. It reproduces the form of the teeth as well as their positions
B. It is quicker and cheaper for the laboratory to produce
C. Metal markers generate streak artefact that obscures the bone
D. Barium is more radiopaque than metal at dental beam energies
E. Metal markers cannot be positioned accurately in a duplicate

400 A fully guided implant at the UL5 is found on the postoperative periapical to sit 2 mm coronal to the planned position, with its axis and mesiodistal position unchanged. The guide was tooth-supported, the drills and the implant mount were those specified by the kit, and the digital plan was checked afterwards and found correct. Which is the most likely explanation?

A. The drill stop was omitted from the final osteotomy drill
B. The guide was not fully seated when the site was prepared
C. The wrong sleeve offset was entered in the planning software
D. The implant was driven home at too high an insertion torque
E. The guide flexed under the pressure of the handpiece

401 A partially dentate patient has an intraoral scan and a CBCT taken for guided placement at the LL6. The remaining teeth in that quadrant carry large metal-ceramic crowns, and the planning software is set to register the two datasets by matching the surfaces of those crowns. Which is the most important consequence?

A. The soft-tissue outline will be inaccurate on the merged file
B. The bone density values will be over-estimated at the site
C. Registration will be inaccurate because the crown surfaces are distorted
D. The guide will not seat because the intraoral scan is distorted
E. The implant length will be mis-scaled in the merged dataset

402 You are assembling the records needed before restoring two implants in a partially dentate patient whose remaining occlusion is unstable and who has lost posterior support on one side. Which of the following must be assessed on articulated study casts rather than on the CBCT?

A. The slide from centric relation to maximum intercuspation
B. The interocclusal space available for the two prostheses
C. The buccolingual position of the mandibular canal at each site
D. A and B only
E. A, B and C

403 Five years after you placed and restored a single implant, a claim is brought alleging that the treatment was inadequately planned and that the patient was not warned of the possible complications. Which of the following would you expect to be able to produce from the clinical record?

A. The justification for each radiograph and a report on its findings
B. A record of the consent discussion, including the alternatives
C. The batch or lot number of the implant that was placed
D. A and B only
E. All of the above

404 Before placing an implant at the UR2 you carry out and record sensibility testing of the UR1 and the UR3. Both are unrestored, uncrowned and asymptomatic, and neither has a history of trauma that the patient recalls, although the UR2 itself was lost in a fall. What is the most important reason for the test?

A. To establish whether either tooth is already non-vital
B. To predict which tooth is most likely to lose vitality
C. To decide whether an immediate placement is appropriate
D. To complete the periodontal charting required at assessment
E. To determine whether a CBCT is justified at this site

405 A CBCT taken to plan an implant at the UR6 shows a smooth, dome-shaped homogeneous opacity about 10 mm high arising from the floor of the right maxillary sinus, beneath the planned site. The ostium is patent, the remainder of the sinus is aerated and the mucosa elsewhere is of normal thickness. The patient has no nasal or sinus symptoms. Residual bone height at the site is 9 mm and the ridge is 8 mm wide. What is the most appropriate course?

A. Refer to an ear, nose and throat surgeon before any surgery
B. Prescribe an antibiotic and a decongestant, then reassess
C. Abandon the implant plan and provide a fixed bridge instead
D. Record and report the finding and proceed with treatment
E. Repeat the CBCT in six months to look for any change

406 A patient is referred for an implant at the UL6 and mentions that a CBCT including the same region was taken at a hospital eight months ago, for an unrelated complaint, but she has not brought it or any report. Your clinical assessment indicates that cross-sectional imaging is genuinely needed. What should you do first?

A. Acquire a new scan, since the earlier one was taken for another purpose
B. Obtain and review the existing images before requesting anything new
C. Take a panoramic radiograph and decide once you have seen it
D. Ask her to describe what she was told about the earlier findings
E. Proceed with a periapical view and place the implant freehand

407 A 55-year-old man needs an implant at the LL6. A recent, correctly positioned panoramic radiograph shows a well-corticated mandibular canal traceable along its whole length, with 15 mm of bone above it at the site, and the ridge is clinically 8 mm wide with no palpable concavity. Which of the following would make cross-sectional imaging necessary in this case?

A. Inability to trace the superior border of the canal on the panoramic
B. The measurement of 15 mm of bone height above the canal
C. His stated wish to see a three-dimensional image of the site
D. A and B only
E. All of the above

408 A CBCT has been justified before placing a single implant at the UR1, where a buccal concavity is suspected clinically and the incisive canal appears close to the planned site on a periapical. The patient is 40 years old and has an otherwise intact, unrestored dentition. Which acquisition is most appropriate?

A. A small field of view at the smallest voxel size the machine offers
B. A small field of view at a voxel size of about 0.2 mm
C. A large field of view at a voxel size of about 0.2 mm
D. A large field of view at the smallest voxel size the machine offers
E. A panoramic radiograph with a radiopaque marker at the site

409 A tooth-supported surgical guide arrives from the laboratory for two implants in the lower left quadrant, together with the printed model on which it was made. Which of the following checks best protects against a positional error at surgery?

A. Seating the guide on the printed model supplied with it
B. Seating it in the mouth and inspecting through the windows
C. Confirming that the resin will tolerate the sterilisation cycle
D. A and B only
E. A, B and C

410 On the CBCT of a 34-year-old woman with a high smile line, the most favourable bone at the UR2 site lies about 3 mm palatal to the position indicated by the diagnostic wax-up and would accept a 4.1 mm implant with no augmentation. Placing the implant where the wax-up indicates would require simultaneous guided bone regeneration. The adjacent teeth are unrestored and the phenotype is thin. Which consideration should govern the implant position?

A. The volume of bone available without any augmentation
B. The morbidity of the additional regenerative procedure
C. The tooth position established on the diagnostic wax-up
D. The axis that permits a screw-retained final restoration
E. The distance from the incisive canal to the planned site

Block P — Surgical anatomy, extended (Q411–440)

411 A 47-year-old man is being planned for an implant at the LR4. On CBCT the mental foramen opens at the level of the second premolar, and successive cross-sections show the inferior alveolar nerve running 3.5 mm anterior to the foramen before turning back on itself to exit. The crest lies 15 mm above the roof of the canal and the ridge is 7 mm wide at its narrowest. Which of the following should determine the posterior limit of the osteotomy?

A. The anterior border of the mental foramen as seen on the panoramic radiograph
B. A point 2 mm anterior to the most anterior extent of the loop on the CBCT
C. A point 2 mm posterior to the anterior border of the mental foramen
D. A point 5 mm anterior to the mental foramen on the panoramic radiograph
E. The mesial surface of the second premolar root on the periapical radiograph

412 A 52-year-old woman is being planned for an implant at the LL6. The CBCT report describes a duplicated mandibular canal: the main canal lies 13 mm below the crest, and a smaller accessory canal runs 4 mm below and lingual to it, rejoining the main canal distal to the site. She has had no previous difficulty with local anaesthesia. Which is the most important consequence for your surgical plan?

A. Anaesthesia obtained by an inferior alveolar block may be incomplete
B. The accessory bundle is purely vascular and carries no neurosensory risk
C. The available implant length is set by the more coronal of the two canals
D. The finding should be confirmed on a panoramic radiograph before planning
E. A wider, shorter implant removes the need to identify either canal

413 A 58-year-old woman has four implants placed between the mental foramina to retain an overdenture. A crestal incision was extended labially with two vertical releasing incisions and the flap raised to the depth of the sulcus, detaching the muscle attachments on the labial aspect of the symphysis. Six weeks later she reports that her chin looks flattened, her lower lip is less competent at rest and her sulcus feels shallow. Sensation is normal. What is the most likely explanation?

A. Detachment of the mentalis muscles with failure of secure re-attachment
B. Bilateral traction neuropraxia of the mental nerves at the foramina
C. Contracture of scar tissue within the two vertical releasing incisions
D. Loss of vestibular depth caused by the periosteal releasing incision
E. Residual post-operative oedema of the labial soft tissues at six weeks

414 Consider these statements about the mandibular incisive canal: (i) it is the anterior continuation of the inferior alveolar canal beyond the mental foramen; (ii) it is identified far more reliably on cross-sectional imaging than on a panoramic radiograph; (iii) injury to it characteristically causes lasting anaesthesia of the lower lip. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

415 A CBCT cross-section at the LR6 shows a ridge 8.5 mm wide at the crest, 13 mm of bone above the canal, and a pronounced lingual concavity whose deepest point lies 6 mm apical to the crest. The crest itself is inclined lingually. Which single modification most reduces the risk of perforating the lingual plate?

A. Elevating a lingual flap and placing a broad retractor against the plate
B. Preparing the final osteotomy with an osteotome rather than a twist drill
C. Selecting a wider implant so that both cortical plates are engaged
D. Angling the osteotomy buccally and accepting a shorter implant
E. Moving the implant one position distally within the same ridge

416 During placement of an implant at the LL6 the lingual plate is perforated by the final drill at a point 10 mm apical to the alveolar crest. There is brisk bleeding into the osteotomy, and over the next hour a firm swelling develops beneath the lower border of the mandible. The floor of the mouth remains at its normal level and the tongue is not displaced. Which space has most likely been entered?

A. The sublingual space, above the mylohyoid attachment
B. The submandibular space, below the mylohyoid attachment
C. The submental space, below the anterior mylohyoid attachment
D. The pterygomandibular space, medial to the ascending ramus
E. The buccal space, superficial to the buccinator attachment

417 A 46-year-old man is having an implant placed at the LL3. Ten minutes after the osteotomy is completed the floor of the mouth begins to swell, the tongue is being displaced upwards and backwards, and he is becoming distressed and finding it difficult to speak. His blood pressure is 148/92 mmHg and his oxygen saturation is 96 per cent on air. Which is the most important immediate action?

A. Apply firm bimanual pressure to the floor of the mouth and observe
B. Place a resorbable haemostatic dressing into the osteotomy and suture
C. Sit the patient upright, give oxygen and review him again in one hour
D. Complete the implant placement so that the fixture tamponades the site
E. Summon emergency assistance and prepare for the airway to be secured

418 Which of the following measures reduce the risk of floor-of-mouth haemorrhage during preparation of an osteotomy in the interforaminal region?

A. Assessing the lingual contour of the ridge on cross-sectional imaging
B. Palpating the lingual plate with a finger while the osteotomy is prepared
C. Directing the osteotomy along the long axis of the alveolar crest
D. A and B only
E. A, B and C

419 A 46-year-old woman requires an implant at the UR1. CBCT shows a nasopalatine canal 7 mm in diameter at its oral opening, and the planned implant surface would lie within 1 mm of the canal wall along most of its length. Bone volume is otherwise adequate. Which statement best explains why an implant that engages the canal is likely to fail?

A. Its contents are neurovascular and fibrous, so bone does not form at that surface
B. The nasopalatine nerve is transected, and denervated bone subsequently resorbs
C. Its walls are dense cortical bone, which undergoes compression necrosis
D. Bleeding from the nasopalatine vessels prevents a stable clot from forming
E. The canal communicates with the nasal cavity, so the site becomes infected

420 A 52-year-old woman needs a subepithelial connective tissue graft 6 mm in height for a recession defect at the UR3. Her palatal vault is shallow, and measurement on the CBCT indicates that the greater palatine neurovascular bundle lies approximately 7 mm from the gingival margin in the premolar and molar region. You intend to leave at least 2 mm of tissue coronal to the incision. Which is the most appropriate donor site?

A. The lateral palate between the distal of the canine and the second molar
B. The lateral palate immediately palatal to the first and second molars
C. The maxillary tuberosity
D. A and C only
E. All of the above

421 While harvesting a connective tissue graft from the left palate, brisk pulsatile bleeding begins at the distal end of the donor site as the incision is extended towards the second molar. The patient is haemodynamically stable and takes no anticoagulant. Which is the most appropriate immediate action?

A. Infiltrate local anaesthetic with adrenaline around the bleeding point
B. Apply firm digital pressure over the site with damp gauze for five minutes
C. Suture a haemostatic dressing beneath a pre-formed clear palatal stent
D. Crush the bone around the greater palatine foramen with a blunt instrument
E. Refer immediately for surgical ligation of the greater palatine artery

422 A CBCT before a planned lateral window sinus elevation at the UL6 shows 5 mm of residual bone height and a bony partition rising 8 mm from the sinus floor immediately mesial to the intended implant site. The membrane is of normal thickness and the ostium is patent. Which statement best describes how the antrostomy should be designed?

A. The septum should be reduced with a bur before the membrane is elevated
B. The septum supports the membrane and reduces the risk of perforation
C. A transcrestal approach is preferable because the septum limits elevation
D. The window should be extended in one piece across the whole septum
E. Two windows should be prepared, one on each side of the septum

423 During preparation of a lateral window in the maxilla, brisk bleeding arises from within the bony wall itself as the bur passes through a small canal approximately 16 mm above the alveolar crest. The bleeding is pulsatile and obscures the field, and the membrane has not yet been elevated. Which vessel is the most likely source?

A. The anastomosis between the posterior superior alveolar and infraorbital arteries
B. The greater palatine artery running forward within its bony canal
C. The sphenopalatine artery crossing the roof of the maxillary sinus
D. The facial artery as it crosses the anterior wall of the maxillary sinus
E. The buccal branch of the maxillary artery within the elevated periosteum

424 During preparation of an osteotomy at the UL1 in a resorbed maxilla, the drill loses resistance at a depth of 15 mm. The patient reports a salty taste and there is a little blood from the left nostril. The residual bone height measured on the pre-operative CBCT was 14 mm. The site is otherwise dry and the ridge is 6 mm wide. What is the most appropriate immediate management?

A. Abandon the site and graft the nasal floor before a second attempt
B. Place the planned implant, engaging the nasal floor for bicortical stability
C. Refer urgently for ear, nose and throat assessment of the perforation
D. Place a shorter implant that remains entirely within the residual bone
E. Prescribe a decongestant and an antibiotic and place four weeks later

425 A 63-year-old man has 4 mm of residual bone beneath the sinus at the UR7 and a bulky maxillary tuberosity. A colleague suggests that instead of grafting you could place a pterygoid implant, angled distally to engage the pterygoid plate. Which of the following are sound reasons for you to decline to undertake this yourself?

A. The maxillary artery and pterygoid venous plexus lie close to the trajectory
B. There are no published survival data for implants placed in this region
C. The apex of a pterygoid implant engages bone of characteristically low density
D. A and B only
E. A, B and C

426 Guided bone regeneration with a particulate graft and a resorbable membrane is planned at the LL6, and the buccal flap must be advanced about 8 mm to achieve tension-free primary closure over the augmented contour. The mental foramen lies well anterior to the flap. Which single manoeuvre most reliably achieves the advancement required?

A. Two divergent vertical releasing incisions extended into the buccal sulcus
B. Coronal advancement of a lingual flap after release of the mylohyoid muscle
C. A horizontal periosteal releasing incision at the base of the buccal flap
D. Extension of the crestal incision distally along the external oblique ridge
E. Horizontal mattress sutures placed in preference to interrupted sutures

427 Which of the following features of flap design protect the blood supply of a buccal mucoperiosteal flap in the mandible?

A. Vertical releasing incisions that diverge as they run apically
B. A flap base narrower than its free margin, to allow greater advancement
C. Split-thickness elevation, leaving the periosteum attached to the bone
D. A and C only
E. A, B and C

428 An implant is to be placed at the LR7, and the crestal incision must be extended distally to gain access. The retromolar pad is bulky, the patient has a broad ascending ramus, and there is 4 mm of keratinised mucosa on the lingual aspect. Which modification of the incision best protects the lingual nerve?

A. Extending the distal incision lingually to recruit the keratinised tissue
B. Placing a vertical releasing incision lingual to the retromolar pad
C. Reflecting a lingual flap and protecting it with a broad periosteal retractor
D. Ending the crestal incision at the distal surface of the second molar
E. Directing the distal extension buccally onto the external oblique ridge

429 A 55-year-old man attends for assessment before an implant at the LR6. The panoramic radiograph shows a well-defined, corticated, ovoid radiolucency about 12 mm across lying below the mandibular canal in the LR6 and LR7 region, close to the lower border. He is asymptomatic, the overlying teeth are vital and there is no expansion. What is the most appropriate management?

A. Refer for an incisional biopsy before any implant treatment is considered
B. Confirm the appearance on cross-sectional imaging and continue planning
C. Review with a repeat panoramic radiograph in twelve months
D. Abandon implant treatment in the lower right quadrant altogether
E. Refer for enucleation of a probable odontogenic keratocyst

430 A CBCT for a planned implant at the LL6 shows 12.5 mm of bone from the alveolar crest to the superior cortex of the mandibular canal, a ridge 7.5 mm wide, and no lingual undercut at the site. The available implant lengths are 8, 10, 11.5, 13 and 15 mm, and you intend to work to the conventional minimum safety margin above the canal. Which implant length is most appropriate?

A. 8 mm
B. 10 mm
C. 11.5 mm
D. 13 mm
E. 15 mm

431 A 34-year-old woman requires an implant at the UR2. The tooth was lost eighteen months ago. CBCT shows a crest 6.5 mm wide with adequate height, but a marked labial concavity beginning about 6 mm apical to the crest, so that the labial plate curves palatally above that level. The planned trajectory follows the long axis of the contralateral crown. Which bony complication is most likely?

A. A labial dehiscence extending from the crest to the level of the apex
B. Perforation of the nasal floor at the apex of the prepared osteotomy
C. Perforation into the incisive canal on the palatal aspect of the site
D. A labial fenestration at the apex with the crestal bone left intact
E. Perforation into the maxillary sinus above the apex of the osteotomy

432 A CBCT taken before a planned sinus floor elevation at the UR6 shows 5 mm of residual bone height and a smooth, dome-shaped opacity 12 mm high arising from the sinus floor, with normal mucosa elsewhere, a patent ostiomeatal complex and a well-aerated sinus above the lesion. The patient has no nasal symptoms and no history of sinus disease. What is the most appropriate course?

A. Refer to an ear, nose and throat surgeon for removal before grafting
B. Prescribe an antibiotic and a nasal decongestant and review in a month
C. Proceed with the planned elevation, recording the finding and the discussion
D. Abandon sinus elevation and place a 6 mm implant in the residual bone
E. Repeat the CBCT in three months to establish whether the lesion is enlarging

433 A palatal flap raised from the greater palatine region can be designed considerably longer and narrower than a buccal flap of comparable dimensions without risk of necrosis. Which statement best explains why?

A. Palatal mucosa is keratinised and is therefore more resistant to necrosis
B. Palatal mucoperiosteum is thick and holds sutures placed under tension
C. The palate is supplied by a dense submucosal capillary plexus alone
D. Palatal wounds heal by secondary intention without flap necrosis
E. The flap carries an axial blood supply from the greater palatine artery

434 A block graft is to be harvested from the mandibular symphysis of a 42-year-old man with a full lower anterior dentition and mental foramina in the usual position. Which of the following are recognised limits on the osteotomy?

A. A margin of at least 5 mm below the apices of the anterior teeth
B. A margin of at least 5 mm mesial to each mental foramen
C. Engagement of the lingual cortex, to obtain a bicortical block

D. A and B only
E. All of the above

435 A 10 mm implant is placed at the LL6. The post-operative radiograph shows the apex 2.5 mm clear of the roof of the mandibular canal, with the implant well centred. There was brisk bleeding from the osteotomy as the final drill was withdrawn, which settled with pressure. The following morning the patient reports numbness of the left lower lip and chin. What is the most likely explanation?

A. Bleeding into the mandibular canal with compression of the nerve
B. Direct transection of the inferior alveolar nerve by the final drill
C. Thermal injury from a blunt drill used with inadequate irrigation
D. Local anaesthetic toxicity following the inferior alveolar block
E. Compression by the implant apex despite the radiographic clearance

436 During implant placement at the UR7 the surgeon notes that the drill advances under almost no pressure and that the walls of the osteotomy feel soft on probing. The CBCT had shown a bulky tuberosity with a thin, incomplete cortex and widely spaced trabeculae, and 11 mm of bone below the sinus floor. Which single modification is most likely to improve primary stability at this site?

A. Countersinking the crestal bone before seating the implant
B. Under-preparing the osteotomy relative to the implant diameter
C. Increasing drill speed to reduce the time spent in contact with bone
D. Selecting a parallel-walled rather than a tapered implant design
E. Tapping the osteotomy with a screw tap before implant insertion

437 A 71-year-old man has an atrophic edentulous maxilla and wants a fixed prosthesis. CBCT shows the incisive foramen opening directly at the crest between the planned UR1 and UL1 positions, with the canal occupying most of the width of the ridge in the midline. Bone at the canine and lateral incisor positions is 7 mm wide and 13 mm high. What is the most appropriate modification of the plan?

A. Place a single implant within the canal after grafting around its contents
B. Reduce the alveolar crest until the foramen is no longer exposed
C. Place short implants either side of the canal at the central incisor sites
D. Enucleate the canal, graft it and place at the central incisor sites at one visit
E. Reposition the implants to the lateral incisor and canine positions

438 A haemorrhage into the sublingual space after perforation of the lingual plate in the anterior mandible can obstruct the airway within minutes, whereas a comparable volume of blood in the buccal soft tissues produces only swelling. Which statement best explains the difference?

A. The sublingual arteries arise from the external carotid and bleed at systemic pressure
B. The mucosa of the floor of the mouth is loose and distends over a wide area
C. The space is bounded by mandible, mylohyoid and tongue, so pressure displaces the tongue
D. The floor of the mouth has no lymphatic drainage to limit the accumulation
E. Blood tracks along the submandibular duct and is aspirated into the pharynx

439 A 66-year-old woman is to have two implants placed in the interforaminal region to retain a mandibular overdenture, and a CBCT has been justified. Which of the following should be recorded from the cross-sectional images before surgery?

A. The position of each mental foramen and the extent of any anterior loop
B. The depth and level of the lingual concavity at each planned site
C. The course of the mandibular incisive canal anterior to the foramen
D. A and B only
E. All of the above

440 During periosteal release of a buccal flap in the posterior maxilla, a lobulated yellow mass herniates through the incision and will not remain reduced. There is no bleeding, the sinus has not been entered, and the flap is otherwise mobile and healthy. What is the tissue, and what should be done?

A. Herniated sinus lining, which should be repaired with a collagen membrane
B. Buccal fat pad, which should be excised so that the flap may be advanced
C. A lipoma, which should be excised and submitted for histopathology
D. Buccal fat pad, which should be replaced and left otherwise undisturbed
E. Herniated parotid tissue, which requires referral for surgical repair

Block Q — Osseointegration, bone biology and healing, extended (Q441–475)

441 A 21-year-old man has a UR1 that was avulsed and replanted at the age of nine. It is now 3 mm infra-occluded, gives a high percussion note, and the periapical radiograph shows loss of the periodontal ligament space with replacement resorption. A colleague proposes six months of orthodontic forced eruption to develop bone and soft tissue before extraction and implant placement. Which statement best explains why this will not work?

A. The ankylosed root has no periodontal ligament through which to transmit force
B. The alveolar crest of the anterior maxilla will not remodel in a vertical direction
C. Extrusion carries soft tissue coronally without bringing any bone with it
D. Replacement resorption would accelerate under a continuous orthodontic load
E. The root would fracture before the necessary force could be developed

442 A 68-year-old man attends with a bridge placed fifteen years ago. Beneath the LR5 pontic the original root was deliberately decoronated and left submerged; the ridge there is full and level with the adjacent crest. At the LR4, extracted at the same appointment, the ridge is visibly collapsed both vertically and buccally. Which best explains the difference between the two sites?

A. The pontic has protected the underlying crest from functional loading
B. Root dentine is replaced by bone through creeping substitution
C. The retained root retains a ligament that supplies its bundle bone
D. The absence of a flap preserved the periosteal supply to the buccal plate
E. Cementum on the retained root has induced new attachment at the crest

443 In a dense mandibular site a surgeon prepares the osteotomy to the full diameter of the implant and taps the threads, so that the implant seats at 25 Ncm. Small voids remain between the thread flanks and the cut bone. A colleague argues that the implant is under-stabilised and should be removed in favour of a wider one. The implant is immobile to finger pressure and correctly positioned. Which statement best describes what will happen in those voids?

A. They fill with fibrous tissue unless the implant is submerged for healing
B. They act as healing chambers in which woven bone forms early
C. Bone bridges only those voids narrower than about 0.1 mm
D. The low seating torque predicts a deeper stability dip during healing
E. Bone apposition begins at the apex and advances coronally through them

444 Four months after a cortico-cancellous block graft was fixed to a deficient ridge, a trephine core taken at implant placement shows tunnels advancing through the cortical component, each with multinucleated cells at its leading end, a vascular loop behind them, and osteoblasts lining the tunnel wall further back. There is no inflammatory infiltrate. Which best describes this structure and its mechanical consequence?

A. A basic multicellular unit, weakening the graft before its strength is restored
B. A resorption lacuna, indicating low-grade infection at the graft interface
C. A Volkmann canal, permitting revascularisation without any resorption
D. A reversal line, marking the limit of osteoinductive activity in the graft
E. A Haversian canal formed by apposition without preceding resorption

445 A physician planning antiresorptive therapy for a patient of yours asks why a drug that acts only on the osteoclast should also, within months, reduce the serum markers of bone formation. She is considering alendronate for osteoporosis and wants to understand the effect on healing before you place an implant. Which statement best explains the observation?

A. Osteoblasts differentiate from the same precursor population as osteoclasts
B. Bisphosphonates bind osteoblast integrins and inhibit their migration
C. Osteocytes cease to secrete sclerostin once resorption is inhibited
D. Remodelling units require an intact periosteum in order to be initiated
E. Osteoclasts release matrix-bound growth factors that recruit osteoblasts

446 A 71-year-old woman has received denosumab 60 mg subcutaneously every six months for four years. She wants an implant at the LR6, and her rheumatologist proposes simply stopping the drug for a year so that surgery can be undertaken safely. She has osteoporosis with two previous vertebral fractures. Which statement provides the strongest objection to that plan?

A. Its effect on RANKL persists for several years after the last injection
B. Discontinuation is followed by a rebound increase in bone resorption
C. Withdrawal provokes the formation of neutralising antibodies to the drug
D. Its skeletal reservoir must be cleared before turnover can normalise
E. The bone mineral density gained is retained once treatment is stopped

447 A patient asks how the bone around an implant "knows" that it is being used, given that the implant has no ligament. You want to describe the mechanism by which functional loading is converted into a change in bone mass around the fixture. Which statement best describes it?

A. Osteoblasts on the bone surface deform and secrete alkaline phosphatase
B. Periosteal fibroblasts respond to tension by releasing morphogenetic proteins
C. Osteocytes sense fluid flow in their canaliculi and modulate sclerostin and RANKL
D. Osteoclast precursors are recruited directly by piezoelectric surface potentials
E. Marrow stromal cells respond to hydrostatic pressure by proliferating

448 An implant is placed in the posterior mandible with an insertion torque of 50 Ncm and an ISQ of 78. The patient is healthy and the site is native, non-grafted bone. A colleague argues that because the implant is so firm at placement it may be restored at any point thereafter. At which period is premature loading most likely to cost the implant, and why?

A. In the first week, while the fibrin clot is the only interface present
B. At two to four weeks, when resorption has outpaced new bone formation
C. At six to eight weeks, as woven bone is being replaced by lamellar bone
D. At three months, when the crestal cortex begins osteonal remodelling
E. At one year, when the interface reaches its final mineral density

449 You are planning an implant at the UL6 and the CBCT software reports a mean grey value of 320 in the region of the intended osteotomy. A colleague states that this places the site firmly in the type 4 category and that the healing period should be doubled on that basis. The scan was taken on a small field of view with a standard protocol. Which statement should most temper that conclusion?

A. Grey values from cone beam imaging are not calibrated Hounsfield units
B. Values below 350 define type 4 bone in every published classification
C. Bone density correlates poorly with insertion torque in the posterior maxilla
D. Lekholm and Zarb classify density from the trabecular pattern on axial slices
E. Medical computed tomography systematically understates density in the maxilla

450 An implant is being placed in the anterior mandible, where the crestal cortex measures 2.5 mm on CBCT and the trabecular bone is dense. The final drill required firm pressure, and the implant stalls at 70 Ncm before it is fully seated. A second implant is to be placed at the contralateral site at the same visit. Which of the following modifications would appropriately reduce the risk of crestal compression necrosis there?

A. Tapping the osteotomy before inserting the implant
B. Countersinking the crestal cortex to the implant's collar diameter
C. Using a graded sequence of sharp drills at reduced speed
D. A and B only
E. All of the above

451 A 68-year-old woman is having two implants placed in the anterior mandible to retain her existing lower denture. They seat at 45 and 40 Ncm with ISQ values of 74 and 72. She has travelled a long distance, declines any period without retention, and immediate loading has been agreed after full discussion. Which single measure most reduces interfacial micromotion during the first eight weeks?

A. Splinting the two implants with a rigid bar
B. Relieving the denture base and relining it with a tissue conditioner
C. Selecting resilient rather than rigid attachment housings
D. Adjusting the occlusion to bilateral balanced contact
E. Instructing her to keep to a soft diet for eight weeks

452 A retrieval study examines implants that failed to integrate and compares them with implants that healed uneventfully, seeking the conditions under which a collagen-rich interface forms rather than bone. All the implants were of the same moderately rough design and none of the sites was infected. Which of the following favour fibrous repair rather than osseointegration?

A. A stable, uninfected gap of 2 mm between implant and bone
B. Relative movement of about 200 micrometres during early healing
C. A necrotic rim produced by drilling above the thermal threshold
D. B and C only
E. A, B and C

453 A colleague drills the final osteotomy in dense cortical bone with a new drill at 1500 revolutions per minute and no irrigation, arguing that the exposure lasts only five seconds and that bone is not damaged over so short a period. He asks you to justify your objection with a figure rather than an opinion. Which statement best supports the case against him?

A. Bone is killed only above 60 °C, whatever the duration of exposure
B. Irrigant delivered below 25 °C prevents any rise above 47 °C
C. Cortical bone dissipates heat within about ten seconds of contact
D. Osteocyte death and resorption follow one minute above about 47 °C
E. Temperature rise is proportional to the square of the drill speed

454 An implant placed at the LR5 eighteen months ago is immobile, shows no peri-implant radiolucency, and standardised radiographs record 0.9 mm of marginal bone loss in the first year of loading and 0.1 mm since. Probing depths are 3 mm without bleeding. The patient has had a persistent area of altered sensation in the lower lip since the day of surgery, which she finds intrusive. How should the implant be classified?

A. Successful, since the radiographic and mobility criteria are both met
B. Failing, because altered sensation indicates progressive loss of integration
C. Successful, since sensory change is a surgical rather than an implant outcome
D. Surviving but not successful, because of the persistent neuropathy
E. Surviving but not successful, because of the first-year bone loss

455 A guided bone regeneration procedure at the UR4 uses a slowly substituted xenograft, which is itself capable of maintaining the space beneath the barrier, and a native collagen membrane that is then fixed at its buccal margin with two tacks. The flap is released and closed without tension. Which principle of regenerative surgery does the fixation of the membrane chiefly serve?

A. Primary closure of the overlying mucoperiosteal flap
B. Angiogenesis into the graft from the underlying bone bed
C. Exclusion of epithelium from the grafted compartment
D. Space maintenance beneath the barrier membrane
E. Stability of the graft and of the clot within it

456 A colleague preparing a lateral ridge augmentation asks why you perforate the cortical plate with a small round bur before placing the particulate graft, having read that some operators do not. The recipient site is a dense, sclerotic buccal plate at the LL5 with no marrow visible at the surface. Which statement best justifies the perforations?

A. They give the graft access to progenitor cells and vessels from the marrow
B. They roughen the surface so that graft particles interlock mechanically
C. They relieve pressure within the cortex as the membrane is tacked
D. They reduce cortical density, so the graft is resorbed more slowly
E. They allow the membrane to be pinned without splitting the plate

457 A cortico-cancellous block graft has been fixed with two positional screws to augment a deficient ridge. The patient asks whether the grafted bone is at its strongest immediately after surgery, when it is solid, or later once it has "joined on". Healing is uneventful and she is a healthy non-smoker. Which statement best describes the mechanical behaviour of the cortical component over the first six months?

A. It gains strength steadily from the moment of fixation onwards
B. It is weakest at fixation and strongest at approximately three months
C. It weakens during the resorptive phase before regaining strength
D. Its strength is unchanged, because a cortical graft is not remodelled
E. It is replaced entirely by new lamellar bone within about six weeks

458 A patient telephones seven days after guided bone regeneration at the UR4 to say that she caught the site with a toothbrush and that it feels "as if it has given way", although the sutures are intact and there is no bleeding, swelling or discharge. You are explaining to your nurse why this is the period of greatest anxiety. When is a surgical wound at its most vulnerable to dehiscence, and why?

A. At 24 hours, before the incision line has epithelialised
B. At three days, when the inflammatory infiltrate is at its maximum
C. At one week, when tensile strength is a small fraction of its final value
D. At three weeks, when collagen cross-linking is being remodelled
E. At six weeks, when the barrier membrane begins to degrade

459 A patient who is a nurse asks what is actually happening in her gum during the second week after implant surgery, having read about the phases of wound healing. The site is closed, clean and uninflamed, and she is healing normally. Which of the following are events of the proliferative phase of healing?

A. Angiogenesis directed by the hypoxic gradient within the wound
B. Migration of epithelium across the incision line
C. Deposition of type III collagen by fibroblasts
D. A and B only
E. All of the above

460 A 58-year-old man attends for implant assessment. His panoramic radiograph shows generalised loss of the lamina dura around the remaining teeth, a diffuse ground-glass trabecular pattern in both jaws, and a well-defined unilocular radiolucency in the left mandibular body that he was unaware of. He is otherwise well, takes no medication, and mentions recurrent renal stones. What is the most appropriate next step?

A. Refer for investigation of possible hyperparathyroidism before proceeding
B. Proceed, since generalised trabecular change does not affect integration
C. Obtain a CBCT to measure bone density before selecting the implant
D. Treat the jaws as type 4 bone and extend healing to six months
E. Refer for a DEXA scan of the femoral neck and lumbar spine

461 A 74-year-old woman with severe osteoporosis takes teriparatide 20 micrograms daily by subcutaneous injection, prescribed after she failed to tolerate two bisphosphonates. She has been on it for eleven months and asks whether it must be stopped before an implant is placed at the LR5, as she was told to stop her previous drug. Which statement about this medication is correct?

A. It suppresses osteoclast function through the RANKL pathway
B. It binds hydroxyapatite and persists in the skeleton for years
C. It carries a risk of osteonecrosis equivalent to zoledronic acid
D. It should be interrupted for three months before oral surgery
E. Given intermittently, it stimulates osteoblastic bone formation

462 A patient taking a long-term antiresorptive asks why the risk of osteonecrosis you have described applies to her jaw when the same drug is intended to protect her hip and spine. You want to explain what is different about the jaws. Which of the following contribute to the localisation of medication-related osteonecrosis to the maxilla and mandible?

A. Their high rate of bone turnover compared with the rest of the skeleton
B. A covering of thin mucosa that separates bone from the oral environment
C. A constant bacterial challenge from the dentition and its supporting tissues
D. A and B only
E. All of the above

463 A 66-year-old man received 66 Gy to a field including the right mandible for a floor-of-mouth carcinoma eight years ago. He is disease-free, has stopped smoking and now asks whether enough time has passed for the risk associated with surgery in that bone to have resolved. He has had no spontaneous exposure of bone. Which of the following are true of the irradiated mandible at this interval?

A. Hyperbaric oxygen restores the vascularity of the irradiated bed
B. The obliterative vascular change is progressive, so risk does not fall with time
C. The risk of osteoradionecrosis is confined to the first two years
D. A and B only
E. A, B and C

464 Twelve months after regenerative surgery for peri-implantitis at the UL5, using surface decontamination and a xenograft beneath a collagen membrane, radiographs show that the intrabony component of the defect has filled and probing depths have fallen from 8 mm to 5 mm with no bleeding. The patient asks whether the implant is now attached to bone as it originally was. Which statement should govern your answer?

A. Radiographic defect fill is direct evidence of re-osseointegration
B. Surface decontamination reliably restores the original bone-to-implant contact
C. Re-osseointegration occurs only on machined implant surfaces
D. Regenerative treatment restores probing depths to their baseline values
E. Re-osseointegration to a contaminated surface is partial and unpredictable

465 An implant placed in the posterior mandible three weeks ago achieved 40 Ncm and an ISQ of 75. The patient is going abroad and asks to have the definitive crown fitted at this visit rather than at the planned eight weeks. The site is healed, the mucosa is healthy and there is no mobility. How is the proposed protocol classified, and what is the principal biological objection?

A. Immediate loading, and it precedes the development of any secondary stability
B. Early loading, and it follows the nadir of the total stability curve
C. Early loading, and it coincides with the nadir of the total stability curve
D. Conventional loading, and no particular biological objection arises
E. Progressive loading, and it requires serial resonance frequency measurement

466 A trainee asks how you can state that an implant is osseointegrated when the only tools available are a radiograph, a probe and a torque driver, and the original definition was histological. The implant in question has been in function for two years and is being reviewed. Which finding best satisfies the accepted clinical definition of osseointegration?

A. Absence of a peri-implant radiolucent line on a periapical radiograph
B. A resonance frequency value above 70 at abutment connection
C. Rigid, asymptomatic fixation maintained during functional loading
D. Direct bone contact over at least 60 per cent of the implant surface
E. An insertion torque exceeding 35 Ncm recorded at the time of placement

467 A UR1 with a vertical root fracture is to be extracted. The buccal plate is intact but thin, the phenotype is thin, and there is a sinus tract with granulation tissue at the apex. You elect to extract, debride and place the implant at six weeks with simultaneous contour grafting, rather than immediately. Which of the following are genuine advantages of waiting for that interval?

A. The resorption of the buccal wall will by then be complete
B. Soft tissue volume increases, allowing tension-free primary closure
C. Primary stability will be greater than at immediate placement
D. A and B only
E. All of the above

468 A colleague grafting an intrabony defect adjacent to an implant site has chosen a demineralised freeze-dried bone allograft in preference to the mineralised form supplied by the same tissue bank, and asks you to explain the difference in behaviour that justifies his choice. The defect is contained by three walls and the site is not infected. Which statement best describes what demineralisation achieves?

A. It exposes bone morphogenetic proteins within the collagenous matrix
B. It removes the antigenic protein that would otherwise provoke rejection
C. It renders the particles more rapidly resorbable by osteoclasts
D. It increases the surface area of each particle for cell attachment
E. It makes the graft more effective at maintaining ridge dimension

469 A patient requiring a small contour graft at implant placement asks why you have suggested harvesting her own bone from the adjacent ramus when a bottled substitute is available and would avoid a second surgical site. She has no objection in principle to either. Which of the following are properties of particulate autogenous bone that a deproteinised bovine xenograft does not share?

A. It contains viable osteogenic cells at the moment of grafting
B. It maintains the grafted ridge dimension more predictably
C. It carries proteins capable of inducing host cells to form bone
D. A and C only
E. A, B and C

470 Six months after extraction of an upper premolar with an intact but thin buccal plate, the ridge has lost about 3 mm of width and 1 mm of height. The socket has filled with bone internally, but the outer surface of the buccal plate has receded. There was no infection and healing was uneventful. Which process best describes the loss of the buccal plate?

A. Remodelling, since resorption and formation are coupled at that surface
B. Creeping substitution of bundle bone by mature lamellar bone
C. Disuse atrophy following the loss of the occlusal load on that tooth
D. Modelling, since resorption occurs there without coupled formation
E. Osteolysis driven by the inflammatory infiltrate within the socket

471 A 54-year-old woman is being planned for an implant at the LL6. Her medication list includes sertraline 100 mg daily, atorvastatin 20 mg daily and amlodipine 5 mg daily, all long-standing and stable. She is a non-smoker with an HbA1c of 39 mmol/mol and healthy periodontal tissues. Which of the following drug classes has been reported in observational studies as associated with an increased rate of implant failure?

A. Selective serotonin reuptake inhibitors
B. Statin therapy
C. Antihypertensive medication
D. A and B only
E. A, B and C

472 A healing abutment is placed at the time of implant surgery at the LR5 and the patient is seen at two weeks for suture removal, at four weeks for a review, and at six weeks for an impression, at which point the healing abutment is removed and replaced twice. She asks how long the cuff of tissue around the abutment takes to become a proper seal. Which best describes the maturation of the peri-implant mucosal barrier?

A. About one week, matching epithelialisation of the incision line
B. About two weeks, matching the removal of the sutures
C. About six to eight weeks, as the connective tissue and epithelium mature
D. About six months, matching completion of bone remodelling at the crest
E. It does not mature until the definitive abutment has been placed

473 A patient has 4 mm of bone height above the mandibular canal at the LR6 and asks for a fixed replacement rather than the short implant you have proposed. A colleague suggests vertical ridge augmentation, and you explain that the predictability of adding height differs greatly from that of adding width. Which factor best accounts for the lower predictability of vertical augmentation?

A. The graft is contained by fewer bony walls and has a smaller contact area
B. The overlying periosteum is less vascular in the mandible than the maxilla
C. Vertical grafts are inevitably loaded by a denture during healing
D. Vertical defects are more often contaminated by residual pathogens
E. Barrier membranes cannot be fixed in the vertical dimension

474 Consider these statements about the curve of total implant stability during the first three months of healing: (i) the primary component falls because compressed and surgically damaged bone must be resorbed; (ii) the rise in the secondary component reflects the formation and progressive mineralisation of new bone at the interface; (iii) a chemically modified hydrophilic surface abolishes the dip altogether. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

475 A colleague tells you that he aims for an insertion torque of at least 70 Ncm at every implant he places, on the grounds that more torque means more contact and therefore better integration. He works largely in the dense anterior mandible and reports early marginal bone loss in several recent cases. Which statement most accurately corrects his reasoning?

A. Insertion torque predicts the bone-to-implant contact achieved at healing
B. Torque below 35 Ncm reliably predicts failure of osseointegration
C. Torque correlates directly with the resonance frequency value obtained
D. Torque above about 50 Ncm in dense bone risks crestal compression necrosis
E. Torque determines the length of the healing period that will be required

Block R — Materials, implant design and surfaces, extended (Q476–515)

476 A manufacturer offers the same implant body in grade 4 commercially pure titanium and in Ti-6Al-4V, and a colleague argues that the alloy must be the better choice because it is roughly twice as strong. The implants are otherwise identical in geometry and in surface treatment, and both are supplied at the same price. What is the principal materials reservation about using the alloy for the implant body?

A. Its yield strength is lower than that of grade 4 commercially pure titanium
B. It does not form a passivating titanium dioxide layer on exposure to air
C. Wear and corrosion release aluminium and vanadium as well as titanium
D. It cannot be given a moderately rough sandblasted and etched surface
E. Its elastic modulus is close to bone, so it flexes appreciably under load

477 A representative describes a new implant made from a titanium alloy containing niobium and zirconium, and states that its "low modulus" will reduce stress shielding and preserve crestal bone. The implant is otherwise conventional in geometry, surface and connection. Which statement most accurately characterises the claim?

A. Beta-phase alloys have an elastic modulus close to that of cortical bone
B. Their modulus is roughly half that of titanium but still far above bone
C. Niobium and zirconium raise the modulus by solid solution strengthening
D. Elastic modulus has no bearing on strain distribution in peri-implant bone
E. Alloying lowers modulus by reducing the stiffness of the oxide layer

478 Retrieved two-piece implants examined after several years of function show discoloured, pitted surfaces on the internal walls of the connection and on the screw shank, while the external surfaces in contact with bone appear unaffected. The joints had been torqued to specification and the restorations were unremarkable. Which best explains the distribution of the damage?

A. The internal surfaces are not passivated during manufacture
B. Saliva cannot enter the joint, so lubrication of the threads is lost
C. The screw is of a different alloy, so a galvanic cell is established
D. Oxygen depletion and a falling pH inside the joint prevent repassivation
E. Titanium dioxide dissolves at the temperatures reached during function

479 A patient with two implant-retained crowns and a high caries rate in her remaining dentition is to be prescribed a high-concentration topical fluoride. Your hygienist has both a neutral sodium fluoride preparation and an acidulated phosphate fluoride gel available and asks whether the choice matters around the implants. Which consideration should determine it?

A. Acidulated fluoride degrades the titanium oxide layer, so a neutral preparation is used
B. Fluoride at any pH is inert towards titanium, so either preparation is suitable
C. Fluoride accelerates the tetragonal to monoclinic transformation of the crowns
D. Acidulated fluoride etches the ceramic crown but does not affect the abutment
E. High-concentration fluoride is avoided because it inhibits osteoblast activity

480 A colleague notes that one manufacturer's hydrophilic implant is supplied in a sealed ampoule of saline while its conventional counterpart, of identical topography, is packaged dry, and asks why an ordinary sandblasted and acid-etched surface should not simply be stored in air and used when needed. Which of the following explain the loss of surface energy that occurs on storage in air?

A. Adsorption of atmospheric hydrocarbons onto the oxide surface
B. Loss of the hydroxylated state of the freshly prepared oxide
C. Progressive thickening of the oxide layer during dry storage
D. A and B only
E. All of the above

481 Two implant surfaces are each quoted by their manufacturers as having an average roughness, Sa, of 1.5 micrometres, yet published comparisons show different early bone responses. Both are subtractive surfaces on grade 4 titanium and both are supplied dry. Which statement best explains how two surfaces of identical Sa can differ?

A. Sa is measured in two dimensions whereas Sdr is measured in three
B. Sa describes the deepest pit and Sdr describes the tallest peak
C. Sa averages height alone, whereas Sdr describes the enlargement of surface area
D. Sa applies to subtractive surfaces and Sdr to additive coatings
E. Sa is measured before sterilisation and Sdr afterwards

482 A patient treated for generalised periodontitis, now stable, needs an implant at the LR6. One system offers a bone-level implant roughened to the top of the platform; another offers a design with a 1.8 mm machined, polished collar. Her mucosal thickness is 2.5 mm and she is in three-monthly supportive care. Which statement best supports choosing the machined collar in this patient?

A. A machined collar attaches to connective tissue through hemidesmosomes
B. A machined collar prevents apical migration of the barrier epithelium
C. A machined collar increases bone-to-implant contact at the crest
D. A machined collar removes the requirement for keratinised mucosa
E. A machined collar retains less biofilm and is easier to clean if exposed

483 A manufacturer's literature states that its chemically modified surface has a water contact angle approaching zero degrees, whereas the conventional version of the same topography exceeds 100 degrees. Both surfaces have an identical Sa and are supplied sterile. What does the contact angle figure tell you about the modified surface?

A. Blood spreads completely across it, so the fibrin clot adheres closely
B. It repels water, keeping it free of contamination until it is placed
C. It has a rougher topography than the conventional surface
D. It carries a thicker oxide layer produced by anodisation
E. It is chemically inert, so plasma proteins do not adsorb to it

484 The laboratory proposes a "highly translucent" zirconia for a monolithic screw-retained crown on a titanium base at the LR6. The patient has marked wear facets and reports night-time grinding, and the opposing dentition is natural. The alternative offered is a conventional zirconia with a small buccal veneer. Which property of the translucent material is most relevant to this decision?

A. Translucent zirconia is glass-infiltrated and therefore inherently weaker
B. Translucency is achieved by lowering the sintering temperature
C. Translucent grades are more susceptible to low-temperature degradation
D. A higher yttria content raises translucency but lowers strength and toughness
E. Translucent grades have larger grains and a higher flexural strength

485 A patient asks how a ceramic can be described as "tough" when ceramics are brittle, having been told that her zirconia crown resists cracking better than the porcelain one it replaces. You want to describe the mechanism by which a crack in yttria-stabilised zirconia is arrested. Which statement describes it correctly?

A. The glassy matrix flows plastically at the tip of the advancing crack
B. Grains at the crack tip transform with a volume increase that compresses it
C. Yttria migrates to the crack tip and re-sinters the fracture surfaces
D. The crack is deflected by leucite crystals dispersed through the matrix
E. Residual compressive stress introduced by glazing closes the crack

486 A laboratory reports that it steam-cleans and then autoclaves its milled zirconia abutments several times during fabrication and before dispatch, as a matter of routine infection control. The abutments are 3Y-TZP, milled and sintered in the laboratory, and are to be delivered on titanium bases. Which consequence of that protocol is most relevant?

A. Autoclaving oxidises the yttria stabiliser and embrittles the surface
B. Autoclaving anneals the surface and removes transformation toughening
C. Moist heat accelerates the surface transformation responsible for ageing
D. Repeated heating causes sintering shrinkage and a loss of marginal fit
E. Steam removes the milled surface layer and exposes subsurface porosity

487 A colleague comparing two ceramic systems for implant crowns notes that one is quoted at a much higher flexural strength than the other and argues that this alone should decide the choice. Both are supplied for monolithic use and both would be bonded to titanium bases. Which of the following explain why a quoted flexural strength predicts clinical performance poorly?

A. Ceramics fail from surface flaws whose size varies between specimens
B. Cyclical loading in a wet environment lowers the stress at which failure occurs
C. Ceramics deform plastically before fracture, absorbing energy first
D. A and B only
E. A, B and C

488 A laboratory proposes a milled polyetheretherketone framework for an implant-supported fixed prosthesis, arguing that its low elastic modulus will absorb occlusal load and protect the implants and the bone. The patient has four implants in a healed mandible, an even occlusion and no parafunction. Which statement most accurately assesses the proposal?

A. A low-modulus framework reduces the peak force transmitted to the bone
B. A low-modulus framework substitutes mechanically for the absent ligament
C. A low-modulus framework reduces the incidence of screw loosening
D. A low-modulus framework raises crestal strain to a physiologically useful level
E. A low-modulus framework flexes without materially reducing the load on bone

489 An abutment on an internal conical connection is to be removed at the LR5 after four years in function to allow the crown to be remade. The retaining screw is loosened normally and lifts out intact, but the abutment cannot be removed by hand or with a pair of tweezers, and the implant and bone are radiographically sound. What is the most likely explanation?

A. The taper is frictionally locked and requires the system's extraction device
B. The screw has fractured within the abutment and is holding it in place
C. Corrosion products have fused the abutment to the internal walls
D. Residual luting cement has entered and set within the connection
E. The implant has rotated within the bone and jammed the connection

490 Two implants with internal conical connections at the LR4 and LR6 receive definitive abutments, which are torqued to specification and impressed at the same appointment. At try-in three weeks later the screw-retained framework does not seat passively, rocking slightly on the mesial abutment, although the laboratory work appears accurate and the abutments are undisturbed. What is the most likely explanation?

A. The impression material has undergone permanent deformation on removal
B. The implants have rotated slightly under the impression coping screws
C. The abutments have settled axially into the tapers since the impression
D. The framework has distorted during casting and requires sectioning
E. The abutment screws have lost preload through embedment relaxation

491 In a series of experimental studies, two-piece implants were placed with the implant–abutment junction at the crest, 1 mm above it and 1 mm below it, and the crestal bone response was measured after healing. The implants were otherwise identical and all healed without infection. Which statement best summarises the finding and its clinical implication?

A. The microgap is sealed by the screw's preload, so its level is immaterial
B. Bone loss depends on the diameter of the microgap rather than its position
C. Bone loss occurs only where the junction lies below the mucosal margin
D. Bone recedes to a level apical to the junction wherever it is placed
E. Bone loss occurs only when the abutment is repeatedly disconnected

492 A colleague argues that once an abutment screw has been torqued to specification the implant and abutment behave as a single solid unit, and that the microgap is therefore of no consequence in a well-made restoration. The joint in question is a conventional internal connection carrying a molar crown with an even occlusion. Which of the following occur at that interface during normal function?

A. Cyclical opening of the joint on the loaded side under lateral force
B. Exchange of fluid and bacteria between the internal cavity and the sulcus
C. Release of metal particles by fretting between the mating surfaces
D. A and B only
E. All of the above

493 A platform-switched bone-level implant was placed at the LR5 in a patient whose vertical mucosal thickness measured 1.5 mm at the crest at the time of surgery. The implant was placed at bone level, restored conventionally and has been well maintained. At one year, radiographs show 1.8 mm of circumferential crestal bone loss, with no bleeding and no suppuration. Which best explains this outcome?

A. Platform switching fails whenever the horizontal mismatch is under 1 mm
B. Where mucosa is thin, bone is lost to establish the attachment despite switching
C. Thin mucosa transmits occlusal load directly to the crestal bone
D. Platform switching is effective only in conjunction with a conical connection
E. Thin mucosa indicates a thin buccal plate, which resorbs independently

494 An implant system offers a variant with fine microthreads over the coronal 2 mm of the body, and the representative states that this will reduce marginal bone loss. Your patient is a former smoker treated for periodontitis, now stable, who is to receive an implant at the UL5 with a thin buccal plate. Which consideration is most relevant to the choice in this patient?

A. Microthreads prevent apical migration of the barrier epithelium
B. Microthreads abolish the remodelling associated with the microgap
C. Microthreads contribute more primary stability than the body threads
D. Microthreads allow a narrower implant to be used in the same site
E. Microthreads add crestal surface area but retain biofilm once exposed

495 An aggressively tapered implant with a cutting apex is placed immediately into the socket of a UR1, whose buccal plate is intact and 1 mm thick. It is positioned against the palatal wall, and insertion torque rises steeply over the last few millimetres to reach 55 Ncm. The socket walls are otherwise intact. What is the principal mechanical risk of relying on this design in this situation?

A. The coronal threads engage first, so insertion torque is limited
B. The tapered form transmits occlusal load to the crest as shear
C. Stability derives from the apical bone alone, risking perforation there
D. The taper prevents the implant from being positioned palatally
E. Tapered implants cannot be placed accurately through a surgical guide

496 During delivery of a crown at the UL6 the abutment screw is dropped, retrieved from the bracket table, wiped and inserted; the assistant notes that it was wet with blood and saline. It is torqued to the manufacturer's specified 35 Ncm with a calibrated driver, and the fit of the crown is verified radiographically. What is the consequence of the contamination for the joint?

A. Contamination alters the friction coefficient, so the preload achieved is unpredictable
B. Contamination raises friction, so more of the torque is converted into preload
C. Contamination lowers friction, so the joint is clamped more securely than intended
D. Contamination is irrelevant, since preload depends only on the torque applied
E. Contamination corrodes the threads and weakens the screw immediately

497 A practice uses two implant systems. A clinician finds that a screwdriver from system A engages the abutment screw of system B and, having no chart to hand, torques it to 35 Ncm, the value he remembers for system A. The screw in system B is specified at 20 Ncm. The crown seats fully and the patient is dismissed. Which statement best describes the problem created?

A. All abutment screws of similar diameter share a common torque specification
B. Applying a higher torque than specified simply increases joint stability
C. Torque values differ only between gold-coated and titanium alloy screws
D. The value is specific to the screw and joint, and exceeding it risks yielding
E. Using a calibrated driver removes the need to consult the specified value

498 An abutment screw retaining a molar crown at the LL6 has fractured twice in two years. On each occasion the fragment was retrieved and a new screw of the correct type fitted with a calibrated driver to the specified torque. The crown is not cantilevered, the fit is passive, the bone level is stable, and the patient has heavy masseters and reports morning jaw stiffness. What does the repeated fracture indicate?

A. The screw is the strongest component, so fracture implies a manufacturing fault
B. The screw is the intended weak link, so repeated fracture signals uncorrected overload
C. The screw fractures because insufficient preload was generated at each delivery
D. Screw fracture indicates loss of osseointegration at the implant apex
E. Screw fracture is a random event requiring only replacement of the component

499 A milled cobalt-chromium framework is to be screwed directly onto titanium multi-unit abutments for a mandibular fixed prosthesis. A colleague is concerned about placing two dissimilar metals in contact in saliva and suggests remaking the framework in titanium. The patient has no history of metal sensitivity. Which of the following are true of this combination?

A. The passive oxide film on titanium keeps galvanic current in this couple low
B. Crevice conditions and fretting at the joints matter more than the galvanic couple
C. The couple causes rapid dissolution of the titanium abutments in service
D. A and B only
E. A, B and C

500 A retrieved implant removed for advanced peri-implantitis shows discoloration and shallow pitting of the exposed threads, and histology of the surrounding tissue demonstrates titanium particles within macrophages. The implant had never been instrumented with metal curettes and no implantoplasty had been performed. Which process best accounts for the appearance of the surface?

A. Galvanic corrosion between the implant body and its abutment screw
B. Dissolution of a plasma-sprayed hydroxyapatite coating applied at manufacture
C. Stress corrosion cracking of the implant neck under cyclical occlusal load
D. Pitting corrosion produced by fluoride in a conventional toothpaste
E. Tribocorrosion, in which mechanical wear and an acidic environment act together

501 A 39-year-old woman requests a zirconia implant, stating that she has been told by a practitioner of complementary medicine that she is allergic to titanium on the basis of a commercial blood test. She has a nickel-related rash from costume jewellery. She is otherwise healthy, and the site at the UR5 is straightforward. Which of the following statements are correct?

A. True type IV hypersensitivity to titanium is rare
B. Patch testing reliably identifies titanium sensitivity
C. Peri-implant inflammation in such patients is usually allergic rather than plaque-related
D. A and B only
E. All of the above

502 A custom abutment milled from titanium by a commercial laboratory arrives polished, in a plastic bag, with an instruction sheet stating that it is ready for insertion. It will be seated transmucosally at the UL4 and left in place permanently. Which statement should determine your handling of it before insertion?

A. Autoclaving alone removes machining and polishing residues from the surface
B. Ultrasonic cleaning is required, since residues impair soft tissue attachment
C. Residues are of no consequence because the abutment is transmucosal
D. Steam cleaning alters the surface energy and should therefore be avoided
E. Only zirconia abutments require cleaning before they are inserted

503 A colleague is choosing between beta-tricalcium phosphate granules and deproteinised bovine bone mineral to graft a small dehiscence at implant placement, and asks how the two materials differ in behaviour once in place. The defect is contained on three sides and will be covered with a resorbable membrane. Which of the following are true of the beta-tricalcium phosphate?

A. It maintains the grafted dimension for longer than the xenograft
B. It is substituted by host bone considerably more rapidly
C. It is osteoinductive as well as osteoconductive
D. A and B only
E. A, B and C

504 A patient who had a sinus floor augmentation with deproteinised bovine bone mineral ten years ago attends for review, and her radiograph shows the grafted compartment still occupied by discrete radiopaque particles rather than by uniform bone. The implants are asymptomatic with stable marginal bone levels and healthy peri-implant tissues. What does this appearance indicate?

A. The particles have been encapsulated in fibrous tissue and never integrated
B. The particles are not resorbable because they contain no calcium phosphate
C. The mineral is substituted extremely slowly and remains radiopaque throughout
D. The appearance reflects reactive sclerosis of the bone surrounding the graft
E. Persisting particles indicate that the graft was never vascularised

505 A colleague proposes a bioactive glass particulate for a contained defect adjacent to an implant, and describes it as behaving differently from the calcium phosphate alloplasts he normally uses. The patient has no objection to any material and the site is uninfected with three bony walls. Which statement best describes how a bioactive glass behaves in the defect?

A. It forms a surface apatite layer and raises the local pH as it dissolves
B. It is osteoinductive, because it contains bone morphogenetic proteins
C. It sets as a rigid mass and resists resorption for several years
D. It is bovine in origin and raises the same consent issues as a xenograft
E. It is radiolucent, so incorporation cannot be assessed radiographically

506 Two particle sizes of the same xenograft are offered for a buccal contour graft at the UR3: a fine grade of 0.25 to 1 mm and a coarse grade of 1 to 2 mm. The defect is a shallow buccal dehiscence that will be covered by a collagen membrane and closed without tension. Which of the following follow from selecting the finer particle size?

A. Improved maintenance of space in an uncontained defect
B. A greater surface area available for cell attachment and turnover
C. A reduced tendency for particles to be displaced during closure
D. A and B only
E. A, B and C

507 A colleague is selecting a resorbable barrier for a lateral augmentation where he expects the flap to be under some tension, and the supplier offers a chemically cross-linked collagen membrane alongside the native product he normally uses, describing it as lasting three times as long. The graft is particulate xenograft and the site is uninfected. Which statement best describes what cross-linking changes?

A. It renders the membrane non-resorbable, so it must be retrieved later
B. It stiffens the membrane sufficiently to maintain space without support
C. It removes the antigenicity of the porcine collagen entirely
D. It prolongs the barrier but worsens integration and increases exposure
E. It allows the membrane to be left deliberately exposed to the mouth

508 A synthetic resorbable barrier of poly-lactic and poly-glycolic acid is proposed for a contained defect, and a colleague asks how its degradation differs from that of a collagen membrane, having read that some patients develop a persistent swelling over such sites weeks after uneventful surgery. The site is uninfected and closure was primary. Which statement is correct?

A. It is degraded by macrophage collagenase over approximately four weeks
B. It is hydrolysed into acidic products that may provoke local inflammation
C. It retains its barrier function for approximately twelve months
D. It integrates with the flap and becomes vascularised as collagen does
E. It must be retrieved at a second procedure at about six weeks

509 A patient with 4 mm of bone above the mandibular canal at the LR6 has been assessed by a specialist colleague, who plans vertical ridge augmentation using a titanium-reinforced high-density PTFE membrane over a mixture of autogenous and xenogeneic particulate. You are asked by the patient why this membrane has been chosen rather than the collagen one used for her earlier buccal graft. Which statement best describes its role?

A. It resorbs slowly enough to avoid the need for a second procedure
B. It seals the wound margin and can tolerate exposure indefinitely
C. It removes the need for a particulate graft beneath it
D. It is indicated because the defect is contained by three bony walls
E. It maintains space in an uncontained defect and must be retrieved

510 You are teaching a colleague the reasoning behind guided bone regeneration and she asks what a barrier membrane actually has to do, having seen products ranging from limp collagen sheets to rigid titanium mesh. The defect under discussion is a two-wall dehiscence at an implant placed in the posterior maxilla. Which of the following are requirements of a barrier membrane?

A. Biocompatibility, with integration into the surrounding host tissue
B. Occlusion of epithelial and soft connective tissue cells from the defect
C. Creation and maintenance of a space beneath it for bone to occupy
D. A and B only
E. All of the above

511 A lithium disilicate crown cemented onto a titanium abutment at the UR4 with a resin-modified glass ionomer fractures through the axial wall eight months after delivery. The occlusion is even with canine guidance and no cantilever, the abutment screw is intact and correctly torqued, and the patient reports no trauma and no parafunction. Which property of the luting agent best explains the failure?

A. Its low film thickness left the crown incompletely seated at delivery
B. Its fluoride release progressively weakened the ceramic fitting surface
C. It absorbs water and expands, generating stress within the ceramic
D. Its elastic modulus is too high, transmitting occlusal load to the ceramic
E. Its acidity etched and roughened the fitting surface of the restoration

512 A crown is to be cemented on an implant abutment and the nurse mixes the luting agent to a thicker consistency than the manufacturer specifies, so that it will not run. The abutment has adequate taper and height, the crown was verified as fully seating at try-in, and the occlusion was correct at that point. Which consideration is most relevant to this alteration?

A. A luting agent must not exceed about 25 micrometres in film thickness or the crown will not seat
B. A thicker film improves retention by increasing the area of the bonded interface
C. Film thickness determines the radiopacity of any cement left in the sulcus
D. Film thickness is governed by the taper of the abutment rather than by the mix
E. A film thicker than 25 micrometres usefully compensates for an inaccurate fit

513 A patient referred to you with peri-implant inflammation at a cemented crown has a periapical radiograph on which no radiopaque material can be seen at the margin, and the referring dentist has concluded that retained cement can be excluded. You are considering which cement to use for a crown of your own on an adjacent implant, and want one whose excess would be visible. Which of the following give a luting cement its radiopacity?

A. The presence of heavy-element fillers such as barium or ytterbium
B. The degree of conversion achieved during polymerisation
C. A refractive index matched to that of the overlying ceramic
D. A and B only
E. All of the above

514 A crown is to be made for an implant at the UR4 in a patient treated for periodontitis. The abutment margin, as designed by the laboratory, would lie 3 mm subgingivally on the palatal aspect and 1.5 mm buccally, in order to follow the mucosal contour. The occlusal space and the implant angulation would permit either retention method. What is the most appropriate response to that design?

A. Use a resin cement of low film thickness and remove excess with an explorer
B. Radiograph the crown after cementation to confirm removal of the excess
C. Use a radiopaque cement and remove any residue at the review appointment
D. Change to a screw-retained design, since cement cannot be retrieved at that depth
E. Ask the laboratory to place the margin deeper so that it follows healthy tissue

515 After delivery of a custom abutment at the LL5, a well-aligned periapical radiograph shows a radiolucent gap of about 0.4 mm between the abutment and the implant platform on the mesial aspect; a second film taken at a different horizontal angle shows the same appearance. The patient is comfortable, the crown seats fully on the abutment and the screw was torqued to specification. What is the most appropriate action?

A. Torque the abutment screw to a higher value to close the gap
B. Accept the appearance, since a small gap is inherent to the joint
C. Repeat the radiograph at a third angle before deciding anything further
D. Remove the abutment, inspect for trapped tissue or debris, and reseat it
E. Replace the abutment screw with a new one and re-torque to specification

Block S — Treatment planning, SAC, consent and referral, extended (Q516–560)

516 A 34-year-old woman lost the UL5 to a vertical root fracture eight months ago. The UL4 carries a large mesio-occluso-distal amalgam with an unsupported palatal cusp and the UL6 has a fifteen-year-old gold crown with recurrent caries at the distal margin; both will need full-coverage restorations whatever else is done. The ridge is 6 mm wide with 10 mm of bone below the sinus floor. She is a healthy non-smoker with a stable periodontium. Which option is most appropriate?

A. An implant-retained crown at the UL5, with new crowns on both neighbours
B. A cantilever resin-bonded bridge from a new crown on the UL6
C. A conventional fixed bridge with retainers on the UL4 and UL6
D. A cobalt-chromium removable partial denture replacing the UL5
E. No replacement, with the two adjacent teeth crowned separately

517 A 69-year-old man is missing the LL6, LL7, LR6 and LR7, extracted over the past decade and never replaced. He has occluding contacts from canine to second premolar on both sides, a plaque score of 14 per cent, no probing depth over 4 mm and no symptoms. The upper second molars are unopposed and have over-erupted by about 3 mm, with exposed dentine and food packing distally. Which finding is the strongest argument against simply accepting the shortened arch?

A. The 3 mm over-eruption of the unopposed upper second molars
B. The concentration of occlusal load onto the premolar units
C. The absence of four occluding premolar pairs in the arch
D. The ten-year interval since the first molars were removed
E. The recognised association between molar loss and joint dysfunction

518 You are planning a single implant at the UR5 in a 57-year-old man. A CBCT taken for the site shows an ill-defined periapical radiolucency about 4 mm across at the apex of the adjacent UR4, root-treated eleven years ago, asymptomatic, with a well-condensed but short root filling. Bone volume at the UR5 is ample and the ridge is 7 mm wide. Which is the most appropriate course?

A. Place the implant and review the UR4 radiographically each year
B. Assess and treat the UR4 before the implant is placed
C. Place the implant, leaving 3 mm between it and the lesion
D. Extract the UR4 and plan implants at both sites
E. Monitor the UR4 for six months, then reconsider the implant

519 A 24-year-old woman has a congenitally absent UL2. The space measures 4.5 mm and orthodontic treatment is planned to open it to 7 mm and correct a 3 mm midline discrepancy. A colleague suggests placing the implant now, at the mesial end of the existing space, so that it can serve as anchorage while the space is opened and the midline corrected. Which is the strongest objection to that plan?

A. Osseointegration would be lost under continuous orthodontic force
B. The ridge is too narrow to accept an implant before space opening
C. Anchorage demands a diameter greater than this site can take
D. The definitive position cannot be known until the roots are finished
E. A provisional crown would be needed throughout the treatment

520 A 49-year-old man completed non-surgical treatment for generalised stage III grade B periodontitis and is re-evaluated at three months before implant planning. Three sites still probe 5 mm and bleed on probing, his full-mouth bleeding score is 14 per cent and his full-mouth plaque score is 18 per cent. He stopped smoking a year ago and attends reliably. Which of these findings fall short of the accepted endpoint of periodontal therapy?

A. The three residual 5 mm sites that bleed on probing
B. The full-mouth bleeding score of 14 per cent
C. The full-mouth plaque score of 18 per cent
D. A and B only
E. A, B and C

521 A 61-year-old man wants an implant at the LR6, lost two years ago. The adjacent LR7 has grade III buccal and lingual furcation involvement, 70 per cent bone loss, grade II mobility and a chronic buccal sinus tract; he has no pain and would like to keep it. The rest of his dentition is stable, he has stopped smoking and he cleans well. Which is the most appropriate course?

A. Place the implant at the LR6 and maintain the LR7 as it is
B. Extract the LR7 and place both implants at the same visit
C. Root-resect the LR7 and place the LR6 implant six weeks later
D. Place the LR6 implant and extract the LR7 at the same visit
E. Extract the LR7 first and plan both spaces once the site has healed

522 A 62-year-old woman has worn complete dentures for fifteen years and cannot control the lower one. She is medically well and cleans effectively. CBCT of the interforaminal region shows 9 mm of bone height and a knife-edge crest 3.5 mm wide at its narrowest, with adequate vertical restorative space; her upper denture is well made and she is happy with it. She wants two implants to retain the lower denture. How is this case best classified?

A. Straightforward surgically and straightforward prosthodontically
B. Straightforward surgically and complex prosthodontically
C. Complex surgically and straightforward prosthodontically
D. Complex surgically and complex prosthodontically
E. Advanced surgically and complex prosthodontically

523 You are grading the restorative aspect of several planned implant cases before deciding which to treat yourself and which to refer. All the patients are medically well with stable periodontal tissues and adequate bone. Which of the following features would place a case in the Advanced prosthodontic category?

A. A planned increase in the occlusal vertical dimension
B. A high lip line exposing the peri-implant mucosal margins
C. An edentulous span of three units in the aesthetic zone
D. A and B only
E. All of the above

524 A 58-year-old woman wants a fixed replacement for the UR6, lost a year ago. CBCT shows 4 mm of residual bone beneath the sinus floor, an intact membrane of normal thickness and no septum. You do not undertake sinus floor elevation, and the options you provide are a cantilever bridge from a new crown on the UR5, a partial denture or no treatment. Which principle most directly determines what you must tell her?

A. Reasonable alternatives must be disclosed even where you do not provide them
B. Disclosure is limited to the treatments available within your own practice
C. The alternatives disclosed are those a responsible body of clinicians would offer
D. Only alternatives with an established evidence base need to be described
E. The patient decides which of the alternatives she wishes to hear about

525 A 63-year-old man attending for a single implant at the LL5 interrupts your explanation of the risks and says, "I don't want to hear any of that — you're the expert, just do whatever you think is right." He has capacity, has read nothing about the procedure, and appears comfortable rather than distressed. The site is straightforward and he has already paid a deposit. What is the most appropriate response?

A. Record his wish not to be informed and proceed on that basis
B. Explore what he does wish to know and give the essential information
C. Provide written information and take his signature without discussion
D. Decline to treat until he has engaged fully with the risk discussion
E. Ask a family member to receive the information on his behalf

526 A 45-year-old man declines the implant, the bridge and the denture you have set out for the LR6, lost six months ago, and simply prefers to leave the space. He has capacity and has understood the discussion. The opposing UR6 is present and has not yet moved, and the LR5 and LR7 are upright and sound. Which of the following must still form part of the consent discussion?

A. A written waiver of your responsibility for the consequences
B. A fixed review date by which he must reconsider the decision
C. The foreseeable consequences of leaving the space untreated
D. A and C only
E. A, B and C

527 A 58-year-old man who smokes 30 cigarettes a day and declines cessation insists on two implants in the posterior mandible. He can repeat back the roughly doubled failure risk and the higher rate of peri-implantitis, his plaque score is 12 per cent, his periodontium is stable and he attends every three months. Your associate suggests that his refusal to stop smoking shows he cannot properly weigh the information. Which principle most directly answers that suggestion?

A. A decision that appears unwise does not of itself demonstrate incapacity
B. Capacity must be reassessed whenever a patient rejects clinical advice
C. Autonomy entitles a patient to receive the treatment they have requested
D. Valid consent requires the patient to accept the recommended option
E. Accepting a known risk transfers responsibility for the outcome

528 An 81-year-old woman in residential care has advanced dementia and lacks capacity for dental decisions. She has no attorney, no deputy and no family. Three retained lower anterior roots are repeatedly ulcerating her lower lip, and her carers report that she pushes her lower denture out and will not wear it. The care home manager asks whether implants would help her keep it in. What is the most appropriate course?

A. Apply to the Court of Protection for authority to extract the roots
B. Extract the roots in her best interests after consulting those who know her
C. Provide implants, since a retained overdenture would end the trauma
D. Take no action, since she cannot consent to any intervention
E. Ask the care home manager to sign a consent form for the extractions

529 A 31-year-old woman brings photographs of a celebrity's smile and asks you to remove her four upper incisors and replace them with implants. The teeth are sound, unrestored, vital and in a Class I relationship with mild rotations and a 1 mm midline diastema. She has capacity, can afford the treatment and says two other dentists have already refused her. She is otherwise well and appears untroubled. What is the most appropriate course?

A. Provide the treatment, having documented her informed request
B. Provide the treatment after obtaining a written psychological opinion
C. Refer her to a colleague who provides full-arch aesthetic work
D. Decline to remove the teeth and discuss non-destructive options
E. Agree in principle and begin with the two central incisors only

530 You are preparing the written estimate for a 47-year-old man who has accepted a single implant crown at the UR5, to be placed in a healed site with no augmentation and restored four months later. He has asked twice what the total cost will be over time and says he has been caught out by add-on charges before. Which of the following should the estimate contain?

A. The fee for the surgical and the restorative stages
B. The cost of supportive maintenance visits and review radiographs
C. The likely cost of replacing components during the life of the restoration
D. A and B only
E. All of the above

531 A 66-year-old man asks about replacing the LR6, lost four years ago. CBCT shows 4 mm of bone above the roof of the mandibular canal and a crestal width of 7 mm. The LR5 and LR7 are sound and unrestored with normal bone levels, the opposing UR6 is in firm contact with the UR7 and has not over-erupted, and he eats a normal diet with no complaint. He is a healthy non-smoker with excellent hygiene. Which is the most appropriate recommendation?

A. No replacement, with review of the opposing tooth
B. Referral for assessment for inferior alveolar nerve repositioning
C. Referral for assessment for vertical ridge augmentation
D. A 6 mm implant with 2 mm of clearance above the canal
E. A conventional fixed bridge from the LR5 to the LR7

532 A 68-year-old man attends with a screw-retained fixed full-arch mandibular prosthesis placed abroad four years ago on five implants. The titanium framework has fractured distal to the left terminal implant, the cantilever is mobile, and he has no documentation of the system used. He has no pain, three sites bleed on probing with 5 mm pockets, and he wants it "fixed today". You place straightforward single implants and restore them. What is the most appropriate course?

A. Remove the fractured cantilever and smooth the framework
B. Order replacement components from the closest compatible manufacturer
C. Remove the prosthesis and fit healing abutments until it is identified
D. Stabilise him symptomatically and refer for restorative assessment
E. Take an impression and have a new framework cast over the fragments

533 A 54-year-old woman wants a fixed replacement for the UL6. CBCT shows 4 mm of residual bone beneath the sinus floor with a septum crossing the intended site and a membrane of normal thickness. She smokes ten cigarettes a day and has a high lip line, though the UL6 is not visible when she smiles. She is otherwise well with a stable periodontium, and you have placed about a hundred straightforward implants. Which of these features, on its own, would justify referring the surgical phase?

A. The 4 mm of residual bone with a septum crossing the site
B. Her habit of ten cigarettes a day
C. Her high lip line
D. A and B only
E. All of the above

534 A 52-year-old man is missing the UR4 and UR5, lost eight years ago. The UR3 and UR6 are sound and unrestored with normal bone levels, the ridge is 7 mm wide and 12 mm high at both sites with no sinus involvement, and his lip line is low. He is a healthy non-smoker with a stable periodontium, no parafunction and a Class I occlusion, and he wants a fixed replacement. Which design is most appropriate?

A. A four-unit conventional bridge from the UR3 to the UR6
B. One implant at the UR4 with a cantilever pontic at the UR5
C. Two implants at the UR4 and UR5 with separate crowns
D. One implant at the UR5 with a cantilever pontic at the UR4
E. Two implants restored as a single splinted two-unit prosthesis

535 A 70-year-old woman is having two interforaminal implants placed to retain a lower complete denture. On a mounted trial set-up the vertical distance from the planned implant platform to the occlusal plane measures 9 mm. She has a resting tremor of both hands, lives alone and manages her upper denture well. She is medically stable and attends regularly. Which attachment design is most appropriate?

A. A milled bar with two distal extensions
B. Individual stud attachments on each implant
C. A round bar with a single clip
D. Magnetic attachments on cast copings
E. A cast bar with two clips and a metal framework

536 A 55-year-old man wants two implants in the lower left quadrant to replace the LL6 and LL7. He has three carious lesions extending into dentine, generalised stage III grade B periodontitis that has not yet been treated, and smokes fifteen cigarettes a day. He is otherwise well, has adequate bone at both sites and is impatient to start, saying he has already waited a year. Which of the following must be completed before implant surgery is undertaken?

A. Treatment of the active carious lesions
B. Periodontal therapy with re-evaluation against defined endpoints
C. Documented cessation of smoking
D. A and B only
E. A, B and C

537 You placed an implant at the UL2 three months ago. At second-stage surgery it is clear that the implant lies about 2 mm too far buccally, two threads are exposed on the labial surface, and the definitive crown will require an over-contoured emergence with a margin that will be visible when she smiles. She has a medium lip line, is delighted with progress so far and has asked no questions. What should you do first?

A. Proceed and manage the contour with a customised abutment
B. Undertake a connective tissue graft and say nothing further
C. Tell her what has happened, apologise, and set out the options
D. Remove the implant and replace it at the correct angulation
E. Obtain an opinion from your indemnity organisation before speaking

538 A 44-year-old woman attends with two implants placed elsewhere eighteen months ago at the UR1 and UR2. Both are positioned too far buccally with visible metal margins and about 3 mm of mid-facial recession. She is distressed, has already complained to the previous practice, and asks you to write a letter stating that the treatment was negligent so that she can recover the fees. What is the most appropriate response?

A. Provide the letter, since the findings speak for themselves
B. Decline to examine her, as the case is now a legal matter
C. Write to the previous dentist requesting an explanation first
D. Tell her that the standard of care was plainly not met
E. Record your findings factually and explain how to raise a complaint

539 A 39-year-old man attends a first consultation for a single implant at the LR5, having driven two hours to see you. He asks you to place the implant this afternoon, saying he has read about the procedure, has taken the day off work and does not want to make the journey again. The site is straightforward, your afternoon list has space and the components are in stock. What is the most appropriate course?

A. Proceed today, having documented a full consent discussion
B. Give him the information and arrange the surgery for another day
C. Proceed today provided he signs a consent form before starting
D. Refer him to a practice nearer home for the treatment
E. Proceed today with a healing abutment, consenting for the crown later

540 A 47-year-old man who smokes twenty cigarettes a day wants a single implant at the LR6 in native, healed bone with no augmentation planned. His periodontium is stable, his plaque score is 15 per cent and he attends regularly. He declines cessation but says he could cut down to ten a day, and asks you what practical difference the smoking makes to him. Which of the following should you tell him?

A. That failure is roughly twice as likely as it would be in a non-smoker
B. That halving his consumption would remove most of the excess risk
C. That a longer course of antibiotics would offset the additional risk
D. A and B only
E. A, B and C

541 A 48-year-old woman wants an implant at the LR6, lost seven years ago. The opposing UR6 has over-erupted by 3 mm; it is sound, unrestored, vital and symptom-free, with normal bone levels and no mobility, and its palatal cusp now occupies most of the restorative space at the LR6. Mounted study casts confirm that a crown of acceptable form and occlusal contact cannot be made without addressing it. Which is the most appropriate first step?

A. Enameloplasty of the UR6 palatal cusp before placing the implant
B. Elective root treatment and a crown on the UR6
C. Extraction of the UR6 and replacement with an implant
D. Orthodontic intrusion of the UR6 before implant placement
E. Placement of the implant with a crown of reduced height

542 A 15-year-old boy has an ankylosed UR1 following avulsion and replantation four years ago. It is now 3 mm infra-occluded, the periapical radiograph shows replacement resorption and the adjacent alveolar crest sits 2 mm coronal to the crest at the contralateral tooth. His standing height increased 4 cm in the past year. His parents ask about an implant. Which is the most appropriate course?

A. Extract the tooth and graft the socket, implanting at eighteen
B. Extract the tooth and provide a resin-bonded bridge until growth ends
C. Decoronate the tooth and provide a bonded pontic until growth ends
D. Surgically luxate the tooth and reposition it orthodontically
E. Build the crown up to the occlusal plane with composite and review

543 A 56-year-old man asks you to extract his LR6 and replace it with an implant. The tooth was root-treated nine years ago and crowned; it is symptom-free, the root filling is well condensed to length with no periapical change, the crown margins are sound with no recurrent caries, there is no mobility and probing depths are 3 mm. He says he has read that implants last longer than root-treated teeth and would rather deal with it now than later. What is the most appropriate response?

A. Agree, since he has capacity and understands the alternatives
B. Agree, provided he signs a disclaimer accepting the risk
C. Refer him for a specialist opinion on the tooth's prognosis
D. Agree, but replace the crown first to test his tolerance of the work
E. Decline, and explain what the survival figures actually compare

544 A 19-year-old woman has a congenitally absent UR2 with 6.5 mm of space between the contact points and 6.2 mm at the alveolar crest, an intact ridge 6 mm wide, and unrestored adjacent teeth with normal bone levels. Serial cephalograms have not been taken and her standing height increased 1 cm over the past year. Her overbite is 2 mm with adequate palatal clearance for a retainer wing. Which features favour a cantilever resin-bonded bridge over an implant at this stage?

A. The unrestored state of the adjacent teeth
B. The evidence that facial growth is not yet complete
C. The mesiodistal space of 6.5 mm between the contact points
D. A and B only
E. All of the above

545 A 58-year-old woman has lost the UR4, UR5 and UR6 to caries over three years. The UR3 and UR7 are sound and unrestored, the ridge is adequate at all three sites with no sinus involvement, and her periodontium is stable. The plan you would recommend is three implants, which she cannot afford; she can meet the cost of one implant now and possibly a second in two years. She is a healthy non-smoker who cleans well. Which plan is most appropriate?

A. Three implants placed now with the fee spread over five years
B. A four-unit conventional bridge from the UR3 to the UR7
C. One implant at the UR6 carrying a two-unit cantilever mesially
D. One implant at the UR4 restored now, with no other replacement
E. A removable partial denture now, with implants staged as funds allow

546 A 41-year-old man works overseas for ten months of the year in a country where the implant system you use is not distributed, returning to the United Kingdom for about four weeks at a time. He wants two implants to replace the LR5 and LR6, is healthy, cleans well, has a stable periodontium and can afford the treatment. The sites are anatomically straightforward. Which is the most important consideration in deciding what to offer him?

A. Who will diagnose and manage a complication while he is abroad
B. Whether both surgical stages can be completed within one visit home
C. Whether he will accept annual rather than six-monthly recalls
D. Whether the prosthesis can be made screw-retained for retrievability
E. Whether his employer's insurance would cover treatment abroad

547 A 58-year-old man wants an implant at the LL7, lost three years ago. The panoramic radiograph shows an unerupted, horizontally impacted LL8 immediately distal to the site, with its crown at the level of the planned implant apex, no associated radiolucency and no symptoms. A small-field CBCT confirms it lies about 1.5 mm from the planned osteotomy and 2 mm above the mandibular canal. Which is the most appropriate course?

A. Place a shorter implant so as not to contact the LL8
B. Place the implant and leave the LL8 undisturbed
C. Remove the LL8 and place the implant once the site has healed
D. Remove the LL8 and place the implant at the same visit
E. Leave the LL8 and site the implant further mesially

548 A 26-year-old woman lost the UR1 and UL1 in an assault a year ago and wears a partial denture. She has a Class III skeletal pattern with a reverse overjet of 4 mm at the remaining incisors, an anterior crossbite and a concave profile that she dislikes and has mentioned twice. The sockets have healed, the ridge is 5 mm wide with adequate height, and she has a high lip line. She asks for implants as soon as possible. What is the most appropriate next step?

A. Place the implants to the existing arch relationship
B. Provide a resin-bonded bridge and review her in five years
C. Augment the ridge and place the implants at four months
D. Arrange a joint orthodontic and orthognathic assessment
E. Refer for a restorative consultant opinion on shade matching

549 A 52-year-old man with capacity declines the implant, bridge and denture options you have set out for a missing LR5 and chooses to leave the space, having understood the discussion. Consider these statements: (i) the option of no treatment need only be raised if the patient asks about it; (ii) the information given and his decision to decline should be recorded in the notes; (iii) a signed refusal form is required before the episode can be closed. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

550 A 49-year-old woman has had four implants placed at three different practices over six years; two failed and were removed. She brings a folder of photographs, a written list of what she wants done and printouts from an online forum, and asks you to remove the two remaining implants and start again from scratch. She has no pain, the two remaining implants have 4 mm probing depths without bleeding, and she has brought no radiographs. What should you do first?

A. Undertake a full independent assessment before recommending anything
B. Remove the two remaining implants, as she has requested
C. Obtain the previous clinicians' records before examining her
D. Refer her for a psychological opinion before considering treatment
E. Agree the plan she has written and confirm it in writing

551 A 60-year-old healthy non-smoker needs a single implant at the LR6. CBCT shows 12 mm of bone above the roof of the mandibular canal and a crestal width of 7 mm, but a pronounced submandibular fossa, so that at the level of the planned apex the lingual plate lies about 3 mm lingual to a line dropped from the crest. Her periodontium is stable, her lip line is low and the opposing UR6 is a sound natural molar in firm occlusion. Which feature most raises the surgical classification of this site?

A. The 12 mm of available bone above the mandibular canal
B. The lingual concavity at the level of the planned apex
C. The crestal width of 7 mm measured at the crest
D. The proximity of the mental foramen 6 mm mesial to the site
E. The opposing natural molar in firm functional occlusion

552 A 50-year-old man is choosing between a single implant crown at the LR5 and a three-unit conventional bridge from the LR4 to the LR6, both of which you have quoted for, and asks whether the implant will "work out cheaper in the long run". Consider these statements: (i) the implant carries the higher initial cost; (ii) both options will incur maintenance costs over their lifetime; (iii) once osseointegration is achieved the implant incurs no further cost. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

553 A 16-year-old girl has a congenitally absent UL2 and 4 mm of crowding in the upper arch. The UL3 has erupted mesially with a pointed cusp tip and a well-formed labial surface, the UL1 is unrestored, her lip line is medium and her standing height is still increasing. Her mother has asked whether an implant can be placed when she turns eighteen. Which is the most appropriate recommendation?

A. A resin-bonded cantilever bridge now, with an implant at twenty-five
B. An implant at eighteen once serial cephalograms confirm growth has ceased
C. A removable partial denture until facial growth is complete
D. Autotransplantation of a premolar into the UL2 site
E. Orthodontic space closure with reshaping of the UL3

554 You are planning two implants to carry a three-unit fixed prosthesis replacing the LR4, LR5 and LR6 in a 61-year-old man. The LR3 is sound. The LR7 is present and has 60 per cent bone loss, grade II mobility, a grade II buccal furcation involvement and a 9 mm distal pocket that bleeds; the remainder of his periodontium is stable, he has stopped smoking and his plaque control is good. Which finding should most alter the planned design?

A. The 9 mm distal pocket on the LR7 that bleeds on probing
B. The overall prognosis of the LR7
C. The grade II mobility of the LR7
D. The grade II buccal furcation involvement of the LR7
E. The soundness of the LR3 at the mesial end of the span

555 You refer a 62-year-old man for placement of two implants in the posterior maxilla and provide the restorative work and the recall yourself. Six months after the crowns are fitted he develops bleeding on probing with 6 mm pockets and 2 mm of radiographic bone loss at one implant, and telephones your practice; the surgeon's clinic tells him it is your responsibility, and you had assumed it was theirs. Which arrangement should have been agreed in writing before the referral?

A. The implant system and the component codes to be used
B. The fee that each clinician would charge for their own phase
C. Which clinician would diagnose and manage peri-implant disease
D. The interval at which review radiographs would be taken
E. Which clinician would obtain consent for the surgical phase

556 A 34-year-old man with a moderate learning disability attends with his support worker to discuss an implant to replace the UR1, lost in a fall. He tells you clearly why he has come, what he dislikes about his denture and that he wants "a tooth that stays in", but becomes confused when you describe the healing period and the surgical risks in your usual terms. He has no attorney and lives semi-independently. What should you do first?

A. Conclude that he lacks capacity and consider his best interests
B. Ask his support worker to consent on his behalf
C. Offer a new denture, which requires no surgical consent
D. Present the information in an accessible form and reassess
E. Refer him to a special care dentistry service for the decision

557 A 17-year-old competitive swimmer lost the UR1 in a collision eight months ago and wears a partial denture she finds humiliating. She is articulate, understands what is proposed and asks for an implant; her parents refuse to agree and want her to wait until she is older. Her standing height increased 1.5 cm over the past year, and a hand-wrist assessment has not been done. The ridge is intact and 6 mm wide. What is the most appropriate course?

A. Place the implant, since at sixteen she may consent for herself
B. Place the implant only with her parents' written agreement
C. Seek a court declaration to resolve the disagreement
D. Refer her to a colleague willing to treat without parental agreement
E. Explain that growth, not consent, is what prevents treatment now

558 A 64-year-old man is being consented for a fixed full-arch mandibular prosthesis on five implants, which a colleague will place and you will restore. He is medically well, cleans effectively, has no parafunction and attends reliably. He tells you he wants to understand what he is committing to for the rest of his life, and asks specifically about the prosthesis rather than the implants. Which statement most accurately describes what he should expect?

A. The prosthesis will need no attention while the implants remain integrated
B. The prosthesis will require periodic removal, repair and eventual replacement
C. Maintenance will be limited to professional hygiene visits twice a year
D. Component replacement will be required only if one of the implants fails
E. The acrylic teeth will need replacing at roughly twenty-year intervals

559 A 57-year-old woman requests an implant at the LL6. She has capacity and is entirely cooperative, but refuses any radiograph of the site, saying that she has had several CT scans during surveillance for a treated breast cancer and will not accept further radiation. Clinically the ridge feels broad and the site is well healed, and she has brought a panoramic radiograph taken elsewhere nine years ago. Which is the most appropriate course?

A. Explain that the assessment cannot be completed and decline to place
B. Place the implant flapless using a guide made from the old radiograph
C. Raise a flap and inspect the bone directly instead of imaging it
D. Take a periapical radiograph only, since the dose is negligible
E. Refer her to a colleague who will accept the limitation

560 A 59-year-old woman has worn an upper complete denture for twelve years and can no longer tolerate palatal coverage since developing a strong gag reflex. CBCT shows 4 mm of residual bone bilaterally in the posterior maxilla with pneumatised sinuses, and 10 mm between the canine eminences. With the denture flange removed her upper lip falls in noticeably. She wants fixed teeth. You place straightforward single implants and have never restored an edentulous arch. What is the most appropriate course?

A. Place four anterior implants and provide a fixed prosthesis
B. Place two anterior implants to retain a horseshoe overdenture
C. Refer for bilateral sinus grafting and a fixed full-arch prosthesis
D. Provide a new complete denture with reduced palatal coverage
E. Refer for a restorative-led assessment of the edentulous maxilla

Block T — Prescribing, local anaesthesia, drug interactions and sedation, extended (Q561–590)

561 A 58 kg woman with no medical history and no regular medication is having two implants placed at the LR5 and LR6 under 2 per cent lidocaine with adrenaline 1:80,000, supplied in 2.2 ml cartridges. The appointment is expected to last ninety minutes and you intend to work within a maximum of 4.4 mg/kg of lidocaine. Disregarding the adrenaline, what is the largest number of whole cartridges you may use?

A. Three cartridges
B. Four cartridges
C. Five cartridges
D. Seven cartridges
E. Nine cartridges

562 A 68-year-old man with stable angina, controlled on bisoprolol and a glyceryl trinitrate spray he uses about once a month, is booked for an implant at the LL5 under local anaesthesia. He last had chest pain three months ago on climbing a hill. A colleague suggests using 3 per cent prilocaine with felypressin so that no adrenaline is given at all. Which is the strongest argument against that substitution?

A. Felypressin is contraindicated in patients taking a beta blocker
B. Felypressin constricts coronary vessels directly at higher doses
C. Prilocaine at 3 per cent gives inadequate pulpal anaesthesia
D. Felypressin provides no haemostasis in the surgical field
E. Prilocaine risks methaemoglobinaemia at the doses required

563 You have raised a small flap to place an implant at the LR6 in a calm 46-year-old man. An inferior alveolar nerve block given fourteen minutes ago has produced profound numbness of the lower lip and chin, and the buccal soft tissues are numb to probing, but he reports sharp pain as the crestal cortex is drilled. He has no relevant medical history. Which is the most appropriate next step?

A. Buccal infiltration of 4 per cent articaine with adrenaline at the site
B. A repeat inferior alveolar nerve block using a fresh cartridge
C. A long buccal nerve block deposited distal to the LR7
D. A Gow-Gates block at a more superior injection point
E. A lingual infiltration adjacent to the LR6

564 Immediately after an inferior alveolar nerve block on the left side, given before implant surgery, a 44-year-old woman cannot close her left eye or raise the left corner of her mouth. Her lower lip is not numb, she is speaking normally, her limbs are moving symmetrically and her blood pressure and pulse are unchanged. She is frightened and asks what has happened. Which explanation and immediate action are correct?

A. A cerebrovascular event; arrange emergency hospital admission
B. Bell's palsy; prescribe high-dose prednisolone and review in a week
C. Intravascular injection of adrenaline; monitor until it settles
D. Deposition within the parotid gland; protect the eye and reassure
E. A pterygomandibular haematoma; apply firm extraoral pressure

565 An implant placed at the LL6 six weeks ago has become acutely infected and must be removed. Despite an inferior alveolar nerve block and supplemental infiltration given with correct technique, anaesthesia is repeatedly inadequate over the inflamed site. Consider these statements: (i) the low pH of inflamed tissue increases the ionised fraction of the anaesthetic; (ii) the increased blood flow of inflamed tissue removes the drug more rapidly; (iii) inflamed nociceptors show altered sodium channel expression and a lowered firing threshold. Which are correct?

A. (i) only
B. (ii) only
C. (i) and (ii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

566 For a long implant procedure with extensive grafting in a 70 kg man you are considering 0.5 per cent bupivacaine with adrenaline for prolonged post-operative analgesia, working to a maximum of 2 mg/kg. He is healthy and takes no medication. Which property of bupivacaine most justifies caution about approaching that limit?

A. Its inactivation depends entirely on plasma cholinesterase
B. Its onset is slow, so an overdose is recognised late
C. Its cardiotoxicity appears at doses close to those causing seizures
D. It causes methaemoglobinaemia more readily than lidocaine
E. It has no antagonist available in a dental practice setting

567 A 71-year-old man weighing 80 kg has well-controlled heart failure, and his cardiologist advises keeping the total adrenaline dose below 0.04 mg for the appointment. He needs ninety minutes of implant surgery at the LL6 and LL7. You stock 4 per cent articaine with adrenaline 1:100,000 and 4 per cent articaine with adrenaline 1:200,000, both in 2.2 ml cartridges. Which plan keeps him within the adrenaline limit while giving the largest volume of solution?

A. Three cartridges of the 1:200,000 solution
B. Two cartridges of the 1:100,000 solution
C. Four cartridges of the 1:200,000 solution
D. One cartridge of the 1:100,000 solution
E. Two cartridges of the 1:200,000 solution

568 A 44-year-old healthy non-smoker with a stable periodontium is having a single implant placed at the LR5 in native, healed bone with no augmentation and no flap extension. She has no allergy and no cardiac history. You are deciding whether to give a single 2 g oral dose of amoxicillin an hour before surgery. Which statement best represents what the randomised evidence supports?

A. It reduces early failure, with about one failure prevented per four patients treated
B. It reduces early failure, with about one failure prevented per twenty-five patients treated
C. It reduces post-operative infection but has no effect on early failure
D. It reduces marginal bone loss measured at twelve months after loading
E. It reduces failure only in sites that have been grafted or augmented

569 You are placing an implant at the UR5 in a healthy 50-year-old with simultaneous guided bone regeneration for a 3 mm buccal dehiscence, using a particulate xenograft and a resorbable collagen membrane. She has no allergies and takes no medication, and the procedure will take about an hour with a full-thickness flap. Which of the following would be appropriate?

A. A single 2 g dose of amoxicillin one hour before surgery
B. A chlorhexidine mouthrinse immediately before the incision
C. A five-day course of amoxicillin 500 mg three times daily afterwards
D. A and B only
E. A, B and C

570 A 61-year-old woman is having a single implant placed at the LR6 in a healed site. Her records document urticaria, facial swelling and wheeze within an hour of taking oral amoxicillin nine years ago, treated in an emergency department. She has taken no penicillin since. You have decided that a single pre-operative prophylactic dose is justified. Which is the most appropriate choice?

A. Cefalexin 2 g orally one hour before the procedure
B. Metronidazole 400 mg orally one hour before the procedure
C. Doxycycline 200 mg orally one hour before the procedure
D. Co-amoxiclav 625 mg orally one hour before the procedure
E. Clindamycin 600 mg orally one hour before the procedure

571 A 55-year-old man develops a spreading infection four days after implant placement at the LL6. He has a diffuse, tender submandibular swelling, trismus limiting opening to 25 mm, difficulty swallowing and a temperature of 38.4°C. His record documents an anaphylactic reaction to penicillin. A colleague proposes metronidazole 400 mg three times daily as the sole antimicrobial while the source is addressed. Which is the strongest objection?

A. It has no useful activity against the aerobic streptococci involved
B. It penetrates cortical bone poorly at conventional oral doses
C. It is bacteriostatic rather than bactericidal in this setting
D. It requires an 800 mg loading dose to reach therapeutic levels
E. It should never be prescribed before a collection has been drained

572 Three days into a course of amoxicillin 500 mg three times daily for a peri-implant abscess at the UL5, which you incised and drained at the outset, a 48-year-old man is worse. The swelling has enlarged and is now diffuse, his temperature is 38.6°C, he can open only 20 mm and he is finding it hard to swallow saliva. He has taken every dose. Which of the following should you do?

A. Change the antibiotic to co-amoxiclav
B. Add metronidazole to the amoxicillin
C. Re-examine him and re-establish drainage of the collection
D. A and C only
E. A, B and C

573 A 46-year-old woman takes lithium carbonate for bipolar affective disorder, with plasma levels monitored every three months and stable for four years. Following uneventful placement of an implant at the UL5 she asks what she can take for discomfort, and mentions that ibuprofen has always suited her. She has no gastric, renal or respiratory history. What is the most important reason for advising against it?

A. Non-steroidal drugs displace lithium from plasma protein binding sites
B. Non-steroidal drugs reduce renal lithium excretion and raise its level
C. Non-steroidal drugs antagonise the mood-stabilising action of lithium
D. Non-steroidal drugs precipitate the fine tremor associated with lithium
E. Non-steroidal drugs increase the risk of lithium-induced hypothyroidism

574 A 71-year-old man takes apixaban 5 mg twice daily for atrial fibrillation and has had two implants placed with flap elevation, managed by missing the morning dose. He telephones that evening in pain, having taken paracetamol with modest benefit. He has no gastric history, normal renal function, and takes no other regular medicine or antiplatelet drug. Which of the following may you recommend?

A. Paracetamol 1 g four times daily, continued regularly
B. Ibuprofen 400 mg three times daily with food
C. Aspirin 600 mg four times daily as required
D. A and B only
E. All of the above

575 A 29-year-old woman who is exclusively breastfeeding a six-week-old baby has had an implant placed at the LR5. Paracetamol alone is not controlling her pain and she has telephoned for advice. She has no gastric, renal or respiratory history, takes no regular medication and is otherwise well. A colleague suggests co-codamol 30/500, two tablets four times daily. Which is the strongest objection?

A. Codeine suppresses lactation through an effect on prolactin release
B. Codeine is contraindicated only in the first six weeks after delivery
C. The paracetamol content would exceed the maximum daily dose
D. Codeine has a poor number needed to treat as a single agent
E. Codeine is contraindicated in breastfeeding because of variable metabolism

576 A 66-year-old woman takes rivaroxaban 20 mg once daily with her breakfast, following a deep vein thrombosis eighteen months ago. She is booked for placement of two implants with flap elevation and simultaneous bone grafting, in the first appointment of the morning at nine o'clock. Her renal function is normal and she takes no other drug affecting haemostasis. What is the most appropriate management of her medication?

A. Take the dose as usual on the morning of surgery
B. Miss the morning dose and take the next one the following morning
C. Omit rivaroxaban for 48 hours before the procedure
D. Miss the morning dose and take an evening dose instead
E. Delay the morning dose and take it once haemostasis is achieved

577 A 58-year-old man takes aspirin 75 mg daily and ticagrelor 90 mg twice daily following an acute coronary syndrome eleven months ago, and his cardiologist has specified twelve months of dual therapy. He needs an implant at the LR5 with a small flap. A colleague states that, as with clopidogrel, his platelets will remain inhibited for a week after any dose is stopped, so there is nothing to be gained by altering anything. Which statement is correct?

A. Ticagrelor and clopidogrel both bind the P2Y12 receptor irreversibly
B. Ticagrelor is a prodrug requiring hepatic activation, as clopidogrel is
C. Ticagrelor binds P2Y12 reversibly, so its effect wanes within days
D. Ticagrelor acts by irreversible acetylation of cyclo-oxygenase-1
E. Ticagrelor requires INR monitoring while dual therapy continues

578 A 79-year-old woman takes dabigatran 110 mg twice daily for atrial fibrillation. Her most recent estimated glomerular filtration rate, recorded six months ago, was 34 ml/min/1.73 m^2, and she has become noticeably frailer since. She is booked for implant placement with flap elevation. Which feature of dabigatran makes her renal function the most important variable in planning her care?

A. It is metabolised by CYP3A4, whose activity falls in renal impairment
B. It is eliminated predominantly unchanged by the kidney
C. It is highly protein bound, so free drug rises in uraemia
D. It is a vitamin K antagonist whose effect is monitored by the INR
E. It has an active metabolite that accumulates in hepatic impairment

579 A 74-year-old man on warfarin for a mechanical mitral valve, with an INR stable at 2.8, develops denture stomatitis beneath the overdenture retained by two implants, together with angular cheilitis. He wears the denture overnight, his hygiene is poor and swabs confirm candidal infection. His anticoagulation must not be disturbed. Which antifungal preparation may be used without altering his anticoagulant control?

A. Miconazole oral gel applied to the fitting surface
B. Fluconazole 50 mg daily for seven days
C. Ketoconazole cream applied to the angles
D. Nystatin oral suspension used as a rinse
E. Miconazole cream applied to the angles alone

580 A 67-year-old woman has taken prednisolone 15 mg daily for four years for giant cell arteritis, together with alendronate 70 mg weekly and calcium with vitamin D. She has never smoked, her HbA1c is 41 mmol/mol and her periodontal tissues are stable. She wants an implant at the LR5 and asks what difference her tablets make. Which of the following does the corticosteroid contribute to this plan?

A. Impaired collagen synthesis with delayed soft tissue healing
B. A shift into the higher-risk group for osteonecrosis of the jaw
C. An increased susceptibility to post-operative wound infection
D. A and B only
E. All of the above

581 A 19-year-old woman is nauseated in recovery after implant surgery under intravenous midazolam sedation and is given metoclopramide 10 mg. Ninety minutes later she develops sustained upward deviation of the eyes, stiffness of the neck and spasm of the jaw muscles. She is fully conscious, frightened, maintaining her own airway, with a pulse of 96 and oxygen saturation of 99 per cent on air. What is the most likely explanation?

A. An acute dystonic reaction caused by dopamine receptor blockade
B. A focal seizure related to residual midazolam and hypoxia
C. Carpopedal spasm and tetany from anxious hyperventilation
D. Local anaesthetic systemic toxicity presenting after a delay
E. Trismus from injection trauma to the medial pterygoid muscle

582 A 58-year-old man with type 1 diabetes becomes drowsy and then unrousable during implant surgery. His capillary glucose is 1.9 mmol/l, he is breathing normally with a patent airway, and there is no intravenous access. He has a long history of alcohol dependence, has eaten nothing since the previous morning, and his liver function is known to be abnormal. Which is the most important limitation of giving intramuscular glucagon here?

A. It cannot be given to a patient who is unconscious
B. Its onset is too slow to be of use in a dental surgery
C. It depends on hepatic glycogen, which may be depleted
D. It is contraindicated in type 1 rather than type 2 diabetes
E. It must be reconstituted with 50 per cent glucose solution

583 A 34-year-old man with known epilepsy, whose seizures are usually well controlled on lamotrigine, has a generalised tonic-clonic seizure four minutes into implant placement at the LL6. You have removed the instruments, protected his head and given oxygen. At five minutes he is still convulsing, and an ambulance has been called. Which of the following should you give?

A. Intramuscular adrenaline 500 micrograms
B. Buccal midazolam 10 mg
C. Oral glucose gel to the buccal sulcus
D. B and C only
E. A, B and C

584 A 63-year-old man with stable angina develops crushing central chest pain twelve minutes into placement of an implant at the UR5. He is pale and clammy, his pulse is 96 and regular and his blood pressure is 118/74 mmHg. He carries his own glyceryl trinitrate spray and asks for it, mentioning that he took a sildenafil tablet the previous evening. Which is the most important consequence for your immediate management?

A. Glyceryl trinitrate should be given sublingually in the usual way
B. Aspirin 300 mg should be withheld until the pain has settled
C. Oxygen should be given regardless of his oxygen saturation
D. Glyceryl trinitrate is withheld because it may cause severe hypotension
E. Adrenaline 500 micrograms should be given intramuscularly

585 You hold a recognised qualification in conscious sedation and provide intravenous midazolam regularly for implant surgery, with a trained team and full monitoring. A colleague asks which of the cases on your list next week fall outside the standard techniques that a general dental practitioner may provide. Which of the following would do so?

A. Intravenous midazolam titrated for a 62-year-old ASA II man
B. Intravenous midazolam combined with intravenous fentanyl
C. Intravenous midazolam for a ten-year-old having an extraction
D. B and C only
E. A, B and C

586 During intravenous midazolam sedation for implant surgery, a 68-year-old man is given supplemental oxygen at 2 litres per minute by nasal cannula. Forty minutes in, his oxygen saturation has read 99 per cent throughout, but his respiratory rate has fallen and his breathing is shallow. He responds only to firm verbal prompting. Which is the most important limitation of relying on pulse oximetry in these circumstances?

A. Supplemental oxygen delays desaturation and masks hypoventilation
B. Pulse oximetry is unreliable in the presence of nail varnish
C. Pulse oximetry reads falsely high when the patient is anaemic
D. Pulse oximetry cannot detect airway obstruction under any conditions
E. Pulse oximetry lags behind the true saturation by about two minutes

587 A 52-year-old woman has taken diazepam 10 mg three times daily for eighteen years, prescribed originally for anxiety and never reviewed. She requires intravenous sedation for implant surgery, and needs 12 mg of midazolam in titrated increments before an adequate end-point is reached, considerably more than you would normally give. She is otherwise fit with a normal airway assessment. Which is the most important implication?

A. Flumazenil should be given at the end to shorten her recovery
B. Her recovery will be as rapid as in a benzodiazepine-naive patient
C. Flumazenil could precipitate acute withdrawal, including seizures
D. Her tolerance means that respiratory depression will not occur
E. She should be converted to inhalation sedation on the day

588 A 45-year-old man arrives for implant placement under intravenous sedation. The cannula is sited and the first 2 mg increment of midazolam has been given when your nurse points out that, although you discussed the treatment fully at a consultation three weeks ago and he agreed to it, the written record of that discussion and his signature were never completed. He is drowsy but rousable and says he is happy to sign now. What is the most appropriate course?

A. Take his signature now, since he is rousable and cooperative
B. Complete the record of the earlier discussion and proceed
C. Reverse the sedation with flumazenil and consent him again
D. Abandon the appointment and rebook it for another day
E. Ask his escort to sign the consent form on his behalf

589 At the end of forty minutes of inhalation sedation with nitrous oxide and oxygen for implant surgery, your colleague turns off the gas, removes the nasal hood immediately and walks the patient through to the recovery area breathing room air. The patient is well, fully responsive and has an uneventful procedure behind her. Which is the most important reason for giving 100 per cent oxygen for several minutes before removing the hood?

A. To reverse the inactivation of vitamin B12 caused by nitrous oxide
B. To clear nitrous oxide from the room and protect the dental team
C. To restore cerebral blood flow after a period of hypocapnia
D. To prevent diffusion hypoxia as nitrous oxide floods the alveoli
E. To reverse the residual analgesic effect before she is discharged

590 Ten days after an inferior alveolar nerve block given with 4 per cent articaine for implant placement at the LR5, a 38-year-old woman still has altered sensation of the left lower lip and the left side of the tongue, with a persistent burning quality and reduced two-point discrimination. The post-operative radiograph shows the implant apex 3 mm above the mandibular canal and the site is healing normally. Which is the most appropriate course?

A. Reassure her that recovery is certain and review her in six months
B. Prescribe a tapering course of corticosteroid to reduce perineural oedema
C. Explore the site surgically and decompress the nerve
D. Repeat the radiograph monthly until sensation has returned
E. Map and record the deficit, explain it, and review at short intervals

Block U — Surgical placement, extended (Q591–650)

591 A 41-year-old woman has a congenitally absent UL2. The mesiodistal space measured between the contact points of the UL1 and UL3 is 7.0 mm, but the roots converge and the interradicular distance at the level of the bone crest is 5.5 mm. The ridge is 6.5 mm wide buccopalatally, the mucosa is thick, and the narrowest implant in the system has a 3.3 mm platform. Which finding most determines whether an implant can be placed?

A. The mesiodistal space of 7.0 mm between the contact points
B. The interradicular distance of 5.5 mm at the bone crest
C. The buccopalatal ridge width of 6.5 mm at the crest
D. The distance from the contact points to the interproximal crest
E. The thickness of the mucosa overlying the edentulous ridge

592 An implant is placed at the UR1 in a healed ridge with intact circumferential bone and a thick phenotype. At the end of surgery the platform lies 5 mm apical to the planned mid-facial mucosal margin instead of the intended 3 mm. The orofacial position is correct, insertion torque was 35 Ncm, and a screw-retained crown is planned. Which consequence is most likely?

A. An abrupt emergence profile with the margin visible through the mucosa
B. Failure of the impression coping to seat fully on the implant platform
C. Additional crestal bone loss as the supracrestal attachment establishes
D. Loss of the papilla shared with the adjacent lateral incisor
E. Fracture of the abutment screw under eccentric anterior loading

593 An implant at the UL1 has been placed with its shoulder 1 mm buccal to the line joining the emergence profiles of the adjacent incisors, leaving 0.4 mm of buccal bone at the platform. The patient has a thin phenotype and a high smile line. Several consequences follow from that position. Which of them is least amenable to prosthetic correction?

A. The screw access channel emerging on the facial surface of the crown
B. The over-contoured cervical third required to mask the position
C. The reduced interproximal space available for the crown to emerge
D. The mid-facial recession that follows resorption of the buccal plate
E. The abutment angulation required to redirect the restorative axis

594 Consider these statements about the comfort and danger zones described for implant position in the anterior maxilla: (i) the apicocoronal danger zone lies more than 3 mm apical to the planned mid-facial mucosal margin; (ii) the orofacial danger zone lies buccal to a line joining the emergence profiles of the adjacent teeth; (iii) the mesiodistal danger zone extends about 1.0 to 1.5 mm from each adjacent root surface. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (ii) and (iii) only
D. (i) and (iii) only
E. (i), (ii) and (iii)

595 A 34-year-old man has lost the UR1 and UR2 to trauma. The span is 15 mm, the ridge is intact and 8 mm wide, and two implants would leave 2.5 mm between the platforms. The alternative is a single implant at the UR1 carrying a cantilevered pontic at the UR2. He has a high smile line and a thick phenotype. What is the strongest argument for the single implant with a cantilever?

A. The crest between two implants 2.5 mm apart would resorb and the papilla be lost
B. A cantilever distributes anterior load more favourably than two splinted units
C. The pontic site can be developed into an ovate form during provisionalisation
D. Splinted adjacent implants show lower survival in the anterior maxilla
E. The single procedure halves the surgical morbidity and component cost

596 A 4.8 mm diameter implant at the LR6 is restored with a 4.1 mm abutment, so that the abutment margin sits about 0.35 mm inside the outer edge of the platform. At two years the marginal bone level is unchanged from the day of placement. Which mechanism best accounts for the reduced crestal remodelling associated with this arrangement?

A. The narrower abutment reduces the bending moment reaching the crestal bone
B. The horizontal offset displaces the microgap and its infiltrate away from the crest
C. The mismatch raises screw preload and abolishes micromovement at the joint
D. The wider platform increases bone-to-implant contact at the crestal cortex
E. The offset permits a thicker band of peri-implant mucosa to form coronally

597 You are placing a single implant at the UL1 in a healed ridge 8 mm wide with no augmentation planned. The adjacent UL2 and UR1 are unrestored and vital, both papillae are intact and full, and the phenotype is thin with a high smile line. Which incision design best protects the existing papillae while giving adequate access?

A. Sulcular incisions around both neighbours with two divergent relieving incisions
B. Sulcular incisions around both neighbours with no relieving incision at all
C. A crestal incision from line angle to line angle, leaving both papillae intact
D. A crestal incision with a single relieving incision at the mid-buccal of the UL2
E. A papilla-base incision at the UL2 with a sulcular extension to the UR1

598 You are raising a buccal flap from the UR4 to the UR6 for implant placement with simultaneous contour augmentation under a resorbable membrane, and coronal advancement will be needed for tension-free closure. Which of the following decisions about the relieving incision would you make?

A. Siting it one tooth away from the margin of the graft
B. Making it divergent, so the base is wider than the free margin
C. Placing it at the mid-buccal of the adjacent tooth for direct access
D. A and B only
E. All of the above

599 A 58-year-old woman needs additional soft tissue volume at the UR2 and you plan to harvest a subepithelial connective tissue graft from the palate. Her palatal vault is shallow. Which anatomical consideration most limits the apical extent of the donor incision?

A. The greater palatine artery running forward in the submucosa of the vault
B. The nasopalatine bundle emerging through the incisive foramen anteriorly
C. The lesser palatine vessels passing backwards to supply the soft palate
D. The palatal branches of the posterior superior alveolar artery in the tuberosity
E. The attachment of the tensor veli palatini tendon around the pterygoid hamulus

600 A buccal flap is to be raised from the LR4 to the LR6 and released for tension-free closure over a titanium-reinforced membrane. CBCT shows the mental foramen at the level of the LR5 apex, 9 mm below the crest, with an anterior loop extending 4 mm mesial to it. Which single measure most reduces the risk of injury to the mental neurovascular bundle?

A. Holding the blade almost parallel to the inner surface of the flap
B. Confining the flap to a crestal incision with no vertical relief
C. Completing the advancement by blunt rather than sharp dissection
D. Raising the flap in a split-thickness plane over the premolar region
E. Identifying the foramen and the emerging bundle before any release

601 An implant is planned at the UR1 and access requires the flap to extend over the root of the UR2, which is unrestored, thin-phenotype, and has a buccal plate measured at 0.6 mm on CBCT. The papillae are intact and the smile line is high. Which flap design over that root is most appropriate?

A. Full thickness, since periosteum must be reflected to expose the bone
B. Full thickness with a periosteal release made at the base of the flap
C. Split thickness, leaving the periosteum on the UR2 root surface
D. Sulcular full thickness with a relief at the distal line angle of the UR2
E. An envelope flap raised from the palatal aspect and turned buccally

602 Guided bone regeneration at the UL4 to UL6 requires coronal advancement of the buccal flap over a resorbable membrane. You have scored the periosteum and wish to confirm before placing the graft that the release is adequate. Which of the following indicate that it is?

A. The flap lies passively 3 to 4 mm beyond the intended closure line
B. The flap can be drawn to the closure line with light forceps traction
C. The flap margin blanches only slightly as the first suture is tightened
D. A and B only
E. A, B and C

603 At the UR4 an implant is placed with a buccal dehiscence, grafted with particulate xenograft and covered with a non-resorbable membrane. The crestal incision was made along the crest and the flap advanced so that the suture line lies directly over the centre of the membrane. Insertion torque was 40 Ncm and the closure is passive. Which is the strongest objection to that arrangement?

A. Braided sutures will wick oral bacteria along their interstices into the graft
B. The suture line is the weakest part of the closure and lies over an avascular device
C. The membrane will be displaced coronally by elastic recoil of the released flap
D. The keratinised band is carried palatally and the vestibular depth is reduced
E. The adjacent papillae cannot be repositioned accurately over a rigid membrane

604 A colleague proposes flapless placement at the LR5 using a 4 mm tissue punch. CBCT shows 9 mm of crestal width with no buccal concavity and 14 mm of bone above the mandibular canal, and a fully guided template has been made. The band of keratinised mucosa measures 2 mm. What is the strongest objection to the plan?

A. The buccal plate cannot be inspected for fenestration during placement
B. Depth judgement is unreliable without a visible bony reference point
C. Coolant cannot reach the cutting tip through the punched mucosal cuff
D. The punch will excise most of the remaining band of keratinised mucosa
E. Primary stability is reduced when the periosteum is not reflected

605 An implant is placed flaplessly and freehand at the UR5 through mucosa that measures 4 mm in thickness, the operator judging depth from the mucosal margin. The post-operative radiograph shows the platform lying 3 mm apical to the interproximal bone crest, rather than at crestal level as planned. Which explanation best accounts for the error?

A. Depth was referenced to the mucosal margin rather than the bone crest
B. The drill advanced beyond its stop in soft cancellous bone
C. The tissue punch removed the crestal tissue used as the reference
D. The implant was seated at an insertion torque above the recommended limit
E. The osteotomy was prepared through a knife-edge crest that fractured

606 Which of the following are prerequisites before a straightforward single implant is placed by a flapless technique?

A. Cross-sectional imaging demonstrating adequate bone volume and contour
B. A band of keratinised mucosa wide enough to survive the punch
C. Confidence that no simultaneous augmentation will be required
D. A and B only
E. All of the above

607 An LR6 with a vertical root fracture is extracted and an implant placed into the socket the same morning, achieving 45 Ncm against the interradicular septum and apical bone. That afternoon a screw-retained provisional crown is fitted in light but definite contact in intercuspation, with no contact in any excursion. The patient uses it normally. How is this protocol best described?

A. Type 1 placement with immediate restoration
B. Type 1 placement with immediate loading
C. Type 1 placement with early loading
D. Type 2 placement with immediate loading
E. Immediate restoration, because the crown is provisional

608 A UR5 is extracted and the socket left to heal without grafting. At fourteen weeks the radiograph shows partial trabecular fill with the socket outline still discernible, and an implant is placed on that day with 40 Ncm and a healing abutment. The definitive screw-retained crown is fitted in full occlusal contact fourteen weeks after placement. Which describes this protocol?

A. Type 2 placement with conventional loading
B. Type 3 placement with early loading
C. Type 3 placement with conventional loading
D. Type 4 placement with conventional loading
E. Type 4 placement with early loading

609 A UR1 with a horizontal root fracture is to be replaced. The buccal plate is thin, the patient has a high smile line, and you elect to place at six weeks rather than immediately. Insertion torque is expected to be adequate either way. What is the principal biological advantage of waiting that interval?

A. Soft tissue healing provides thicker, more abundant mucosa for coverage
B. The buccal plate has completed its resorption and its contour is final
C. Bone fill of the socket supplies the primary stability that is lacking on the day
D. Any apical infection resolves spontaneously without antibiotic treatment
E. The mucosal phenotype thickens enough to mask a titanium abutment

610 Which of the following are genuine advantages of early placement at four to eight weeks over immediate placement in the anterior maxilla?

A. Increased soft tissue volume allowing tension-free coverage of a graft
B. Resolution of apical or marginal infection before the implant is placed
C. Preservation of the buccal bone wall, which is no longer resorbing
D. A and B only
E. A, B and C

611 A 47-year-old non-smoker has an implant placed at the UR4 in a ridge healed for two years. Insertion torque was 38 Ncm and the ISQ 73, there is no parafunction, and the opposing dentition is natural. A 3 mm buccal dehiscence was found on flap elevation and has been grafted with xenograft under a resorbable membrane. He asks for the crown to be fitted this week. Which factor most argues against immediate loading?

A. The insertion torque of 38 Ncm at the time of placement
B. The maxillary rather than mandibular location of the site
C. The simultaneous guided bone regeneration at the buccal aspect
D. The unsplinted single-unit design of the planned restoration
E. The resonance frequency reading of ISQ 73 at placement

612 A patient with a failing lower dentition asks whether all his remaining teeth can be removed and a fixed provisional bridge fitted on four implants on the same day, having read that this is now routine. He is a healthy non-smoker with generalised periodontitis and reduced bone height posteriorly. Which statement best represents the position you should put to him?

A. The protocol lacks evidence and should not be offered to any patient
B. The protocol carries the same risk as immediate loading of a single implant
C. The protocol requires each implant to exceed 70 Ncm at the time of insertion
D. The protocol is contraindicated whenever teeth are extracted at the same visit
E. The protocol is well documented but exceeds general practice competence

613 An implant is placed at the LL5 in a healed ridge with 40 Ncm and a healing abutment is fitted. The patient is reviewed at ten days, the mucosa is quiet, the ISQ reads 74, and a screw-retained definitive crown is delivered in full occlusal contact at that visit. How is the loading protocol best described?

A. Immediate loading, since the crown was fitted within two weeks of surgery
B. Early loading, since the crown was fitted between one week and two months
C. Conventional loading, since undisturbed healing preceded the restoration
D. Immediate restoration, since the crown was fitted before osseointegration
E. Early restoration, since the crown was fitted well before three months

614 An implant has been placed at the LR5 in a ridge healed for two years and the patient wishes to be restored within the first week. Which of the following must be satisfied before that is agreed?

A. Primary stability at or above the customary 35 Ncm threshold
B. Absence of uncontrolled parafunctional activity
C. Absence of any requirement for simultaneous augmentation
D. A and B only
E. All of the above

615 An implant is placed in the dense anterior mandible through an osteotomy prepared to the manufacturer's full sequence, including a countersink and a tap. The motor records an insertion torque of 15 Ncm, but the resonance frequency reading taken immediately afterwards is ISQ 76. The operator is disconcerted by the disagreement. Which statement best explains the two findings?

A. Resonance frequency analysis systematically overestimates stability in cortical bone
B. The countersink has reduced the area of bone in contact with the implant surface
C. The transducer was read buccolingually, which gives the higher of the two values
D. Tapping removed the frictional interference while the bone itself remains stiff
E. The implant lies within a pre-existing socket and is fibrously encapsulated

616 An implant placed in the posterior maxilla records an insertion torque of 30 Ncm and an ISQ of 58 at surgery. The patient has no parafunction, wears no denture, the site was not augmented, and the mucosa is thick with an ample keratinised band. What is the most appropriate protocol?

A. Immediate restoration out of occlusion at forty-eight hours
B. Early loading at six weeks following a repeat resonance reading
C. Conventional loading after at least three months of undisturbed healing
D. Immediate loading with the crown splinted to the adjacent implant
E. Removal and replacement with an implant of greater diameter

617 An implant at the LR6 is reviewed at one year. It is asymptomatic, the mucosa is healthy with no bleeding on probing, and the patient reports no problem, but the resonance frequency reading has fallen from ISQ 75 at placement to ISQ 62, and the radiograph shows 2.5 mm of crestal bone loss. Which best explains the fall in the reading?

A. Loss of crestal bone has lengthened the unsupported portion of the implant
B. Secondary stability is always lower than the primary stability it replaces
C. The transducer has been screwed to the abutment rather than to the implant
D. Mucosal inflammation reduces the measured stiffness of the interface
E. The oxide layer of a hydrophilic surface loses stiffness as it matures

618 Which of the following raise the insertion torque recorded at placement without any change in the density of the bone at the site?

A. Preparing the osteotomy to less than the implant's core diameter
B. Selecting an implant with a tapered body and an aggressive thread form
C. Omitting the countersink so that the platform engages the crestal cortex
D. A and B only
E. All of the above

619 An implant at the LL3 in dense anterior mandibular bone will not seat fully. With 1.5 mm still to go the ratchet reads 80 Ncm, the osteotomy was prepared to the manufacturer's final drill without a countersink or a tap, and the operator increases the force applied. Which is the most likely immediate mechanical consequence?

A. Elastic elongation of the abutment screw beyond its yield point
B. Deformation of the internal connection or fracture of the insertion post
C. Cold welding of the implant surface to the wall of the osteotomy
D. Stripping of the threads that the implant has cut in the cortical bone
E. Delamination of the roughened surface layer from the implant body

620 A colleague prepares implant osteotomies with a straight surgical handpiece at 2500 revolutions per minute, arguing that a faster drill cuts more cleanly and spends less time in contact with the bone. He uses a graded drill sequence, sharp drills and copious external irrigation. Which is the strongest objection to that practice?

A. The drill is more likely to wander from the intended axis at that speed
B. Available torque at that speed is insufficient to cut dense cortical bone
C. Frictional heating rises with rotational speed for a given rate of advance
D. External irrigation cannot keep pace with the volume of debris produced
E. The speed lies outside the range specified by the implant manufacturer

621 A 13 mm osteotomy is being prepared through a heavily corticated crest in the anterior mandible. Which of the following genuinely reduce the temperature reached at the wall of the osteotomy?

A. Withdrawing the drill intermittently so that coolant reaches the cutting tip
B. Directing a greater volume of coolant onto the shank of the rotating drill
C. Raising the rotational speed so that the pass is completed more quickly
D. A and B only
E. All of the above

622 An implant was placed in the dense anterior mandible through a 13 mm osteotomy taken to depth in a single continuous pass, with irrigation twice interrupted by the assistant. At four weeks the patient reports a dull persistent ache, the mucosa is intact, the implant is not mobile, and the radiograph shows a narrow radiolucent halo surrounding the implant with a faint denser line beyond it. What is the most likely explanation?

A. Early peri-implantitis following contamination of the osteotomy at surgery
B. Physiological remodelling of the crestal cortex to the level of the first thread
C. Fibrous encapsulation caused by micromotion under a provisional restoration
D. A ring sequestrum of thermally necrosed bone separating from viable bone
E. Compression necrosis of the crest from an excessive insertion torque

623 In the posterior maxilla the pilot drill advances under its own weight with almost no resistance. You omit the final drill, undersizing the osteotomy by 0.4 mm, and the implant seats at 45 Ncm. The ridge is 8 mm wide and there is 12 mm of bone below the sinus floor. Which statement best describes what has been achieved?

A. Greater bone-to-implant contact, which will shorten the healing period required
B. Greater mechanical interference at the interface, with the bone no stiffer than before
C. Greater compression of the crestal cortex, which will raise the ISQ proportionately
D. Greater primary stability, which of itself justifies loading within the first week
E. Greater lateral density, converting the site from type 4 bone to type 2 bone

624 CBCT for an implant at the LL3 shows a thick cortex enclosing dense homogeneous trabeculae with almost no marrow space, and at surgery the pilot drill cuts slowly with a chalky feel and produces little bleeding. Which modification to the protocol is most appropriate?

A. Undersizing the osteotomy and omitting the countersink to raise interference
B. Increasing drill speed and reducing the applied load to shorten the contact time
C. Countersinking the crest and tapping the osteotomy before the implant is inserted
D. Selecting a tapered implant with an aggressive thread form for the same osteotomy
E. Reducing the implant length so that less cortical bone has to be cut through

625 A fully guided protocol is used at the LR6, every drill passing through a metal sleeve 5 mm long seated 2 mm above the crest. Which of the following features of that arrangement increase the risk of thermal injury to the bone?

A. Exclusion of external coolant from the cutting tip by the sleeve
B. Conduction of heat from the drill shank into the overlying mucosa
C. A requirement to run guided drills at higher rotational speeds
D. A and B only
E. A, B and C

626 CBCT before placement at the UR6 showed 9 mm of bone below the sinus floor and a 9 mm implant was planned. During preparation the drill loses resistance at 8 mm and the depth gauge moves freely beyond that point without a bony floor being felt. There is no bleeding and the patient reports nothing. What is the most appropriate course?

A. Place the planned 9 mm implant and graft particulate bone around the crest
B. Place a 6 mm implant short of the perforation and allow conventional healing
C. Abandon the site, close the flap and reassess with repeat imaging at three months
D. Complete a transcrestal elevation and graft through the existing osteotomy
E. Close the site and refer for a lateral window approach at a later visit

627 At the LR5 the final drill is inadvertently taken 0.5 mm wider than planned and the 4.1 mm implant, 8.5 mm long, spins at 8 Ncm as its platform reaches the crest. The ridge measures 6.5 mm wide, CBCT shows 10 mm of bone above the mandibular canal, and no wider implant of any length is available. What is the most appropriate course?

A. Advance the implant a further 1.5 mm to engage bone closer to the canal
B. Fit a cover screw and allow six months of undisturbed submerged healing
C. Order a wider implant and reschedule the placement for the following week
D. Splint the implant to the LR4 with a bonded composite provisional restoration
E. Remove the implant, graft the osteotomy and re-enter after four months

628 After placement at the UR3 a fenestration of the buccal plate is found at the apical third, with three threads exposed over a defect 4 mm in height. The implant is stable at 40 Ncm, the crestal bone is intact circumferentially, the mucosa is thick and the patient does not smoke. What is the most appropriate management?

A. Remove the implant and reposition the osteotomy further palatally
B. Reverse the implant out by two threads so the exposure is reduced
C. Graft the exposed threads and cover the defect with a resorbable membrane
D. Leave the defect, since apical fenestrations become quiescent under periosteum
E. Submerge the implant without grafting and re-enter the site at six months

629 The tip of a 2 mm pilot drill fractures during preparation at the LL6 and remains within the osteotomy, its upper end visible about 3 mm below the crest under the raised flap. The patient is comfortable, bleeding is minimal and the field is clear. What is the most appropriate management?

A. Leave the fragment in situ, since titanium alloy is inert within bone
B. Place the planned implant over it so the fragment is contained apically
C. Close the flap and refer for retrieval under general anaesthesia
D. Take a periapical radiograph and review the patient in one week
E. Retrieve it under direct vision, widening the osteotomy if necessary

630 An implant at the LR6 has been placed with its long axis inclined 40 degrees buccally to the planned prosthetic axis. The bone is intact, insertion torque was 40 Ncm, and at two weeks the patient is asymptomatic with a healing abutment in place. Which statement best describes the restorative position?

A. An angled screw channel abutment will redirect the access to the fossa
B. A cemented crown on a stock straight abutment will conceal the deviation
C. A cast custom abutment can redirect the axis without altering emergence
D. The deviation exceeds the correction that angled components can provide
E. The screw will emerge lingually, which is restorable in a molar position

631 An implant is placed immediately into an intact UR1 socket. Which of the following bear on the decision whether to graft the residual buccal gap?

A. The horizontal dimension of the gap at the level of the implant shoulder
B. The insertion torque achieved as the implant engaged apical bone
C. The length of the implant selected for the site
D. A and B only
E. A, B and C

632 An implant placed immediately at the UL1 leaves a horizontal gap of 1.2 mm at the shoulder against a buccal plate measured at 0.7 mm. The phenotype is thin and the smile line high. A colleague advises leaving the gap ungrafted, on the grounds that defects under 2 mm fill with bone spontaneously. What is the strongest objection to that advice?

A. Gaps of less than 2 mm do not fill spontaneously in the anterior maxilla
B. Spontaneous fill restores bone but not the buccal contour that will be lost
C. The blood clot will be displaced by the healing abutment during function
D. Bone fill requires a membrane wherever the buccal plate is under 1 mm thick
E. Grafting the gap is what raises the insertion torque to an acceptable level

633 An implant is placed at the LR6 in a one-stage protocol with 40 Ncm and no augmentation. The peri-implant mucosa measures 4 mm in thickness from the bone crest, and the tallest healing abutment on the tray has a transmucosal height of 3 mm. Which consequence follows from fitting it?

A. The abutment will be loaded directly by the opposing dentition during healing
B. The implant will behave as a submerged implant and integrate more rapidly
C. The mucosal seal will form against the implant platform rather than the abutment
D. The abutment screw will loosen as the mucosa exerts lateral pressure on it
E. The abutment will be overgrown by mucosa and require a second exposure

634 After extraction of the UR4 the socket is grafted for ridge preservation and you are choosing between advancing the buccal flap for primary closure and closing the orifice with a free epithelialised punch graft from the palate. The buccal band of keratinised mucosa measures 3 mm and the vestibule is shallow. Which is the strongest argument against flap advancement?

A. It leaves the suture line under residual tension directly over the graft
B. It requires a periosteal release close to important neurovascular structures
C. It displaces the mucogingival junction and narrows the keratinised band
D. It prevents the graft being retained beneath a crossed mattress suture
E. It delays epithelialisation of the socket orifice by several further weeks

635 A 62-year-old woman has an implant placed at the UL4 with 35 Ncm and no augmentation. She wears an acrylic partial denture whose saddle rests directly over the site and she is unwilling to be without it in public. Which of the following would you do at the end of surgery and immediately afterwards?

A. Fit a cover screw and submerge the implant beneath the closed flap
B. Relieve the fitting surface generously and line it with a soft material
C. Rebase the denture over the site in self-curing hard acrylic resin
D. A and B only
E. All of the above

636 You are closing a coronally advanced flap over a particulate graft and a resorbable membrane at the UR4. The mucosa is thin and mobile, the release is passive, and the closure must hold for two weeks. Which needle and suture combination is most appropriate?

A. A reverse-cutting needle with 5/0 polytetrafluoroethylene monofilament
B. A taper-point needle with 3/0 braided silk on a half-circle body
C. A reverse-cutting needle with 3/0 braided polyglactin 910
D. A taper-point needle with 4/0 chromic gut on a three-eighths circle
E. A straight cutting needle with 5/0 plain gut for rapid absorption

637 A grafted site has been closed with 5/0 polytetrafluoroethylene monofilament. At the one-week review the tissues are healthy and the margins are apposed, but three of the knots have come undone and their suture ends lie free in the sulcus. Which property of the material best explains this?

A. Its elasticity, which allows the loops to lengthen as swelling develops
B. Its hydrophobicity, which prevents tissue ingrowth from securing the knot
C. Its low coefficient of friction, which requires additional knot throws
D. Its shape memory, which returns the strand to its packaged configuration
E. Its inertness, which prevents an inflammatory seal forming around the knot

638 A single implant is placed at the UL1 through a crestal incision that leaves both papillae intact, with simultaneous contour grafting on the buccal aspect. The flap needs to be held slightly coronally so that the graft is covered, and the phenotype is thin. Which suturing arrangement best achieves that without compressing the interdental tissues?

A. Interrupted sutures passed through each papilla to the palatal flap
B. A sling suture passed around the adjacent teeth to suspend the flap
C. A continuous locking suture run along the length of the crestal incision
D. Horizontal mattress sutures placed 5 mm apical to the incision line
E. A figure-of-eight suture drawn across the crest between the neighbours

639 A colleague maintains that implant placement demands full theatre discipline — sterile gowns, extensive draping, a dedicated operating room — and that this matters more than any drug given. You are discussing straightforward placement of a single implant in a healed site in a healthy patient. Which statement best represents the evidence?

A. Comparative studies show similar survival under clean and under sterile conditions
B. Sterile conditions approximately halve the rate of early failure compared with clean
C. Neither asepsis nor antibiotic prophylaxis influences the rate of early failure
D. Antibiotic prophylaxis is effective only where sterile technique is not observed
E. The advantage of sterile technique is confined to sites that have been augmented

640 Your surgical motor can deliver coolant either from a bag of sterile saline through a single-use tubing set, or from the dental unit's water supply through the handpiece in the same way as for restorative work. The unit's waterlines are treated periodically according to the manufacturer's schedule. Which is the principal reason for using the sterile line?

A. Unit water is delivered at a lower flow rate than a pressurised saline bag
B. Unit water is not chilled and therefore cools the cutting tip less effectively
C. Unit water contains residual disinfectant that is toxic to osteoblasts
D. Unit water cannot be delivered through internally irrigated drills at all
E. Unit waterlines harbour biofilm that would be flushed into the osteotomy

641 Which of the following measures are supported by evidence as reducing early implant failure after routine placement in a healed site?

A. A 0.2 per cent chlorhexidine mouthrinse immediately before surgery
B. A single 2 g dose of amoxicillin given one hour before surgery
C. A five-day course of amoxicillin begun on the day of surgery
D. A and B only
E. A, B and C

642 During preparation at the UR1 the drill is directed slightly palatally and at 10 mm brisk bright bleeding wells from the osteotomy. The flap is otherwise dry, the patient is comfortable, and the pre-operative CBCT reviewed on the screen shows the incisive canal lying about 2 mm palatal to the planned axis. What is the most appropriate immediate action?

A. Apply local pressure, reassess the axis and redirect the osteotomy buccally
B. Seat the planned implant into the canal so that the vessel is tamponaded
C. Pack the canal with bone wax and then place the implant as planned
D. Abandon the site, close the flap and refer for exploration of the canal
E. Prescribe tranexamic acid mouthwash and review the patient the next day

643 A general dentist with a special interest is performing a lateral window sinus floor elevation when brisk pulsatile bleeding begins from the bony margin of the window and obscures the field. CBCT reviewed afterwards shows an intraosseous vessel crossing the lateral wall about 16 mm above the crest. The patient is otherwise well and takes no anticoagulant. What is the most appropriate immediate management?

A. Apply topical adrenaline-soaked gauze against the sinus membrane
B. Ligate the vessel through an extension of the bony window superiorly
C. Compress the bleeding point and burnish the surrounding bone onto the vessel
D. Abandon the procedure and transfer the patient urgently to hospital
E. Continue the elevation and rely on the graft to tamponade the bleeding

644 Bleeding wells persistently from within a mandibular osteotomy at the LR5 and obscures the field. Which of the following are appropriate measures?

A. Sustained direct pressure with a saline-moistened swab
B. Placement of a resorbable oxidised cellulose or collagen haemostat
C. Packing of the osteotomy with bone wax before the implant is placed
D. A and B only
E. A, B and C

645 A UR1 requires extraction. CBCT shows the root lying against the buccal plate along its whole length, with the apex perforating that plate over 2 mm, and 4 mm of bone apical to the socket. The remaining buccal plate measures 0.6 mm, the phenotype is thin and the smile line high. The patient is a healthy non-smoker. What is the most appropriate approach?

A. Immediate placement with gap grafting and a screw-retained provisional
B. Immediate placement with a connective tissue graft and submerged healing
C. Ridge preservation of the socket with late placement into a healed ridge
D. Early placement at four to eight weeks after soft tissue healing, with contour grafting
E. Extraction followed by referral for a block graft before any implant is considered

646 Which of the following must be present before an immediately placed anterior maxillary implant is provisionalised on the day of surgery?

A. Insertion torque sufficient to resist functional micromotion
B. An intact buccal plate that the implant surface does not contact
C. A provisional relieved of contact in intercuspation and in excursion
D. A and B only
E. All of the above

647 An implant placed at the UL5 with a simultaneous buccal graft under a non-resorbable membrane was submerged. Insertion torque was 40 Ncm and healing has been entirely uneventful, with no exposure of the incision line. The patient telephones at ten weeks asking when the abutment can be fitted. What is the most appropriate interval before second-stage surgery?

A. Six weeks, given the high insertion torque achieved at placement
B. Eight weeks, as for a one-stage protocol in the posterior mandible
C. Three months, as for an ungrafted implant in the posterior maxilla
D. Six months, to allow maturation of the regenerated bone before exposure
E. Twelve months, to allow complete substitution of the graft particles

648 At second-stage surgery at the UR2 the implant is well integrated and correctly positioned, but there is a 2 mm concavity of the buccal soft tissue, the phenotype is thin and the abutment is likely to shine through. The band of keratinised mucosa measures 5 mm and the palatal mucosa is thick. Which procedure best addresses the deficiency?

A. A free epithelialised graft harvested from the palate and placed buccally
B. An apically positioned split-thickness flap to widen the keratinised band
C. Exposure with a tissue punch and a wide-diameter healing abutment
D. A free gingival graft placed at the level of the mucogingival junction
E. A pedicled de-epithelialised palatal roll flap turned to the buccal aspect

649 Immediately after the osteotomy at the LR5 is dried with a three-in-one syringe, the patient's right cheek swells rapidly over about a minute and the tissue crackles under the fingers. She is comfortable, breathing normally with a clear chest, has no urticaria, no wheeze and a normal blood pressure. What is the most appropriate management?

A. Give intramuscular adrenaline and call for emergency ambulance transfer
B. Stop the procedure, reassure her, prescribe antibiotics and review closely
C. Complete the placement as planned and prescribe an oral antihistamine
D. Incise over the swelling to decompress the tissue and release the air
E. Apply firm external pressure to the cheek and continue with the surgery

650 While a healing abutment is being transferred at the LL6 it slips from the driver and disappears posteriorly. The patient coughs several times and then says she feels entirely normal. Her oxygen saturation is 98 per cent, her chest is clear on auscultation, she has no stridor, and neither the component nor any fragment can be found in the mouth, on the drapes or on the floor. What is the most appropriate action?

A. Reassure her and ask her to check her stools over the following few days
B. Take periapical and panoramic radiographs to locate the missing component
C. Arrange urgent chest and abdominal radiography with hospital assessment
D. Prescribe an antibiotic and review her at the practice the following morning
E. Ask her to attend her general practitioner should she become breathless

Block V — Minor hard and soft tissue augmentation, extended (Q651–700)

651 A 46-year-old non-smoker had a 3 mm labial contour deficiency at the UR1 grafted with deproteinised bovine bone mineral beneath a native collagen membrane. At the five-year review the labial profile is unchanged, and a small-volume CBCT taken for an unrelated reason shows discrete radiopaque particles still lying within the labial plate. Which property of the material best explains their persistence?

A. High-temperature processing yields large apatite crystallites of low solubility
B. Its retained organic matrix resists enzymatic degradation
C. Its particles are too dense for osteoclasts to attach to
D. Its calcium-to-phosphate ratio favours dissolution over cellular resorption
E. The overlying collagen membrane shields it from remodelling

652 A staged horizontal augmentation is re-entered at six months and a trephine core is taken before the osteotomy is prepared. Histology shows newly formed lamellar bone in direct contact with residual graft particles that occupy rather less than a third of the specimen, with no inflammatory infiltrate and no multinucleate giant cells at the particle surfaces. Which material was most probably used?

A. Autogenous cortical chips
B. Beta-tricalcium phosphate
C. Calcium sulphate
D. Demineralised freeze-dried bone allograft
E. Deproteinised bovine bone mineral

653 A supplier tells you that the mineralised freeze-dried bone allograft he sells is regulated in the same way as any other bone substitute and that its origin need not be raised with patients because processing removes all cellular material. Which of the following apply to this material?

A. Its human origin must be disclosed to the patient before it is used
B. Its osteoinductive potential is equivalent to that of autogenous bone
C. It is exempt from batch traceability once it has been processed
D. A and B only
E. All of the above

654 You can harvest either cortical shavings from the external oblique ridge with a bone scraper, or cancellous bone from the maxillary tuberosity, to combine with a xenograft for a contour augmentation at the UL3 in a 44-year-old. You choose the cortical shavings. Which property best justifies that choice?

A. They revascularise more rapidly and are remodelled sooner
B. They contain more osteoprogenitor cells per unit volume
C. They are osteoinductive whereas cancellous bone is not
D. They resorb more slowly and hold the augmented volume longer
E. They can be taken without creating a separate surgical site

655 Consider these statements about the tissue regenerated beneath a barrier membrane six to eight months after a composite graft of autogenous chips and a low-substitution xenograft: (i) it contains a higher proportion of woven bone than adjacent native bone; (ii) the bone-to-implant contact achieved in it is lower than in native bone; (iii) residual xenograft particles persist within it, unremodelled. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

656 A 44-year-old woman will accept no material of animal or human origin for a small contour augmentation at the UR3, and you are choosing between beta-tricalcium phosphate and a bioactive glass. Which statement correctly distinguishes the two?

A. Beta-tricalcium phosphate is converted in situ to hydroxyapatite and persists indefinitely
B. Bioactive glass recruits mesenchymal cells and is therefore osteoinductive
C. Beta-tricalcium phosphate supplies the rigidity needed to support a collagen membrane
D. Bioactive glass is prepared from marine coral and is not strictly synthetic
E. Bioactive glass develops a surface apatite layer that bonds chemically to bone

657 A colleague proposes using recombinant human bone morphogenetic protein-2 on an absorbable collagen sponge to reconstruct a 5 mm vertical defect at the LR6 in your practice, arguing that it removes the need for a donor site. What is the strongest objection to that proposal?

A. It is osteoconductive rather than osteoinductive and cannot bridge a defect
B. It is a potent osteoinductive agent whose dose-related swelling demands a specialist setting
C. It is inactivated by the collagen carrier within the first twenty-four hours
D. It acts only where autogenous bone is added to supply osteoprogenitor cells
E. It carries the transmission risk associated with all recombinant human proteins

658 Before placing a particulate graft for a staged horizontal augmentation at the LL4 to LL6, you perforate the buccal cortex in several places with a small round bur under copious irrigation. Which effect best justifies the manoeuvre?

A. It increases the surface area of bone in contact with the graft particles
B. It releases bone morphogenetic proteins held within the cortical matrix
C. It opens the marrow spaces, admitting vessels and osteoprogenitor cells to the graft
D. It roughens the cortex so that the membrane is less easily displaced
E. It reduces cortical density so that the later osteotomy is easier to prepare

659 A 4 mm labial contour deficiency at the UR2 has no bony walls to contain a graft, and you are planning how to keep the barrier from collapsing onto the underlying cortex. Which of the following would achieve that?

A. A titanium-reinforced barrier fixed with pins
B. A slowly substituted particulate filling the space beneath the membrane
C. A native collagen membrane sutured tightly across the defect
D. A and B only
E. A, B and C

660 A staged augmentation at the LR5 is re-entered at six months. The flap healed without any dehiscence, but beneath it lies a thin layer of soft tissue that curettes away easily from the cortex, and almost no width has been gained. The membrane had not been fixed, and the patient wore an unrelieved tissue-borne partial denture throughout healing. Which principle has most clearly failed?

A. Primary closure of the flap
B. Angiogenesis within the graft
C. Space maintenance beneath the barrier
D. Stability of the graft and the clot
E. Asepsis during handling of the graft

661 A grafted socket at the UR5 is covered with a dense polytetrafluoroethylene membrane that is tucked beneath the flap margins, deliberately left exposed to the mouth and secured with a crossed suture, so that keratinised tissue is not advanced across the ridge. Which of the PASS principles is being set aside here, and what substitutes for it?

A. Primary closure, replaced by the impermeability of the barrier itself
B. Space maintenance, replaced by the intact walls of the socket
C. Stability, replaced by the crossed suture over the membrane
D. Angiogenesis, replaced by the blood supply of the socket walls
E. Primary closure, replaced by a course of systemic antibiotics

662 You need a barrier that will still be functioning at four months over a staged horizontal augmentation, and you are choosing between a native porcine collagen membrane and a chemically cross-linked one. Which statement best describes the trade-off between them?

A. Cross-linking stiffens the membrane enough to maintain space unaided
B. Cross-linking makes the membrane cell-occlusive, which the native form is not
C. Cross-linking prolongs barrier function but an exposure heals less readily
D. Cross-linking improves integration with the overlying flap
E. Cross-linking allows the membrane to be left deliberately exposed

663 An expanded polytetrafluoroethylene membrane placed over a staged augmentation at the UL4 becomes exposed over about 2 mm at four weeks. There is no suppuration, the patient is comfortable, and she has been using chlorhexidine gel at the site as instructed. What is the most appropriate management?

A. Chlorhexidine gel and weekly review until removal at six months
B. Remove the membrane now and preserve the bone already formed
C. Trim the exposed portion and advance the flap over the remainder
D. Prescribe amoxicillin for a week and review the site
E. Cover the exposure with a free gingival graft

664 You intend to fix a barrier membrane buccally with bone tacks over a staged augmentation at the LL4 and LL5 in a 55-year-old. Which of the following are genuine hazards of tack placement at this site?

A. Injury to the mental nerve where it emerges and loops forward
B. Damage to the root of an adjacent tooth
C. Perforation of the lingual plate into the sublingual space
D. A and B only
E. A, B and C

665 At re-entry after a staged augmentation, new bone is present centrally over the defect, but a band of soft tissue extends beneath what had been the periphery of the membrane, over bone that was sound at the time of grafting. The flap had healed without dehiscence and the graft was a low-substitution xenograft. Which technical error best explains the finding?

A. The graft was overpacked at the margins of the defect
B. The membrane was trimmed to the outline of the defect rather than overlapping sound bone
C. The membrane resorbed before regeneration was complete
D. The flap was closed under residual tension
E. The graft particles were too small to support the barrier

666 A 57-year-old non-smoker has been edentulous at the LR6 for nine years. CBCT shows 5 mm of bone above the superior border of the mandibular canal, a crest 3.5 mm wide, and the crest lying 5 mm apical to the bone level at the LR5. He is periodontally healthy with excellent plaque control and wants a fixed replacement. A colleague suggests titanium mesh with a composite graft. What is the most appropriate course?

A. Undertake mesh-supported vertical and horizontal augmentation
B. Place a 6 mm implant in the residual bone and splint it to a second
C. Refer for assessment for combined vertical and horizontal reconstruction
D. Stage a horizontal augmentation and reassess the height afterwards
E. Provide a cantilevered bridge from the LR5 and review

667 Two implants are placed at the same visit in a 48-year-old. At the UR3 there is an apical fenestration measuring 4 mm by 3 mm with 5 mm of intact bone coronal to it; at the UR4 there is a crestal dehiscence 5 mm deep running from the bone crest apically along the implant. Both are grafted with the same material and covered by the same membrane. Which defect will regenerate more completely, and why?

A. The dehiscence, because the crest supplies more osteoprogenitor cells
B. The dehiscence, because it lies closer to the periosteum of the flap
C. Both equally, since the outcome depends on the barrier rather than the defect
D. The fenestration, because it is enclosed by bone at every margin
E. The fenestration, because it lies further from the oral environment

668 A patient asks why you are unwilling to rebuild the 5 mm of lost ridge height at the LL6 and are proposing a short implant instead. Which statement best represents the current evidence?

A. Vertical augmentation reliably gains 5 mm but requires a second surgical site
B. Short implants show comparable survival with fewer complications
C. Short implants survive less well but avoid the morbidity of grafting
D. Vertical augmentation is contraindicated in the posterior mandible
E. Short implants are indicated only where the opposing arch is a denture

669 Before extracting the UR4 for a vertical root fracture you review the CBCT. The buccal plate measures 0.6 mm over the mid-root and the palatal plate 1.8 mm, and both are continuous. The patient is a 45-year-old non-smoker with a medium smile line who asks what will happen to the ridge if the socket is simply left to heal. Which change is most likely?

A. Marked vertical collapse of the buccal crest as well as loss of width
B. Loss of width from the palatal aspect, the buccal crest being preserved
C. Negligible change, since the plate is intact and undamaged
D. Loss confined to the coronal 2 mm of the buccal wall
E. Equal resorption of the buccal and palatal walls

670 A patient is being consented for alveolar ridge preservation at the UR5 with a xenograft and a collagen matrix, with implant placement planned at four months. Which of the following may fairly be claimed for the procedure?

A. It reduces horizontal ridge reduction by roughly half
B. It maintains the ridge dimensions essentially unchanged
C. It limits the mean horizontal loss to less than 1 mm
D. A and B only
E. All of the above

671 The UR2 is to be extracted in a 34-year-old with a thin phenotype and a high smile line. The labial soft tissue margin sits level with the contralateral tooth, but a probe passes 7 mm apically along the labial root surface without meeting bone, and CBCT confirms the labial plate is absent over its coronal two-thirds. What does this configuration imply for the plan?

A. Immediate placement with simultaneous contour grafting is the treatment of choice
B. The soft tissue is intact but the bone is not, so staged reconstruction is safer
C. Both soft tissue and bone are deficient, so orthodontic extrusion is required
D. The socket is intact and no augmentation will be needed
E. The tooth should be retained until the soft tissue margin has receded

672 Consider these statements about placing an implant at the UR1 eight weeks after extraction rather than immediately: (i) the soft tissue volume available for flap management has increased; (ii) any residual pathology in the socket has resolved; (iii) resorption of the labial plate is complete by this stage. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

673 At the UL5 the ridge is broad palatally and thin buccally. At surgery you find that the implant will reach 35 Ncm only if it is placed about 2 mm palatal to the planned position and angled palatally; in the prosthetically correct position it reaches 15 Ncm and would leave a 6 mm buccal dehiscence. The patient is a healthy non-smoker. What should you do?

A. Place it where it is stable and graft the resulting buccal defect
B. Place it in the prosthetic position at 15 Ncm and graft simultaneously
C. Remove the implant, graft the ridge and return in six to nine months
D. Place a narrower implant in the prosthetic position and graft
E. Place it in the prosthetic position and plan an angulated screw channel

674 You are deciding whether to graft at the same visit as implant placement at the UR4 rather than staging the augmentation. Which of the following conditions must be satisfied before doing so?

A. The implant can be placed in the prosthetically correct position
B. The implant is primarily stable within native bone
C. The flap can be closed without a periosteal releasing incision
D. A and B only
E. A, B and C

675 Two staged horizontal augmentations are carried out on the same morning in two healthy non-smokers of similar age, with the same flap design and the same defect dimensions. Which single difference would most extend the period before the site can be re-entered for implant placement?

A. Fixation of the membrane with two tacks rather than with sutures
B. Use of a cross-linked rather than a native collagen membrane
C. Use of a periosteal releasing incision to obtain closure
D. Placement of the graft in the maxilla rather than the mandible
E. Use of a slowly substituted xenograft rather than autogenous bone alone

676 At re-entry six months after a staged augmentation at the UL5, the regenerate is firm at its base but the crestal 2 mm is soft and crumbles as the pilot drill engages, with a little fibrous tissue coming away on the flutes. The width at the base of the defect is adequate and the flap healed uneventfully. What is the most appropriate course?

A. Under-prepare the osteotomy and place an implant of wider platform
B. Place a shorter implant seated 2 mm deeper into the firm bone
C. Curette the soft crestal tissue, re-graft and re-enter in three to four months
D. Place the implant and extend the unloaded healing period to six months
E. Abandon implant treatment at this site and offer a bridge

677 A 4 mm wide maxillary ridge at the UL4 shows, on the cross-sectional CBCT slice, buccal and palatal cortices each about 1 mm thick with a clear cancellous layer between them, and 13 mm of height with no vertical deficiency. Which feature makes this site suitable for ridge expansion?

A. The intervening cancellous layer allows the buccal plate to be out-fractured in greenstick fashion
B. The maxillary cortex is thicker and therefore resists a complete fracture
C. The buccal plate has a dual blood supply from bone and from periosteum
D. Maxillary bone remodels quickly, so the expanded gap closes within weeks
E. Maxillary bone tolerates a full-thickness flap without devascularising the segment

678 A 3 mm wide knife-edge ridge at the LR5 shows on CBCT two cortical plates in direct contact with no cancellous bone between them, 14 mm of height above the mandibular canal and no vertical deficiency. Which finding most strongly contraindicates ridge expansion at this site?

A. The crestal width of 3 mm
B. The absence of a cancellous layer between the cortices
C. The mandibular rather than maxillary location
D. The height of 14 mm above the canal
E. The absence of attached mucosa over the crest

679 During a ridge expansion at the UL4 and UL5 the buccal segment fractures completely at its apical extent and becomes freely mobile, though it remains attached to the buccal periosteum along its length. No implant has yet been placed. What is the most appropriate management?

A. Remove the fragment and convert the case to a staged horizontal augmentation
B. Place the implants so that they stabilise the mobile segment
C. Discard the fragment and place narrower implants palatally
D. Reposition and fix the segment, graft the gap and close without placing implants
E. Close the site without grafting and re-enter at six months

680 CBCT at the UR4 in a 41-year-old non-smoker shows a crest 4.5 mm wide, 13 mm of height, a thin soft tissue phenotype and a high smile line. There is no vertical deficiency and the adjacent teeth are sound. Which plan is most appropriate?

A. Place a 3.3 mm implant without augmentation
B. Place a 4.1 mm implant with simultaneous contour grafting
C. Stage a horizontal augmentation and place the implant at six months
D. Expand the ridge and place the implant at the same visit
E. Provide a resin-bonded bridge and review the ridge annually

681 During harvest of a connective tissue graft from the lateral palate at the level of the second premolar in a 52-year-old on no anticoagulant, brisk pulsatile bleeding begins from the apical extent of the wound and does not settle with gauze pressure held for thirty seconds. What is the most appropriate immediate management?

A. Firm digital pressure for five minutes, then a horizontal mattress suture apical to the bleeding point
B. Electrosurgical cautery applied to the base of the wound
C. Oxidised cellulose packed into the wound and the patient discharged with a stent
D. Infiltration of the area with further adrenaline-containing local anaesthetic
E. Exploration to identify and ligate the greater palatine artery at its foramen

682 Ten days after harvest of a connective tissue graft from the lateral palate, a patient reports that the palate on that side feels numb from behind the canine back to the soft palate. The donor wound has epithelialised, is not painful, and the recipient site has healed well. What is the most likely explanation and outlook?

A. Nasopalatine nerve injury, with permanent numbness of the anterior palate
B. Greater palatine nerve injury, with recovery expected over weeks to months
C. Lesser palatine nerve injury, affecting the soft palate alone
D. Local anaesthetic neuropraxia, which should have resolved within a day
E. Middle superior alveolar nerve injury from the infiltration used

683 A free gingival graft about 1 mm thick is harvested and sutured buccal to an implant at the LL6 to widen the keratinised mucosa. It takes uneventfully with no necrosis, but at three months the band of keratinised tissue is appreciably narrower than the graft that was placed. Which process best explains this?

A. Primary contraction on removal, which is greater in thin grafts
B. Central necrosis, because diffusion could not sustain the graft
C. Loss of the lamina propria that determines keratinisation
D. Displacement by muscle pull before revascularisation
E. Secondary contraction during healing, which is greater in thin grafts

684 You want the most volume-stable autogenous graft available to thicken the buccal tissue at an implant at the UL2, in a patient whose palatal vault is of average depth and whose third molars are absent. Which harvest is most likely to deliver it?

A. A subepithelial connective tissue graft raised through a trap-door flap
B. A xenogeneic collagen matrix placed beneath a tunnelled flap
C. A free gingival graft sutured onto the buccal surface
D. A de-epithelialised free gingival graft taken as superficial lamina propria
E. A pedicled palatal roll flap rotated buccally

685 A patient who has read about "graft-free" soft tissue procedures refuses a palatal donor site and asks about xenogeneic collagen matrices. Which of the following may fairly be claimed for them?

A. They avoid a second surgical site and shorten operating time
B. They produce volume gain equal to that of autogenous connective tissue
C. They remove the need for an apically positioned flap when widening keratinised tissue
D. A and B only
E. A, B and C

686 An implant at the LR6 has 1 mm of keratinised mucosa buccally, mobile mucosa reaching the margin, and bleeds when the patient brushes it, which she finds uncomfortable. Bone levels have been stable for three years and the crown is screw-retained and well contoured. She asks whether grafting will "save the implant". Which is the most accurate statement to give her?

A. It will reliably reduce plaque accumulation, bleeding and discomfort at the site
B. It will arrest any future marginal bone loss at the implant
C. It will prevent peri-implantitis from developing at this implant
D. It will increase the thickness of the mucosa rather than its width
E. It will remove the need for supportive care at this implant

687 An implant crown at the UR2 placed elsewhere shows 2 mm of buccal mucosal recession with exposed titanium. The implant lies about 1 mm buccal to the ideal position, the phenotype is thin, the papillae are intact and the tissues are not inflamed. The patient wants the metal covered and asks what can be achieved. What should you tell her?

A. A coronally advanced flap with a connective tissue graft achieves complete coverage in most cases
B. Coverage is predictable so long as the interproximal papillae remain intact
C. The recession will resolve once the emergence profile of the crown is reduced
D. Coverage is less predictable than at a tooth and complete coverage cannot be promised
E. An enamel matrix derivative will improve the result as it does at teeth

688 A patient is being consented for harvest of a palatal connective tissue graft and is anxious about pain afterwards. Which of the following measures reduce donor-site morbidity?

A. A vacuum-formed palatal stent worn during the first week
B. A single-incision rather than a trap-door harvest
C. A five-day course of systemic antibiotics
D. A and B only
E. All of the above

689 At second-stage surgery at the UR5 there is a 1 mm buccal contour deficiency over an otherwise well-positioned implant. The palatal mucosa is notably thick, the keratinised band is adequate, and the patient is unwilling to accept a separate donor wound. Which technique is most appropriate?

A. A de-epithelialised pedicled palatal roll flap tucked buccally
B. A free gingival graft from the palate placed on the buccal aspect
C. A xenogeneic collagen matrix beneath a coronally advanced flap
D. An apically positioned flap to redistribute the keratinised tissue
E. A subepithelial connective tissue graft harvested from the tuberosity

690 To obtain tension-free closure over a staged augmentation at the LR4 and LR5 the buccal flap is advanced about 6 mm using a periosteal releasing incision, and the wound heals uneventfully. Which of the following should be anticipated as a consequence of that advancement?

A. A permanent reduction in the blood supply to the graft
B. Coronal displacement of the mucogingival junction and a shallower vestibule
C. Compression of the graft particles beneath the advanced flap
D. B and C only
E. A, B and C

691 An implant crown is planned at the UL2 adjacent to a natural UL1. On the periapical radiograph the distance from the proposed contact point to the crest of interproximal bone measures 7 mm. The patient has a high lip line and asks whether the papilla will fill the space. What is the most accurate answer?

A. The papilla fills completely in almost every case at this distance
B. The papilla fills completely in about half of such cases
C. The papilla fills completely in about a quarter of such cases
D. The papilla fills once the provisional crown has been contoured
E. The distance from contact point to bone does not predict papilla fill

692 Two adjacent implants are planned at the UR1 and UR2 in a 38-year-old with a high smile line and a thin phenotype. Which of the following statements about the soft tissue between two adjacent implants are correct?

A. A horizontal separation of less than 3 mm increases crestal bone loss between them
B. The soft tissue height achievable between two implants is lower than between a tooth and an implant
C. Splinting the two crowns increases the height of the interimplant papilla
D. A and B only
E. A, B and C

693 A black triangle 3 mm high is present between an implant crown at the UL1 and the natural UL2 four months after the definitive crown was fitted. The tissues are healthy and non-inflamed, the contact point to bone crest distance is 7 mm, and the patient is distressed by the appearance. What is the most appropriate management?

A. Surgical papilla reconstruction with a pedicle graft from the palate
B. A connective tissue graft placed through a tunnel beneath the papilla
C. Remake the crown with the contact point extended apically
D. Orthodontic extrusion of the UL2 to bring the interproximal bone coronally
E. Reassurance that the papilla will fill spontaneously within eighteen months

694 You are planning a transcrestal sinus floor elevation with simultaneous implant placement at the UR6. CBCT shows 6.5 mm of residual bone height, a membrane 1.5 mm thick, no septum in the site, and a sinus floor that slopes steeply, the angle between the medial and lateral antral walls at the intended site measuring about 30 degrees. Which of these findings most increases the risk of membrane perforation?

A. The residual bone height of 6.5 mm
B. The steeply sloping floor at the intended site
C. The membrane thickness of 1.5 mm
D. The absence of a septum in the site
E. The site being a first molar rather than a premolar

695 During a transcrestal elevation at the UL6 with 6 mm of residual bone height, the osteotome suddenly loses resistance and a Valsalva manoeuvre produces bubbling at the osteotomy. No graft material has yet been introduced and the patient is comfortable under local anaesthesia. What is the most appropriate course?

A. Continue, placing a collagen sponge to seal the tear before grafting
B. Place the implant without any graft and review at three months
C. Abandon the elevation and the graft, close the site and reassess in about two months
D. Convert to a lateral window in order to repair the membrane under direct vision
E. Graft with a xenograft and prescribe an antibiotic and a decongestant

696 CBCT taken before a planned transcrestal elevation at the UR6 shows 7 mm of residual bone height and a flat sinus floor, but also a mucosal thickening of about 12 mm filling the floor of the antrum, with the region of the ostium appearing obstructed. The patient reports no nasal symptoms and has no history of sinus disease. What should you do?

A. Proceed, since a thickened membrane resists tearing better than a thin one
B. Proceed under antibiotic and decongestant cover
C. Refer for an ear, nose and throat opinion before any elevation is undertaken
D. Proceed but graft with autogenous bone rather than with a xenograft
E. Proceed and warn the patient of an increased risk of post-operative sinusitis

697 An uneventful transcrestal sinus elevation with simultaneous implant placement has been carried out at the UR5 in a healthy 54-year-old. Which single post-operative instruction is most important in protecting the elevated membrane and the graft?

A. Avoid blowing the nose, and sneeze with the mouth open, for about two weeks
B. Use a chlorhexidine mouthrinse twice daily for two weeks
C. Continue a nasal decongestant for six weeks
D. Avoid chewing on that side until the implant is restored
E. Sleep with the head raised on two pillows for the first week

698 You plan a contour augmentation at the UR1 using deproteinised bovine bone mineral beneath a porcine collagen membrane in a 39-year-old who has asked nothing about the materials to be used. What must the consent discussion include?

A. The bovine and porcine origin of the materials, and the alternatives available
B. The batch numbers of both materials, recorded and given to the patient
C. A statement that all organic material is removed, so origin is immaterial
D. The theoretical transmission risk, but only if the patient raises the subject
E. Confirmation that a synthetic alternative would perform equally well

699 Five augmentation procedures are being planned for five different patients in your practice. Which of them should be referred rather than undertaken by a general dentist with a special interest in implant dentistry?

A. Guided bone regeneration of a 5 mm buccal dehiscence at a primarily stable implant
B. Transcrestal sinus elevation where residual bone height is 5 mm and the floor slopes
C. Alveolar ridge preservation of a socket whose walls are intact
D. A connective tissue graft to thicken the buccal mucosa at a single implant
E. Ridge expansion of a 4 mm maxillary crest with a cancellous interspace

700 A 58-year-old man has 8 mm of bone above the mandibular canal at the LR6 and a ridge 7 mm wide. He has read about bone grafting and asks you to build the ridge up so that a longer implant can be used, believing that a longer implant will last longer. He is a non-smoker with excellent plaque control. Which reply best reflects the principles that should govern augmentation?

A. Augmentation is justified only where it is needed to place the implant in the prosthetically correct position
B. Longer implants have better long-term survival, so his request is a reasonable one
C. Augmentation would be justified if he accepts the additional cost and the second surgical site
D. Augmentation should be offered because it will improve the crown-to-implant ratio
E. Augmentation is contraindicated in the posterior mandible in any circumstances

Block W — Restoring implants, occlusion and the laboratory, extended (Q701–760)

701 A laboratory compares two intraoral scanners by scanning a reference model carrying six implant analogues twenty times with each. The datasets from scanner A deviate from the reference by a mean of 61 micrometres, and the individual scans lie within 12 micrometres of one another. The datasets from scanner B deviate by a mean of 24 micrometres, but the individual scans are scattered across 88 micrometres. Which statement best describes the two instruments?

A. Scanner A has the better precision and scanner B the better trueness
B. Scanner A has the better trueness and scanner B the better precision
C. Scanner A is the more accurate because its scans agree with each other
D. Scanner B is the more accurate because its mean deviation is smaller
E. Neither trueness nor precision can be judged from bench data of this kind

702 An implant at the LL6 was placed 3 mm subcrestally and the peri-implant mucosa is 4 mm thick. When the scan body is seated, only about 3 mm of its cylindrical portion projects above the mucosa and is visible to the scanner. The definitive screw-retained crown returns with the access channel emerging through the mesiobuccal cusp instead of the central fossa. Which best explains the error?

A. Soft tissue contact along the scan body distorted the recorded surface
B. The axis was computed from a short length of cylinder, magnifying angular error
C. The scanner could not resolve the matt polymer surface at that depth
D. The library file assumes a fixed depth of the platform below the mucosa
E. Stitching error accumulated along the length of the scan path

703 You are taking implant-level impressions in two cases on the same morning: a single implant at the UR5 between two sound natural teeth, and five implants across an edentulous mandible that diverge from one another by up to 20 degrees. A colleague argues that impression copings should be splinted with resin in every case, because splinting can only improve accuracy. Which circumstances are supported by the evidence as benefiting from splinting?

A. Multiple implants recorded across an edentulous span
B. Marked divergence between adjacent implants
C. A single implant flanked by natural teeth
D. A and B only
E. All of the above

704 A 62-year-old woman requires an implant-level impression at the LR6. Her maximum interincisal opening is 26 mm following long-standing temporomandibular disorder, and she has a pronounced gag reflex. Your practice has no intraoral scanner. A colleague suggests an open-tray technique with a custom tray. What is the principal reason that technique is impractical here?

A. The coping screw cannot be unscrewed through the tray aperture at this opening
B. The impression material cannot be syringed around a coping of that length
C. The tray and its projecting copings need more vertical clearance than she has
D. The coping is likely to be displaced by the tongue during seating of the tray
E. The set impression cannot be recovered elastically over a long parallel coping

705 A full-arch maxillary reconstruction on six implants is being recorded, and the laboratory offers photogrammetry, in which coded markers screwed to the implants are photographed from multiple angles and their relative positions computed. The technique reports very small deviations across long spans. Which is the principal limitation you must plan around?

A. Deviation increases sharply once more than four implants are recorded
B. The technique cannot resolve implants that diverge by more than 15 degrees
C. Accuracy falls where the distance between adjacent markers exceeds 20 mm
D. It records only implant positions, so soft tissue must be captured separately
E. The markers must be splinted with resin before the images are captured

706 Before syringing impression material around five splinted open-tray copings in an edentulous mandible, you take periapical radiographs of each. Four interfaces appear closed. At the LL6 a radiolucent line about 0.2 mm wide runs uniformly across the whole width of the coping–implant interface, with no wedge shape and no tapering. What should you do?

A. Proceed, since a uniform line of this width indicates beam angulation
B. Remove the coping, clear the platform, reseat it and repeat the film
C. Tighten the coping screw further and repeat the radiograph
D. Proceed, and have the discrepancy corrected on the master cast
E. Substitute a coping of a different length at that implant

707 A colleague states that intraoral scanning has now been shown to be more accurate than conventional impressions for all implant work, and proposes to abandon splinted open-tray impressions altogether, including for a ten-unit screw-retained framework. Which statement best represents the current evidence?

A. Scanning is more accurate than conventional impressions at every span length
B. Conventional impressions remain more accurate even for single implant crowns
C. The two are comparable over short spans, and scanning deviates more across an arch
D. Neither method achieves a deviation small enough for a passive framework
E. The differences reported between the methods have no clinical consequence

708 A ten-unit maxillary framework produced from an intraoral scan of six implants in an edentulous arch will not seat, and you are reviewing what went wrong. The scan was taken with the patient reclined, the mucosa was not retracted, the sulci had been bleeding after healing abutment removal, and the scan was interrupted twice for the patient to rinse and then resumed. Which of the following increased the deviation?

A. A long span of mobile mucosa with no fixed landmarks between scan bodies
B. Blood and saliva lying on the surfaces of the scan bodies
C. Interruption of the scan and its resumption after re-registration
D. A and B only
E. All of the above

709 Six implants have been placed in an edentulous maxilla for a screw-retained fixed prosthesis. The two posterior implants on each side were tilted to avoid the sinuses, and the axes of adjacent implants diverge by up to 30 degrees. The laboratory asks for the impression to be taken at multi-unit abutment level rather than implant level. Which is the principal advantage in this particular case?

A. The definitive abutments are placed once and never subsequently removed
B. The abutment copings are shorter and easier to accommodate in the tray
C. Fewer components are required than in an implant-level workflow
D. The abutments provide a common path of insertion despite the divergence
E. The prosthetic screws may be torqued to a higher value than implant screws

710 An interocclusal record is required for an edentulous maxilla restored with six implants, opposing a natural mandibular dentition. A trainee proposes an acrylic base resting on the mucosa carrying a wax rim softened in a water bath, closed at the vertical dimension of the patient's existing well-worn denture. Which of the following would reduce error in the record?

A. A record base screwed to the implants rather than resting on the mucosa
B. A softened wax rim that allows the patient to close fully into it
C. Recording the relationship at the height of the existing denture
D. A and B only
E. A, B and C

711 A full-arch mandibular framework produced from an intraoral scan of five implants fails the one-screw test at try-in with a visible gap at the terminal abutment. The scan bodies were undamaged, correctly seated and taken from the manufacturer's own range, and the field was dry. Before rescanning, which modification is most likely to reduce the deviation?

A. Scanning in a single continuous sweep beginning at a terminal implant
B. Joining the scan bodies with a rigid resin bar before the arch is scanned
C. Fitting scan bodies of larger diameter at each of the five implants
D. Drying and retracting the mucosa with cheek retractors during the scan
E. Taking three shorter scans and merging them within the software

712 A ten-unit screw-retained framework is being assessed at try-in. The screws all reach the manufacturer's torque value without difficulty, the patient reports no discomfort at any stage, and the framework appears fully seated when every screw is tightened. Which of the following would demonstrate that the framework does not seat passively?

A. A gap opening at the far abutment when only one terminal screw is tightened
B. The torque value reached at each screw during final tightening
C. The patient's report of discomfort as the screws are tightened
D. A and B only
E. A, B and C

713 A 57-year-old woman treated five years ago for generalised periodontitis, now stable, had a crown cemented at the UL5 on a custom abutment with the margin 2 mm below the mucosal margin palatally. At review four years later there is suppuration, a 7 mm pocket and 3 mm of bone loss. A periodontally healthy colleague of hers, restored in the same way at the same visit, has no problem. Which best explains the difference in outcome?

A. Resin cements are more retentive and therefore harder to remove completely
B. A deep margin prevents residual cement from being seen on a radiograph
C. Retained cement produces peri-implant disease far more often in periodontitis patients
D. Cement-retained crowns show lower survival than screw-retained crowns overall
E. Cement remnants act as a nidus for calculus deposition on the titanium surface

714 An implant at the UR1 emerges about 20 degrees labial to the ideal axis, and a restoration using an angulated screw channel abutment is planned so that the access opening lies palatally. The laboratory asks you to confirm the components required. Which of the following are true of angulated screw channel systems?

A. They redirect the screw access channel by up to about 25 degrees
B. They require a dedicated screw and a matching angled driver
C. They permit a higher torque than a straight access channel
D. A and B only
E. All of the above

715 A colleague routinely cements definitive implant crowns with a provisional zinc oxide-based cement, arguing that this keeps the restoration retrievable without the compromises of screw retention. His crowns have shallow, supramucosal margins and he checks them at every recall. What is the strongest argument against the practice?

A. Partial loss of the seal is silent and permits bacterial colonisation of the margin
B. Provisional cements are radiolucent and cannot be identified on a radiograph
C. Repeated recementation wears the abutment and progressively reduces retention
D. A crown that decements may be swallowed or, rarely, inhaled by the patient
E. Provisional cements have a greater film thickness and raise the occlusion

716 A bone-level implant at the UR1 is correctly angulated, so that screw access emerges just palatal to the incisal edge. The platform lies 3.5 mm below the mid-facial mucosal margin because the site was augmented and the implant placed deep. You are choosing between a custom zirconia abutment carrying a cemented crown and a zirconia crown bonded extra-orally to a prefabricated titanium base. Which is the principal advantage of the second option in this case?

A. It allows the emergence profile to be individualised more precisely
B. It removes cement from the peri-implant sulcus altogether
C. It maintains a titanium-to-titanium interface at the implant connection
D. It reduces the risk of fracture of the zirconia at the margin
E. It permits the restoration to be removed without an access channel

717 A trainee has been taught that screw retention is always preferable and asks you to justify the cemented crowns in the practice. The case in front of you is a molar implant whose axis emerges through the mesiobuccal cusp, in a patient with no aesthetic concern and 7 mm of interocclusal space. Which of the following are genuine advantages of cement retention?

A. The cement layer compensates for a gross framework misfit
B. The occlusal surface is left intact where access would emerge on a cusp
C. The restoration can be retrieved without being destroyed
D. A and B only
E. A, B and C

718 A screw-retained crown has been torqued at the LR6 and the access channel is to be sealed. A colleague places a cotton wool pellet over the screw head and covers it with composite. Which is the principal objection to the cotton?

A. Cotton fibres bond into the composite and weaken the seal at its base
B. Cotton cannot be removed cleanly when the restoration is later retrieved
C. Cotton absorbs fluid and supports bacterial growth within the channel
D. Cotton does not protect the screw head from the overlying composite
E. Cotton compresses over time and allows the composite plug to loosen

719 A 49-year-old man has had recurrent peri-implant mucositis at the UL4 despite excellent plaque control and a screw-retained crown with a shallow, cleansable emergence. Each time the crown is removed there is a foul odour from the internal cavity of the implant, which has an external hexagon connection. Which feature of a connection most limits bacterial leakage at the interface?

A. A long conical friction fit between the abutment and the implant wall
B. A platform switch of 0.5 mm at the abutment margin
C. An abutment screw torque above the manufacturer's specified value
D. A silicone sealing ring seated within the internal cavity
E. An external hexagon with a machined flat-to-flat mating surface

720 An implant crown at the UR2 has been fitted and the patient is unhappy that a dark triangle remains between the UR1 and the crown. The mucosa is healthy, the crown contour is correct, and a paralleling periapical shows the interproximal bone crest on the distal of the UR1 lying 7 mm below the contact point, with the bone at the implant's mesial aspect 1 mm below its platform. Which of the following determine the height of that papilla?

A. The distance from the contact point to the bone crest on the UR1
B. The interproximal bone level at the mesial aspect of the implant
C. The width of keratinised mucosa on the buccal aspect of the crown
D. A and B only
E. A, B and C

721 A colleague has read that an emergence angle greater than 30 degrees is associated with peri-implantitis, and proposes to remake a well-functioning crown on a tissue-level implant at the LR6 whose measured emergence angle is 38 degrees. The tissues are healthy, there is no bleeding on probing and the bone level is unchanged over four years. Which statement best represents the published finding?

A. The association applies equally to bone-level and tissue-level designs
B. The association was demonstrated for bone-level implants and not for tissue-level
C. The association applies only where the restoration is cement-retained
D. The association becomes significant only beyond about 45 degrees
E. The association reflects the convexity of the profile rather than its angle

722 An implant with a 5.0 mm platform has been placed at the LR6 with the platform only 1 mm below the mid-buccal mucosal margin. The crown must be about 10.5 mm wide buccolingually to meet the adjacent teeth and the opposing dentition. Which consequence follows most directly from the position at which the implant has been placed?

A. The abutment will be too short to retain a cemented crown adequately
B. The screw access channel will emerge outside the central fossa
C. The transition must occur within 1 mm, forcing a steep emergence profile
D. The mucosal margin will migrate apically during the first year of function
E. The proximal contacts will sit too far from the interproximal bone crest

723 An implant at the UL1 has been placed by a colleague with its platform 4 mm below the mid-facial mucosal margin, in a patient with a thick phenotype. The implant is otherwise well positioned in all three dimensions and is stable. You are planning the restoration and explaining the implications to the patient. Which of the following follow from that depth?

A. Cement cannot be removed reliably from the sulcus if cement retention is used
B. Probing depths at the implant will be deep even when the tissues are healthy
C. The restoration must carry a long transmucosal section to reach the mucosa
D. A and B only
E. All of the above

724 A screw-retained provisional at the UR1 has been modified three times over twelve weeks, and the tissues are healthy with a thick phenotype. You must decide whether the emergence profile is ready to be recorded for the definitive restoration. Which observation best indicates that it is?

A. The provisional has now been in place for at least three months
B. The papillae completely fill both interproximal embrasures
C. The mucosa no longer bleeds when the provisional is unscrewed
D. The margin has been unchanged at successive visits and blanching settles quickly
E. The patient is satisfied with the appearance of the provisional crown

725 A conditioned emergence profile at the UR1 is being transferred conventionally. The provisional is unscrewed, secured to an analogue held in putty, and an index is made; the provisional is returned to the mouth and an impression coping is seated in the index and the void filled with flowable composite. The definitive crown arrives with a subgingival contour that supports the tissue well on the palatal aspect but is deficient labially. Which is the most likely cause?

A. The flowable composite contracted as it polymerised within the index
B. The putty index distorted as the provisional was withdrawn from it
C. The coping was seated before the flowable composite had been dispensed
D. The provisional was reseated in the mouth before the index had fully set
E. The coping was seated in the index in a different rotational position

726 A 32-year-old woman with a thin, scalloped phenotype has an implant at the UR1. The provisional's critical contour was made markedly convex at the time of second-stage surgery in order to push the mucosal margin apically to match the UL1. At four weeks the mid-facial margin has receded 1.5 mm beyond the intended position and the mucosa there looks grey. What is the most likely explanation?

A. The titanium abutment is showing through mucosa that has become thinner
B. The provisional has been removed repeatedly, disturbing the mucosal seal
C. The implant has been placed too far labially within the alveolar ridge
D. Pressure from the over-convex critical contour has produced mid-facial recession
E. A connective tissue graft placed at the time of surgery has failed to integrate

727 A full-arch fixed mandibular prosthesis on five implants is being designed for a 70-year-old man with moderate manual dexterity, who will be reviewed four-monthly. The technician asks how the intaglio surface and the retention should be specified so that the prosthesis can be maintained. Which of the following improve cleansability?

A. A convex intaglio surface in light contact with the mucosa
B. Screw retention allowing removal for professional cleaning
C. A ridge-lap extension covering the crest of the residual ridge
D. A and B only
E. All of the above

728 A technician offers three custom abutments for a UL5: a cast gold alloy abutment, a machined titanium abutment and a zirconia abutment. The margin will lie 0.5 mm below the mucosa, the phenotype is thick, and there is no aesthetic demand at this site. A trainee asks whether the material makes any difference to the soft tissues. Which statement best represents the evidence?

A. All three permit an equivalent soft tissue attachment at the transmucosal surface
B. Gold alloy is associated with an apical shift of the attachment and crestal bone loss
C. Zirconia provokes more inflammation than titanium at the transmucosal surface
D. The attachment depends on surface roughness rather than on the material itself
E. Material affects the attachment only where the abutment margin is submucosal

729 At a six-month review an implant crown at the LR6 marks heavily and broadly when 40 micrometre articulating paper is used, and the patient reports no symptoms. Shimstock foil 12 micrometres thick pulls freely through the contact on the crown, while it is firmly held at the LR5 and LR7 in light closure. Which is the correct interpretation?

A. The crown is in premature contact and requires occlusal adjustment
B. The intensity of the paper mark indicates the force at that contact
C. The contact on the crown is lighter than the adjacent teeth, as intended
D. The crown has intruded relative to the two adjacent natural teeth
E. Shimstock of this thickness is too thin to assess an implant restoration

730 A 44-year-old man requires an implant crown at the UR1. He has a class II division 2 incisor relationship with an overbite of 6 mm and an overjet of 1 mm, an intact unworn dentition and no parafunction. The implant is well positioned and the crown height from the platform will be 11 mm. Which feature of this occlusion most threatens the restoration?

A. The absence of posterior disclusion during protrusion
B. The steep guidance angle generating a large horizontal force
C. The reduced interocclusal space available for restorative material
D. The likelihood of an anterior interference during lateral excursion
E. The tendency of the lower incisors to overerupt against the crown

731 A 55-year-old woman with an implant crown at the LR6 reports that the tooth "catches" when she chews on the left. Examination shows a contact on the mesiolingual cusp of the implant crown during left lateral excursion, with the right side disoccluded, and the crown holds shimstock in intercuspal position more firmly than its neighbours. Why is the excursive contact she has developed particularly damaging?

A. It applies a large horizontal force with no proprioceptive protection
B. It loads the implant during the closing stroke of every masticatory cycle
C. It generates a compressive force directed along the implant long axis
D. It transfers load preferentially to the terminal implant of the segment
E. It occurs at a point where the elevator muscles are at their weakest

732 A full-arch fixed mandibular prosthesis on six implants is being fitted opposite an intact natural maxillary dentition. A trainee proposes relieving every unit of the prosthesis by 25 to 30 micrometres, as she has been taught for implant-protected occlusion, so that the prosthesis contacts only on firm clenching. Which statement best answers her proposal?

A. The relief should be doubled, since no unit of the prosthesis has a ligament
B. Even, simultaneous contacts should be provided across the whole prosthesis
C. The relief applies only where the opposing arch is also implant-borne
D. Contacts should be confined to the premolar region of the prosthesis
E. Occlusal contacts should be established only in the anterior segment

733 A 47-year-old man has bilateral masseteric hypertrophy, wear facets that his diet does not explain and a partner who reports night-time grinding. He needs two implant crowns in the lower right quadrant. Consider these statements: (i) sleep bruxism is associated with an increase in technical complications; (ii) sleep bruxism is associated with a marked reduction in implant survival; (iii) providing a hard occlusal splint should form part of the treatment plan. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

734 A colleague specifies acrylic occlusal surfaces on a full-arch implant-supported prosthesis, explaining that the material will absorb occlusal force and protect the implants from overload, and that this is why he never uses ceramic in such cases. Which statement best answers that reasoning?

A. Acrylic reduces the peak force reaching the implant by about one third
B. Acrylic transmits less force to the implant but wears more rapidly
C. Acrylic protects the opposing natural dentition from abrasive wear
D. Acrylic allows the occlusion to be adjusted more readily at review
E. The occlusal material has little effect on the force transmitted to the implant

735 A 26-year-old man had an implant crown placed at the UR1 at the age of 20 following trauma. He now attends because the crown looks short and its incisal edge sits about 1 mm above those of the adjacent incisors, and the mucosal margin appears more coronal than at the contralateral tooth. The crown is sound, the screw is tight and the bone level is unchanged on a paralleling periapical. Which is the most likely explanation?

A. The crown has worn appreciably at its incisal edge over six years
B. The implant has intruded into the bone under occlusal loading
C. The abutment screw has settled progressively into the connection
D. The adjacent natural dentition has continued to erupt around it
E. The peri-implant mucosa has become hyperplastic around the crown

736 A patient with an implant crown at the LR6 placed four years ago attends with food packing between the crown and the LR5, and floss passes without resistance. Consider these statements: (i) open proximal contacts adjacent to implant restorations affect roughly a third of cases; (ii) they occur more often at the mesial than the distal contact; (iii) they arise because the implant continues to erupt with the dentition. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

737 An implant crown at the LR6 carries a single occlusal contact lying 3 mm buccal to the long axis of the implant. You intend to narrow the occlusal table so that the contact lies 1 mm buccal to the axis instead, with the magnitude of the applied vertical force unchanged. By what proportion does the bending moment at the implant platform fall?

A. One quarter
B. One half
C. Two thirds
D. Three quarters
E. It is unchanged, since the force is unaltered

738 A 58-year-old woman has an implant crown at the LL6 fitted three weeks ago. Shimstock is held firmly at the crown in light closure and pulls freely at the adjacent LL5 and LL7. She says the bite feels entirely comfortable and that she has noticed nothing. Which best explains why her report cannot be relied upon?

A. Loss of periodontal mechanoreceptors raises the active tactile threshold
B. An osseointegrated implant transmits force directly into cortical bone
C. The masticatory muscles adapt to an altered contact within a few days
D. The opposing natural tooth intrudes and masks the premature contact
E. Bone has a lower elastic modulus than the periodontal ligament

739 An edentulous mandible has been restored with four implants for a fixed prosthesis. The anteroposterior spread, from the centre of the most anterior implant to the line joining the distal aspects of the terminal implants, measures 8 mm. The patient wants first molars, which would require a cantilever of about 15 mm distal to each terminal implant. He is a non-smoker with a natural maxillary dentition and no parafunction. What is the most appropriate response?

A. Proceed with 15 mm, since the opposing arch will be adjusted to reduce load
B. Limit the cantilever to about 12 mm and accept a shorter dental arch
C. Place a fifth implant anteriorly to increase the anteroposterior spread
D. Splint the prosthesis more rigidly so that the effective spread is increased
E. Retain the 15 mm cantilever but narrow the occlusal table of the molars

740 A 61-year-old man is having the LR4, LR5 and LR6 replaced, and bone volume allows implants at any of the three sites. Two implants are affordable. The choice lies between implants at the LR4 and LR6 with a pontic at the LR5, and implants at the LR4 and LR5 with a cantilever pontic at the LR6. Occlusal loading and bone quality are comparable at all three sites. Which is preferable, and for what reason?

A. Implants at the LR4 and LR6, because the pontic lies between the two supports
B. Implants at the LR4 and LR5, because anterior bone is of better quality
C. Implants at the LR4 and LR6, because the pontic can then be narrowed
D. Implants at the LR4 and LR5, because the cantilever length can be limited
E. Either arrangement, because rigid splinting equalises load in both designs

741 A 6 mm implant has been placed at the LR6 to avoid a sinus elevation, and the definitive crown will have a height of 13 mm from the platform, giving a crown-to-implant ratio of about 2.2 to 1. The patient has a full natural opposing dentition, no parafunction and good bone quality. Which measure most reduces the bending moment applied at the implant platform?

A. Splinting the crown to the adjacent natural LR5 with a rigid connector
B. Narrowing the occlusal table of the crown buccolingually
C. Eliminating all excursive contacts on the implant crown
D. Selecting a monolithic zirconia crown rather than a metal-ceramic one
E. Torquing the abutment screw to the highest value the system permits

742 Two 8.5 mm implants have been placed at the LR6 and LR7 in a healthy 54-year-old non-smoker with no parafunction and good bone quality. The technician asks whether the crowns should be splinted, and a colleague asserts that splinting always reduces marginal bone loss and should be routine. Which statement best represents the evidence?

A. Splinting reduces marginal bone loss by about 0.5 mm at five years
B. Splinting reduces implant failure rates in the posterior mandible
C. Splinting is contraindicated because it obstructs interproximal cleaning
D. Bone levels are similar, and splinting is chosen for mechanical reasons
E. Splinting is indicated only where the two implants diverge appreciably

743 Three implants are planned at the LR5, LR6 and LR7 in a ridge of ample width. A colleague proposes displacing the middle implant 2 mm buccally relative to the other two, on the grounds that the resulting tripod arrangement will distribute occlusal load between the three implants more favourably than a straight line. Which statement best represents the position?

A. Offsetting reduces the bending moments at each implant by about one third
B. Offsetting is essential wherever the implants are shorter than 8 mm
C. Offsetting improves load distribution only where the crowns are splinted
D. Offsetting distributes the load evenly between the three implants
E. Offsetting has not been shown to reduce load and worsens the emergence

744 A three-unit implant-supported prosthesis with a zirconia framework and a veneering ceramic has chipped twice in eighteen months, once at a marginal ridge and once at a buccal cusp. The occlusion has been checked and corrected, the framework supports the veneer evenly to within a millimetre of the finished contour, and the screws are sound. Which laboratory factor is most likely to be responsible?

A. The veneering ceramic was fired at too high a temperature for the framework
B. The restoration was cooled too rapidly after the final ceramic firing
C. The zirconia framework was air-abraded before the veneer was applied
D. The thermal expansion coefficients of framework and veneer were mismatched
E. The zirconia framework was sintered at an inadequate temperature

745 A monolithic zirconia crown is planned for an implant at the LR7 in a 46-year-old man with sleep bruxism and heavy masseters, with no aesthetic demand at this site. The technician offers a conventional 3 mol% yttria-stabilised zirconia and a highly translucent multilayer grade, and recommends the translucent material because it looks better. Which consideration should most influence the choice?

A. Translucent grades have equivalent strength and superior optical properties
B. Translucent grades are tougher because of their larger grain size
C. A higher cubic phase content raises translucency and lowers strength
D. The strength difference between the grades matters only in thin sections
E. Translucent grades are more susceptible to low-temperature degradation

746 A monolithic zirconia crown bonded to a prefabricated titanium base at the UL4 has debonded twice in a year, each time leaving the base and screw intact and correctly torqued. The technician air-abraded the titanium with 50 micrometre alumina at low pressure and used a resin cement containing 10-MDP, and the titanium base has a gingival height of 2.5 mm. Which factor most increases the risk of a further debond?

A. The short bonding height of the titanium base at 2.5 mm
B. The use of a resin cement containing 10-MDP for the bond
C. Air abrasion of the titanium surface before the cement was applied
D. Screw retention of the assembly rather than cementation intraorally
E. The crown being monolithic zirconia rather than veneered ceramic

747 A patient attends because the crown at the UL4 moves. It is a monolithic zirconia crown bonded to a titanium base and screw-retained. When you remove the access seal, the screw is found to be tight and reaches the manufacturer's torque value without turning further, and a paralleling periapical shows no gap at the implant–abutment interface and an unchanged bone level. What is the most likely explanation?

A. The abutment screw has fractured immediately below its head
B. The implant has lost osseointegration but remains clinically stable
C. The internal threads of the implant have stripped under load
D. The zirconia crown has debonded from the titanium base beneath it
E. The proximal contacts have opened, permitting the crown to move

748 A 68-year-old man has worn a metal-acrylic full-arch fixed mandibular prosthesis on four implants for six years, opposing a complete maxillary denture. He attends for review and asks what is likely to need attention over the next few years, having so far had no problems. Which of the following are recognised technical complications of this type of prosthesis?

A. Fracture of the acrylic denture teeth
B. Wear of the occlusal surfaces over time
C. Separation of the acrylic from the metal framework
D. A and B only
E. All of the above

749 An immediate full-arch provisional is to be made on four implants in an edentulous mandible, two of them tilted, with a distal cantilever of 10 mm on each side. The provisional will be worn for about six months while the tissues mature. The patient has heavy masseters. Which specification most reduces the risk of the provisional fracturing?

A. A thicker layer of acrylic processed over the prosthetic cylinders
B. A rigid metal bar incorporated within the acrylic provisional
C. Milled polymethyl methacrylate rather than conventional acrylic
D. A hard occlusal splint to be worn over the provisional at night
E. Reduction of the buccolingual width of the occlusal table

750 A trainee has learned that an abutment screw torqued to 35 Ncm stores a clamping force, and asks what actually determines whether the joint between abutment and implant begins to open during function. The restoration is a single molar crown with a well-centred occlusal contact and no cantilever. Under which condition does the joint begin to separate?

A. When the applied occlusal force exceeds the yield strength of the screw
B. When the delivered torque falls below the value the manufacturer specifies
C. When the tensile force at the joint exceeds the residual preload in the screw
D. When the coefficient of friction at the thread flanks falls during function
E. When the settling effect has removed all elongation stored in the screw

751 A full-arch framework is being delivered on multi-unit abutments. The manufacturer specifies 30 Ncm for the abutment screws that fix the abutments to the implants, and 15 Ncm for the prosthetic screws that fix the framework to the abutments. A trainee asks why the second value is only half the first, since both are titanium alloy screws in the same assembly. Which is the best explanation?

A. The prosthetic screw is longer and therefore stretches further at a given torque
B. The framework would distort if the prosthetic screws were torqued more highly
C. The abutment screw is shielded from occlusal load by the abutment above it
D. The prosthetic screw is smaller and is designed to be the component that fails
E. The prosthetic screw engages a shallower thread cut in a softer alloy

752 An abutment screw is being torqued with a beam-type wrench calibrated last month. The operator holds the handle at its very end with one hand, steadies the beam near its midpoint with a finger of the other hand to keep the assembly still, applies the load smoothly, and reads the scale from directly above. Which of these handling features most alters the torque actually delivered?

A. Applying the load smoothly rather than in a single rapid movement
B. Steadying the implant and abutment with the free hand during tightening
C. Using the wrench after repeated cycles of steam sterilisation
D. Steadying the beam with a finger part-way along its length
E. Reading the scale from directly above rather than obliquely

753 A ten-unit framework shows a gap of about 60 micrometres at the terminal abutment when the opposite terminal screw is tightened. The operator elects to close it by tightening all the screws to the specified value, after which the framework appears fully seated and the radiograph shows closed interfaces. What happens to the clamping force at the screw that closed the gap?

A. It is unchanged, because torque alone determines the preload achieved
B. It rises, because that screw has been made to elongate further than the others
C. It falls, because the mating surfaces are not in full contact with one another
D. It rises, because friction is lower across an incompletely seated interface
E. It falls, because part of the screw's elongation is spent closing the gap

754 A 53-year-old man has had an abutment screw fracture at the LR6, and you are considering what predisposed him to it before restoring the implant again. His crown has a 6 mm distal cantilever, the screw had loosened twice in the preceding two years and been retightened each time, and the same screw had been in service since the crown was fitted five years ago. Which of the following increase the risk of abutment screw fracture?

A. Repeated loosening that has been retightened without correcting a cause
B. A distal cantilever on the restoration
C. Reuse of a screw that has been torqued and untorqued repeatedly
D. A and B only
E. All of the above

755 An abutment screw has fractured at the LL5, leaving the apical fragment within the internal threads with its fractured surface just below the platform. The implant is osseointegrated and the mucosa is healthy. You have applied fine ultrasonic tips in an anticlockwise direction under magnification with irrigation for fifteen minutes, without the fragment moving. What is the most appropriate next step?

A. Cut a slot across the fragment with a small round bur and use a flat-bladed driver
B. Irrigate with a penetrating solvent and leave the site for a week before retrying
C. Use the manufacturer's retrieval kit with its reverse-cutting retrieval instrument
D. Restore the implant with a crown cemented to the conical connection instead
E. Remove the implant with a counter-torque device and place a replacement

756 A patient collects a definitive implant crown made by an external laboratory on a custom titanium abutment, and asks for the documentation that accompanies it, explaining that she keeps papers for all her implanted devices. You check what the laboratory has supplied with the work. Which of the following must appear on the statement provided with a custom-made device of this kind?

A. The UKCA mark applied to the finished restoration
B. The name of the prescriber and data identifying the patient
C. The GDC registration number of the technician who made it
D. A and B only
E. A, B and C

757 A five-unit screw-retained framework will not seat, and on enquiry the laboratory tells you that it did not use the implant manufacturer's own interface components but a compatible third-party equivalent, which it has used for two years without telling you. The patient is in the chair and the framework has been returned for remaking. Which is the principal concern arising from what you have been told?

A. The finished prosthesis no longer qualifies as a custom-made device
B. You cannot state which components were used or trace them subsequently
C. The manufacturer's specified torque values may no longer be applicable
D. The laboratory has acted in breach of its contract with the practice
E. The restoration cannot properly be described to the patient as an implant crown

758 An edentulous mandible is to be restored with a two-implant overdenture for a 74-year-old woman with a resorbed ridge and limited manual dexterity. On the mounted casts the distance from the implant platforms to the planned occlusal plane measures 9 mm anteriorly, and the denture must retain a labial flange for lip support. Which design is most appropriate?

A. A stud attachment on each implant with a matrix in the denture base
B. A milled bar with two short distal extensions and a machined fit
C. A round bar joining the implants with a single clip in the base
D. A magnetic attachment on each implant beneath the denture base
E. A fixed prosthesis carried on the two anterior implants alone

759 A 72-year-old man is being consented for a two-implant mandibular overdenture with stud attachments. He is delighted by the prospect and asks what will be involved once it is finished, saying that his current denture has needed nothing in eight years. Which of the following should be included in your explanation of the maintenance he should expect?

A. Periodic replacement of the retentive inserts as they wear
B. Relining of the denture base as the posterior ridges resorb
C. Possible fracture of the acrylic base over the attachment housings
D. A and B only
E. All of the above

760 Two implants placed in the anterior mandible for an overdenture diverge from one another by about 35 degrees on the mounted casts, one being angled markedly distally. The ridge is favourable, the restorative space is generous at 14 mm, and the patient has good dexterity. Which is the strongest argument for a bar rather than two independent stud attachments?

A. A bar distributes occlusal load more evenly to the underlying mucosa
B. A bar allows a shorter labial flange to be used on the denture
C. A bar provides a single path of insertion despite the divergence
D. A bar generates greater retention than two stud attachments
E. A bar reduces the frequency with which attachments need maintenance

Block X — Complications, peri-implant disease, maintenance and governance, extended (Q761–800)

761 A 66-year-old man attends six years after an implant was placed at the UR4. It was placed deliberately deep to manage a thin phenotype, and probing depths are 6 mm circumferentially. There is no suppuration, no discomfort, and the bone level lies 3 mm below the platform, unchanged from the film taken at prosthesis fitting. His plaque score is 11 per cent. Which finding, were it present, would exclude a diagnosis of peri-implant health?

A. A probing depth that had increased to 7 mm since the last visit
B. A radiographic bone level 4 mm apical to the implant platform
C. Bleeding on gentle probing at one of the six recorded sites
D. A complete absence of keratinised mucosa on the buccal aspect
E. A full-mouth plaque score recorded above 25 per cent

762 A 54-year-old woman has an implant crown at the LL5 fitted three years ago. All six sites bleed on gentle probing, one suppurates, probing depths are 6 mm, and the mucosa is swollen. You hold the periapical taken at prosthesis fitting, which shows bone at the level of the platform, and today's paralleling film shows the bone 2.5 mm apical to the platform. Which finding establishes the diagnosis of peri-implantitis rather than mucositis?

A. The probing depth of 6 mm at all six recorded sites
B. The suppuration from one of the peri-implant sites
C. The bleeding on gentle probing at every site examined
D. The bone level 2.5 mm apical to the baseline radiograph
E. The three years that have elapsed since the prosthesis was fitted

763 A patient transfers to you with an implant at the LR6 placed eight years ago. No radiograph was taken when the prosthesis was fitted, but a periapical exists from the one-year review showing bone at the first thread. There is now bleeding and suppuration, a 7 mm pocket, and the bone lies 3 mm apical to that level. Which reference should be used to judge whether bone has been lost beyond initial remodelling?

A. The 6 mm probing threshold, since no film exists from the fit appointment
B. The one-year film, because initial remodelling is complete by that stage
C. Today's film alone, since bone loss cannot be quantified without a fit-time film
D. The surgical record of how far the platform was placed below the crest
E. The pre-operative cone beam examination, which shows the original crest

764 You are teaching a trainee the clinical signs that support a diagnosis of peri-implantitis. Consider these statements: (i) bleeding on gentle probing must be present before the diagnosis can be made; (ii) suppuration may be found at a site that does not bleed on probing; (iii) an increase in probing depth compared with previous examinations supports the diagnosis. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only
E. (i), (ii) and (iii)

765 A colleague argues that peri-implantitis progresses slowly and steadily in the way that periodontitis does, and that annual review with a radiograph every two years is therefore sufficient for his implant patients, several of whom have a periodontitis history. Which statement best answers his position?

A. Peri-implantitis tends to progress in a non-linear, accelerating pattern
B. Peri-implantitis progresses more slowly than periodontitis at matched sites
C. The rate of progression can be predicted from the bone loss at diagnosis
D. Progression is linear but begins earlier than it does in periodontitis
E. Rates of progression cannot be estimated from the available cohort data

766 A patient who has read about implants asks why published figures for peri-implantitis vary so widely, quoting a range from about 1 to 47 per cent that she has found. She wants to know which figure applies to her. Which explanation best accounts for the spread of the reported estimates?

A. Differences in the implant surfaces used in the studies contributing data
B. Differences in the length of follow-up between the contributing cohorts
C. Differences in the case definition and bone-loss threshold applied
D. Differences between reporting at patient level and at implant level
E. Differences between the geographical populations that were studied

767 A practice is deciding when to concentrate its radiographic and clinical surveillance of implant patients, and a colleague suggests that resources should be directed at the ten-year point, since peri-implantitis is a late complication of long service. Which statement best represents the evidence on when peri-implantitis begins?

A. It becomes apparent after ten years in most implants that are affected
B. It becomes apparent within the first year in most implants affected
C. Onset is distributed evenly across the years of function
D. Onset is commonly within the first three years of function
E. Onset cannot be dated because baseline radiographs are rarely available

768 You are explaining to a patient why you propose four-monthly maintenance after her implants are restored, and she asks which of her characteristics actually predict trouble. She is a lifelong non-smoker with treated stage III periodontitis, currently stable, 2 mm of keratinised mucosa at each planned site and a heavily worn dentition. Which of the following are established risk factors or indicators for peri-implantitis?

A. A history of treated periodontitis
B. Absence of a band of keratinised peri-implant mucosa
C. Occlusal overload of the implant restoration
D. A and B only
E. A, B and C

769 A trainee has been taught that smoking and diabetes are established risk factors for peri-implantitis in the same way that a periodontitis history is, and asks whether she should tell patients so. The patient in front of you smokes ten a day and has an HbA1c of 55 mmol/mol. Which statement best represents the position taken by the 2017 consensus?

A. Both are established risk factors, supported by consistent evidence
B. The evidence for both remains inconsistent, although both are plausible
C. Smoking is established as a risk factor and diabetes is not
D. Diabetes is established as a risk factor and smoking is not
E. Neither has been examined in enough studies to permit a judgement

770 A 61-year-old man has bleeding, suppuration and a 6 mm pocket at a cement-retained crown fitted four years ago, with the margin about 2 mm submucosally. You suspect residual cement. He asks how often cement is really the cause, having read that it is a fashionable explanation. Which statement is best supported by the published evidence?

A. Excess cement causes disease only where it has been present for over a year
B. Excess cement is found at about one fifth of implants with signs of disease
C. Excess cement is detectable on a periapical radiograph in most cases
D. Removal of the excess cement rarely resolves the peri-implant inflammation
E. Excess cement was present at most diseased implants examined endoscopically

771 A 58-year-old woman has bleeding, suppuration and a 6 mm pocket at the UR5, restored four years ago with a crown cemented on a custom abutment whose margin lies 2 mm below the mucosa palatally. A paralleling periapical shows 2 mm of bone loss and no radiopaque material at the margin. Her plaque control is good and the crown contour is cleansable. What is the most appropriate next step?

A. Debride the pocket with titanium curettes and reassess at one month
B. Place a local antimicrobial in the pocket and review at six weeks
C. Take a cone beam examination to define the defect and look for material
D. Remove the crown and abutment to inspect and clean the sulcus directly
E. Raise a flap for surgical access and decontamination of the surface

772 An implant at the LL6 has a 7 mm pocket with bleeding and suppuration and 3 mm of bone loss. The crown has a convex submucosal contour with an emergence angle of about 45 degrees that the patient cannot clean, and her interdental brush will not pass. You are planning cause-related therapy before deciding about surgery. Which of the following add clinically meaningful benefit to submucosal debridement in this situation?

A. A course of adjunctive systemic antibiotics
B. Correction of the prosthesis so that the site can be cleaned
C. A locally delivered antimicrobial placed in the pocket
D. A and B only
E. All of the above

773 A 63-year-old man with peri-implantitis at the LR6 has completed non-surgical therapy and prosthetic correction, and a 7 mm pocket with bleeding persists. You are consenting him for surgical access and he asks what the operation is likely to achieve. Which account best represents the outcomes reported in longer-term series?

A. Disease resolution in over 90 per cent, with recurrence uncommon thereafter
B. Disease resolution in about three quarters, with recurrence rarely seen
C. Disease resolution in about half, with recurrence common in the remainder
D. Disease resolution in under a fifth, so surgery is seldom worth undertaking
E. Resolution rates cannot be quoted because the definitions used differ

774 A flap has been raised at the LL6 for peri-implantitis and implantoplasty is planned on the exposed threads. The site is not visible when the patient smiles, the implant is 4.8 mm in diameter and stable, and the defect is a buccal dehiscence with no containing walls. A trainee asks what implantoplasty is intended to achieve. Which statement best describes it?

A. It renders the exposed surface cleansable, at the cost of exposing the implant body
B. It restores the bone level by smoothing the threads and allowing reattachment
C. It removes the need for chemical or mechanical decontamination of the surface
D. It is contraindicated in implants of less than 4 mm diameter because of fracture risk
E. It should be combined with a particulate graft to fill the residual defect

775 A flap is raised at the LL6 after non-surgical therapy has failed. The defect proves to be a buccal dehiscence extending 5 mm apically from the platform, with the buccal plate absent and no bony walls remaining on that aspect; the mesial, distal and lingual bone is intact and at the level of the platform. The implant is stable, the site is not visible on smiling and the prosthesis is screw-retained and cleansable. Which approach is best supported?

A. Reconstructive therapy with a bone substitute and a resorbable membrane
B. Reconstructive therapy using autogenous bone chips from the adjacent ridge
C. Access surgery with decontamination and primary closure over the defect
D. Resective surgery with implantoplasty and an apically positioned flap
E. Explantation of the implant and grafting of the site for later replacement

776 A contained four-walled intrabony defect at the UL5 has been decontaminated and grafted with a bone substitute, and the patient asks whether the bone will now "stick back on" to the implant as it did originally. She is a non-smoker with excellent plaque control and has been enrolled in three-monthly supportive care. Which answer is most accurate?

A. Re-osseointegration occurs predictably once the surface has been decontaminated
B. Radiographic defect fill improves, but re-osseointegration is not predictable
C. Re-osseointegration occurs only where a barrier membrane has been used
D. Radiographic fill is unchanged by grafting a defect of this configuration
E. Re-osseointegration depends chiefly on which graft material has been selected

777 A 59-year-old woman has a contained circumferential intrabony defect at the LR5 that has been decontaminated and grafted. Her crown is cement-retained with a submucosal margin and a convex emergence that her interdental brush will not pass, and she has attended once in four years. You are planning the whole episode of care rather than the operation alone. Which of the following are necessary parts of the treatment?

A. Thorough decontamination of the exposed implant surface at surgery
B. Removal or modification of the prosthesis so that the site can be cleaned
C. A programme of supportive care at short intervals after healing
D. A and B only
E. All of the above

778 A 62-year-old man has an implant at the UR2 placed elsewhere eight years ago. It sits about 3 mm labial to the adjacent root positions with two threads exposed through a thin mucosa, there is 6 mm of radiographic bone loss with suppuration, and he displays his gingival margins fully when he smiles. There is 1 mm of keratinised mucosa. Which finding most strongly indicates that the implant should be removed rather than treated?

A. The 6 mm of radiographic bone loss around the implant
B. The suppuration from the peri-implant sulcus on probing
C. The labial malposition of the implant within the ridge
D. The high lip line displaying the gingival margins on smiling
E. The single millimetre of keratinised mucosa at the site

779 A 66-year-old woman is to have an implant removed from the LL6. It has 5 mm of circumferential bone loss with suppuration, but it is not mobile and the apical portion appears to remain in contact with bone on the radiograph. You intend to use a counter-torque device rather than a trephine in order to preserve the ridge for a replacement. Which factor most determines whether that approach will succeed?

A. The extent of osseointegration remaining at the apical portion of the implant
B. The diameter of the implant measured at the level of the platform
C. The type of internal connection engaged by the counter-torque instrument
D. The thickness of the buccal plate remaining over the coronal threads
E. The length of time that has elapsed since the implant was first placed

780 A 57-year-old man with four implants and a treated periodontitis history attends for supportive care at four-monthly intervals. He is well motivated, his plaque score is 14 per cent, and his last radiographs were taken eighteen months ago and were stable. Which of the following belong in the supportive care appointment?

A. Probing of all surfaces of each implant with recording of bleeding
B. Assessment of the prosthesis for stability and for access for cleaning
C. Reinforcement of his individual cleaning technique at each site
D. A and B only
E. All of the above

781 During supportive care you are removing biofilm from two exposed threads and a machined titanium abutment at the LL6, a year after surgical treatment of peri-implantitis. The site bleeds at one point with a 5 mm pocket and a stable bone level. Your nurse offers four air-polishing powders. Which choice is appropriate, and for what reason?

A. Sodium bicarbonate, because its particles dissolve rapidly in the sulcus
B. Erythritol, because its low abrasivity spares the titanium surface
C. Aluminium oxide, because it removes adherent calculus most efficiently
D. Calcium carbonate, because it buffers the acidity of the mature biofilm
E. Silica, because it restores the polish of the titanium oxide layer

782 A 49-year-old woman with peri-implant mucositis at two implants was debrided with implant-safe instruments and given individual oral hygiene instruction six weeks ago. At review her plaque score has fallen from 41 to 18 per cent, but two of the twelve peri-implant sites still bleed on gentle probing. Probing depths are 4 mm and unchanged, and there is no suppuration. What is the most appropriate course?

A. Accept this as the expected residual inflammatory response around implants
B. Prescribe a chlorhexidine gel for local application at the two bleeding sites
C. Take a periapical radiograph to exclude early bone loss at those two sites
D. Re-instrument and re-instruct, and continue reviewing until bleeding resolves
E. Refer for surgical access at the two sites that continue to bleed

783 A 52-year-old woman had an implant placed at the LR5 yesterday in an appointment lasting ninety minutes, with a flap extended forward past the mental foramen and held with a retractor. The pre-operative cone beam examination and the post-operative periapical both show 4 mm of bone between the implant apex and the roof of the mandibular canal. She telephones today with numbness of the lower lip and chin on that side and no pain. What is the most likely explanation?

A. Direct transection of the inferior alveolar nerve by the twist drill
B. Compression of the inferior alveolar nerve by the implant within the canal
C. Traction on the mental nerve from prolonged retraction of the flap
D. Intraneural deposition of local anaesthetic solution at the mental foramen
E. Haematoma within the mandibular canal compressing the neurovascular bundle

784 The same patient is seen the following morning. The affected area is mapped, photographed and recorded, sensory testing is documented, and you are satisfied that the injury arose from retraction rather than from the implant. She is distressed and asks whether something went wrong. What is your obligation?

A. Tell her at once what has happened, apologise, explain the plan and record it
B. Await the outcome at six weeks before discussing the cause of the injury
C. Notify your indemnity organisation and take advice before speaking to her
D. Record the event and report it to the regulator within twenty-eight days
E. Explain that an apology cannot be offered until the question of liability is settled

785 Four weeks after that appointment the numbness is unchanged, and she reports that the area feels unpleasant when touched. Repeat examination confirms an area of altered sensation over the lower lip and chin, and the implant is known from two radiographs to lie 4 mm from the canal. Which of the following should form part of your management now?

A. Mapping and photographing the affected area and recording the sensory findings
B. Referral for specialist assessment without further delay
C. A repeat cone beam examination to re-measure the implant's relation to the canal
D. A and B only
E. A, B and C

786 During placement of an implant at the UR6 with 4 mm of residual bone below the sinus floor, and after a transcrestal elevation, the implant is lost superiorly and a periapical shows it lying within the maxillary sinus. The patient is comfortable, there is no bleeding of consequence and the crestal access is about 4 mm across. What is the most appropriate course?

A. Leave the implant in place, since titanium is inert within the sinus cavity
B. Refer promptly for retrieval, providing antibiotic cover in the interim
C. Retrieve it through the crestal osteotomy with a fine suction tip
D. Close the site and review it with a radiograph in three months' time
E. Enlarge the crestal osteotomy laterally and retrieve the implant directly

787 Six days after a transcrestal sinus floor elevation with a particulate graft and simultaneous implant placement at the UL6, a 55-year-old man returns with unilateral purulent nasal discharge, pain over the left cheek, nasal obstruction and a temperature of 37.9 degrees Celsius. The implant is stable and the crestal wound is intact with no discharge. What is the most appropriate management?

A. Reassure him, as these features are expected after sinus floor elevation
B. Prescribe a nasal decongestant alone and review him in one week
C. Remove the implant immediately and curette the graft from the sinus
D. Take a cone beam examination before deciding on any treatment
E. Start antibiotics and a decongestant and arrange ENT assessment

788 You are consenting a patient for a single implant crown and want to describe the technical problems that may arise over the years, having already covered implant survival. Consider these statements: (i) screw loosening is the commonest technical complication of single implant crowns; (ii) chipping of veneering ceramic is a common technical complication of implant-supported fixed prostheses; (iii) fracture of the abutment or screw occurs in about one case in ten by five years. Which are correct?

A. (i) only
B. (i) and (ii) only
C. (ii) and (iii) only
D. (i) and (iii) only
E. (i), (ii) and (iii)

789 A 70-year-old woman is being consented for a two-implant mandibular overdenture with stud attachments, and asks what the prosthesis will need over the coming years. Her existing complete denture has needed nothing since it was made. Which account best represents the published cohort data?

A. Most patients require attachment maintenance within the first five years
B. Attachment maintenance is required in roughly one patient in ten
C. Maintenance is largely confined to relining of the denture base
D. Maintenance events become rare once the attachments have bedded in
E. Maintenance requirements cannot be predicted from the published cohorts

790 A 55-year-old man has had a second abutment screw fracture at the LR6, eighteen months after the first, and the fragment has been retrieved intact on both occasions. The implant is osseointegrated, the mucosa is healthy, and the crown has a 6 mm distal cantilever and a working-side excursive contact. He asks you to fit a new screw and a new crown so that he can get on with his life. What is the most important step?

A. Replace the screw with a new one and torque it to the specified value
B. Convert the restoration to cement retention on a new custom abutment
C. Remake the crown in a monolithic material to reduce its brittleness
D. Identify and correct the mechanical cause before restoring the implant again
E. Remove the implant and replace it with one of wider platform diameter

791 A manufacturer issues a field safety notice concerning a lot of abutments distributed over a two-year period, asking that affected patients be identified and reviewed. Your records include the reference and lot number of every component fitted. Which of the following should you now do?

A. Search the records for patients treated with components from that lot
B. Contact those patients and arrange review appointments for them
C. Record the notice and the action taken in the practice governance file
D. A and B only
E. All of the above

792 A 47-year-old woman attends with a mobile crown at the LR6 five years after placement. The implant body has fractured circumferentially just below the platform; the apical portion remains integrated. The components were within their expiry date, the torque values used were as specified, and the crown had no cantilever and a light, well-centred occlusal contact. Besides treating her, what action is required?

A. Notify the General Dental Council that a device has failed in service
B. Report the incident to the MHRA and to the manufacturer, retaining the parts
C. Return the fractured components to the laboratory for their analysis
D. Inform the patient's general medical practitioner of the device failure
E. Record the failure in the practice's annual audit of implant survival

793 A 60-year-old man has a 7 mm pocket with bleeding and suppuration at the LR6 and 3 mm of bone loss on a paralleling periapical. You are considering a small-field cone beam examination to define the buccal extent of the defect before deciding between resective and reconstructive surgery. Which consideration should most influence that decision?

A. Cone beam imaging should be obtained routinely before peri-implant surgery
B. Cone beam imaging quantifies buccal bone loss more accurately than a periapical
C. Metal artefact limits assessment adjacent to the implant, so a periapical comes first
D. Cone beam imaging is justified because it delivers less dose than two periapicals
E. Cone beam imaging should be reserved for implants placed in the posterior maxilla

794 A screw-retained crown is being fitted at the LL6 in a healthy 44-year-old. Seating has been confirmed clinically, the occlusion is adjusted, the access channel is sealed and the patient is happy. A trainee asks whether any radiograph is needed now, given that a film was taken at abutment try-in a fortnight ago and was satisfactory. What is the correct practice?

A. A paralleling periapical at prosthesis fitting, retained as the baseline
B. A panoramic radiograph at fitting to provide a whole-mouth record
C. A periapical only where complete seating of the abutment is in doubt
D. A small-field cone beam examination to record the buccal bone level
E. No further film, since the surgical radiograph records the bone level

795 You have diagnosed peri-implantitis at the UL5 in a patient who has attended irregularly. You have charted the site, compared today's radiograph with the baseline, examined the prosthesis and discussed the options and their likely adverse effects, including recession and exposed threads. You now write the record. Which of the following must it contain?

A. A six-point probing chart recording bleeding and suppuration at each site
B. The comparison of today's radiograph against the baseline film held
C. The insertion torque recorded at the time the implant was placed
D. A and B only
E. A, B and C

796 A patient reviewing her records under a subject access request asks you to remove an entry from three years ago recording that she smoked fifteen cigarettes a day, saying that she had stopped by then and that the note is unfair to her. The entry was made contemporaneously and reflects what she reported at the time. What is the correct response?

A. Delete the entry, since the data are hers and she has objected to them
B. Explain that an addendum recording her disagreement can be added
C. Amend the original entry and initial the change with today's date
D. Decline and advise her to complain to the Information Commissioner
E. Delete the entry from the record but retain a copy in a separate file

797 Three weeks after an implant crown was fitted at the UR2 you receive a written complaint about its appearance, addressed to the practice. You have seen the patient once since fitting and consider the shade and contour a reasonable match, and the tissues are healthy. You have already telephoned your indemnity organisation. What is the correct handling of the complaint itself?

A. Respond only once your indemnity organisation has approved a draft reply
B. Respond in writing within twenty-eight days as the GDC requires
C. Offer a refund at the outset in order to prevent the matter escalating
D. Decline to correspond while there remains a possibility of a claim
E. Acknowledge it promptly and follow the practice's complaints procedure

798 A trainee associate asks you what the duty of candour actually requires of her personally, having heard that it applies only to hospitals and only where a patient has been seriously harmed. She has just placed a slightly malpositioned implant which she believes she can restore acceptably. Which statement best describes her position?

A. The duty arises only where the harm caused is moderate or severe
B. The duty requires an admission of legal liability once harm has occurred
C. The duty applies to every registrant, whatever the degree of harm
D. The duty is discharged by making a full entry in the clinical record
E. The duty applies only to providers registered with the Care Quality Commission

799 Following the nerve injury described earlier in this practice, the team meets to consider what should change. The operator is experienced, the cone beam examination had been reported, and the injury arose from prolonged retraction during a long appointment. The practice has no written protocol for flap design or appointment length in the interforaminal and premolar regions. Which of the following belong in the practice's response?

A. A significant event analysis leading to a change in the operating protocol
B. A disciplinary review of the operator's competence to place implants
C. Notification of the incident to the General Dental Council by the practice
D. A and B only
E. All of the above

800 A 63-year-old woman with four implants restored eight years ago is emigrating and asks what she should take with her so that her new dentist can look after them. She has been maintained four-monthly, has had one episode of treated mucositis, and her prosthesis is a screw-retained fixed bridge on two of the implants with single crowns on the others. What is the most appropriate provision?

A. A copy of the radiograph taken at fitting and the laboratory docket
B. A letter confirming that her implants are healthy and well maintained
C. Her implant record card alone, since it identifies the components used
D. A summary of components, occlusal scheme and history, with her radiographs
E. A referral to a colleague in her new country for reassessment on arrival